

Mars 2020 (M2020)

Software Interface Specification

Interface Title: **Camera Instrument Experiment Data Record (EDR) and Reduced Data Record (RDR) Data Products**

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Mars 2020 Project

Software Interface Specification (SIS)

Camera Experiment Data Record (EDR) and Reduced Data Record (RDR) Data Products

Version 0.5

DRAFT

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CHANGE LOG

DATE	SECTIONS CHANGED	REASON FOR CHANGE	REVISION

OPEN ISSUE ITEMS

REVISION	OPEN ISSUE	CLOSED

TABLE OF CONTENTS

Change Log	i
Open Issue Items for Closure	ii
List of Figures	vii
List of Tables	viii
Acronyms and Abbreviations	ix
1. INTRODUCTION	
1.1 Purpose and Scope	
1.2 Contents	
1.3 Constraints and Applicable Documents	
1.3.1 Relationships with Other Interfaces	
2. INSTRUMENT OVERVIEW	
2.1 Engineering Camera Instrument Suite	
2.1.1 Hazard Avoidance Camera (Hazcam)	
2.1.2 Navigation Camera (Navcam)	
2.1.3 Cache Camera (CacheCam)	
2.2 SuperCam Remote Micro Imager (RMI)	
2.3 Mastcam-Z Camera Suite	
2.3.1 Mastcam-Z Left (MCZ-L)	
2.3.2 Mastcam-Z Right (MCZ-R)	
2.4 SHERLOC Camera Suite	
2.4.1 WATSON	
2.4.2 Autofocus and Context Imager (ACI)	
2.5 PIXL Micro Context Camera (MCC)	
2.6 MEDA Sky Camera (SkyCam)	
2.7 EDL Camera Suite	
2.7.1 Parachute Uplook Camera (PUC)	
2.7.2 Descent Stage Downlook Camera (DDC)	
2.7.3 Rover Uplook Camera (RUC)	
2.7.4 Rover Downlook Camera (RDC)	
3. GENERAL DATA PRODUCT OVERVIEW	
3.1 Data Processing Levels	
3.2 Product Label and Header Descriptions	
3.2.1 Overview of Labels	
3.2.2 PDS and ODL Labels	
3.2.2.1 PDS Local Data Dictionary	
3.2.2.2 Keyword Length Limits	
3.2.2.3 Data Type Restrictions	
3.2.2.4 Interpretation of N/A, UNK, and NULL	
3.2.2.5 PDS/ODL Label Constructs “Class”, “Object” and “Group”	
3.2.2.6 PDS/ODL Image Object	
3.2.2.7 PDS/ODL Table Objects and “.FMT” Files	
3.2.3 VICAR Label	
3.2.4 Mapping of PDS/ODL and VICAR Labels	
3.3 Binary Data Storage Conventions	
3.3.1 Bit and Byte Ordering	

4. EDR PRODUCT SPECIFICATION	
4.1 EDR General Processing	
4.2 EDR Product Types	
4.2.1 Image EDRs.....	
4.2.1.1 Standard Image EDR	
4.2.1.2 Image Sampling Types	
4.2.1.2.1 “Full Frame” EDR.....	
4.2.1.2.2 “Subframe” EDR	
4.2.1.2.3 “Downsampled” EDR	
4.2.1.2.4 “Thumbnail” EDR	
4.2.2 Image Support EDRs	
4.2.2.1 “Row Summation” EDR.....	
4.2.2.2 “Column Summation” EDR	
4.2.2.3 “Reference Pixel” EDR.....	
4.2.2.4 “Histogram” EDR	
4.2.4 Engineering EDRs.....	
4.2.4.1 “NavMap” EDR	
4.3 EDR Product Format	
4.3.1 Engineering Camera Suite	
4.3.2 SuperCam RMI.....	
4.3.3 Mastcam-Z Camera Suite.....	
4.3.4 SHERLOC Camera Suite	
4.3.5 PIXL Micro Context Camera.....	
4.3.6 MEDA Sky Camera	
4.3.7 EDL Camera Suite	
4.4 EDR Product Structure.....	
4.4.1 Engineering Camera Suite	
4.4.2 SuperCam RMI.....	
4.4.3 Mastcam-Z Camera Suite.....	
4.4.4 SHERLOC Camera Suite	
4.4.5 PIXL Camera Suite.....	
4.4.6 MEDA Sky Camera	
4.4.7 EDL Camera Suite	
4.5 EDR Product Validation	
5. *** RDR PRODUCT SPECIFICATION ***THIS SECTION TBD AND HERE FOR FUTURE REF***	
5.1 RDR General Processing	
5.1.1 Image RDRs.....	
5.1.1.1 Common Processing	
5.1.1.1.1 Geometrically Corrected Images (Linearization).....	
5.1.1.1.2 Image Overlays.....	
5.1.1.1.3 Filled Images.....	
5.1.1.1.4 Color Processed Images.....	
5.1.1.1.5 Rover Volume Exclusion Image Masks	
5.1.1.1.6 Masked Images.....	
5.1.1.1.7 EDR-like RDRs	
5.2 RDR Product Types.....	
5.2.1 Image RDRs.....	
5.2.1.1 “Inverse LUT” RDR.....	
5.2.1.2 “Radiometrically Corrected” RDR.....	
5.2.1.2.1 CHEMRAD Method	

5.2.1.2.2	MIPLRAD Method	
5.2.1.3	“Disparity” RDR	
5.2.1.3.1	Stereo Pair Matching Method	
5.2.1.4	“XYZ” RDR	
5.2.1.4.1	Rover Mask RDR	
5.2.1.5	“Range” RDR.....	
5.2.1.6	“Surface Normal” (UVW) RDR	
5.2.1.7	“Surface Roughness” RDR	
5.2.1.8	Slope RDRs.....	
5.2.1.8.1	“Slope” (nominal) RDR.....	
5.2.1.8.2	“Slope Rover Direction” RDR	
5.2.1.8.3	“Slope Heading” RDR	
5.2.1.8.4	“Slope Magnitude” RDR	
5.2.1.8.5	“Slope Northerly Tilt” RDR.....	
5.2.1.8.6	“Solar Energy” RDR	
5.2.1.9	Arm Reachability RDRs	
5.2.1.9.1	“Arm Reachability” RDR	
5.2.1.9.2	“Arm Preload” RDR.....	
5.2.1.10	Color RDRs	
5.2.1.11	“Photometry” RDR	
5.2.1.12	Terrain Map RDRs.....	
5.2.1.12.1	“Per-XYZ” Terrain RDRs	
5.2.1.12.2	“Unified” Terrain RDRs.....	
5.2.1.13	Mosaic RDRs.....	
5.2.1.13.1	Overview of Mosaics in General.....	
5.2.1.13.2	Mosaicking Method	
5.2.1.13.3	Mosaic Ancillary Files.....	
5.2.1.13.4	“Cylindrical Projection Mosaic” RDR.....	
5.2.1.13.5	“Camera Point Perspective Mosaic” RDR	
5.2.1.13.6	“Cylindrical-Perspective Projection Mosaic” RDR.....	
5.2.1.13.7	“Polar Projection Mosaic” RDR.....	
5.2.1.13.8	“Vertical Projection Mosaic” RDR.....	
5.2.1.13.9	“Orthographic Projection Mosaic” RDR	
5.2.1.13.10	“Orthorectified Projection Mosaic” RDR.....	
5.2.1.13.11	Non-image Mosaics	
5.2.1.14	“Anaglyph” RDR	
5.3	RDR Product Format	
5.3.1	Image RDRs.....	
5.4	RDR Product Structure	
5.5	RDR Product Validation	
6.	STANDARDS USED IN GENERATING PRODUCTS.....	
6.1	File Naming Standards.....	
6.1.1	EDR and Single-frame RDR Filename	
6.1.2	**** TBD **** Mosaic RDR Filename.....	
6.1.3	**** TBD ****Unified Terrain Mesh RDR Filename	
	PDS Standards	
6.2	Time Standards.....	
6.3	Coordinate Frame Standards.....	
6.3.1	Rover Navigation (Rover Nav) Frame.....	
6.3.2	Rover Mechanical (Rover Mech) Frame	

6.3.3 Local Level Frame.....	
6.3.4 Site Frame.....	
6.3.5 RSM Frame.....	
6.3.6 Arm Frames	
7. APPLICABLE SOFTWARE.....	
7.1 Utility Programs.....	
7.2 Applicable PDS Software Tools.....	
7.3 Software Distribution and Update Procedures.....	
APPENDICES	
A APID NAMES	
B Label Keyword Definitions, Valid Values, Sources	

LIST OF FIGURES

Figure 1.1- M2020 XXXXXXXYY	
Figure 1.2- M2020 XXXXXXXYY	
Figure 1.3- M2020 XXXXXXXYY	
Figure 1.4- M2020 XXXXXXXYY	
Figure 1.5- M2020 XXXXXXXYY	
Figure 1.6- M2020 XXXXXXXYY	
Figure 1.7- M2020 XXXXXXXYY	
Figure 1.8- M2020 XXXXXXXYY	

LIST OF TABLES

Table 1.1 - M2020 XXXXXXXYY	
Table 1.2 - M2020 XXXXXXXYY	
Table 1.3 - M2020 XXXXXXXYY	
Table 1.4 - M2020 XXXXXXXYY	
Table 1.5 - M2020 XXXXXXXYY	
Table 1.6 - M2020 XXXXXXXYY	
Table 1.7 - M2020 XXXXXXXYY	
Table 1.8 - M2020 XXXXXXXYY	

ACRONYMS AND ABBREVIATIONS

ACA	Adaptive Caching Assembly
ACI	Autofocus and Context Imager
APID	Application Process Identifier
APSS	Activity Planning and Sequencing Subsystem
ASCII	American Standard Code for Information Interchange
ASTTRO	Advanced Science Targeting Tool for Robotic Operations
ATLO	Assembly, Test, Launch and Operations
BITCAR	Bit-Carousel
BU	Body-Unit (SuperCam)
CAHV	Center, Axis, Horizontal, Vertical (camera model)
CAHVOR	Center, Axis, Horizontal, Vertical, Optical, Radial (camera model)
CAHVORE	Center, Axis, Horizontal, Vertical, Optical, Radial, Entrance (camera model)
CBLO	Core Bit Lock Out
CCD	Charged Coupled Device
CCBU	Chemistry Camera Body Unit
CCMU	Chemistry Camera Mast Unit
CEDL	Cruise Entry Descent and Landing
CMOS	Complementary Metal Oxide Semiconductor
SHA	Collection and Handling for Interior Martian Rock Analysis
CNES	Centre National d'Etudes spatiales (French Space Agency)
CODMAC	Committee on Data Management and Computation
CSV	Comma-separated-value
DDC	Descent Stage Downlook Camera
DEA	Digital Electronics Assembly
DEM	Digital Elevation Map
DN	Digital Number
DOY	Day of Year
DP	Data Product (telemetry)
DPO	Data Product Object
DTE	Direct to Earth
DVT	Data Validity Time
ECAM	Engineering Camera
EECAM	Enhanced Engineering Camera
EDL	Entry, Descent and Landing
EDR	Experiment Data Record
EHA	Engineering, Housekeeping & Accountability (EH&A)
EM	Engineering Model

EMD	Earth Metadata file (".emd")
EPDU	End-of-Product PDU
ERT	Earth Received Time
FDD	Functional Design Document
FDR	Fundamental Data Record
FEI	File Exchange Interface
FGICD	Flight-Ground ICD
FHAZ	Front Hazard Avoidance Camera
FM	Flight Model
FOV	Field of View
FPGA	Field Programmable Gate Array
FSW	Flight Software
GSFC	Goddard Space Flight Center
GDS	Ground Data System
GSE	Ground Support Equipment
Hazcam	Hazard Avoidance Camera
HGA	High Gain Antenna
IC	Inlets Cover
ICD	Interface Control Document
ICER	Image compression algorithm (not an acronym)
ID	Identification
IDPH	Instrument Data Product Header
IDS	Instrument Data Systems
IFOV	Instantaneous Field of View
ILUT	Inverse Lookup Table
IPE	Integrated Planning and Execution (MS element)
IRAP	Institut de Recherche en Astrophysique et Planétologie
ISIS	Integrated Software for Imagers and Spectrometers
IVP	Inertial Vector Propagation
JPEG	Joint Photographic Experts Group (compression)
JPL	Jet Propulsion Laboratory
LANL	Los Alamos National Laboratory
LIBS	Laser-Induced Breakdown Spectrometer (SuperCam)
LOCO	LOW-COMplexity, LOSSless COMpression
Mastcam-Z	Mastcamera-Zoom
MCZ	Mastcam-Z
MER	Mars Exploration Rover
MIC	Microphone
MIPL	Multimission Instrument Processing Laboratory

MOS	Mission Operations System
MPCS	Mission data Processing and Control Subsystem
MPDU	Metadata Protocol Data Unit
MPF	Mars Pathfinder
MS	Mission System
MU	Mast Unit (SuperCam)
M2020	Mars 2020 Rover
M2020ICE	M2020 Operations InterfaCE
MSSS	Malin Space Science Systems
NASA	National Aeronautics and Space Administration
Navcam	Navigation Camera
OCS	Operations Cloud Store
ODL	Object Description Language
OPGS	Operations Products Generation Subsystem
PDC	Parachute Downlook Camera
PDS	Planetary Data System
PDU	Protocol Data Unit
PPDU	Product Data Protocol Data Unit
PRT	Platinum Resistance Thermometer
PSDD	Planetary Science Data Dictionary
PUC	Parachute Uplook Camera
RA	Robotic Arm
RCE	Rover Compute Element
RDC	Rover Downlook Camera
RDR	Reduced Data Record
RHAZ	Rear Hazard Avoidance Camera
RMI	Remote Micro-Imager (SuperCam)
ROI	Region of Interest
RSM	Remote Sensing Mast
RTO	Real Time Operations (MS element)
RUC	Rover Uplook Camera
RSVP	Rover Sequencing and Visualization Program
SAPP	Surface Attitude, Positioning and Pointing
SCID	Spacecraft ID
SCLK	Spacecraft Clock
SCM	Spacecraft Configuration Manager
SCS	Sample Cache System
SFDU	Standard Format Data Unit
SHA	Sample Handling Arm

SHERLOC	Scanning Habitable Environments with Raman and Luminescence for Organics and Chemicals
SIS	Software Interface Specification
SOH	State of Health (SuperCam)
SOL	Mars Solar Day
SOWG	Science Operations Working Group
SPaH	Sample Processing and Handling
SPICE	Spacecraft, Planet, Instrument, C-matrix, Events kernels
SRAM	Static Random Access Memory
SuperCam	Super Camera Instrument
SwRI	Southwest Research Institute
TBD	To Be Determined
TBU	Testbed Unit
TDS	Telemetry Delivery Subsystem
UDR	Unprocessed Data Record
USGS	United States Geological Survey
VCID	Virtual Channel Identifier
VICAR	Video Image Communication and Retrieval
WATSON	Wide Angle Topographic Sensor for Operations and eNginneering
WEB	Warm Electronic Box

1. INTRODUCTION

1.1 Purpose and Scope

The purpose of this Data Product Software Interface Specification (SIS) is to provide consumers of M2020 instrument Experiment Data Record (EDR) and Reduced Data Record (RDR) data products with a detailed description of the products and how they are generated, including data sources and destinations. Content in this document supports EDR and RDR data products generated by the Instrument Data Subsystem (IDS) for the following instruments:

- Engineering Camera (ECAM) suite:
 - a. Navigation Cameras (Navcams)
 - b. Hazard Avoidance Cameras (Hazcams)
 - c. Sample Caching System Camera (CacheCam)
- SuperCam Remote Micro-Imager (RMI)
- Mastcam-Z (MCZ) instrument suite:
 - a. Mastcam-Z Left (MCZ-L)
 - b. Mastcam-Z Right (MCZ-R)
- SHERLOC camera suite
 - a. SHERLOC-WATSON
 - b. SHERLOC-ACI
- PIXL Micro Context Camera (MCC)
- MEDA Skycam
- EDL camera suite
 - a. Parachute Uplook Cameras (PUC)
 - b. Descent Stage Downlook Camera (DDC)
 - c. Rover Uplook Camera (RUC)
 - d. Rover Downlook Camera (RDC)
- Landing Visual System (LVS)
- Helicopter camera suite
 - a. Scout camera
 - b. Navigation camera

The users for whom this SIS is intended include IDS, the Activity Planning and Sequencing Subsystem (APSS), users and developers of Science Operations Analysis Software (SOAS), member scientists of the project's Science Operations Working Group (SOWG) who will analyze the data, and other scientists in the general planetary science community.

In this document, the EDR data product is the raw, uncalibrated, uncorrected image data acquired by the M2020 instrument. It may include decompression if there was data product compression performed onboard the rover by the instrument. Within the group of camera instruments, the full frame image EDR data products are identical in format, except for some product label differences.

The RDR data products described in this document are limited to camera instruments only, and are derived directly from one or more image EDR or image RDR data product(s). They are comprised of radiometrically decalibrated and/or camera model corrected and/or geometrically altered (including reprojected) versions of the raw camera data, as well as higher-order processed results (i.e. XYZ points and surface normal).

1.2 Contents

This Data Product SIS describes how the EDR data product is acquired by the M2020 camera and how it is processed, formatted, labeled, and uniquely identified, and how the image RDR data product is derived from image EDR or image RDR data products. The document discusses standards used in generating the product and software that may be used to access the product. The EDR and RDR data product structure and organization is described in sufficient detail to enable a user to read the product. Finally, examples of composite EDR/RDR labels are provided, along with the definitions of the keywords in the label.

1.3 Constraints and Applicable Documents

This SIS is meant to be consistent with the contract negotiated between the M2020 Project and the M2020 Principal Investigators (PI) for the Engineering cameras, the SuperCam RMI, the Mastcam-Z cameras, the Sherloc-Imaging (WATSON) and ACI cameras, the PIXL Micro Context Camera, and the MEDA Sky Camera, in which reduced data records and documentation are explicitly defined as deliverable products. Because this SIS governs the specification of data products used during mission operations, any proposed changes to this SIS must be impacted by all affected software subsystems observing this SIS in support of operations (e.g., APSS, IDS, SOAS).

Product label keywords may be added to future revisions of this SIS. Therefore, it is recommended that software designed to process EDRs and RDRs specified by this SIS should be robust to (new) unrecognized keywords.

This Data Product SIS is responsive to the following M2020 documents:

1. Mars 2020 CEDL Engineering Camera Imaging Functional Design Description (FDD), "Baseline Release, Rev A", Daniel Petrizzo, JPL D-95849, <Date>
2. Mars 2020 Surface Engineering Camera Imaging Functional Design Description (FDD), "Baseline Release, Rev A", Daniel Petrizzo, JPL D-96164, <Date>
3. Mars 2020 Mares Helicopter System Functional Design Description (FDD), "Baseline Release, Rev A", Daniel Petrizzo, JPL D-XXXXX, <Date>
4. Mars 2020 CEDL and Surface SuperCam Functional Design Description (FDD), "Baseline Release, Rev A", Ivair Gontijo, JPL D-95868, <Date>
5. Mars 2020 CEDL and Surface Mastcam-Z Functional Design Description (FDD), "Baseline Release, Rev A", Zach Bailey, JPL D-95850, <Date>
6. Mars 2020 CEDL Surface SHERLOC Functional Design Description (FDD), "Baseline Release, Rev A", Zach Bailey, JPL D-95867, <Date>
7. Mars 2020 CEDL and Surface MEDA Functional Design Description (FDD), "Baseline Release, Rev A", Christina Diaz, JPL D-95851, <Date>
8. Mars 2020 CEDL and Surface PIXL Functional Design Description (FDD), "Baseline Release, Rev A", Joan Ervin, JPL D-95857, <Date>
9. Mars 2020 CEDL Camera Functional Design Description (FDD), "Baseline Release, Rev A", Danielle Nuding, JPL D-95844, <Date>

10. Mars 2020 Flight-Ground Interface Control Document (FGICD), "Volume 1, Downlink, Rev A, Version 1.0", Biren Shah, JPL D-95521, October 3, 2017

Additionally, this SIS is also consistent with the following Planetary Data System documents:

11. Planetary Science Data Dictionary Document, Version 1.81, November 24, 2010.
12. Planetary Science Data M2020 Local Data Dictionary, Version 1.0, January 15, 2013.
13. Planetary Data System Archive Preparation Guide, Version 1.4, JPL D-31224, April 1, 2010.

Finally, this SIS makes reference to the following documents for technical background information:

14. A System for Extracting Three-Dimensional Measurements from a Stereo Pair of TV Cameras, Y. Yakimovsky and R. Cunningham, January 7, 1977.
15. Camera Calibration, D. Gennery, JPL IOM 347/86/10, February 5, 1986.
16. Sensing and Perception Research for Space Telerobotics at JPL, D. Gennery et al., *Proceedings of the IEEE Intern. Conf. on Robotics and Automation*, March 31 - April 3, 1987.
17. Camera Calibration Including Lens Distortion, D. Gennery, JPL D-8580, May 31, 1991.
18. Algorithm for Using CAHV to Determine SGI Graphics Viewpoint and Perspective, B. Bon, JPL IOM 3472-91-057, August 6, 1991.
19. Inclusion of Old Internal Camera Model in New Calibration, D. Gennery, JPL IOM 386.3-94-001, February 22, 1994.
20. "Least-Squares Camera Calibration Including Lens Distortion and Automatic Editing of Calibration Points", Calibration and Orientation of Cameras in Computer Vision, D. Gennery, ISBN 3-540-65283-3, 2001.
21. Computations for Generalized Camera Model Including Entrance, Part 1 and Part 2, D. Gennery, unpublished, May 23, 2001.
22. Generalized Camera Calibration Including Fish-Eye Lenses, D. Gennery, JPL D- 03-0869, 2002.
23. Issues with Linearization, R. Deen, JPL Docushare Collection 2700, File 75670, 2003.
24. Mastcam-Z Multispectral Imaging on the Mars Science Laboratory Rover: Wavelength Coverage and Imaging Strategies at the Gale Crater Field Site, J.F. Bell III et al., *43rd Lunar and Planetary Science Conference*, 2012.
25. Anderson, R.C., et al., Mars Science Laboratory Participating Scientists Program Proposal Information Package, December 14, 2010.
26. Deen, R.G. and J.J. Lorre (2005), Seeing in Three Dimensions: Correlation and Triangulation of Mars Exploration Rover Imagery, submitted to 2005 IEEE International Conf. on Systems, Man, and Cybernetics, Waikoloa, Hawaii.
27. Mars Exploration Rover (MER) Project ICER User's Guide, Aaron Kiely, MER 420-8-0538, JPL D-22103, January 5, 2004.

1.3.1 Relationships with Other Interfaces

Changes to this EDR/RDR data product SIS document affect the following products, software, and/or documents.

Table 1.3.1 - Product and Software Interfaces to this SIS

Name	Type P = product S = software D = document	Owner
MIPL database schema	P	MIPL (JPL)
M2020 Camera Instrument EDRs • Cachecam • Hazcam • Navcam • Supercam RMI • Mastcam-Z • SHERLOC-WATSON • SHERLOC-ACI • MEDA Skycam • PIXL • EDL Cam • LVS • Helicopter	P	MIPL (JPL)
RSVP	S	RSVP Dev Team (JPL)
ASTTRO	S	ASTTRO Dev Team (JPL)
RGIS	S	RGIS Dev Team (JPL)
Mars Program Suite • m2020edrgen • m2020autoloco • marsautotie • marsautotie2 • marsbias • marsbrt • marsc2uvw • marscahv • marscheckcm • marschkovl • marscolor • marscoordtrans • marscor2 • marscor3 • marscorr • marscsv • marsdebayer • marsdispcompare • marsdispinvert • marsdispwarp • marsecorr • marserrdisp • marserror • marsfakedisp • marsfidfinder • marsfilter • marsint • marsinterp	S	MIPL (JPL)

Name	Type P = product S = software D = document	Owner
• marsinverter • marsinvrangle • marsjplstereo • marsmap • marsmask • marsmcauley • marsmesh • marsmos • marsnav • marsnav2 • marsortho • marsproj • marsprojfid • marsrad • marsrange • marsreach • marsrefmesh • marsrelabel • marsrescale • marsrough • marsseedgen • marsslope • marstie • marstiexyz • marstile • marsunlinearize • marsunmosaic • marsuvw • marsuvwproj • marsuvwrot • marsxyz • marsxyzmerge		

2. INSTRUMENT OVERVIEW

In this section, overviews are provided for the all of M2020 camera instruments mentioned in section 1.1. The M2020 rover instrument payload includes 25 individual cameras. The main differences between these instruments are in the optics, mounted position, and articulation methods. The cameras are color, except for the SHERLOC-ACI, PIXL MCC, and MEDA SkyCam. Of the 25 cameras, there are 5 sets of stereo pairs and 15 single cameras, as listed in Table 2.1 below.

Table 2.1 - Tabulation of M2020 Cameras

M2020 Camera Instrument	Location	Number
Navcam	Stereo pair on RSM	2
Front Hazcam	Stereo pair at front of WEB	4
Rear Hazcam	Stereo pair at rear of WEB	2
CacheCam	Monoscopic inside WEB	1
SuperCam RMI	Monoscopic on RSM	1
Mastcam-Z	Stereo pair on RSM	2
SHERLOC-WATSON	Monoscopic on robotic arm turret	1
SHERLOC-ACI	Monoscopic on robotic arm turret	1
MEDA SkyCam	Monoscopic on rover deck	1
PIXL MCC	Monoscopic on robotic arm turret	1
EDL PUC	Monoscopic on top of backshell	3
EDL DDC	Monoscopic on bottom of descent stage structure	1
EDL RUC	Monoscopic on top of rover deck	1
EDL RDC	Monoscopic on side of WEB	1
LVS	TBD	1
Helicopter Navigation Cam	TBD	1
Helicopter Scout Cam	TBD	1
	Total	25

There are several color capable cameras onboard the rover and they all utilize a Bayer filter/pattern, consisting of repeated squares of red (R), green (G), and blue (B) filters, as shown in figure 2.1.

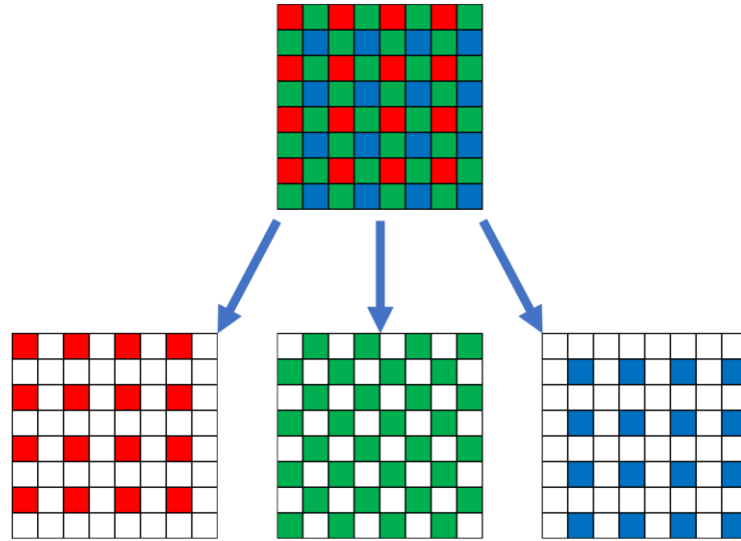


Figure 2.1 – Example Bayer filter pattern

INSERT M2020 Rover with camera locations here

Figure 2.2 – M2020 Camera Locations

2.1 Engineering Camera Instrument Suite

Each Engineering Camera (ECAM) is composed of a detector and electronics within a single mechanical housing, specifically: 1) an optical lens assembly, 2) a CMOS image sensor and 3) electronics that includes, a Field Programmable Gate Array, memory, and a legacy interface electronics to the rover.). All of the EECAMs use identical 5120 x 3840, 20 megapixel global shutter CMOS image sensors with a Bayer filter/pattern as shown in figure 2.1. Because of the electronics commonality, image data from all engineering cameras are functionally equivalent. The cameras have an external film heater that warms up the electronics to above the minimum operating temperature of -55 deg. C. The rover provides supply voltage of +5.5 V to the cameras. Each camera weighs less than 750 grams and uses approximately 4.5 Watts of power.

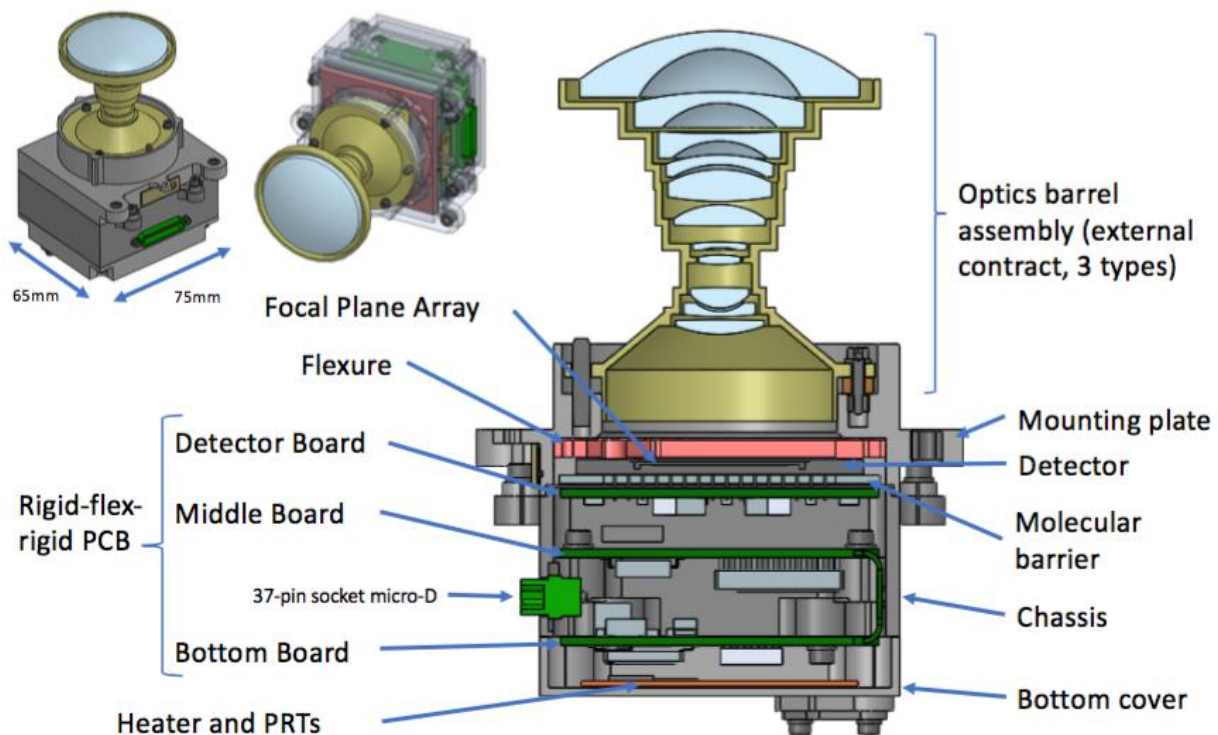


Figure 2.3 Engineering Camera Hardware Design

Insert Image of Camera Detector Readout

Figure 2.4 - Acquisition and Readout of Engineering Camera Image Data

2.1.1 Hazard Avoidance Camera (Hazcam)

The Hazard Avoidance Cameras (Hazcams) are two stereo pairs of engineering cameras with fish-eye lenses at 16 cm (front) and 10 cm (rear) baseline separation mounted at both the front and rear ends of the Warm Electronics Box (WEB). Hazcam assembly includes 2 cameras with a fixed Red 275 nm bandpass filter (identical to the Navcams).

The Hazcams provide imaging primarily of the near field (< 5 m) both in front of and behind the rover. These cameras will be used to determine safe driving directions for the rover and provide for on-board hazard detection using stereo data to build range maps. They also support science operations for selecting near field target and robotic arm operations.

Hazcam optics characteristics useful in the analysis of EDR and RDR products are described in Table 2.1.1, with Hazcam locations shown in Figure 2.1.1.1 – 2.1.1.3 below:

Table 2.1 - Hazcam Operational Characteristics

Characteristic	Value
Field of View (FOV)	134 x 110 deg
Baseline Stereo Separation	25 cm for front, 93.4 cm for rear
Angular Resolution	0.46 mrad/pixel at center
Spectral Bandpass	400 - 675 nm
Focal Length	13.8 mm
f/number	12
Depth of Field	0.8 m - infinity
Best Focus	1.0 m

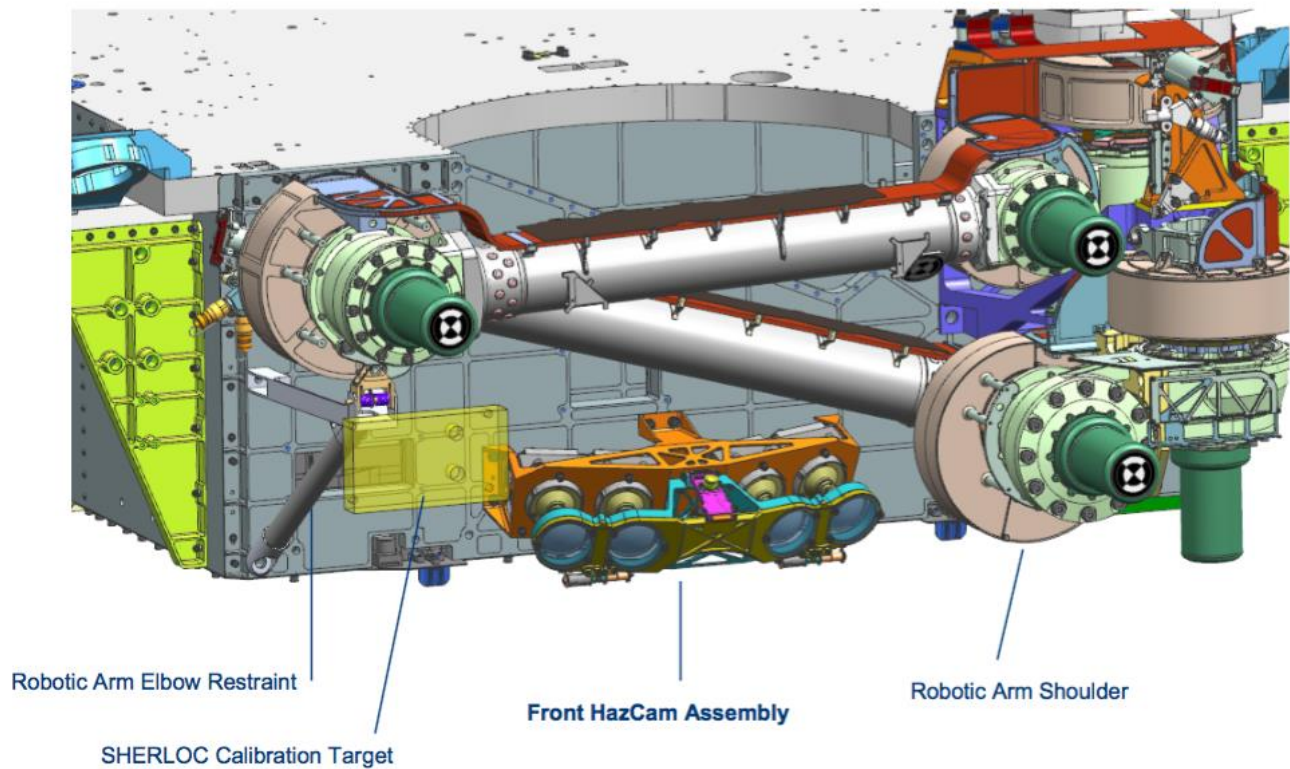


Figure 2.5 – Front Hazcam locations

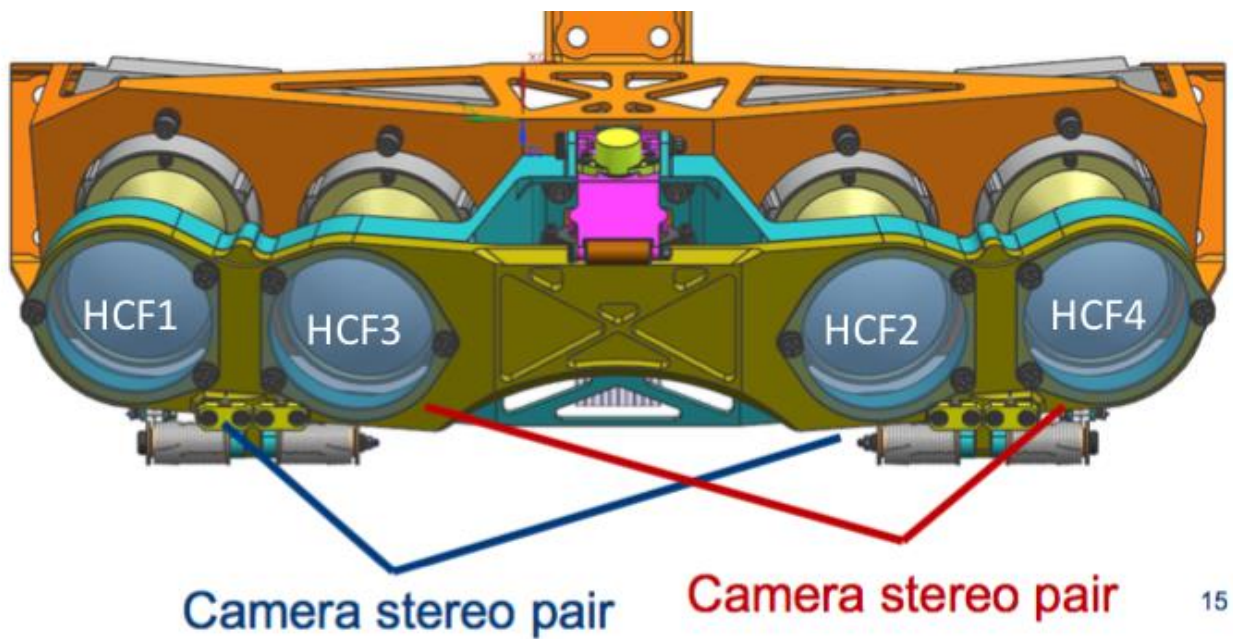


Figure 2.6 – Front Hazcam stereo pair mapping

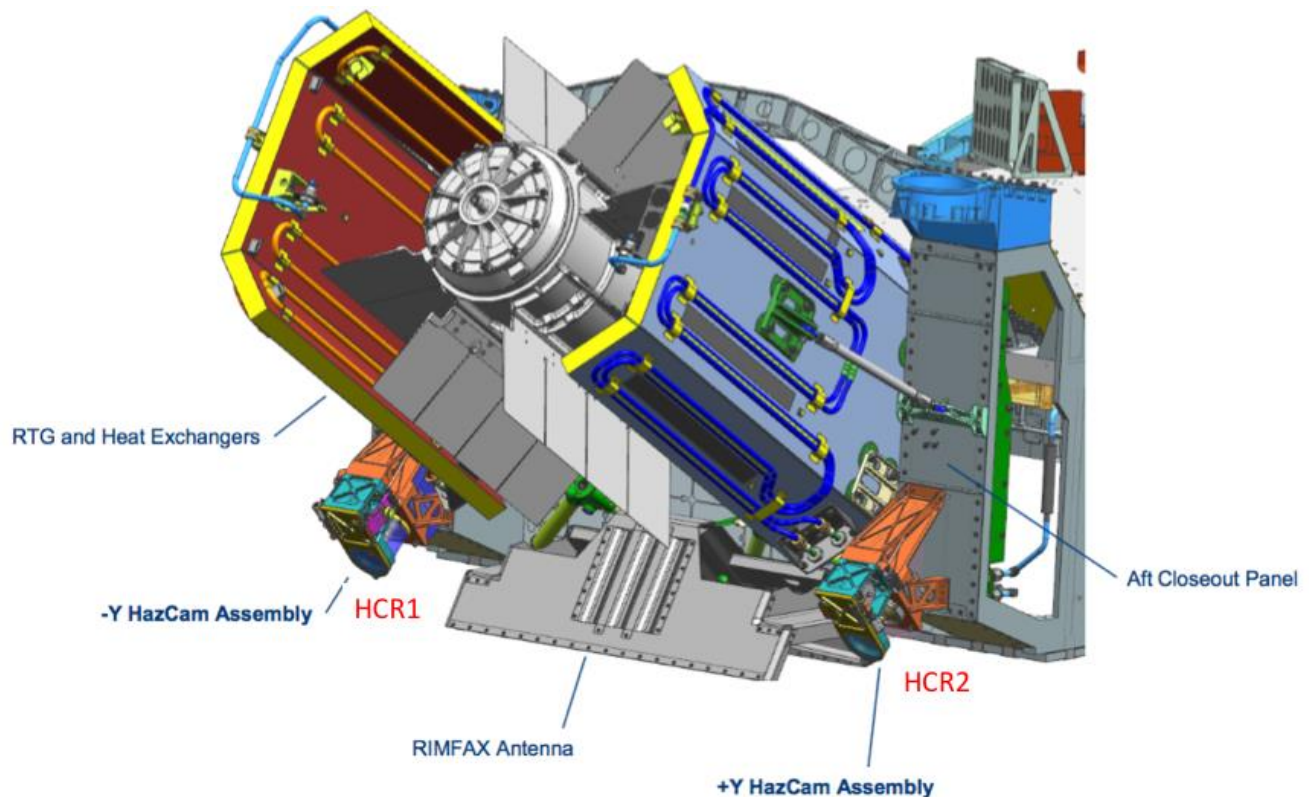


Figure 2.7 – Rear Hazcam location

2.1.2 Navigation Camera (Navcam)

The Navigation Camera (Navcam) is two mast-mounted stereo pairs of engineering cameras at 42.4 cm baseline separation with a spectral bandpass at approximately 275 nm. It will primarily be used for navigation purposes and general site characterization (360° panoramic images and targeted images of interest, including terrain not viewable by the Hazcams).

The cameras are boresighted with the Mastcam-Z, and Navcam images will also be used for Science target selection and analysis.

Navcam optics characteristics useful in the analysis of EDR and RDR products are described in Table 2.1.2, with Navcam field of view shown in Figure 2.1.2 below:

Table 2.2- Navcam Operational Characteristics

Characteristic	Value
Field of View (FOV)	95° x 71°
Baseline Stereo Separation	42.4 cm
Angular Resolution	0.32 mrad/pixel at center
Spectral Bandpass	400 - 1000 nm ??
Focal Length	19.2 mm
f/number	12

Characteristic	Value
Depth of Field	1.5 m - infinity
Best Focus	3.5 m

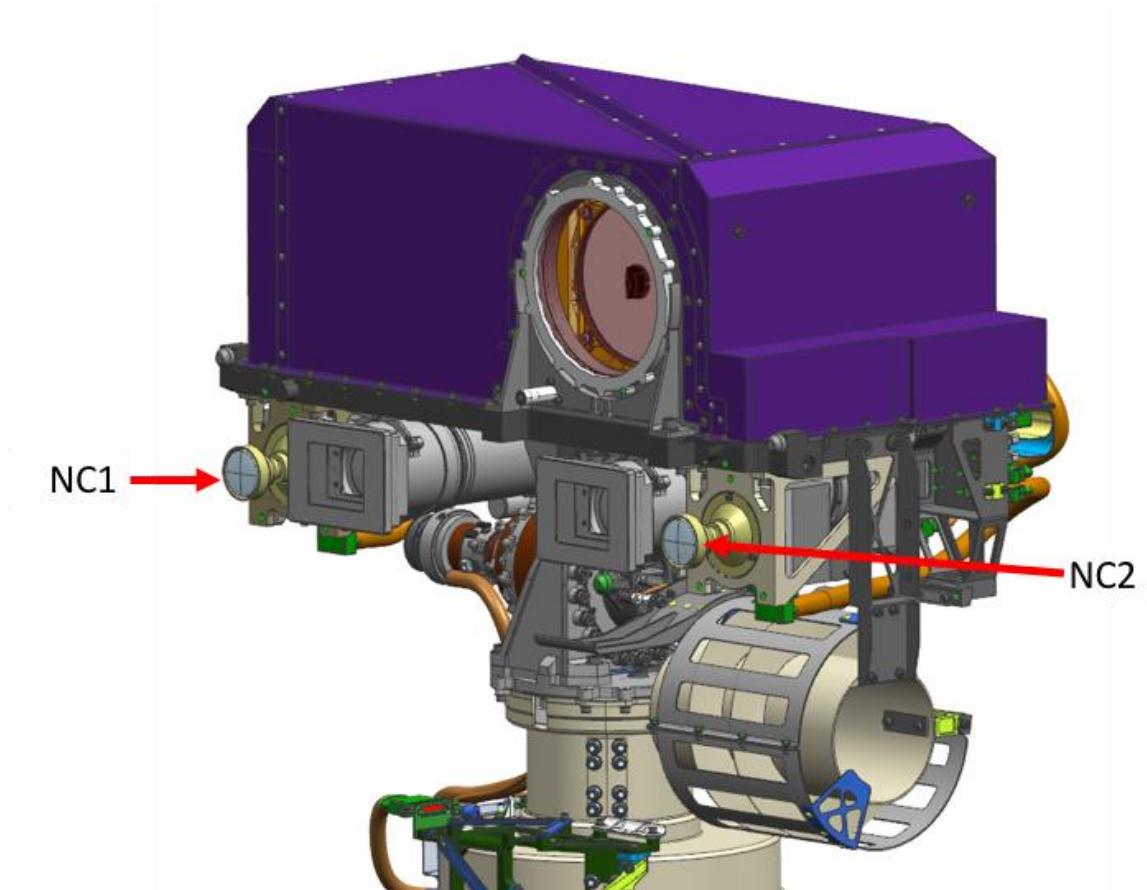


Figure 2.8 – Navcam location

2.1.3 Sample Caching System Camera (CacheCam)

The Navigation Camera (Navcam) is two mast-mounted stereo pairs of engineering cameras at 42.4 cm baseline separation with a spectral bandpass at approximately 275 nm. It will primarily be used for navigation purposes and general site characterization (360° panoramic images and targeted images of interest, including terrain not viewable by the Hazcams).

The cameras are boresighted with the Mastcam-Z, and Navcam images will also be used for Science target selection and analysis.

Navcam optics characteristics useful in the analysis of EDR and RDR products are described in Table 2.1.2, with Navcam field of view shown in Figure 2.1.2 below:

Table 2.3 - CacheCam Operational Characteristics

Characteristic	Value
Field of View (FOV)	° x °
Angular Resolution	mrad
Spectral Bandpass	TBD
Focal Length	TBD
f/number	f/8
Depth of Field	TBD
Best Focus	TBD

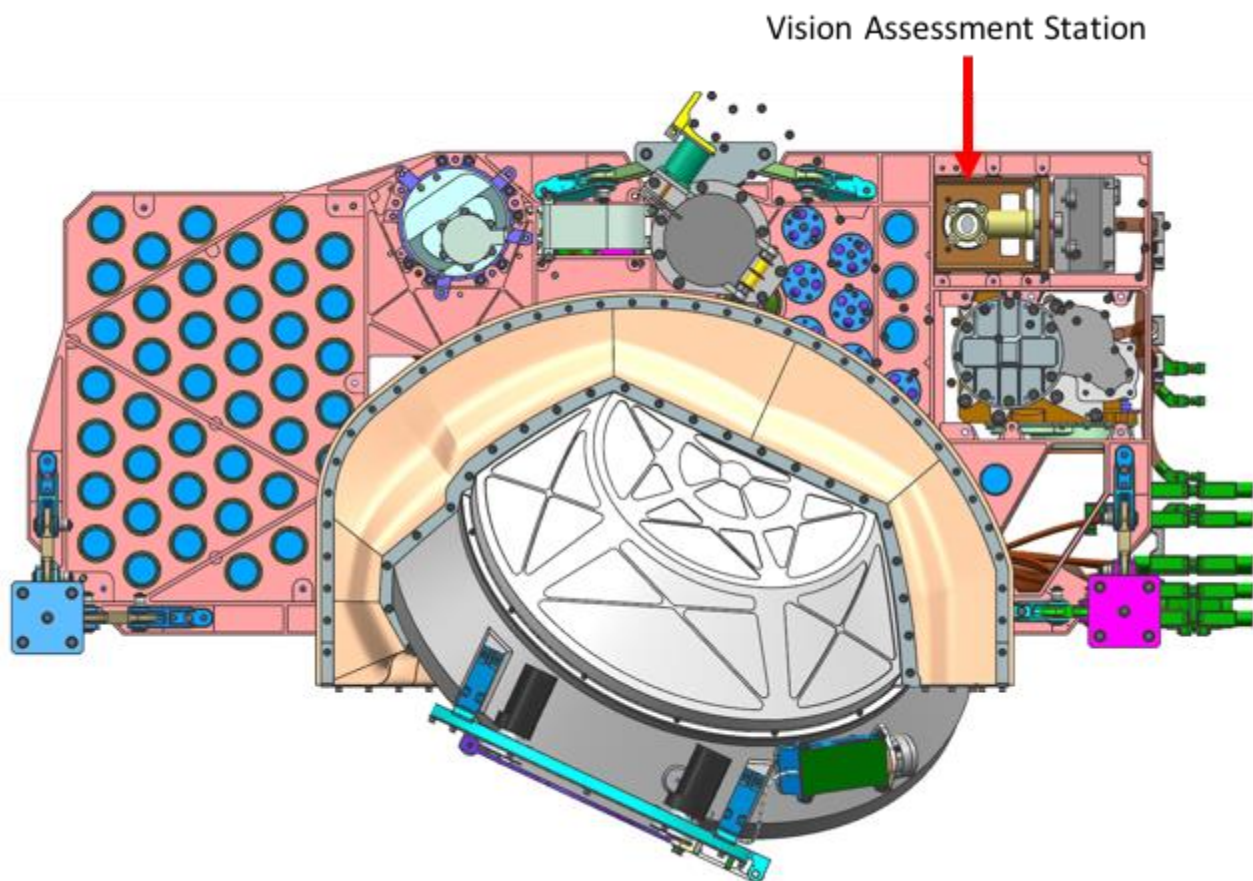


Figure 2.9 – CacheCam location

2.2 SuperCam RMI

The SuperCam RMI camera subsystem is packaged onboard the rover in two instrumentation units: the Mast Unit affixed to the RSM (figure 2.2.1), and the Body Unit housed within the rover WEB. The RMI camera provides high-resolution color context images of the LIBS sampling area as well as imaging rock morphologies and distant features. It includes an adjustable focus capability with 2048 x 2048 CMOS color detector (Bayer Filter). Raw RMI images are 13-bit (coded on 16-bits) and are equivalent to 4 CFAs of 1024 x 1024 pixels in bands red, green1, green2, and blue (RGGGB).

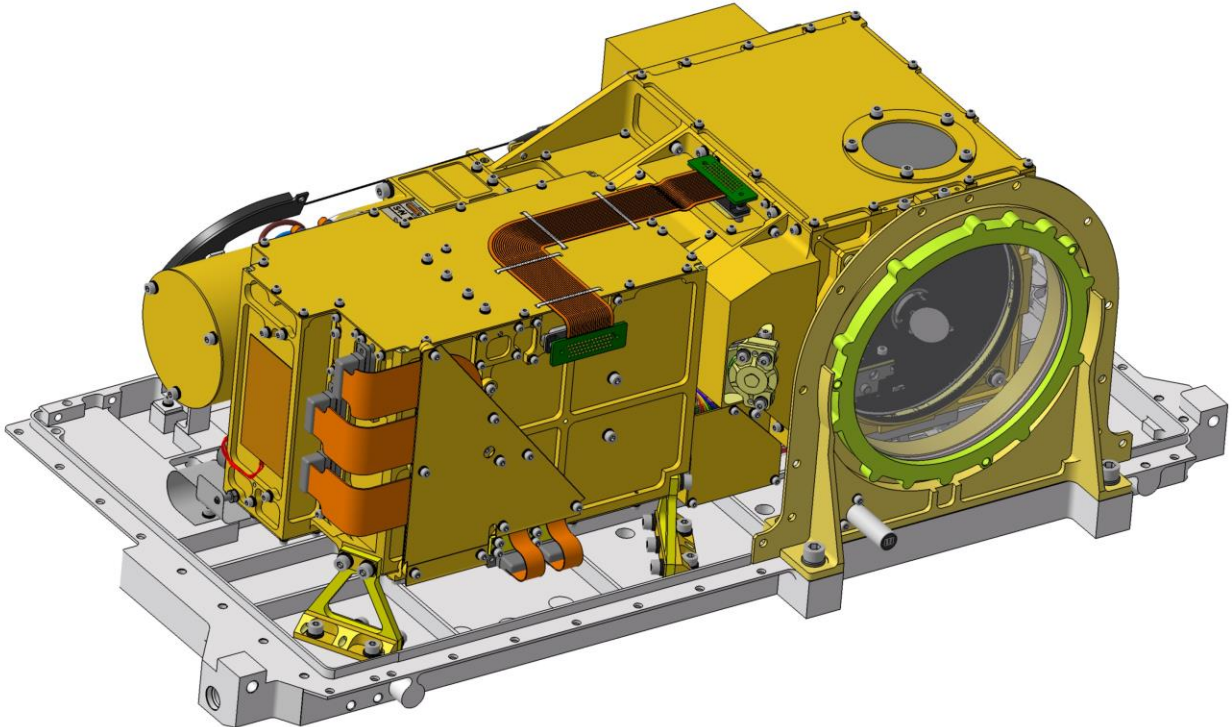


Figure 2.10 – SuperCam Mast Unit

RMI optics characteristics useful in the analysis of EDR and RDR products are described in Table 2.2.1 below:

Table 2.4 - RMI Operational Characteristics

Characteristic	Value
Field of View (FOV)	° x °
Spatial Resolution	TBD
Angular Resolution	μrad
Spectral Bandpass	TBD
Focus Range	TBD

Characteristic	Value
Number of Spectral Filters	TBD

RMI image data return compression modes are listed in Table X.X below:

Table 2.5 - RMI Compression Modes

Mode	Description
0	No compression
1	ICER
2	JPEG

The camera acquisition of the scene and subsequent onboard storage and readout of image data is illustrated in Figure X.X below:

Insert Image Readout graphic here

Figure 2.11 - Acquisition and Readout of RMI Camera Image Data

2.3 Mastcam-Z Camera Instrument Suite

2.3.1 Mastcam-Z

Mastcam-Z is comprised of a pair of color-capable focusable stereo cameras (“eyes”) mounted on the rover’s Remote Sensing Mast (RSM). Each camera has identical zoom/focus capabilities and set of spectral filters. Together, they can acquire images of up to 1600 x 1200 pixels and are capable of video. They acquire color via Bayer-pattern filters on the CCD, but also have selectable filters for science/geology applications. For more details on Bayer pattern filters, see Section 4.3.3.1.

Mastcam-Z optics characteristics that are useful in the analysis of EDR and RDR products are described in Table X.X below:

Table 2.7 – Mastcam-Z Operational Characteristics

Characteristic	Left Eye	Right Eye
Field of View (FOV)	Wide: ~23° x 18° Narrow: ~6° x 5°	Wide: ~23° x 18° Narrow: ~6° x 5°
Baseline Stereo Separation	24 cm	24 cm
Toe-in	1.25°	1.25°

Characteristic	Left Eye	Right Eye
Spatial Resolution	1 mm @ 1 – 2 m 3 – 4 mm @ 100 m	1 mm @ 1 – 2 m 3 – 4 mm @ 100 m
Angular Resolution	Wide: 266 μ rad Narrow: 74 μ rad	Wide: 266 μ rad Narrow: 74 μ rad
Focal Length	28 – 100 mm	28 – 100 mm
f/number	Wide: f/8 Narrow: f/10	Wide: f/8 Narrow: f/10
Focus Range	1 – 2 m – infinity	1 – 2 m – infinity
Number of Spectral Filters	8 plus Bayer pattern	8 plus Bayer pattern

Each Mastcam-Z camera has an 8-position filter wheel. One of the positions is a broadband filter for use with the Bayer color capability of the CCD. One additional filter per eye has a neutral density coating to provide direct solar imaging capability in two colors. The remaining filters are used for science imaging.

Thirteen of the sixteen filters provide color imaging capability in eleven unique wavelengths from 400 to 1100 nm, two of the remaining filters have neutral density coatings to provide direct solar imaging capability in two colors, and one filter wheel position has been left empty to provide for maximum broadband light sensitivity. The spectral bandwidths [Ref 28] are described in Table X.X below:

Table 2.8 - Mastcam-Z Spectral Filters and Bandpasses

Filter Position	Left Eye Wavelength ($\pm$ Bandwidth), $\lambda_{\text{eff}} \pm \text{HWHM (nm)}$	Right Eye Wavelength ($\pm$ Bandwidth), $\lambda_{\text{eff}} \pm \text{HWHM (nm)}$
0	590 $\pm$ 88 (Broadband)	590 $\pm$ 88 (Broadband)
	640 $\pm$ 44 (Bayer filter Red)	640 $\pm$ 44 (Bayer filter Red)
	554 $\pm$ 38 (Bayer filter Green)	554 $\pm$ 38 (Bayer filter Green)
	495 $\pm$ 37 (Bayer filter Blue)	495 $\pm$ 37 (Bayer filter Blue)
1	445 $\pm$ 10	805 $\pm$ 10
2	527 $\pm$ 10	867 $\pm$ 10
3	600 $\pm$ 10	908 $\pm$ 10
4	676 $\pm$ 10	937 $\pm$ 10
5	751 $\pm$ 10	975 $\pm$ 10
6	805 $\pm$ 21	1013 $\pm$ 20
7	590 $\pm$ 88 ND6	880 $\pm$ 10 ND5

The Mastcam-Z filter response profiles, including those of the RGB Bayer filters, are plotted below for the left and right camera eyes. Figure X.X [Ref Y] shows the Bayer filter profiles in red, green and blue when the filter wheel is set to Filter 0 (broadband IR cutoff shown as dashed line).

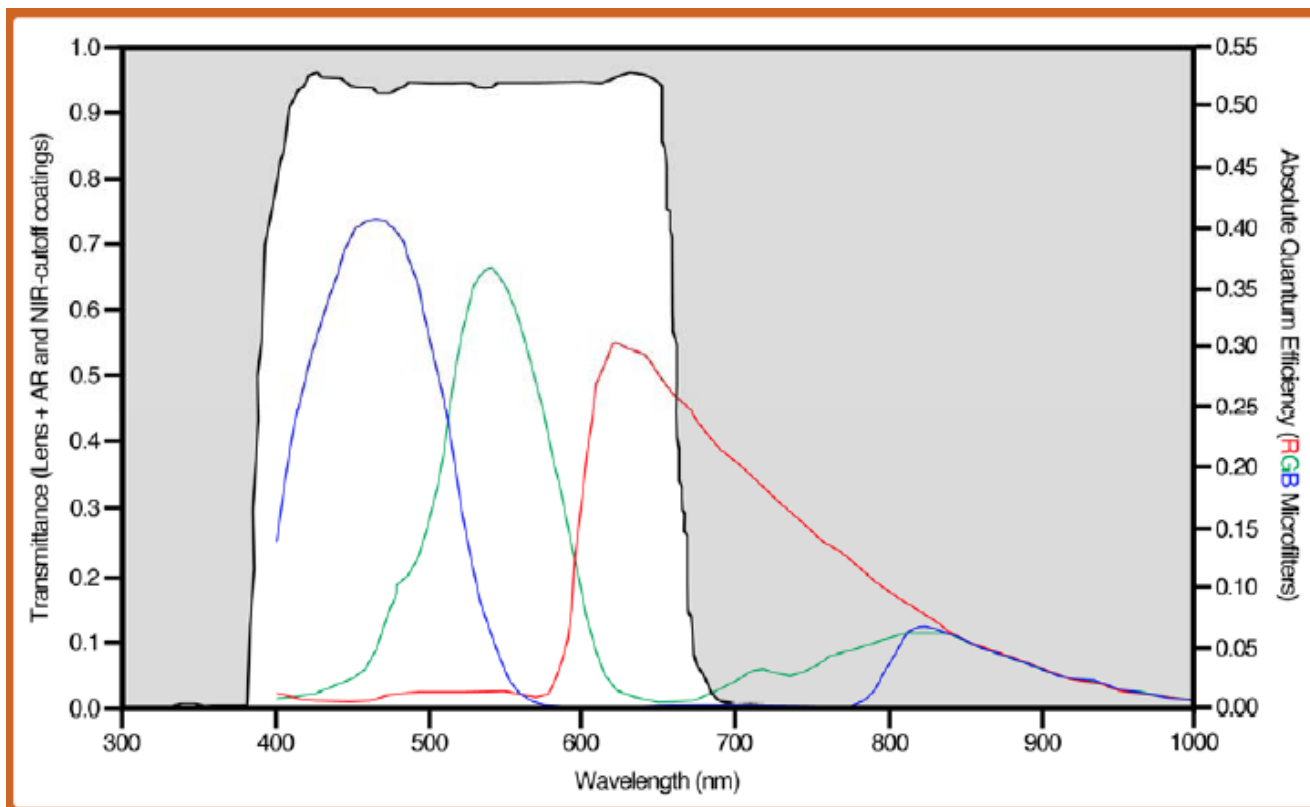


Figure 2.9 - Mastcam-Z Filter Profiles, Filter Wheel Set to Filter 0

Figure X.X [Ref Y] shows the Bayer filter profiles in red, green and blue when the filter wheel is set to a nonzero number. Additionally, the quantum efficiency of the CCD detector is graphically depicted as a dashed line. The Bayer filter array and CCD quantum efficiency profiles are not shown to scale.

Mastcam-Z image data return compression modes are listed in Table 2.9 below:

Table 2.9 – MCZ Image Compression Modes

Mode	Description	Factor
0	No compression or some form of lossless	1
1	Predictive Lossless Bayer Huffman Encoding	X
2	Raw Raster	X
3	JPEG Discrete Cosine Transform (DCT)	X

2.4 SHERLOC Camera Suite

2.4.1 SHERLOC-WATSON

The SHERLOC-WATSON camera is a focusable color camera located on the turret at the end of the M2020 robotic arm and is a build-to-print copy of the MSL MAHLI instrument. The instrument acquires images of up to 1600 by 1200 pixels with a color quality equivalent to that of consumer digital cameras using a Bayer pattern. For details on Bayer pattern filters, see Section 4.3.3.1. It is also capable of video. SHERLOC-WATSON optics characteristics useful in the analysis of EDR and RDR products are described in Table 2.10 below.

Table 2.10 - SHERLOC-WATSON Operational Characteristics

Characteristic	Value
Field of View (FOV)	34.0 - 39.4° deg diagonal
Spatial Resolution	15 $\mu\text{m}/\text{pixel}$ at 25 mm distance
Angular Resolution	0.3 - 0.34 mrad/pixel
Spectral Wavelength $\pm$ Bandwidth ($\lambda_{\text{eff}} \pm \text{HWHM}$)	590 $\pm$ 88 nm (Broadband)
	640 $\pm$ 44 nm (Bayer filter Red)
	554 $\pm$ 38 nm (Bayer filter Green)
	495 $\pm$ 37 nm (Bayer filter Blue)
Focal Length	18.3 - 21.3 mm
f/number	9.8 - 8.5
Depth of Field	1.6 mm - >4800 mm
Focus Range	20.5 mm - infinity
Number of Spectral Filters	1 plus Bayer pattern on CCD

Note that the spatial resolution in Table X.X measures the working distance, which is not the same as the distance from the camera model (C) point. Spatial resolution may be calculated by:

$$\text{Pixel scale } (\mu\text{m}/\text{pixel}) = 6.9001 + [3.5201 * \text{Working Distance (cm)}]$$

Note also that the spectral bandpasses in Table X.X are approximate and use the Left Mastcam-Z bandpasses as the representative values.

2.4.2 Autofocus and Context Imager

The SHERLOC-ACI is co-boresighted with the laser illumination and spectral collection path. Light from the target (indirect sunlight or LED illumination) is collected by the autofocus objective and directed to the context imager by the foreoptics.

Table 2.11 – SHERLOC-ACI Operational Characteristics

Characteristic	Value
Field of View (FOV)	34.0 – 39.4° diagonal
Spatial Resolution	15 μm /pixel at 25 mm distance
Angular Resolution	0.3 - 0.34 mrad/pixel
Spectral Wavelength	500 $\pm$ 50 nm (Broadband)
Depth of Field	1.6 mm - >4800 mm
Number of Spectral Filters	0 (monochrome detector)

2.5 PIXL Micro Context Camera

The PIXL Micro Context Camera (MCC) consists of a Data Processing Unit (DPU), a Camera Head Unit (CHU) with Structured Light Illuminators (SLIE), Flood Light Illuminator Electronics, and Flood Light Illuminator (FLI). The DPU is mounted within the rover body and all of the optical components are mounted on the turret of the robotic arm.

CCD 7.95mm x 6.45mm. 752x580 pixels, each pixel is 8.6 x 8.3 μm . 12mm optical lens. FLI consists of 4 color channels R, G, B and UV. Each channel has 6 LEDs and each color can be controlled individually. Green LEDs can control individual LEDs.

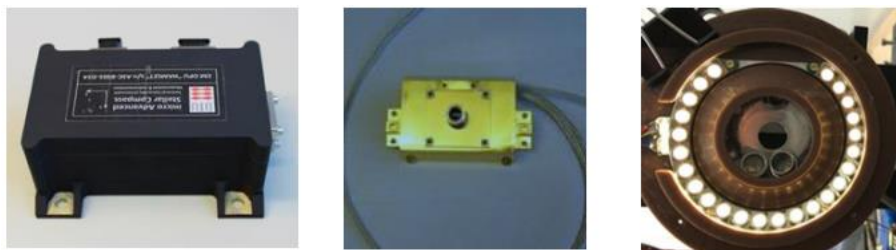


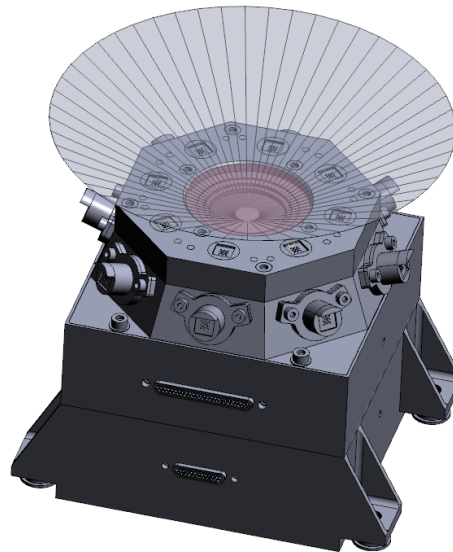
Figure 2.10 DPU, CHU, FLI and two SLIs mounted with CHU.

2.6 MEDA Sky Camera

The MEDA Skycam is a build-to-print copy of the MER and MSL Engineering Cameras. The camera is mounted in the center of the Radiation and Dust Sensor (RDS) and also includes a 120° FOV Baffle (ring on lens) that will be used to block sunlight at specific times during the Martian Sol.

Table 2.11 – Skycam Operational Characteristics

Characteristic	Value
Field of View (FOV)	124 x 124 deg
Angular Resolution	2.1 mrad/pixel at center
Spectral Bandpass	600 - 800 nm
Focal Length	5.58 mm
f/number	15
Depth of Field	0.1 m - infinity
Best Focus	0.5 m

**Figure 2.11 RDS-Skycam with FOV**

2.7 EDL Camera Suite

2.7.1 Parachute Uplook Camera (PUC)

FLIR (Formerly Point Grey) Chameleon3

2.7.2 Descent Stage Downlook Camera (DDC)

TBD

2.7.3 Rover Uplook Camera (RUC)

TBD

2.7.4 Rover Downlook Camera (RDC)

TBD

2.8 Lander Visusal System

TBD

2.9 Helicopter imaging suite

2.9.1 Scout

TBD

2.9.2 Navigation

TBD

3. GENERAL DATA PRODUCT OVERVIEW

ADD NOTE about how the data is received from DSN and recreated in EDRGEN. APID LIST TABLE.

3.1 Data Processing Levels

This documentation uses the “Planetary Data System Standard 4” (PDS4) data level numbering system. The camera instrument operations data products referred to in this document as EDRs are considered “Raw”. The EDRs are to be reconstructed from “Telemetry”, which are the telemetry packets within the project specific Standard Formatted Data Unit (SFDU) record.

Camera instrument operations data products referred to in this document as RDRs are considered “Partially Processed”, “Calibrated”, or “Derived”. The RDRs may be reconstructed from “Raw” data, or processed from other RDR data levels.

Refer to Table 3.1 for a breakdown of the PDS4 data processing levels.

Table 3.1 – PDS4 Processing Levels for Instrument Experiment Data Sets

Processing Level for PDS4 Archive	Operations Data Product Name	Description
Telemetry	n/a	An encoded byte stream used to transfer data from one or more instruments to temporary storage where the raw instrument data will be extracted. PDS does not archive telemetry data.
Raw	EDR (Experiment Data Record, heritage term based on MSL mission)	Original data from an instrument. If compression, reformatting, packetization, or other translation has been applied to facilitate data transmission or storage, those processes will be reversed so that the archived data are in a PDS approved archive format.
TBD	FDR (Fundamental Data Record)	Data that have been processed beyond the raw stage, but which have not yet reached calibrated status. (MSL ILUT product.)
Partially Processed	RDR (Reduced Data Record, heritage term based on MSL mission)	Data that have been processed beyond the raw stage, but which have not yet reached calibrated status. These are more highly processed products.
Calibrated	RDR	Data converted to physical units, which makes values independent of the instrument.
Derived	RDR	Results that have been distilled from one or more calibrated data products (for example, maps, gravity or magnetic fields, or ring particle size distributions). Supplementary data, such as calibration tables or tables of viewing geometry, used to interpret observational data should also be classified as “derived” data if not easily matched to one of the other three categories.

3.2 Product Label and Header Descriptions

3.2.1 Overview of Labels

There are up to three different sets of product labels (metadata) that can be associated with an EDR/RDR file, to be described subsequently in this section: a) detached PDS (version 4) label, b) attached ODL (Object Description Language) [Ref 4] label, and c) attached VICAR (Video Communications and Retrieval) label. While formats and keywords may differ, all three can be reliably used to extract metadata.

***** NEED TO UPDATE for PDS3 no longer being supported *****

The primary label supporting operations is the attached ODL label. ODL is the format used by PDS3 labels. Therefore, from a syntactic point of view, they look identical to PDS labels (with the exception of the initial keyword and other item omissions and/or name differences) and can be processed using most PDS tools. The difference is that the keywords are not validated and are not guaranteed to be in the PDS Data Dictionary. The reason for the ODL label is because the PDS standards approval process typically is out of phase with operations delivery schedules; keywords cannot be approved and appear in the Data Dictionary within the timeline needed in operations. Every attempt is made to keep the ODL and Archive Attributes the same, but there are some discrepancies. The most common difference is that many Archive Attributes have a "M2020:" prefix to indicate a local data dictionary, while the ODL keywords do not.

In the case of image EDR/RDR files only, a secondary operations label exists that is the attached VICAR label. This is used by MIPL/IDS software to operate on the images. Again, the keywords for the most part match the ODL keywords; the difference is mostly formatting and syntax.

The primary label from the archive perspective is the detached PDS4 label. This is a separate file with the same base name as the image file, with a ".LBL" extension. The detached label references the EDR/RDR filename via a keyword. This label is fully compliant with all PDS archive standards.

3.2.2 PDS and ODL Labels *** Need to update for PDS3 no longer supported: DOUG working on write-up for ALL SIS'*****

As implied in the previous section, EDRs and RDRs described in this document, with the exception of the IDS Terrain RDR, have an attached ODL label and a detached PDS label. Each institution is responsible for converting PDS-formatted products to be compatible with their own software systems (such as VICAR, IDL, ISIS, etc.).

Because the PDS and ODL formats are identical, they are described together here, with differences in content noted as such.

Per PDS standards, the detached PDS label starts with the keyword assignment:

PDS_VERSION_ID = PDS3

The attached ODL label, which contains keywords that may not be compliant to PDS standards, starts with the keyword assignment:

ODL_VERSION_ID = ODL3

This is how the two label types are distinguished. With the exception of this keyword, any PDS-format reader that does not validate against the PDS Data Dictionary should be able to read the ODL labels.

A PDS/ODL label is object-oriented and describes the objects in the data file. The PDS/ODL label contains keywords for product identification, along with the data object definition containing descriptive information needed to interpret or process the data in the file. The PDS spreadsheet object is used to describe the format of the products.

PDS/ODL label statements have the form of "keyword = value". Each label statement is terminated with a carriage return character (ASCII 13) and a line feed character (ASCII 10) sequence to allow the label to be read by many operating systems. Pointer statements with the following format are used to indicate the location of data objects in the file:

^object = location

where the carat character (^, also called a pointer) is followed by the name of the specific data object. The location is the 1-based starting record number for the data object within the file. Alternatively, it could be the 1-based byte location within the file if it includes a <bytes> unit tag. The PDS detached label includes the filename as part of the pointer:

^object = (filename, location)

Pointers are used to define the locations of the binary instrument data itself (^IMAGE for image data), the VICAR header in the case of images (^IMAGE_HEADER), and the ancillary binary data (SuperCam only, e.g. ^SCAM_SOH_DPO).

3.2.2.1 PDS Local Data Dictionary

The PDS label contains many keywords with a "M2020:" prefix. These indicate keywords that are defined in the M2020 local data dictionary, rather than the primary PDS data dictionary. The ODL label does not contain these prefixes.

3.2.2.2 Keyword Length Limits

All PDS/ODL keywords are limited to 30 characters in length (Section 12.7.3 in PDS Standards Reference) [Ref 14], not including the "M2020:" prefix. Therefore, software that reads M2020 PDS/ODL labels must be able to ingest keywords up to 30 characters in length.

For image RDR-producing institutions wishing to accommodate the VICAR mapping (see Section 3.2.2) of PDS/ODL keywords that use a <unit> tag after the value, such keywords must be limited to 24 characters in length. Otherwise, those keywords will not be transcoded from the PDS/ODL label into a VICAR label.

3.2.2.3 Data Type Restrictions

In order to accommodate VICAR dual-labeled files, 16-bit data must be stored as signed data. Unsigned 16-bit data are not supported. 12-bit unsigned data from the cameras are stored in a 16-bit signed value. 8-bit data are unsigned.

3.2.2.4 Interpretation of N/A, UNK, and NULL

During the completion of data product labels or catalog files, one or more values may not be available for some set of required data elements. In this case PDS provides the symbolic literals "N/A", "UNK",

and “NULL”, each of which is appropriate under different circumstances. As a note, if any one of these three symbolic literals are used in place of a keyword value that is normally followed by a Unit Tag(s) (e.g., “<value>”), the Unit Tag(s) is removed from the label.

- “N/A” (“Not Applicable”) indicates that the values within the domain of this data element are not applicable in this instance. For example, a data set catalog file describing NAIF SPK kernels would contain the statement:

INSTRUMENT_ID = "N/A"

because this data set is not associated with a particular instrument.

“N/A” may be used as needed for data elements of any type (e.g., text, date, numeric, etc.).

- “UNK” (“Unknown”) indicates that the value for the data element is not known and never will be. For example, in a data set comprising a series of images, each taken with a different filter, one of the labels might contain the statement:

FILTER_NAME = "UNK"

if the observing log recording the filter name was lost or destroyed and the name of the filter is not otherwise recoverable.

“UNK” may be used as needed for data elements of any type.

- “NULL” is used to flag values that are *temporarily* unknown. It indicates that the data preparer recognizes that a specific value should be applied, but that the true value was not readily available. “NULL” is a placeholder. For example, the statement:

DATA_SET_RELEASE_DATE = "NULL"

might be used in a data set catalog file during the development and review process to indicate that the release date has not yet been determined.

“NULL” may be used as needed for data elements of any type.

Note that all “NULL” indicators should be replaced by their actual values prior to final archiving of the associated data.

3.2.2.5 PDS/ODL Label Constructs “Class”, “Object” and “Group”

The PDS has designed a set of formal and informal constructs for labeling data products. In the PDS realm, “formal” implies a standardized design or set of rules that provides a protocol across multiple data products (e.g., multiple flight missions) for PDS validation tools, and involves a rigorous approval process. “Informal” implies a less rigorous process by which the construct meets PDS approval. For both formal and informal constructs, the PDS member keywords must be defined in the *Planetary Science Data Dictionary* (PSDD) [Ref 11] or the *Planetary Science Data M2020 Local Dictionary* [Ref 12]. ODL label keywords need not be in the data dictionaries. For the EDRs and RDRs described in this document, the PDS/ODL label includes the following “formal” and “informal” constructs:

- *Class* - The Class construct is informal and resides in a PDS/ODL label as a grouping of keywords that are thematically tied together. Classes are usually

preceded by a label comment, although it is not required. PDS/ODL label comments are character strings bounded by “/* */” characters.

In the M2020 Camera PDS/ODL label a Class of keywords will be preceded by a comment string as follows:

```
/* comment string */
keyword      = keyword value
keyword      = keyword value
```

- *Object* - The Object construct is formal and is a set of standard keywords used for a particular data product. In the PSDD, each Object definition lists the elements required to be present each time the Object is used in a product label. The PSDD also provides a list of additional, optional keywords that are frequently used in the Object. Any element defined in the PSDD may be included as an optional element in any Object definition, at the discretion of the data preparer.

In the M2020 Camera PDS/ODL label an Object's set of keywords is specified as follows:

```
OBJECT        = Object identifier
keyword       = keyword value
keyword       = keyword value
END_OBJECT    = Object identifier
```

- *Group* - The Group construct can be either a formal or informal grouping of keywords that are not components of a larger Object. Group keywords may reside in more than one Group within the label.

The Group construct is further described in section 12.4.5 of the PDS Standards Reference, "Object Description Language Specification and Usage: GROUP Statement".

In the M2020 Camera PDS/ODL label, a Group's set of keywords is specified as follows:

```
GROUP         = Group identifier
keyword       = keyword value
keyword       = keyword value
END_GROUP     = Group identifier
```

3.2.2.6 PDS/ODL Image Object

An IMAGE object is a two-dimensional array of values, all of the same type, each of which is referred to as a *sample*. IMAGE objects are normally processed with special display tools to produce a visual representation of the samples by assigning brightness levels or display colors to the values. An IMAGE consists of a series of lines, each containing the same number of samples.

The required IMAGE keywords define the parameters for simple IMAGE objects:

- LINES is the number of lines in the image.
- LINE_SAMPLES is the number of samples in each line.
- SAMPLE_BITS is the number of bits in each individual sample.
- SAMPLE_TYPE defines the sample data type.

The IMAGE object has a number of keywords relating to image statistics. These keywords will be present in all EDRs. In RDRs, they are optional, and if they are present, they must be updated to reflect the current statistics of the image (often they will be omitted for the sake of computational efficiency). Note that the VICAR label never contains these keywords; see section 3.2.2. The statistics keywords are:

- MEAN
- MEDIAN
- MAXIMUM
- MINIMUM
- STANDARD_DEVIATION
- CHECKSUM

Many variations on the basic IMAGE object are possible with the addition of optional keywords and/or objects. The “^IMAGE” keyword identifies the start of the image. Recommended image formats are described and illustrated in Reference 4, Appendix A.19.

3.2.2.7 PDS/ODL Table Objects and “.FMT” Files

In lieu of the binary Image Data Product Header (IDPH) Data Product Object (DPO) contained in the Engineering Camera and Mastcam-Z EDR products, SuperCam EDRs include binary DPOs that contain ancillary information about the EDR. Of these, the primary carrier of such information is the Ancillary DPO. Refer to Figure 4.4 (Diagrams B and C) for a visual schematic of the SuperCam EDR structures, wherein the DPO information is noted as “auxiliary” data. Some of the data from these DPOs are pulled out into standard PDS/ODL label items. Section 4.4.2 discusses the SuperCam EDR structure and how the DPOs reside within the EDR.

The ^pointer at the beginning of the label points to the data itself. There is also a PDS/ODL Object in the label, which defines the contents and structure of the data. How the structure is described is defined by the PDS standards documents [Ref 14], so it is not fully defined here.

Of particular note is that most of the structure is offloaded to a separate format (.FMT) file via a ^STRUCTURE keyword within the object. The various “.FMT” files supporting the SuperCam EDR labels are described explicitly in Appendix D.

An example of the label object is shown below:

```
OBJECT          = TABLE
NAME            = "SCAM_SOH_DPO_TABLE"
COLUMNS        = 20
ROWS            = 80
ROW_BYTES       = 46
INTERCHANGE_FORMA = "BINARY"
```

```

DESCRIPTION          = "SuperCam State of Health Structure"

/* This format file defines columns 1-43, the first part of the SCIDATA object. */

^STRUCTURE           = "SCAM_SOH_DPO_TABLE.FMT"

/* This container includes columns 44, 45, and 46. This set of columns may */
/* be repeated up to 20 occurrences. */

OBJECT               = CONTAINER
  NAME                = "SCAM_SOH_TO_RCE_CONTAINER"
  START_BYTE          = 129
  BYTES               = 100
  REPETITIONS         = 20 /* or any number between 1 and 20 */
  DESCRIPTION         = "SuperCam SOH to RCE structure."
  Includes arrays Scam_DPU_SOH_struct and Scam_MU_SOH_struct."

/* This format file defines columns 44, 45, and 46, the last part */
/* of the SCIDATA object. */

^STRUCTURE           = "SCAM_SOH_TO_RCE_CONTAINER.FMT"

END_OBJECT           = CONTAINER
END_OBJECT           = TABLE

```

3.2.3 PDS4 Label

TBD

3.2.4 VICAR Label

For all image EDR data products and MIPL produced image RDR data products, an embedded VICAR label follows the ODL label and is pointed to by the PDS/ODL pointer “^IMAGE_HEADER”. The VICAR label is also organized in an ASCII, “keyword = value” format, although there are only spaces between keywords (no carriage return/line feeds as in PDS). The information in the VICAR label is an exact copy of the information in the PDS label as defined in the next section. The reader is referred to the VICAR File Format document for details of the format, which is available at the URL “http://www-mipl.jpl.nasa.gov/external/VICAR_file_fmt.pdf”. The following text is an excerpt which describes the basic structure:

A VICAR file consists of two major parts: the labels, which describe what the file is, and the image area, which contains the actual image. The labels are potentially split into two parts, one at the beginning of the file, and one at the end. Normally, only the labels at the front of the file will be present. However, if the EOL keyword in the system label (described below) is equal to 1, then the EOL labels (End Of file Labels) are present. This happens if the labels expand beyond the space allocated for them. The VICAR file is treated as a series of fixed-length records, of size RECSIZE (see below). The image area always starts at a record boundary, so there may be unused space at the end of the label, before the actual image data starts.

The label consists of a sequence of “keyword=value” pairs that describe the image, and is made up entirely of ASCII characters. Each keyword-value pair is separated by spaces. Keywords are strings, up to 32 characters in length, and consist of uppercase characters, underscores (“_”), and numbers (but should start with a letter). Values may

be integer, real, or strings, and may be multiple enclosed in parentheses (e.g. an array of 5 integers, but types cannot be mixed in a single value). Spaces may appear on either side of the equals character (=), but are not normally present. The first keyword is always LBLSIZE, which specifies the size of the label area in bytes. LBLSIZE is always a multiple of RECSIZE, even if the labels don't fill up the record. If the labels end before LBLSIZE is reached (the normal case), then a 0 byte terminates the label string. If the labels are exactly LBLSIZE bytes long, a null terminator is not necessarily present. The size of the label string is determined by the occurrence of the first 0 byte, or LBLSIZE bytes, whichever is smaller. If the system keyword EOL has the value 1, then End-Of-file Labels exist at the end of the image area (see above). The EOL labels, if present, start with another LBLSIZE keyword, which is treated exactly the same as the main LBLSIZE keyword. The length of the EOL labels is the smaller of the length to the first 0 byte or the EOL's LBLSIZE. Note that the main LBLSIZE does not include the size of the EOL labels. In order to read in the full label string, simply read in the EOL labels, strip off the LBLSIZE keyword, and append the rest to the end of the main label string.

Note that the EOL labels will not appear in archive products.

A binary header may appear in between the VICAR label and the image, containing arbitrary binary data that are not interpreted by VICAR. The number of records in this header is defined by the VICAR system keyword NLB. This binary header area is where the SuperCam DPOs are stored.

3.2.5 Mapping of PDS/ODL and VICAR Labels

In the cases of the attached ODL and VICAR labels, information content is identical, by definition. ODL and VICAR labels may be used interchangeably, for any purpose in the mission. Any software that modifies one label must also modify the other, or remove them entirely. This is often most easily accomplished by stripping off one of the headers, processing the remaining label as desired locally, and then running a conversion tool to re-create the missing header. Such tools will be provided by MIPL.

In the case of the detached PDS label, information content is not necessarily identical to the ODL label, and therefore not necessarily identical to the VICAR label when applicable. It is intended that a PDS label can have reduced content, but not increased content, in comparison to the ODL and VICAR labels. Appendix **TBD** tracks the label keywords that are not present in the PDS label.

It is important to note that only product labels containing the label keyword assignment "PDS_VERSION_ID = PDS3" are valid PDS products.

The mapping between PDS and ODL keywords is straightforward, and keyword names are usually the same. However, there are some keyword name differences. Appendix A shows a label in ODL format, and identifies differences in ODL and Archive Attribute names. For space reasons in this document, the corresponding VICAR label is omitted from this document, but it is required. The mapping rules are as follows:

- For mapped PDS, ODL and VICAR (if applicable) keywords, values are identical in all cases with the exception of differences mandated by the file format itself, such as quoting rules. See the respective PDS and VICAR documents for details, but in general, PDS/ODL uses double quotes (") while VICAR uses single quotes (').
- For ODL and VICAR label keywords, with the exception of those defining the file format itself (described below), names are identical in both cases.

- For PDS labels, resident keywords map 1-to-1 with ODL keywords, but not all ODL keywords have a mapping to Archive Attributes. ODL keywords not represented in PDS labels are listed in Appendix **TBD**.
- For PDS labels, keyword names are usually identical to keywords in the ODL label, but not necessarily so. Name differences between keywords are identified in Appendix A.
- Any ODL label group maps 1-to-1 to a VICAR property set with the same name (Group name == property set name). All contained keywords are identical in both cases. The GROUP and END-GROUP keywords are omitted from the VICAR label; PROPERTY keywords are used instead (as per the VICAR file format definition).
- For PDS labels, groups map 1-to-1 with the ODL label, but group names can differ. Name differences between groups are identified in Appendix A.
- Any set of PDS/ODL keywords not in a group (in PDS terms, a class) is identified by an introductory comment (e.g. `/* IDENTIFICATION DATA ELEMENTS */`). Such classes map 1-to-1 to a VICAR property set. The name of the VICAR property set and the name of the PDS introductory comment map as follows:

Table 3.2.4.1 – PDS/ODL Class to VICAR Property Set Mappings

PDS/ODL Class Comment	VICAR Property Set Name
<code>/* FILE DATA ELEMENTS */</code>	special case, see below
<code>/* POINTERS TO DATA OBJECTS */</code>	special case, see below
<code>/* IDENTIFICATION DATA ELEMENTS */</code>	IDENTIFICATION
<code>/* TELEMETRY DATA ELEMENTS */</code>	TELEMETRY
<code>/* HISTORY DATA ELEMENTS */</code>	PDS_HISTORY_PARMS
<code>/* COMPRESSION RESULTS */</code>	COMPRESSION_PARMS

- For VICAR labels, PDS/ODL comments (i.e., `/* string */`) are stored in a keyword named "PDS_COMMENT". This keyword appears in the VICAR property containing the elements immediately following the comment. When converting from VICAR to PDS/ODL, the comment is placed immediately before the group or class. Blank lines should surround the comment. Note that with IDS-generated EDR and RDR data products, multiple comment lines in a Group are not supported.
- The PDS/ODL objects IMAGE_HEADER and IMAGE, as well as the keywords in `/* FILE DATA ELEMENTS */` and the ^IMAGE and ^IMAGE_HEADER pointers (in `/* POINTERS TO DATA OBJECTS */`) in the table above, do not map directly to VICAR. They all describe the layout of the file and the image data. The VICAR equivalent for all of these items is the VICAR System label. Information maps between these in a straightforward way. It should be trivial to construct a VICAR system label and the above-referenced PDS entities after referring to the respective file-format-definition documents. Note that the `/* FILE DATA ELEMENTS */` and `/* POINTERS TO DATA OBJECTS */` comments are constant and so are not mapped to PDS_COMMENT keywords in the VICAR label. They are inserted automatically as part of the system label conversion process.

- The statistics-related keywords in the PDS/ODL IMAGE object are MEAN, MEDIAN, MAXIMUM, MINIMUM, STANDARD_DEVIATION, and CHECKSUM. These keywords are never transferred to the VICAR label. For VICAR -> PDS/ODL conversion, they can be computed from the image, or simply omitted from the PDS/ODL image (for RDRs only - EDRs require them).
- A few remaining items in the PDS/ODL_IMAGE object are treated specially. The FIRST_LINE, FIRST_LINE_SAMPLE, INVALID_CONSTANT, and MISSING_CONSTANT keywords are transferred to the VICAR IMAGE_DATA property set.
- Any PDS/ODL keyword with a *<unit>* tag after the value is transferred to the VICAR label without the unit tag. A VICAR keyword with the same name, but with "__UNIT" (two underscores) appended to the end, is added with the value of the unit. So for example, the PDS/ODL keyword "EXPOSURE_TIME = 1.5 <s>" would translate to two VICAR keywords: "EXPOSURE_TIME = 1.5" and "EXPOSURE_TIME__UNIT = s". Note that because of this, any PDS/ODL keyword that can support a unit is limited to 24 characters. If there is more than one value (an array), a unit is associated with each. In this case, the "__UNIT" VICAR keyword becomes multi-valued also, with each unit copied in sequence. If one of the elements does not have a unit (but others do), the corresponding entry is "N/A" (which is not copied to the PDS/ODL label). So for example, PDS/ODL "CONTRIVED_ANGLE = (1.2 <rad>, 22.0, 54.1 <deg>)" would map to VICAR "CONTRIVED_ANGLE = (1.2, 22.0, 54.1)" and "CONTRIVED_ANGLE__UNIT = (rad, N/A, deg)".
- The VICAR history label is omitted from the PDS header
- PDS/ODL TABLE objects (used to describe the SuperCam DPOs) map to VICAR property sets with each table, or container within the table (subobject), being a separate property set. The name of the property set matches the name of the table object (OBJECT = name). Certain keywords are handled specially, as described in the following table:

Table 3.2 – Referencing PDS/ODL Objects in VICAR Label

VICAR Label	PDS / ODL Meaning
PDS_OBJECT__TYPE=TABLE	Type of PDS/ODL object.
PDS_OBJECT__PTR=recnum	^pointer = (recnum + pointer to header area)
PDS_OBJECT__OFFSET=offset	Byte offset within record for pointer, if needed.
PDS_OBJECT__LOC=BINARY HEADER	Tells where the object is for proper PTR adjustment.
PDS_OBJECT__CHILD=(xxx,yyy)	Names of subobjects (PDS/ODL containers).
PDS_OBJECT__PARENT=aaa	Name of parent object.

Note: PDS_OBJECT__PTR is a 1-based record number within the binary header, so it must be adjusted to be relative to the start of the file for the ODS/ODL pointer.

- PDS/ODL ^pointer objects other than ^IMAGE and ^IMAGE_HEADER that point to tables or containers map to PDS_OBJECT__* keywords as described in the table above. Any remaining pointers, such as ^STRUCTURE, map to VICAR by removing the leading caret and appending "__PTR". Thus "^STRUCTURE" becomes "STRUCTURE__PTR" in VICAR.

3.3

3.4 Binary Data Storage Conventions

EDR and RDR data products for M2020 camera image are stored as binary data. For the image EDRs, the data formats include rescaled 8-bit integers stored in an unsigned byte, as well as 10-bit to 13-bit integers stored in signed 16-bit integers. The PDS and VICAR labels are stored as ASCII text.

3.4.1 Bit and Byte Ordering

The ordering of bits and bytes is only significant for instrument and binary header data; all other labeling information is in ASCII.

For non-byte instrument data, which includes 8-bit unsigned shorts, 16-bit signed shorts, 32-bit signed ints, and 32- and 64-bit IEEE floating-point numbers, the data may be stored in either Most Significant Byte first ("big-endian", as used by e.g. Sun computers and Java), or Least Significant Byte first ("little-endian", as used by e.g. Linux and Windows computers). In a EDR/RDR product, the instrument data can have only one ordering, but it is dependent on the host platform where the data was processed. Binary header data, applicable to SuperCam, can have a different ordering than the instrument data. This follows both the PDS/ODL and VICAR file format conventions.

For Engineering Camera, SuperCam RMI, Mastcam, and SHERLOC image data, the PDS/ODL label carries keyword `SAMPLE_TYPE` in the IMAGE Object to define which ordering is used in the file. The VICAR label carries keywords `INTFMT` and `REALFMT` in the System portion of the label to define the ordering. Both of these file formats specify that bit 0 is the least significant bit of a byte. See the respective PDS and VICAR file format definition documents.

Table 3.3 - M2020 Image EDR/RDR and Bit Ordering

Address	MSB-first	LSB-first
n	most significant byte	least significant byte
n+1	next	next
n+2	next	next
n+3	least significant byte	most significant byte

4. EDR PRODUCT SPECIFICATION

M2020 instrument EDRs and RDRs described in this document will be generated by JPL's Multimission Instrument Processing Laboratory (MIPL) under the IDS subsystem of the M2020 GDS Realtime Operations (RTO) element. Other RDRs TBD described in this document will be generated by the <Instrument Team(s) TBD>.

The EDR consists of unprocessed experiment data stored in binary format. The EDR structures defined in this document vary depending on instrument, with all containing attached metadata labels.

For M2020, the camera image EDR are the fundamental instrument data archive product. They will be generated as "raw" uncalibrated data within an automated pipeline process managed by MIPL under IDS at JPL. The size of an image EDR data product varies per instrument and is dependent on several factors (sub-framing, downsample, compression, etc.). The total estimated volume of image EDRs over the course of the nominal 2-year M2020 mission is approximately TBD Gigabytes.

4.1 EDR General Processing

The EDR processing begins with the reconstruction of packetized telemetry data resident on the TDS by the Mission data Processing and Control Subsystem (MPCS) into a binary ".dat" data product and associated ".emd" Earth meta-data file. The data product and meta-data are written by MPCS to the Operations Cloud Store (OCS) and messages are generated on a Java Message Server (JMS) bus, where they are ingested by MIPL's EDR generator "m2020edrgen" and processed with SPICE kernels provided by NAIF. The EDR will be generated within TBD seconds after the notification describing the OCS location of the respective the binary data product and associated Earth meta-data file has been received by the IDS pipeline system. This data flow is illustrated in Figure 4.1, and is elaborated subsequently in this section.

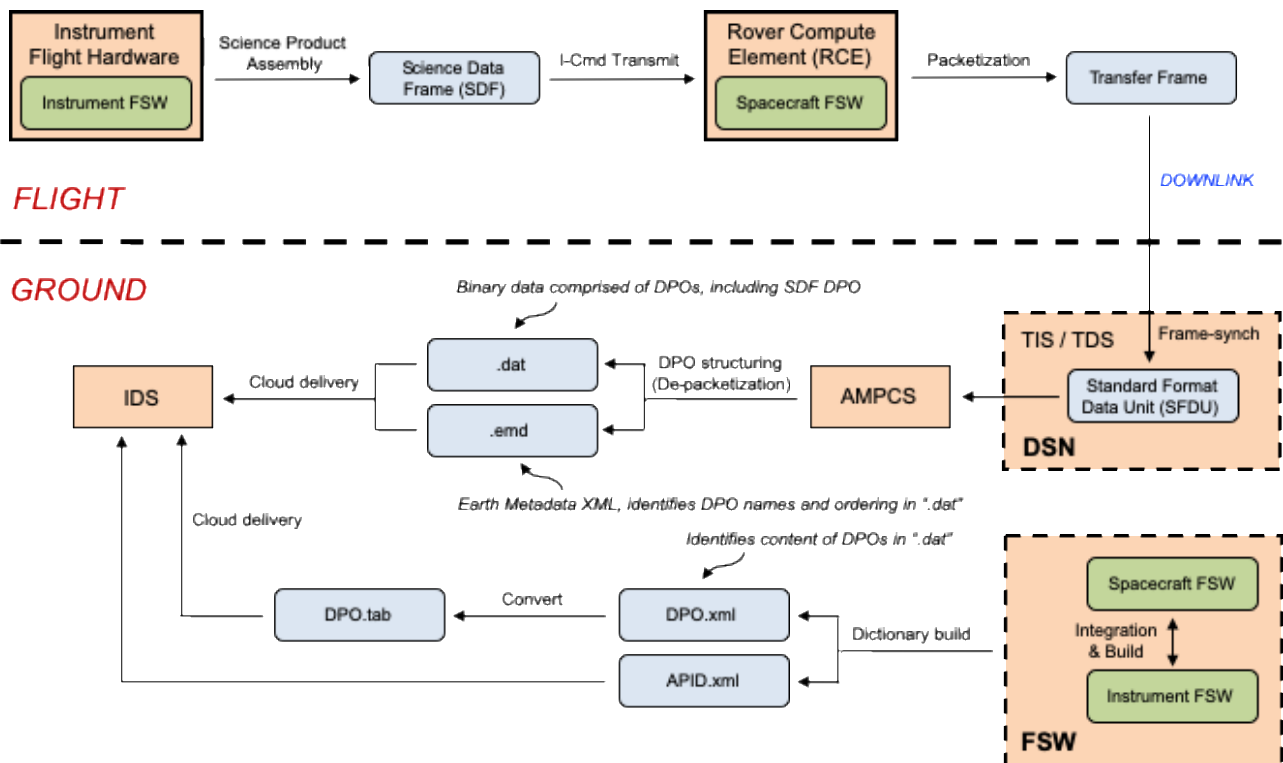


Figure 4.1 – Instrument Data Flow to IDS

In all EDR cases, missing packets will be identified and reported for retransmission to the ground as “partial datasets”. Prior to retransmission, the missing EDR data will be filled with zeros. The EDR data will be reprocessed only after all “partial datasets” are retransmitted and received on the ground. In these cases, the EDR version will be incremented so as not to overwrite any previous EDR versions. The EDR data product will be placed into FEI for distribution and to facilitate the archiving process.

4.2 EDR Product Types

Descriptions for the various EDR product types are provided in this section. They are broken down into two groupings: a) Image and b) Image Support. Refer also to Table 4.3 for a mapping between the source M2020 instrument and the EDR product type.

4.2.1 Image EDRs

4.2.1.1 Standard EDR

All of the imaging instruments create standard image EDRs. These images are uncompressed, raster-format data products. They are nominally 10 to 13-bit products, stored as 16-bit signed integers. If 12-to-8 bit scaling (called “companding”) is performed onboard, then pixels are decomanded as part of the EDR generation process. ECAM, SuperCam RMI, MCZ, SHERLOC-WATSON, and EDL cameras may consist of one or three bands, depending on instrument readout mode and any on-board color processing. PIXL MCC, MEDA Skycam, and SHERLOC-ACI are all

one band (grayscale). (LVS, Heli TBD.)

The ECAM Standard EDR is a special case because the detector can only be readout in tiles with a maximum size of 1280 lines by 960 samples. This results in an individual “Tile” EDR and a “Reconstructed” EDR, comprised of the individual “Tiles” readout from a single exposure.

4.2.1.1.1 Tile EDR

Individual image tile, as read out from ECAM detector, stored as lowest 12-bits of 16-bit signed integers. Nominal tile-size will be 1280x960 pixels. If 12-to-8 bit companding was performed onboard, the image pixels will be decompanded back to 12-bits and stored as lowest 12-bits of 16-bit signed integers. This EDR type only applies to ECAM.

4.2.1.1.2 Reconstructed EDR

Individual tiles from a single-exposure are combined together, reconstructing part or all of the original image acquired by the instrument. The reconstructed image will not necessarily be a “full” image and its geometry and color are dependent on what was saved on-board the Rover and sent back to Earth. This EDR type only applies to ECAM.

4.2.1.2 JPEG EDR

Images compressed using the JPEG algorithm for downlink. The JPEG EDR contains this JPEG as originally downlinked; it has not been uncompressed or recompressed. This type is not produced unless JPEG compression used onboard (i.e. no ground compression). It may be a color (three band) or grayscale (single band) JPEG. This EDR type applies to ECAM, SuperCam, MCZ, SHERLOC, PIXL, EDL cameras. (LVS, HELI TBD)

4.2.1.3 Z-stack EDR

Z-stack images are created onboard by combining images taken at different focus settings. The result is a best-focus image with a much greater depth of field. Each pixel contains data from the source image(s) that are in best focus (potentially interpolated between neighboring images). This image may be color or a single band. This EDR type applies to SuperCam, MCZ, and SHERLOC.

NOTE: The metadata regarding vehicle state for Z-stack images reflects the state of the vehicle at the time the Z-stack product itself was created. This is generally not the same as the vehicle state at the time the imagery was acquired. Most importantly, this means the arm state and mast state, and thus camera model, for these images do not properly describe the image. The metadata must be obtained from the thumbnail or full frame of one of the images that went into making the Z-stack. Determining the proper image to use is beyond the scope of this document, but can often be inferred by inspection of the available data.

4.2.1.4 Depth Map EDR

As part of Z-stack processing, a Depth Map image may also be produced. This 8-bit, single-band EDR indicates which image the best-focus data came from. Pixels with a digital number (DN) of 0 indicate the first image in the Z-stack, while a DN of 255 indicates the last, with linear scaling between. For example, with a 5-image stack and expected DNs of 0, 64, 128, 192, and 255, a DN of 96 would indicate the depth is halfway between images 2 and 3. This EDR type applies to SuperCam, MCZ, and SHERLOC.

NOTE: The metadata regarding vehicle state for Depth Map images reflects the state of the vehicle at the time the Depth Map product itself was created. This is generally not the same as the vehicle state at the time the imagery was acquired. Most importantly, this means the arm state and mast state, and thus camera model, for these images do not properly describe the

image. The metadata must be obtained from the thumbnail or full frame of one of the images that went into making the Depth Map. Determining the proper image to use is beyond the scope of this document, but can often be inferred by inspection of the available data.

4.2.1.5 Video EDR

The Video EDR is identical in format to the standard image EDR. It represents a single frame of a video sequence. The EDR type is separate in order to better distinguish video frames from still frames. This EDR type only applies to MCZ.

4.2.1.6 Recovered EDR

An Recovered EDR is an EDR for which the metadata is unavailable. It existed in camera memory but for whatever reason the onboard data product describing it was lost or deleted. Such products have extremely limited metadata, with a lot of "UNK" (unknown) values. This EDR type applies to MCZ and SHERLOC.

4.2.1.7 Image Sampling Types

Except where otherwise noted, in general, the image types described above can be sent in one of three pixel sampling types: 1) Full-frame, 2) Subframe, or 3) Downsampled. They may also be sent as a Thumbnail, in addition to (or instead of) the three aforementioned formats.

4.2.1.7.1 Full Frame EDR

Full-frame EDRs contain the entire contents of the imaging detector at full resolution.

4.2.1.7.2 Subframe EDR

Sub-frame EDRs are a subset of rows and columns of the full-frame image. They can be thought of as a window on the detector, with the same resolution but smaller coverage area.

4.2.1.7.3 Downsampled EDR

Downsampled EDRs contain a smaller version of the image, resulting in reduced resolution of the same coverage area. Downsampling can be done via one of three methods: 1) nearest neighbor pixel averaging, 2) pixel averaging with outlier rejection, or 3) computing the median pixel value. Note that downsampling can be applied to subframes.

4.2.1.7.4 Thumbnail EDR

Thumbnails are a reduced-resolution version of the original image, sent in addition to, or instead of, the original image. They apply to all image EDRs. The main purpose of a Thumbnail EDR is to provide an image summary using a very low data volume compared to the original image. Decisions about downlinking the original image can be made using the Thumbnail. For the engineering cameras and RMI, Thumbnails are produced relative to the full-frame image, even if the product is downsampled or subframed. For Mastcam-Z and SHERLOC camera images, thumbnails are of the subframe region only, and not the full-frame.

4.2.2 Image Support EDRs

4.2.2.1 Row Summation EDR

A row summation EDR is the summing of the rows of a full-frame or subframed image and returning the results. The EDR is a nx1 array of 32-bit integers (whose length is equal to the image height) where the DN value of the ith element is the value of the sum of all the pixels in the ith row. Applicable to the Engineering Camera instrument suite.

4.2.2.2 Column Summation EDR

A column summation EDR is the summing of the columns of a full-frame or sub-framed image and returning the results. The EDR is a 1xn array of 32-bit integers (whose length is equal to the image width) where the DN value of the ith element is the value of the sum of all the pixels in the ith column. Applicable to the Engineering Camera instrument suite.

4.2.2.3 Reference Pixel EDR

The onboard detector array has "Reference" dark pixels located before and after the image data. For the SuperCam RMI, there are 12 "pre-Reference" lines before the image lines and 12 "post-Reference" lines after the image lines. Also, there are 16 "pre-Reference" pixels before the image data in each row. Applicable to the SuperCam RMI.

4.2.2.4 Histogram EDR

The histogram EDR is a 32-bit integer array storing the histogram of the image. A 1x256 or 1x4096 array will be returned. Applicable to the Engineering Camera instrument suite.

4.2.2.5 Status Report EDR

TBD

4.2.2.6 MCC Key Point Translation Relative Navigation (TRN) EDR

TBD

4.2.2.7 MCC Structured Light Illuminator Estimate (SLI) EDR

TBD

4.2.2.8 MCC Centroids Image EDR

TBD

4.2.2.9 MCC Circle Centroid EDR

TBD

4.2.2.10 MCC Region of Interest (ROI) EDR

TBD

4.2.2.11 NavMap EDR

The NavMap products are engineering products used to assess the performance of the rover's on-board navigation software. All 3-char product identifiers starting with "N" (form: Nxx) are NavMap products.

The list of NavMap products and their short descriptions are provided in Table 4.3. Most, but not all, are images of some sort in the standard VICAR/ODL/PDS format with varying data types, sizes, and numbers of bands. Because these are detailed engineering products, no further description of them is provided in this document. They will not be archived.

4.2.3 Image Support EDRs

4.2.3.1 State of Health EDR

The SOH EDR is comprised of binary metadata describing the health and safety of SuperCam RMI. They are stored in 16-bit unsigned integer format. **TBD**

4.2.3.2 Status Report EDR

TBD

4.3 EDR Product Format

Description of EDR product formats in this section will be by instrument suite: a) Engineering Cameras, b) SuperCam RMI, c) Mastcam-Z, d) SHERLOC cameras, e) PIXL MCC, f) MEDA SkyCam, g) EDL cameras, h) LVS, i) Helicopter.

The EDR will be formatted according to this SIS, following the general terms of labeling and bit ordering previously discussed in Sections 3.2 and 3.3, respectively. This section details the specifics of a variety of formats across all images. The various EDR formats and their data sizes, across all instrument suites, are listed in Table 4.1 and discussed subsequently in this section:

Table 4.1 – List of EDR Types and Formats

Type	Product Identifier	Size (Lines x Samples)	Format (bits)	Instruments	Description
Full frame	EDR	• ECAM (Tile): 976 x 1296 • SCAM-RMI: 2048 x 2048 • MCZ: 1200 x 1648 • SHERLOC-WATSON: 1200 x 1648 • SHERLOC-ACI: 1200 x 1648 • PIXL MCC: 752x580 • MEDA Skycam: 1024 x 1024 • EDL Cams: N x M • LVS: N x M • Heli: N x M	16-bit signed integer	ECAM, SCAM-RMI, MCZ, SHERLOC-WATSON, PIXL MCC, MEDA Skycam	Nominal full sized, full resolution data product. If “12 to 8-bit” scaling was performed, the 12-bit data has been unscaled back to 12 bits, stored as the last 12 bits of a 16-bit integer.

Type	Product Identifier	Size (Lines x Samples)	Format (bits)	Instruments	Description
Subframe	EDR	Variable size	16-bit signed integer	ECAM, SCAM-RMI, MCZ, SHERLOC-WATSON, PIXL MCC, MEDA Skycam	Same format as Full Frame, but only a selected row (line) and/or column sub-frame is read back. The bit scaling rules described for the Full Frame case above also apply here.
Downsampled	EDR	Variable size	16-bit signed integer	ECAM, MEDA Skycam	Images are converted to smaller images via a) nearest neighbor pixel averaging, b) pixel averaging with outlier rejection, or c) computing the median pixel value. The bit scaling rules described for the Full Frame case above also apply here.
Thumbnail	EDR	Variable size	16-bit signed	ECAM, SCAM-RMI, MCZ, SHERLOC-WATSON, PIXL MCC, SkyCam, EDL	This data product is a spatially sized down version of an existing Full Frame, so is less than full size and less than full resolution. The bit scaling rules described for the Full Frame case above also apply here.
Companded (bit-scaled)	ECM	Variable size	8-bit unsigned	All cameras	Contains companded data either as originally downlinked or JPEG decompressed. It may be a color or grayscale.
JPEG	EJP	Variable size	8-bit unsigned	ECAM, MCZ, SHERLOC-WATSON, PIXL MCC, SkyCam, EDL	Contains JPEG-compressed data as originally downlinked. It may be a color or grayscale (single band) JPEG.
Z-stack	EZS	Variable size	8-bit unsigned	MCZ, SHERLOC-WATSON, RMI	A best-focus Z-stack image, from a combination of images at different focus settings, with a much greater depth of field. Typically SHERLOC-WATSON data. It may be color or a single band.
Depth map	EDM	Variable size	8-bit unsigned	MCZ and SHERLOC-WATSON	Part Z-stack processing. This single-band image indicates which image the best-focus data came from. DN of 0 indicates first image in the Z-stack, 255 is the last, with linear scaling between.
Video frame	EVD	Variable size	8-bit unsigned	MCZ and SHERLOC-WATSON	Identical in format to the Standard image EDR. It represents a single frame of a video sequence.
Imaging Support EDRs					
Row Summed	ERS	N lines x 1 sample	32-bit unsigned	ECAM, SkyCam	Array of 32-bit integers whose length is equal to image height, wherein the DN value for the Jth

Type	Product Identifier	Size (Lines x Samples)	Format (bits)	Instruments	Description
					element equals the sum of all pixels in the Jth row.
Column Summed	ECS	1 line x N samples	32-bit unsigned	ECAM	Array of 32-bit integers whose length is equal to image width, wherein the DN value for the Jth element equals the sum of all pixels in the Jth column.
Reference Pixel	ERP		16-bit unsigned		Dark pixels bookending (pre- and post-) image pixels during serial register readout. There are “pre-” Reference and “post-” Reference pixels.
Histogram	EHG	Variable size	32-bit unsigned	ECAM	DN histogram computed from image can have either 256 or 4096 bins, each capable of holding count values of up to 4,194,304.
IDPH only	EID	1 line x 1 sample	8-bit unsigned or 16-bit signed integer	ECAM	IDPH (Image Data Product Header) only, with no image data. The data is formatted as a 1x1 image with a 0 pixel value. This product is generated when the cameras are commanded to acquire a picture (for example, to pre-point the RSM), but no image data are requested from the camera. These types of products are intended to serve as metadata only and are typically used to identify RSM pre-point activities.
Status Report	E48 E49	Variable size		PIXL	
Key Point TRN	ESO	Variable size		PIXL	
SLI Estimate	ESF	Variable size		PIXL	
Centroid Image	ESA	Variable size		PIXL	
Circle Centroid	ESC	Variable size		PIXL	
Image ROI	ESR	Variable size		PIXL	
State of Health	SOH	Variable size		SCAM	
Navigation Map EDRs					
Navigation Map	TBD	TBD	TBD	TBD	Multiple types TBD

4.3.1 Engineering Camera Instrument Suite

The binary content of the Engineering Camera image EDR data product is a copy of the scene that had been projected onto the camera instrument’s CMOS detector and readout from the camera

memory buffer, as shown in Figure 2.X previously. The image data will be decoded and decompressed into two different EDR products: 1) Tile EDR and 2) Reconstructed EDR.

For the Engineering Camera image EDR, there are two possible formats of telemetered image data. In the first case, 12-bit data scaled onboard to 8-bit via a “12 to 8-bit” Lookup Table (LUT) or, by bit shifting, will be downlinked as 8-bit data and unscaled (decompressed) back to 12-bits and stored as the 12 lowest bits of the signed 16-bit integer. In the second case, 12-bit data without onboard LUT scaling or bit shifting will be downlinked as 12-bit data and stored “as is” in the 12 lowest bits of the signed 16-bit integer. The binary data may be returned in uncompressed or compressed form. Data returned as compressed are JPEG, ICER, or LOCO encoded and will be decompressed as part of the EDR processing.

4.3.2 SuperCam RMI EDR

The binary content of the RMI image EDR product is a copy of the scene that had been projected onto the camera instrument’s CMOS detector onboard and read out from the camera memory buffer, as shown in Figure X.x.Y.Y previously. Depending on the size of the image, the RMI science data could require up to 3 transfers from the instrument image buffer to the flight system. The flight system will concatenate the data together into one data product to be sent to Earth. The image data, telemetered on a variable number of bits (up to 13-bits uncompressed), will be decoded in single frame form and stored in the lowest 10 to 13-bits of a 16-bit integer as the image EDR. Data returned as compressed are ICER (color) or JPEG (color) encoded and will be decompressed as part of the EDR processing.

4.3.3 Mastcam-Z

The data from the MCZ cameras can come down in one of four different ways: color or grayscale JPEG, losslessly-compressed, or uncompressed. These methods apply across all of the image EDR types (standard image, Z-stack, depth map, video frame) and geometry types (full-frame, subframe, thumbnail).

4.3.3.1 Bayer Pattern

The Mastcam-Z cameras create images of 1648x1200 pixels. Each pixel on the CCD has an individual red (R), green (G), or blue (B) filter arranged in a Bayer pattern, which consists of repeated squares of pixels:

(R,G)
(G,B)

This allows an RGB color picture to be taken in a single exposure. Additional geology filters are available, which interact with the Bayer filters in complex ways. See Section 2.3 for an overview or the MCZ camera documentation for details.

Of the 1648 pixels per line, only 1608 are photoactive pixels. The line structure is shown in Figure 4.3.3.1 below, and is broken down into the following:

- 2 dark pixels from the end of the previous line (“P” in Figure 4.3.3.1)

- 1 invalid **ADC** pipeline pixel from the interline time (“A” in Figure 4.3.3.1)
- 4 isolation pixels (“I” in Figure 4.3.3.1)
- 16 dark pixels (“D” in Figure 4.3.3.1)
- 1608 photoactive RGB pixels (“R”, “G”, “B” in Figure 4.3.3.1)
- 17 dark, isolation, and overscan pixels (“X” in Figure 4.3.3.1)

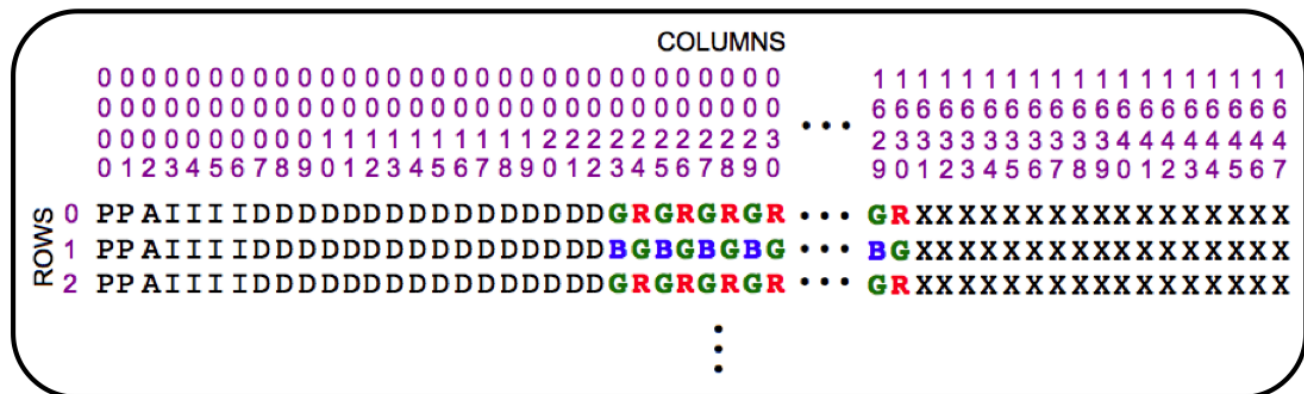


Figure 4.3.3.1 – Mastcam-Z RGB in Bayer Pattern Layout on CCD

Because there are an odd number of starting dark pixels, the first valid pixel (23) on even lines will be G (GRGRGR...) while on odd lines it will be B (BGBGBG...).

4.3.4 SHERLOC Camera Instrument Suite

The data from the SHERLOC cameras can come down in one of four different ways: color or grayscale JPEG, losslessly-compressed, or uncompressed. These methods apply across all of the image EDR types (standard image, Z-stack, depth map, video frame) and geometry types (full-frame, subframe, thumbnail).

4.3.4.1 WATSON

The SHERLOC-WATSON camera creates images similar to Mastcam-Z. See section 4.3.3.1 for details.

4.3.4.2 ACI

The SHERLOC-ACI camera creates images similar to WATSON, but there is no CFA on the detector and all images will be grayscale.

4.3.5 PIXL MCC

RAW – raw bit-mapped image in DP category 62

JPEG – JPEG compressed image in DP category 61

ROI – region of interest data in DP category 60

Circle Centroid Circle Centroid Struct in DP category 59

Centroids – SLI Centroid Image in Dp category 58

SLI – SLI Estimate Struct in DP category 57

TRT – Terrain Relative Translation estimate in DP category 56

4.3.6 MEDA Skycam

TBD

4.3.7 EDL Camera Suite

TBD

4.3.8 LVS

TBD

4.3.9 Helicopter

TBD

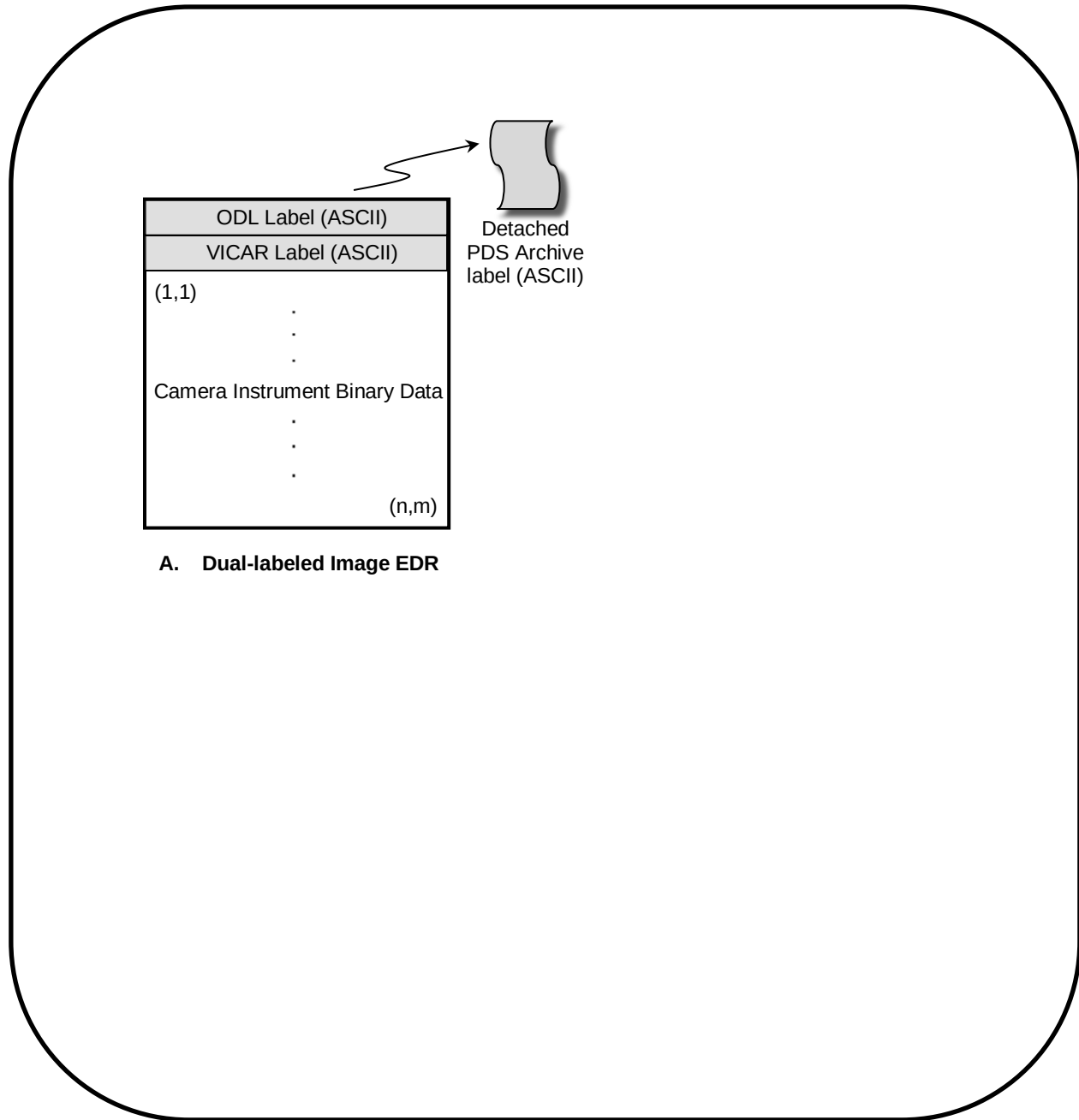
4.4 EDR Product Structure

As in the previous section “EDR Product Format”, description of EDR product structures in this section will be by instrument suite: Engineering Cameras, SuperCam RMI, MCZ Cameras, SHERLOC Cameras, PIXL MCC, MEDA SkyCam, EDL Cameras, LVS, Helicopter Cameras.

This section details the specifics of a variety of EDR structures, taking into account the concept of product labels and the product’s binary content described previously in Sections 3.2 and 3.3, respectively.

Figure 4.4 shows the format of an EDR: an image EDR that has a VICAR label wrapped by an ODL label, Diagram A. Shown with detached ASCII PDS-compliant labels for archive purposes.

For a description of the PDS and ODL labels, see Section 3.2.2, and for a description of the VICAR Label, see Section 3.2.3, and for a mapping between PDS/ODL and VICAR, see Section 3.2.4.



4.4.1 Engineering Camera Instrument Suite EDR

4.4.1.1 Cruise EDR

All image EDRs for the Engineering Camera instrument suite have the same structure, containing the four attributes shown in Diagram A of Figure 4.4. They are listed below. The first three attributes comprise the image file and consist of an integral number of “records”, where the record size is the

size needed to contain one line of image data. The fourth attribute is an external (detached) label file for archive purposes:

- 1) An operations label in ODL (ASCII) format
- 2) An operations label in VICAR (ASCII) format
- 3) An $n \times m$ block of binary image data with the origin at the upper left pixel in line (row) 1, sample (column) 1.
- 4) A detached PDS4-compliant label, in a separate file. Its availability to users will not coincide with the delivery of the EDR to the ODS file system; it will be delivered later with the release of the Archive Volume by PDS.

4.4.1.2 Surface EDR

All image EDRs for the Engineering Camera instrument suite have the same structure, containing the four attributes shown in Diagram A of Figure 4.4. They are listed below. The first three attributes comprise the image file and consist of an integral number of “records”, where the record size is the size needed to contain one line of image data. The fourth attribute is an external (detached) label file for archive purposes:

- 1) An operations label in ODL (ASCII) format
- 2) An operations label in VICAR (ASCII) format
- 3) An $n \times m$ block of binary image data with the origin at the upper left pixel in line (row) 1, sample (column) 1.

A detached PDS4-compliant label, in a separate file. Its availability to users will not coincide with the delivery of the EDR to the ODS file system; it will be delivered later with the release of the Archive Volume by PDS.

4.4.2 SuperCam RMI EDR

4.4.2.1 Cruise EDR

All Cruise image EDRs for the RMI have the same structure, containing the four attributes in Diagram A of Figure 4.4. They are listed below. The first three attributes comprise the image file and consist of an integral number of records, where the record size is the size need to contain one line of image data. The fourth attribute is an external (detached) label file for archive purposes.

- 1) An operations label in ODL (ASCII) format
- 2) An operations label in VICAR (ASCII) format
- 3) An $n \times m$ block of binary image data with the origin at the upper left pixel in line (row) 1, sample (column) 1.
- 4) A detached PDS4-compliant label, in a separate file. Its availability to users will not coincide with the delivery of the EDR to the ODS file system; it will be delivered later with the release of the Archive Volume by PDS.

4.4.2.2 Surface EDR

4.4.2.2.1 Image EDR

All Surface image EDRs for the RMI have the same structure, containing the four attributes in Diagram A of Figure 4.4. They are listed below. The first three attributes comprise the image file and consist of an integral number of records, where the record size is the size need to contain one line of image data. The fourth attribute is an external (detached) label file for archive purposes.

- 1) An operations label in ODL (ASCII) format
- 2) An operations label in VICAR (ASCII) format
- 3) An $n \times m$ block of binary image data with the origin at the upper left pixel in line (row) 1, sample (column) 1.

- 4) A detached PDS4-compliant label, in a separate file. Its availability to users will not coincide with the delivery of the EDR to the ODS file system; it will be delivered later with the release of the Archive Volume by PDS.

4.4.3 Mastcam-Z EDR

4.4.3.1 Cruise EDR

All Cruise image EDRs for MCZ have the same structure, containing the four attributes in Diagram A of Figure 4.4. They are listed below. The first three attributes comprise the image file and consist of an integral number of records, where the record size is the size need to contain one line of image data. The fourth attribute is an external (detached) label file for archive purposes.

- 1) An operations label in ODL (ASCII) format
- 2) An operations label in VICAR (ASCII) format
- 3) An $n \times m$ block of binary image data with the origin at the upper left pixel in line (row) 1, sample (column) 1.
- 4) A detached PDS4-compliant label, in a separate file. Its availability to users will not coincide with the delivery of the EDR to the ODS file system; it will be delivered later with the release of the Archive Volume by PDS.

4.4.3.2 Surface EDR

All Surface image EDRs for MCZ have the same structure, containing the four attributes in Diagram A of Figure 4.4. They are listed below. The first three attributes comprise the image file and consist of an integral number of records, where the record size is the size need to contain one line of image data. The fourth attribute is an external (detached) label file for archive purposes.

- 1) An operations label in ODL (ASCII) format
- 2) An operations label in VICAR (ASCII) format
- 3) An $n \times m$ block of binary image data with the origin at the upper left pixel in line (row) 1, sample (column) 1.
- 4) A detached PDS4-compliant label, in a separate file. Its availability to users will not coincide with the delivery of the EDR to the ODS file system; it will be delivered later with the release of the Archive Volume by PDS.

4.4.4 SHERLOC Camera Suite EDR

4.4.4.1 Cruise EDR

All Cruise image EDRs for SHERLOC have the same structure, containing the four attributes in Diagram A of Figure 4.4. They are listed below. The first three attributes comprise the image file and consist of an integral number of records, where the record size is the size need to contain one line of image data. The fourth attribute is an external (detached) label file for archive purposes.

- 1) An operations label in ODL (ASCII) format
- 2) An operations label in VICAR (ASCII) format
- 3) An $n \times m$ block of binary image data with the origin at the upper left pixel in line (row) 1, sample (column) 1.
- 4) A detached PDS4-compliant label, in a separate file. Its availability to users will not coincide with the delivery of the EDR to the ODS file system; it will be delivered later with the release of the Archive Volume by PDS.

4.4.4.2 Surface EDR

All Surface image EDRs for SHERLOC have the same structure, containing the four attributes in Diagram A of Figure 4.4. They are listed below. The first three attributes comprise the image file and consist of an integral number of records, where the record size is the size need to contain one line of image data. The fourth attribute is an external (detached) label file for archive purposes.

- 1) An operations label in ODL (ASCII) format
- 2) An operations label in VICAR (ASCII) format
- 3) An $n \times m$ block of binary image data with the origin at the upper left pixel in line (row) 1, sample (column) 1.
- 4) A detached PDS4-compliant label, in a separate file. Its availability to users will not coincide with the delivery of the EDR to the ODS file system; it will be delivered later with the release of the Archive Volume by PDS.

4.4.5 PIXL MCC EDR

4.4.5.1 Cruise EDR

4.4.5.1.1 Image EDR

All Cruise image EDRs for PIXL have the same structure, containing the four attributes in Diagram A of Figure 4.4. They are listed below. The first three attributes comprise the image file and consist of an integral number of records, where the record size is the size need to contain one line of image data. The fourth attribute is an external (detached) label file for archive purposes.

- 5) An operations label in ODL (ASCII) format
- 6) An operations label in VICAR (ASCII) format
- 7) An $n \times m$ block of binary image data with the origin at the upper left pixel in line (row) 1, sample (column) 1.
- 8) A detached PDS4-compliant label, in a separate file. Its availability to users will not coincide with the delivery of the EDR to the ODS file system; it will be delivered later with the release of the Archive Volume by PDS.

4.4.5.2 Surface EDR

4.4.5.2.1 Image EDR

All Surface image EDRs for PIXL have the same structure, containing the four attributes in Diagram A of Figure 4.4. They are listed below. The first three attributes comprise the image file and consist of an integral number of records, where the record size is the size need to contain one line of image data. The fourth attribute is an external (detached) label file for archive purposes.

- 1) An operations label in ODL (ASCII) format
- 2) An operations label in VICAR (ASCII) format
- 3) An $n \times m$ block of binary image data with the origin at the upper left pixel in line (row) 1, sample (column) 1.
- 4) A detached PDS4-compliant label, in a separate file. Its availability to users will not coincide with the delivery of the EDR to the ODS file system; it will be delivered later with the release of the Archive Volume by PDS.

4.4.5.2.2 Status Report EDR

4.4.5.2.3 Key Point Translation Relative Navigation (TRN) EDR

4.4.5.2.4 Structured Light Illuminator (SLI) Estimates EDR

4.4.5.2.5 Centroids Image EDR

4.4.5.2.6 Circle Centroid EDR

4.4.5.2.7 Image ROI EDR

4.4.6 MEDA SkyCam EDR

4.4.6.1 Cruise EDR

All Cruise image EDRs for MEDA SkyCam have the same structure, containing the four attributes in Diagram A of Figure 4.4. They are listed below. The first three attributes comprise the image file and consist of an integral number of records, where the record size is the size need to contain one line of image data. The fourth attribute is an external (detached) label file for archive purposes.

- 1) An operations label in ODL (ASCII) format
- 2) An operations label in VICAR (ASCII) format
- 3) An $n \times m$ block of binary image data with the origin at the upper left pixel in line (row) 1, sample (column) 1.
- 4) A detached PDS4-compliant label, in a separate file. Its availability to users will not coincide with the delivery of the EDR to the ODS file system; it will be delivered later with the release of the Archive Volume by PDS.

4.4.6.2 Surface EDR

All Surface image EDRs for MEDA SkyCam have the same structure, containing the four attributes in Diagram A of Figure 4.4. They are listed below. The first three attributes comprise the image file and consist of an integral number of records, where the record size is the size need to contain one line of image data. The fourth attribute is an external (detached) label file for archive purposes.

- 1) An operations label in ODL (ASCII) format
- 2) An operations label in VICAR (ASCII) format
- 3) An $n \times m$ block of binary image data with the origin at the upper left pixel in line (row) 1, sample (column) 1.
- 4) A detached PDS4-compliant label, in a separate file. Its availability to users will not coincide with the delivery of the EDR to the ODS file system; it will be delivered later with the release of the Archive Volume by PDS.

4.4.7 EDL Camera EDR

TBD

4.4.8 LVS EDR

TBD

4.4.9 Helicopter EDR

TBD

4.5 EDR Product Validation

Validation of the M2020 EDRs will fall into two primary categories: automated and manual. Automated validation will be performed on every EDR product produced for the mission. Manual validation will only be performed on a subset.

Automated validation will be performed as a part of the archiving process and will be done simultaneously with the archive volume validation. Validation operations performed will include such

things as verification that the checksum in the label matches a calculated checksum for the data product (i.e., that the data product included in the archive is identical to that produced by the real-time process), a validation of the PDS syntax of the label, a check of the label values against the database and against the index tables included on the archive volume, and checks for internal consistency of the label items. The latter include such things as verifying that the product creation date is later than the earth received time, and comparing the geometry pointing information with the specified target. As problems are discovered and/or new possibilities identified for automated verification, they will be added to the validation procedure.

Manual validation of the images will be performed both as spot-checking of data through-out the life of the mission, and comprehensive validation of a sub-set of the data (for example, a couple of days' worth of data). These products will be viewed by a human being. Validation in this case will include inspection of the image or other data object for errors (like missing lines) not specified in the label parameters, verification that the target shown / apparent geometry matches that specified in the labels, verification that the product is viewable using the specified software tools, and a general check for any problems that might not have been anticipated in the automated validation procedure.

***** RDR SECTION FOR FUTURE REFERENCE ONLY *****

5. RDR PRODUCT SPECIFICATION

RDR data products described in this document will be generated by MIPL personnel using the Mars Suite of VICAR image processing software within IDS at JPL and the SuperCam Science Team using team-developed software at Los Alamos National Laboratory (LANL) in New Mexico, Institut de Recherche en Astrophysique et Planétologie (IRAP) and Centre National d'Etudes spatiales (CNES) in France. The RDRs produced will be “processed” data. The input will be one or more image or spectrum EDR or RDR data products and the output will be formatted according to this SIS. Additional metadata may be added by the software to the product label.

There may be multiple versions of image and spectrum RDRs. The RDR data product will be placed into FEI for distribution.

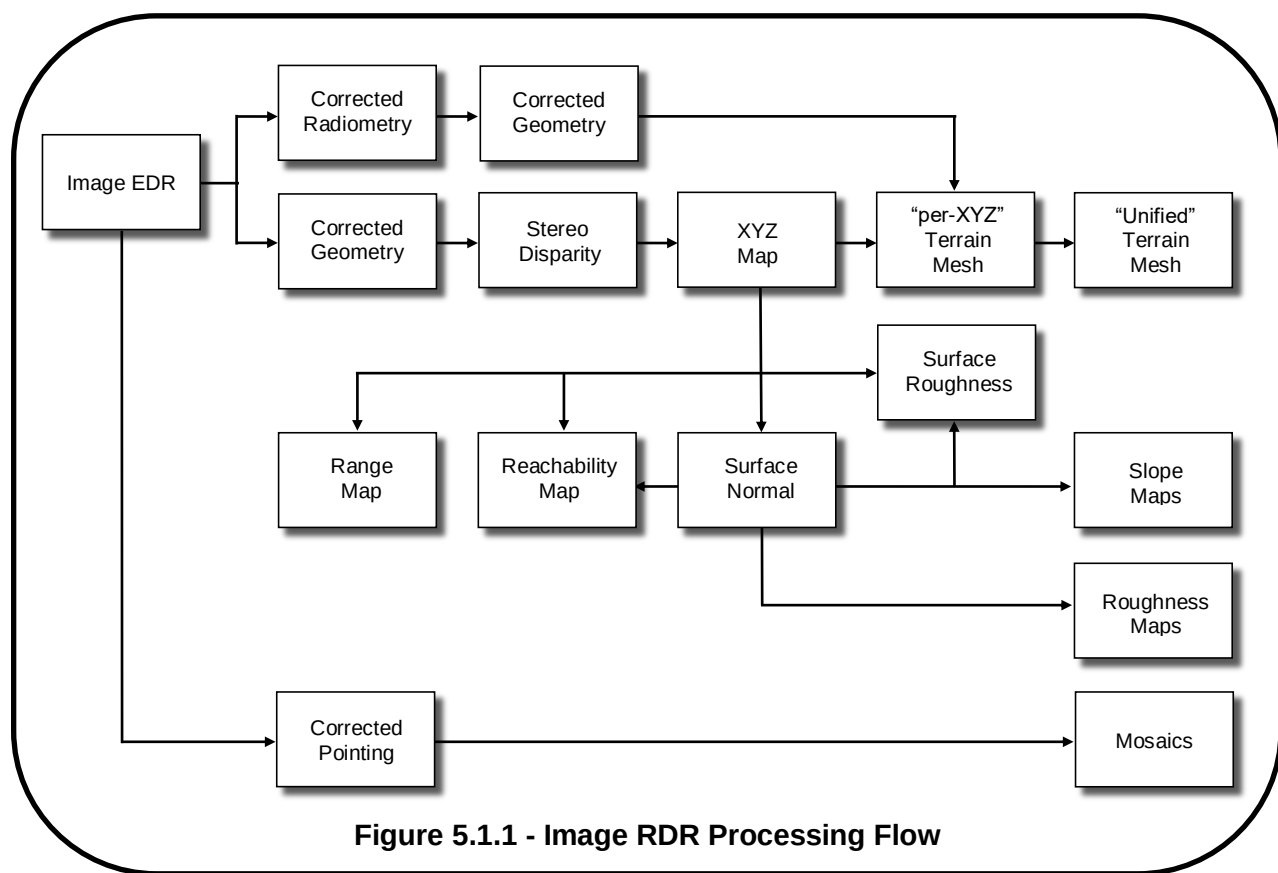
5.1 RDR General Processing

Processing is different for each image RDR type, as described in this section. In general, each RDR process inherits the metadata from its source EDR/RDR, modifying a portion of the metadata as necessary to reflect the subsequent output RDR.

5.1.1 Image RDRs

Image RDR data products described in this section will be generated by the EDR/RDR data product pipeline operating under the IDS at JPL. Pipeline processing will include

For Engineering Camera and MCZ camera data, and to a lesser degree SuperCam RMI data, RDRs will be processed using the Mars Suite subset of the VICAR image processing software developed at MIPL. The various image RDR interfaces in the VICAR processing flow are shown in Figure 5.1.1, and their descriptions are described subsequently in Section 5.2.



5.1.1.1 Common Processing

Although the various Engineering Camera and MCZ camera RDR product types are specific in nature, certain types of processing are often common to different RDR base files. These are described below.

5.1.1.1.1 Geometrically Corrected Images (Linearization)

EDRs and single-frame RDRs are described by a camera model. This model, represented by a set of vectors and numbers, permit a point in space to be traced into the image plane, and vice-versa. Linearization mode is indicated by the “geom” flag in the filename.

EDR camera models are derived by acquiring images of calibration targets with known geometry at a fixed azimuth/elevation. The vectors representing the model are derived from analysis of this imagery. These vectors are then translated and rotated based on the actual pointing of the camera to represent the conditions of each specific image. The resulting vectors make up the “camera model” for the EDR.

The Navcam, SuperCam, and MCZ cameras use a CAHVOR model, while the Hazcams use a more general CAHVORE model. Both model types are nonlinear and involve some complex calculations to transform line/sample points in the image plane to XYZ positions in the scene. To simplify this, the images are “warped”, or reprojected, in a process often called “linearization”, such that they can be described by a linear CAHV model. See Figure 5.1.1.1.1 for a visual comparison between a Front Hazcam image EDR (left) and the Front Hazcam “linearized” image RDR (right).



Figure 5.1.1.1.1 - Image EDR (left) vs “Linearized” Image RDR (right)

This linearization process has several benefits:

- 1) It removes geometric distortions inherent in the camera instruments, with the result that straight lines in the scene are straight in the image.
- 2) It aligns the images for stereo viewing, known as epipolar alignment. Matching points are on the same image line in both left and right images, and both left and right models point in the same direction.
- 3) It facilitates correlation, allowing the use of 1-D correlators in some cases.
- 4) It simplifies the math involved in using the camera model.

However, it also introduces some artifacts in terms of scale change and/or omitted data (see the references).

The linearized CAHV camera model is derived from the EDR's camera model by considering both the left and right eye models and constructing a pair of matched linear CAHV models that conform to the above criteria. For details on this algorithm see the references.

There are two types of linearization, indicated in the filename. For the nominal case, each image is linearized using a virtual camera model constructed to indicate what the other eye's model would look like at the same pointing. This allows each image to be processed independently, without the need to find the stereo partner (or even acquire the image), yet provides the same results as if the match had been performed.

However, this virtual partner is not always appropriate. In some cases the stereo pair is composed of images acquired at different pointings (often needed for very close-range mast work in order to achieve sufficient overlap), different positions (for an arm-based camera such as SHERLOC-WATSON), different rover locations (long-baseline stereo, used to resolve distances far greater than a normal stereo pair can do), different conditions (such as different focus or zoom on the Mastcam-Z), or even different instruments. These cases require linearizing with the actual stereo partner.

Regardless of linearization type, the image is then projected, or warped, from the CAHVOR/CAHVORE model to the CAHV model. This involves projecting each pixel through the EDR camera model into space, intersecting it with a surface (which matters only for the CAHVORE-based Hazcams and is a 1m radius sphere centered on the camera), and projecting the pixel back through the CAHV model into the output image.

C - The 3D position of the entrance pupil

A - A unit vector normal to the image plane pointing outward (towards C)

H - A vector pointing roughly rightward in the image; it is a composite of the orientation of the CCD rows, the horizontal scale, the horizontal center

V - A vector pointing roughly downward in the image; it is a composite of the orientation of the CCD columns, the vertical scale, the vertical center, and A.

If P is a point in the scene then the corresponding image locations x and y can be computed from:

$$x = \frac{(P - C) H}{(P - C) A}$$

$$y = \frac{(P - C) V}{(P - C) A}$$

For details on the camera model math and calibration and more description of the CAHV-model family, see references [Ref 18] through [Ref 27]. Note that the X and Y positions above are 0-based coordinates; i.e., the coordinate (0,0) is the center of the upper-left pixel. This is different than the common PDS convention of 1-based coordinates, where (1,1) is the center of the upper-left pixel, used elsewhere in this document.

5.1.1.1.2 Image Overlays

Many image RDR types represent some quantity other than intensity of light, such as XYZ or slope. The value at each pixel indicates the measurement of the quantity at the corresponding point in the original image. These types can be overlaid on a background of the EDR or other intensity image, using color coding to represent the RDR value. See Figures 5.2.1.4 (XYZ overlay), 5.2.1.5 (Range overlay) and 5.2.1.9.1 (Arm Reachability overlay) for examples. This gives a visualization of the RDR in context of the scene. Overlays are generally indicated by a product type ending with “O” in the filename.

5.1.1.1.3 Filled Images

Many RDR types do not achieve full coverage, e.g. the correlator is unable to find a solution at every point or the XYZ point failed the various filters. These “holes” are preserved in the RDRs using some value to indicate no solution (see the MISSING_CONSTANT labels). For ops work, it is critical to know where the holes are, so they are preserved in all nominal RDRs. Some RDRs created for purposes other than operations have these holes filled in using an interpolation mechanism. These are referred to as “filled” RDRs. Filled RDRs are generally indicated by a product type ending with “F” in the filename (but not all trailing “F”s mean Filled).

5.1.1.1.4 Color Processed Images

The MCZ cameras routinely produce color images via their Bayer-pattern CCD's. The de-Bayering is typically done onboard but can also be done on the ground if not. In any case, RDRs that contain image data, such as ILT, RAD, RAS, etc., are produced by pulling the three bands apart into separate files, processing each independently, and then combining them into color again. Both the color image and one or more of the individual bands are kept as end products, as specified in the Config flag (3rd character) in the filename.

RDRs related to image geometry (e.g. disparity, XYZ, all downstream products) are created using only the Green band extracted from the color image. The Green band is used because there are two green pixels per Bayer cell, as opposed to one for red and blue, leading to a higher resolution image.

Note that not all MCZ images are color; in particular some of the geology filters for the Mastcam-Z produce single-band images, which are processed like any other single-band image.

5.1.1.1.5 Rover Volume Exclusion Image Masks

For the purposes of Terrain Mesh RDR generation, IDS will create “Rover Volume Exclusion Mask” files that can be applied to several types of RDR (primarily XYZ). They are used to filter out rover features from generated terrain products, as well as the horizon and other undesirable features. They are single-band, byte files corresponding to the source image, where 255 indicates the corresponding pixel should be removed, or 0 indicates the pixel should remain in the output. These mask files typically have an “M” as the first character of the product type.

5.1.1.1.6 Masked Images

The exclusion masks, or other masks, can be combined with the RDR to create a Masked image. The contents are identical to the source RDR (most often, XYZ) except that where the mask is 255, the value is set to the value specified by MISSING_CONSTANT. Masked files typically have an “M” as the third character of the product type (exception: masked ARM is called ARK).

5.1.1.1.7 EDR-like RDRs

The LIN and BAY RDRs are exactly like EDRs except they have had linearization or de-Bayer processing done to them. This processing can be done to any image and does not change the product type code; the config, geom, and/or samp fields change instead. The LIN and BAY product type codes exist simply to ensure that the “EDR” product type code is not used for any RDRs.

5.1.2 Spectroscopy RDRs

Spectroscopy data products derived from the SuperCam LIBS spectrum EDR will be RDRs generated by software tools used and developed by SuperCam instrument team members at LANL. To generate a LIBS spectroscopy RDR product, the spectral data are extracted from the binary EDR file and converted to an ASCII format. Multivariate analyses is performed on the spectral data, and elemental abundances are reported using partial least squares regression (PLS).

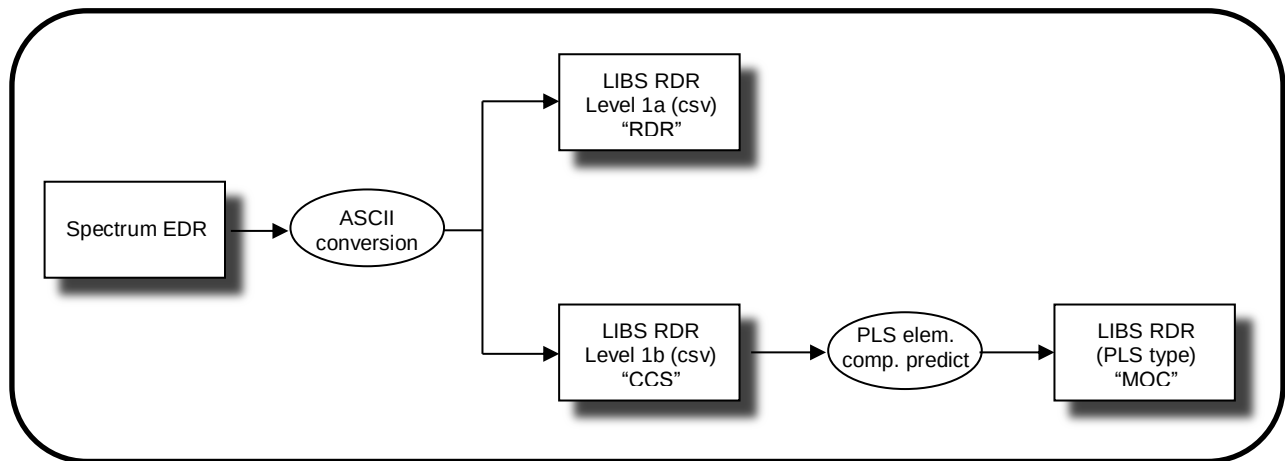


Figure 5.1.2 - Spectroscopy (LIBS) RDR Processing Flow

5.2 RDR Product Types

Descriptions for the various RDR product types are provided in this section. Refer also to Tables 5.4.1 and 5.4.2 for a concise list of the RDR product types.

5.2.1 Image RDRs

5.2.1.1 “Inverse LUT” RDR ***Deprecated as the EDR is now decomanded**

The ILT RDR is produced by IDS to reconstruct the original 12-bit pixels generated by the camera sensors. If the EDR is already in 12-bit format, ILT is simply a copy. However, if the EDR is in “8-bit” format (see Section 4.4) as a result of onboard “12 to 8-bit” scaling using a Lookup Table (LUT) or bit shifting, then an Inverse LUT (ILUT) is to be used to rescale the 8 lowest bits to the 12 lowest bits in the 16-bit signed integer.

The ILC type has had additional processing to remove certain instrument artifacts. At launch for M2020, this type is not used. However, operational experience with MER shows that over time, cameras degrade, with noise appearing due to radiation exposure and other effects. ILC is thus a placeholder for processing such as despiking, which removes this kind of noise, or interpolation of dead pixels.

The ILP type represents an ILT or ILC that has been pointing-corrected (see the Mosaic section) or re-localized based on ground analysis. It is expected that a special processing flag will be set for any products employing this type.

5.2.1.2 “Radiometrically Corrected” RDR

There are 3 different kinds of radiometrically corrected products. “RA*” have been corrected to absolute radiance units of $W/m^2/nm/steradian$. “RI*” products have been corrected for instrument effects only, and are in units of DN. “IO*” products are radiance factor (I/F) and are dimensionless. Within each kind, the values may be represented as floating-point values, or scaled to integers for ease of manipulation.

Independent of type, there are multiple methods of performing radiometric correction, distinguished by the `RADIOMETRIC_CORRECTION_TYPE` keyword. The two described here are CHEMRAD and MIPLRAD.

5.2.1.2.1 CHEMRAD Method

This refers to partial radiometric correction of SuperCam RMI image data performed using an image processing toolkit developed in IDL by SuperCam Science Team members at IRAP and CNES in France. It is deemed “partial” in the sense that the raw image data are not calibrated to physical radiance units. Although the calibration portion of radiometric correction is not performed for the input EDR, other corrections are applied. These corrections are a function of the image geometry, exposure time (milliseconds), sensor temperature (deg Celsius), and target distance (meters). The latter three are extractable from the SOH portion of the EDR file.

The main IDL procedure is `RMI_PROCESSING.pro`, which can apply successive corrections to RMI EDR image defects by incorporating, in a specific order, several lookup tables (LUTs) constructed during ground calibrations. The processing flow is shown in Figure 5.2.1.2.1.

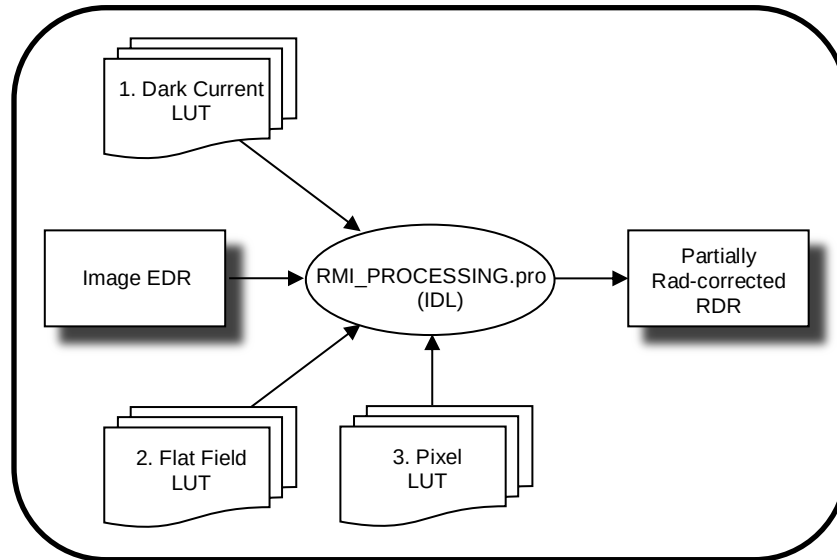


Figure 5.2.1.2.1 - Partial Radiometric Correction for SuperCam RMI

During the ground calibrations, several lookup tables were built with the constant values necessary to compute the corrections; they are included with the `RMI_PROCESSING.pro` code. These lookup tables will need to be revised with in-flight calibrations at a frequency to be defined.

The code applies corrective image processing steps in the reverse order of the apparition of the defects in the image, since one effect can influence the subsequent ones. In order, the corrections are as follows:

1. Subtract the estimated dark current image (temperature, time). This step incorporates a dark current LUT.
2. Iteratively (row by row) subtract the smearing drag (time) due to the CCD.
3. Iteratively (row by row) subtract the ghost image that is linked to the LIBS telescope and optics.
4. Divide by the estimated flat field that is linked to the LIBS telescope and optics. This step incorporates a flat field LUT.
5. Correct for known bad pixels by replacing them with median of neighboring pixels. This step incorporates a pixel LUT.
6. Apply a mask corresponding to the footprint of the telescope on the useful part of the CCD detector.
7. Enhance the contrast of the corrected image using histogram equalization.

The resulting RDR product is only partially radiometrically corrected and is not calibrated into physical radiance units. Note that the default flat field LUT is derived from a sky observation on Sol 32. Data acquired previous to Sol 32 were reprocessed using the Sol 32 flat field LUT.

Effects not accounted for in the correction process include:

- a) Non-linearity - Each pixel of the CCD is a potential well, in which electrons are trapped. The number of trapped electrons is an affine function of the lighting, but above a certain threshold the linearity is broken (saturation). At room temperatures, at gain 12, the non-linearity starts around 700 counts, and saturation is reached around 850 counts, i.e. an integration of about 10 ms. This response function of the detector is a complex function of the temperature and can vary from pixel to pixel. It has not been calibrated since the primary objective of the RMI is to give a qualitative context of the LIBS experiment.
- b) Direct Light - Some light from the far field can reach the detector without passing by the telescope mirrors, despite the baffle on the secondary mirror. It can be seen as extra light in the bottom part of some images, and sometimes as the focused picture of the distant scene. The diffuse light is taken into account in the determination of the reference flat field, and thus rather well corrected, but the focused distant scene cannot be corrected. However, this rare and faint, except in the corners that are removed when the telescope mask is applied.

5.2.1.2.2 MIPLRAD Method

This refers to radiometric correction of any camera instrument data systematically performed by MIPL (IDS at JPL) to meet tactical time constraints imposed by rover planners. The exception is SuperCam RMI data, which were processed only using the CHEMRAD method described in the previous section. The resulting rad-corrected RDRs are integrated into terrain mesh products used for traverse planning. This method is typically less precise than the methods used by the science teams.

MIPLRAD is a first-order correction only and should be considered approximate. MIPLRAD first backs out any onboard flat field that was performed. It then applies the following corrections: flat field, exposure time, temperature-compensated responsivity. The result is calibrated to physical units of $W/m^2/nm/sr$. The actual algorithm and equations used for MIPLRAD are shown below. Not every step applies to each camera type. Each correction is applied in sequence, to every pixel:

1. For the Engineering Camera instrument suite only, if on-board flat-fielding has been applied, it is backed out according to the parameters in FLAT_FIELD_CORRECTION_PARM, which is described in Appendix F and defines $ff(x,y)$.

$$\text{output}(x,y) = \text{input}(x,y) / ff(x,y)$$

2. For the flat-field adjustment, the x and y coordinates are adjusted based on downsampling and subframing to find the corresponding pixel in the flat field, then the DN is divided by the flat field value (obtained by pre-launch calibration):

$$\text{output}(x,y) = \text{input}(x,y) / \text{flat_field}(x',y')$$

3. Exposure time is then removed. Exposure time comes from EXPOSURE_DURATION, converted to seconds:

$$\text{output}(x,y) = \text{input}(x,y) / \text{exposure_time}$$

4. The temperature responsivity is removed next. The temperature of the CCD is determined from INSTRUMENT_TEMPERATURE using the following rules, where the first valid temperature found (0.0 is ignored as no-reading, and $>50C$ is considered broken) is the one used:

- a) Use the CCD temp of said camera
- b) Use the CCD temp of left/right partner

- c) Use the CCD temp of alternate A/B side same-eye camera
- d) Use the CCD temp of alternate A/B side left/right partner
- e) Use the CCD temp of “similar” camera (other Hazcam, other mast-mount) in a-d order
- f) Use the CCD temp of any camera
- g) Use electronics temp instead of CCD in a-f order

The temperature is combined with parameters R0, R1, and R2, which were derived from ground calibration and come from the flat field parameter file (see Appendix F) according to the following formula:

$$\text{output}(x,y) = \text{input}(x,y) * (R0 + R1*\text{temp} + R2*\text{temp}*\text{temp})$$

5. Finally, the result is (optionally) converted to integers using the RADIANCE_OFFSET and RADIANCE_SCALING_FACTOR keywords:

$$\text{output}(x,y) = (\text{input}(x,y) - \text{RADIANCE_OFFSET}) / \text{RADIANCE_SCALING_FACTOR} + 0.5$$

Note that the engineering cameras were not well calibrated radiometrically. Specifically, only flat fields were obtained, not temperature coefficients. Since they are build-to-print copies of the MER engineering cameras, the MER temperature responsivity parameters are used. For MCZ, no temperature compensation is applied by the MIPLRAD method.

5.2.1.3 “Disparity” RDR

A Disparity file contains 2 bands of 32-bit floating point numbers in the Band Sequential order (line, sample). Alternatively, line and sample may be stored in separate single-band files.

The parallax, or difference measured in pixels, between an object location in two individual images (typically the left and right images of a stereo pair) is called the “disparity”. Disparity files contain these disparity values in both the line and sample dimension for each pixel in the reference image. This reference image is traditionally the left image of a stereo pair, but could also be the right image. The geometry of the Disparity image is the same as the geometry of the reference image. This means that for any pixel in the reference image the disparity of the viewed point can be obtained from the same pixel location in the Disparity image.

There are three types of disparity. In the primary type (DS*), the values in the Disparity image are the 1-based coordinates of the corresponding point in the non-reference image. Thus, the coordinates in the reference image are the same as the *coordinates* in the Disparity image, and the matching coordinates in the stereo partner image are the *values* in the Disparity image. Disparity values of 0.0 indicate no valid disparity exists, for example due to lack of overlap, correlation failure, or parallax occlusion. This value is reflected in the MISSING_CONSTANT keyword. Such holes are rather common depending on the scene,

Disparity images can also be “delta” disparity (DD*), which measures the relative offset between coordinates in the two images. This is what most imaging scientists mean by disparity. These products are not produced in the nominal pipeline, but can be produced as special products. Missing values are flagged by the value specified in MISSING_CONSTANT, typically 0.0.

The third type is a “first-stage” disparity (DF*). This is a by-product of the two-stage MIPL correlation procedure [Ref 30] and represents the intermediate step between stages. It contains coordinate values, as in the primary disparity.

For each of the three primary types, several kinds of file can be produced: normal, line, sample, raw, error metric, and grid. Not all kinds apply to all types.

Normal disparity files contain 2 bands of floating-point numbers in (line, sample) order using the Band Sequential format. The line and sample components may be stored in separate single-band files.

Raw files contain the results before doing left->right and right->left reconciliation. They should be considered an intermediate result.

Disparity error metric files contain information about the quality of the correlation match. The tools to create them have not yet been developed; they are listed here as a placeholder for future expansion. Thus the format has not yet been determined as of this writing.

Grid overlays are an aid to visualization that may be created on occasion. These files are single-band byte images showing how a regular grid is distorted by the disparity matches (which is itself an indication of the terrain).

5.2.1.3.1 Stereo Pair Matching Method

Inherent in the designed operation of the stereo cameras is time-synchronization in the acquisition of left and right images intended for stereo processing. So, the SCLK timestamps of the respective left and right image acquired as a stereo pair onboard will be used to automatically identify them as a stereo image pair during nominal ground data processing.

Occasionally, stereo pairs will need to be processed that were not acquired simultaneously. This could be due to a sequence error, or for special operations such as re-pointing the mast between frames or driving the rover (long-baseline stereo). These off-nominal stereo pairs will be identified manually during ops, and processed as necessary. The special processing flag will be used to identify these. Additionally, most will use the "Actual" linearization method rather than "Nominal".

5.2.1.4 "XYZ" RDR

An XYZ file contains 3 bands of 32-bit floating point numbers in the Band Sequential order. Alternatively, X, Y and Z may be stored in separate single-band files as a X Component RDR, Y Component RDR and Z Component RDR, respectively. The single component RDRs are implicitly the same as the XYZ file, which is described below. XYZ locations in all coordinate frames for M2020 are expressed in meters unless otherwise noted.

The pixels in an XYZ image are coordinates in 3-D space of the corresponding pixel in the reference image. This reference image is traditionally the left image of a stereo pair, but could be the right image as well. The geometry of the XYZ image is the same as the geometry of the reference image. This means that for any pixel in the reference image the 3-D position of the viewed point can be obtained from the same pixel location in the XYZ image. The 3-D points can be referenced to any of the M2020 coordinate systems (specified by DERIVED_IMAGE_PARMS Group in the PDS label).

Most XYZ images will contain "holes", or pixels for which no XYZ value exists. These are caused by many factors such as differences in overlap, correlation failures, and the failure of a result to meet quality checks in the XYZ program. Realize this list of factors is only representative, and not exhaustive. Holes are indicated by X, Y, and Z all having the same specific value. This value is defined by the MISSING_CONSTANT keyword in the IMAGE object. For the XYZ RDR, this value is (0.0,0.0,0.0), meaning that all three bands must be zero (if only one or two bands are zero, that does not indicate missing data). Note that "0.0,0.0,0.0" is technically a legal value, but could occur at most once in an image and will rarely occur at all. The value is based on legacy software from previous

missions and is compatible with current mission software. Additionally, it is extremely unlikely that the value will be zero, which cannot be in the Site origin (0,0,0) is a single pixel value in the image would have

An XYZ Error float product X, Y and Z are aligned, which generally be completely correct. It is a 3-band product, each of the bands are axis-aligned, which will be correct.

XYZ files can be used to create a rover mask. ZZO for Z-band. For more detail, see Section 5.

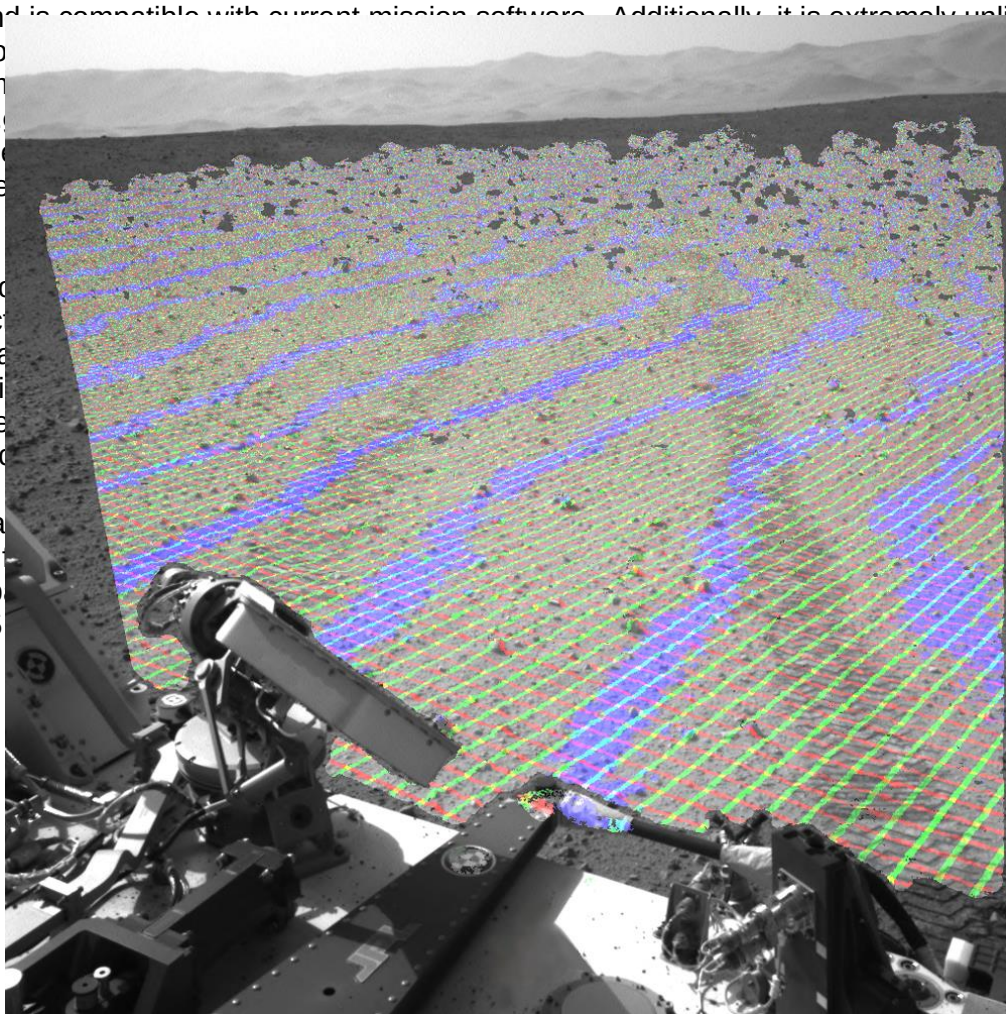


Figure 5.2.1.4 - XYZ Data Masked and Overlaid onto Image EDR

5.2.1.4.1 Rover Mask RDR

The MXY (mask for XYZ) file is a special kind of mask file, called a rover mask. This file is intended to mask off the rover, so that XYZ points correlated on the rover do not show up in terrains. It is created using a low-fidelity volumetric model of the rover (with some margin around the rover components), which is articulated based on the telemetered joint positions of the arm, wheels, and suspension. This model is then projected into the image to create a mask. The articulation means the mask will reflect the arm and wheel positions, minimizing the amount of good terrain that is masked off. If joint positions are not available, a "swept volume" is used, meaning the mask covers all possible positions of the articulating device. The HGA is always modeled as a swept volume.

Rover masks can also be generated using predicted arm joint positions; this is the basis of the "SuperCam finder" mosaics produced during operations. Such predicted-state masks are marked using the special processing flag field in the filename.

Associated with the MXY mask image is an XML file with the same base name. This file contains a polygonal representation of the articulated low-resolution volumetric model that is used to make the rover mask (before it is projected into the image). The format of this file is not described completely here, but it is straightforward XML, consisting of a large number of polygons in XYZ space.

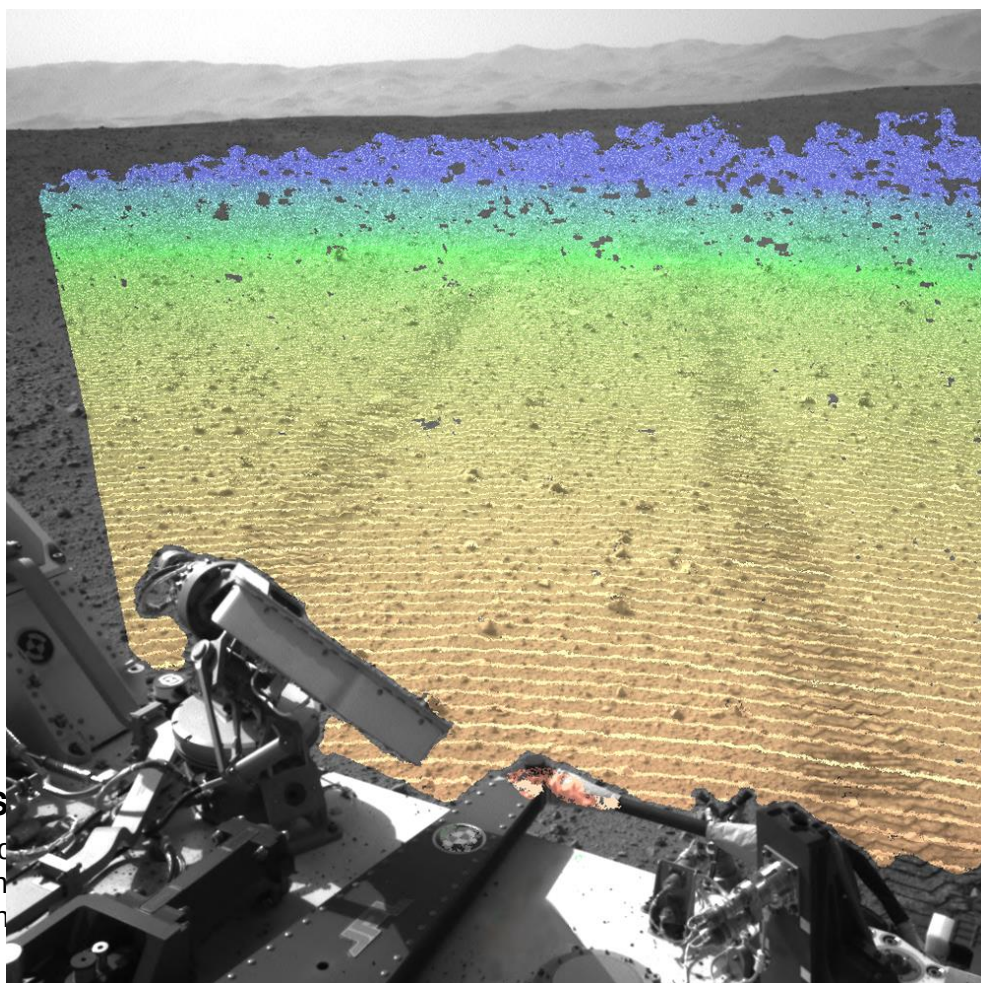
5.2.1.5 "Range" RDR

A Range (distance) file contains 1 band of 32-bit floating point numbers.

The pixels in a Range image represent Cartesian distances from a reference point (defined by the RANGE_ORIGIN_VECTOR keyword in the PDS label) to the XYZ position of each pixel (see XYZ RDR). This reference point is normally the camera position as defined by the C point of the camera model. A Range image is derived from an XYZ image and shares the same pixel geometry and XYZ coordinate system. As with XYZ images, range images can contain holes, defined by MISSING_CONSTANT. For M2020, this value is 0.0.

The Range Error metric (RNE) gives the estimated error of the XYZ point in meters for each pixel. Like XYE, it is a 3-band float product, whose three bands define the error ellipsoid (in meters). However, for RNE the bands are interpreted differently, in a way that more naturally represents the underlying error mechanism. The first band is the error in the downrange direction - radially away from the camera. This is the primary error for any stereo ranging system. The other two bands contain the cross-range error, orthogonal to each other and to the downrange vector direction. Band

2 (first crossrange) is coplanar with the camera stereo baseline and as such can generally be thought of as the "horizontal" crossrange direction, with band 3 the vertical, for common stereo geometries. Range products can become masked (RNM), as well as filled (RNA) and overlaid (RNO), as shown in Figure 5.2.1.5 below.



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The pixels in a UYW image correspond to the pixels in an XYZ file, with the same image geometry. However, the pixels are interpreted as a unit vector representing the normal to the surface at the point represented by the pixel. U contains the X component of the vector, V the Y component,

and W the Z component. The vector is defined to point out of the surface (e.g. upwards for a flat ground). The unit vector can be referenced to any of the M2020 coordinate systems (specified by the DERIVED_IMAGE_PARAMS Group in the PDS label).

Most UVW images will contain "holes", or pixels for which no UVW value exists. These are caused by many factors such as differences in overlap, correlation failures, and insufficient neighbors to compute a surface normal. Holes are indicated by U, V, and W all having the same specific value, defined by MISSING_CONSTANT as (0.0,0.0,0.0).

Two special kinds of surface normal products are defined based on MER experience. These are not expected to be used during nominal M2020 ops, but are included in case they become needed. The UVP type projects the surface normal onto an arbitrary plane in space, so the unit vectors always lie parallel to the plane. The UVT type provides the angle between the surface normal and the same arbitrary plane in a single band.

The UVW product contains the surface normal resulting from analyzing a small patch of pixels, which is appropriate for arm work. In contrast, the UVS product contains the surface normal resulting from analyzing a much larger patch of pixels (comparable to the size of the rover), which is appropriate for driving slope determination. Specific patch sizes are operational tuning parameters which have not yet been set, but will be provided in the history portion of the embedded VICAR label.

Surface normals can be filled (UVA) and overlaid (UVO). Separate U,V,W files can also be filled.

5.2.1.7 “Surface Roughness” RDR

The roughness maps, RUD (for the Drill) and RUT (for the Dust Removal Tool (BITCAR)) contain surface roughness estimates at each pixel in the image, along with a "goodness" flag indicating whether the roughness meets certain criteria.

For each pixel, the surface normal defines a reference plane. XYZ pixels in the area of interest are gathered, and the distance to the plane is computed. Outliers are thrown out. For the remainders, the minimum and maximum distances from the plane are found. Roughness is defined as the distance between this min and max (thus, is peak-to-peak variation within the area along the normal vector).

Two roughnesses are potentially computed. The first is an overall measurement containing all points within a radius of the central pixel. This is used for the BITCAR, and for the drill body. The second, used only for the drill, contains points within a ring between two radii. This is used for the drill stabilizer bars, and is not used for the BITCAR.

In each case, the computed roughnesses are compared to thresholds, which determine whether the point is "good" or not. The potential values of this goodness state are:

- 0.0 = No solution
- 1.0 = Both ring and overall roughnesses exceed thresholds
- 2.0 = Overall roughness (only) exceeds its threshold
- 3.0 = Ring roughness (only) exceeds its threshold
- 4.0 = Roughnesses within threshold (i.e. point is good)

Values of 1.0 and 3.0 appear only in the RUD product, since there is no ring for RUT.

The files are thus 2-band (RUT) or 3-band (RUD) float images, with the first band being the state, the second band being the overall roughness, and the third band being the ring roughness. For the second and third bands, 0.0 does not indicate a missing value (unlike most other products). Rather, 1.0 is used, as specified in MISSING_CONSTANT.

The default parameters and thresholds for these products are shown in Table 5.2.1.7 (all values in meters):

Table 5.2.1.7 - Surface Roughness Params and Thresholds

Param / Threshold	Product Type	
	RUD	RUT
Overall radius	0.075 m	0.035 m
Overall threshold	0.04 m	0.005 m
Ring inner radius	0.06 m	n/a
Ring outer radius	0.075 m	n/a
Ring threshold	0.015 m	n/a

5.2.1.8 Slope RDRs

The Slope-related RDRs represent aspects of the slope of the terrain as determined by stereo imaging. The Slope Map is derived from the UVS product, which contains the rover-sized surface normal at every point. There are several slope types, each of which can additionally be overlaid on a background.

In the equations below, u , v , and w are values from the UVS file, while x , y , and z are values from the XYZ file.

5.2.1.8.1 “Slope” (nominal) RDR

The SLP (SLO) type contains the slope in degrees for each pixel.

$$\text{slope} = \frac{180}{\rho} \arctan \left(\frac{w}{\sqrt{u^2 + v^2}} \right)$$

5.2.1.8.2 “Slope Rover Direction” RDR

The SRD (SRO) type contains the component of the slope (in degrees) that was facing the rover, i.e. if the rover went radially outward from its current position, this indicates the climb or descent. In the formula below, R is the rover’s position.

$$V = \frac{[x - R_x \quad y - R_y]}{\sqrt{(x - R_x)^2 + (y - R_y)^2}}$$

$$\text{srd} = - \frac{180}{\rho} \arctan \left(\frac{V_x u + V_y v}{-w} \right)$$

5.2.1.8.3 “Slope Heading” RDR

The SHD (SHO) type contains the direction of the slope as a clockwise angle from north, in degrees. Use the 4-quadrant form of arctangent to get a full 360-degree range.

$$\text{slope_heading} = \frac{180}{\rho} \tan^{-1} \frac{v}{u}$$

5.2.1.8.4 “Slope Magnitude” RDR

The SMG (SMO) type contains the magnitude of the normal unit vector projected onto the horizontal plane. It is directly related to sin(slope).

$$\text{slope_mag} = \sqrt{u^2 + v^2}$$

5.2.1.8.5 “Slope Northerly Tilt” RDR

The SNT (SNO) type contains the component of the slope in degrees that points north.

$$\text{northerly_tilt} = \frac{180}{\rho} \sin^{-1}(u)$$

5.2.1.8.6 “Solar Energy” RDR

The SEP (SEO) type is included as a placeholder. It is not used for M2020.

5.2.1.9 Arm Reachability RDRs

The Arm Reachability maps contain information about whether or not the instruments on the Arm can "reach" (contact or image) the object or location represented by each pixel in the scene, and how hard they can push ("preload"). They are derived from the XYZ and Surface Normal (UVW) products.

The geometry of the reachability maps match the reference XYZ, and Surface Normal (UVW) images, in that each pixel in the file directly corresponds to the pixel at the same location in the other products.

For the arm reachability map products, pixels with a DN of 0 denote an area where an arm instrument is unable to make contact (see Tables 5.2.1.9.1.1 and 5.2.1.9.1.2). To avoid ambiguity in the operational data store and archive, all arm reachability products are generated and archived, regardless of whether the reachability values are 0 or not. Consequently, there are arm reachability products comprised entirely of pixels with DN 0, that is, with no pixels indicating reachability. These products help document the decision process employed by the M2020 project when identifying contact science targets.

5.2.1.9.1 “Arm Reachability” RDR

The reachability map (ARM) encodes information for each of the 5 arm instruments in each of the 8 possible arm configurations, for a total 40 values per pixel. It is stored as a 5-band image of 16-bit integers in standard Band Sequential order. Each band represents one of the 5 arm instruments in the order defined by INSTRUMENT_BAND_ID. Within each band, the 16-bit integer contains 2 bits of information for each of the 8 configurations (in the order defined by CONFIGURATION_BIT_ID)

packed into the 16-bit value. The first mode in CONFIGURATION_BIT_ID is in the high-order 2 bits of the integer. The two bits represent 4 states: not reachable (0), and three levels of reachability, with 3 being the most easily reachable. Reachability is determined by checking if the contact, approach, and overdrive positions are reachable using the computed surface normal and 3 additional normals differing by a "tweak" angle from the computed one. Reachability level 3 indicates all checks pass, while 1 and 2 indicate that some did not and may require extra attention from the arm operator in order for the instrument to safely reach the point.

Table 5.2.1.9.1.1 - Arm Reachability Bit Assignments Per Configuration

16-bit Integer Bit Order (15=MSB, 0=LSB)							
15 - 14	13 - 12	11 - 10	9 - 8	7 - 6	5 - 4	3 - 2	1 - 0
Shoulder Out, Elbow Up, Wrist Up	Shoulder Out, Elbow Up, Wrist Down	Shoulder Out, Elbow Down, Wrist Up	Shoulder Out, Elbow Down, Wrist Down	Shoulder In, Elbow Up, Wrist Up	Shoulder In, Elbow Up, Wrist Down	Shoulder In, Elbow Down, Wrist Up	Shoulder In, Elbow Down, Wrist Down

Table 5.2.1.9.1.2 - Arm Reachability Values

Value		Description
Decimal	Binary	
0	00	Not reachable
1	01	Reachability quality 1
2	10	Reachability quality 2
3	11	Reachability quality 3 (best)

Note that reachability maps may include masks (MAR), be masked (ARK), and be overlaid (ARO) as shown in Figure 5.2.1.9.1 below.

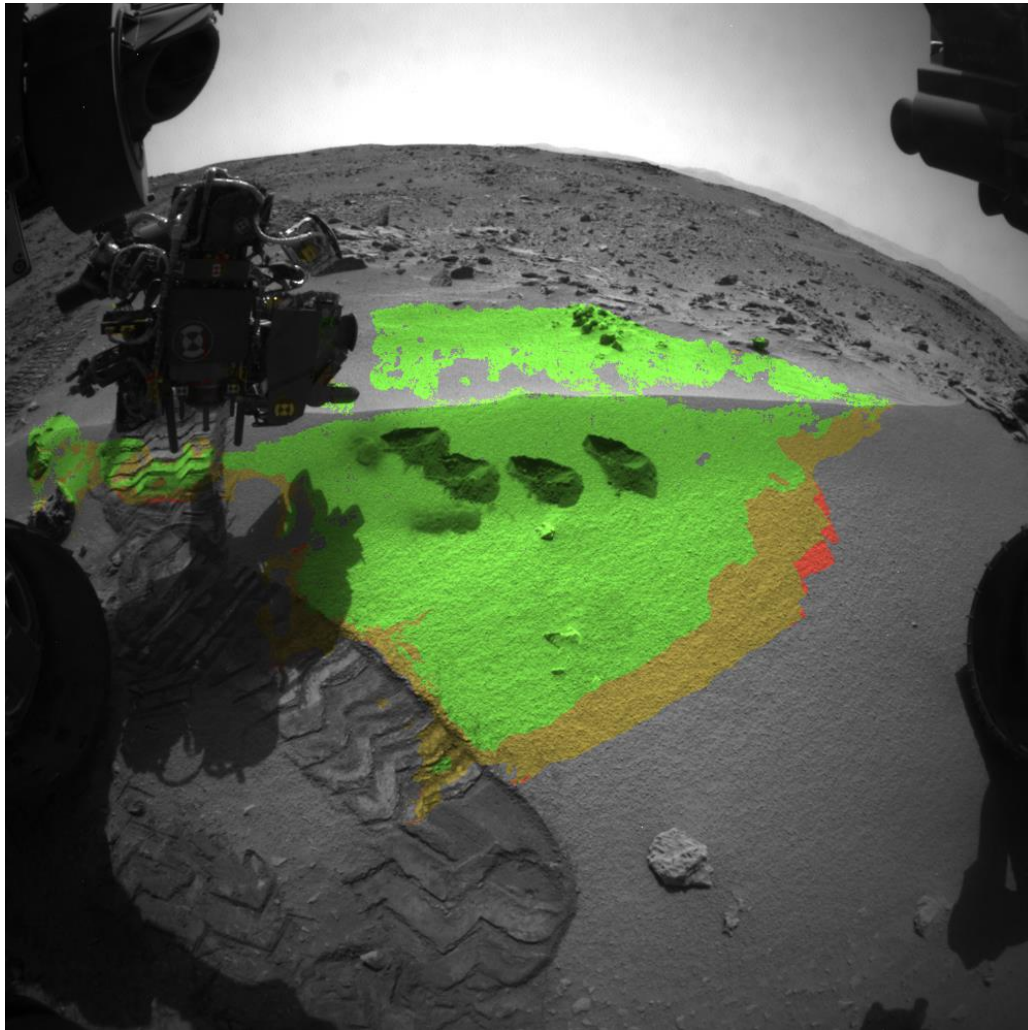


Figure 5.2.1.9.1 - Arm Reachability Qualities Color-coded, Overlaid onto Image EDR

5.2.1.9.2 “Arm Preload” RDR

The ARP type indicates the minimum and maximum preload values (in Newtons) that can be applied by the Drill instrument at the point represented by the pixel. This is a 2-band 16-bit signed integer products, where the bands represent (minimum, maximum) preloads (also defined by INSTRUMENT_BAND_ID).

5.2.1.10 Color RDRs TBD

The ECAM, SuperCam RMI, MCZ, SHERLOC, and EDL cameras contain a Bayer pattern of color filters on the detector. This is a repeating pattern of 4 pixels where each “cell” contains one red, two green, and one blue pixel (see Figure 4.3.3.1). This allows acquisition of color without using the filter wheel. In order to be generally useful, the image must normally be “de-Bayered” in one of several ways. The method of de-Bayering is indicated by the “config” field in the product filename, with

additional support from the “samp” field. Any time de-Bayering results in an image size change, the camera model must be adjusted to match.

For color images, the Bayer cells are typically extracted to separate color bands. This is indicated by values of “R”, “G”, or “B” for the “color” field. If the image is half-size in both dimensions as compared to the original (e.g. the RGB cells are simply extracted), then the “samp” field displays “TBD” to indicate Bayer subsampling. If the pixels are interpolated back to full size, the “samp” field is unchanged. Green pixels, being twice as numerous, present special challenges. For a full-size image, the “color” code is “TBD”. For Bayer-subsampled images, several cases are possible. If only the upper or lower green pixels are used from each cell, then the “color” values are “A” or “D”. If both are combined, the value is “TBD”. If both are present, so the image is twice as big in one dimension than the other, the value is “D”. Regardless of the extraction mode, if all three colors are merged back into a single 3-band image, the value is “F”.

For non-color images taken using geology filters, there are two additional options. Individual cells can be extracted as described above. They can also be averaged in a 2x2 pattern (potentially taking into account responsivity of the filter) to create one pixel per Bayer cell. This is indicated with a “color” value of “TBD”. Or, each pixel can be corrected for the combined responsivity of the Bayer and geology filters; this is indicated with a value of “TBD”.

Note that some de-Bayering is done onboard. This is reflected using the “color” code. 2x2 averaging or subsampling simply results in a downsampled or subsampled image, while JPEG creation results in a single 3-band EDR image. Thus for all EDRs, the “color” field represents the Bayer state.

5.2.1.11 “Photometry” RDR

The IEP type contains incidence, emission, and phase angles for each pixel for use in photometry work. It is a 3-band float product derived from UVW in the order (incidence, emission, phase). It can also be filled (IEA).

5.2.1.12 Terrain Map RDRs TBD

Terrain models are high level products which are derived using XYZ files and the corresponding image files. The XYZ files contain point clouds: sets of vertices in a specific coordinate system. The corresponding image files are used to obtain intensity or color information for each vertex in the point cloud. The terrain models are generated by triangulating point clouds using volume based surface extraction. The original image is used as a texture map to add detail and color to the polygonal surface representation. Terrain models are stored in Open Inventor binary format. Image textures are stored in SGI RGB format. Height maps (i.e., digital elevation maps, or DEMs) are also produced and used by the Rover Sequencing and Visualization Program (RSVP) for tasks which require simple and fast lookup such as rover settling. Height Maps are stored in VICAR format.

5.2.1.12.1 “Per-XYZ” Terrain RDRs TBD

For every XYZ RDR created, the following terrain products are generated and follow the Single-frame RDR filename convention (see Section 6.1.1):

- a) (*.tar) - A collection of tiles representing spatial subdivision of a point cloud. Each tile is a separate file within the tar-file. Each tile contains vertices that defines terrain in multiple Level of Details (LOD). From the vertices, triangles are striped for rendering efficiency. Note that tar-files are not used for unified mesh creation and though self-contained, serve only as intermediate products.

- b) (*.iv) - Open Inventor terrain representation of a point cloud defined in the XYZ RDR. It's a single file generated by combining all tiles contained in the tar-file described above and storing it as an Open Inventor binary file. It constitutes the per-XYZ mesh product.
- c) (*.rgb) - Image file in SGI RGB format that is used as the Texture Map for the per-XYZ mesh product.
- d) (*.ht) - Height Map (DEM) derived from the XYZ RDR, stored as an image file. It's in VICAR format but is not fully compliant to VICAR label specifications. It has 3 bands. Band 1 contains actual height data. Band 2 fills areas for which there is no actual data using interpolation. Band 3 provides metric of how close a pixel value is to the actual data.

Example:

Given the point cloud XYZ RDR named NLA_412403715XYZLF0060000NCAM15000M1.IMG and the corresponding image NLA_412403715RASLF0060000NCAM15000M1.IMG, the following files are created:

- NLA_412403715RASLF0060000NCAM15000M1.tar - Collection of vertices tiles in Open Inventor ASCII format representing spatial subdivision of the point cloud.
- NLA_412403715RASLF0060000NCAM15000M1.iv - Concatenation of all tiles into one Open Inventor file in binary format.
- NLA_412403715RASLF0060000NCAM15000M1.rgb - Texture Map image in SGI image format.
- NLA_412403715RASLF0060000NCAM15000M1.ht - Height Map image, with the following label items defining the spatial extent of the Height Map:

NL = 512	(number of lines)
NS = 442	(number of samples)
X_AXIS_MINIMUM = -39.906654	
Y_AXIS_MINIMUM = -8.1579	
MAP_SCALE = 0.087151	(Resolution at which Height Map has been generated)

5.2.1.12.2 “Unified” Terrain RDRs *TBD*

Just as individual images can be combined into image mosaics, per-XYZ meshes can be combined into unified terrain meshes. These are the ultimate terrain products used by rover planners during tactical operations. Per-XYZ polygonal surfaces are generated using XYZ RDRs defined in a Site frame. The tool RSVP extracts Site information from the unified mesh product filename to render terrains into the proper locations for rover traverse and arm placement applications, as shown in Figures 5.2.1.12.2.1 and 5.2.1.12.2.2.

The terrain products listed below follow the unified mesh product filename convention (see Section 6.1.3), which differs from the Single-frame filenames carried by the per-XYZ terrain products:

- a) (*.iv) - Open Inventor file in ASCII format that contains references to all individual binary per-XYZ Open Inventor files
- b) (*.mod) - ASCII file that contains references to all corresponding individual per-XYZ “.ht” Height Maps files.

Example:

Assuming the generation of multiple per-XYZ mesh products described and exemplified in Section 5.2.1.12.1, the following unified mesh files are created:

- N_L0168_RASLF_006_0000_AUTOGENM1.iv - Collection of references to Open Inventor

files including NLA_412403715RASLF0060000NCAM15000M1.iv

- N_L0168_RASLF_006_0000_AUTOGENM1.mod - Collection of references to Height Maps including N_L0168_RASLF_006_0000_AUTOGENM1.ht

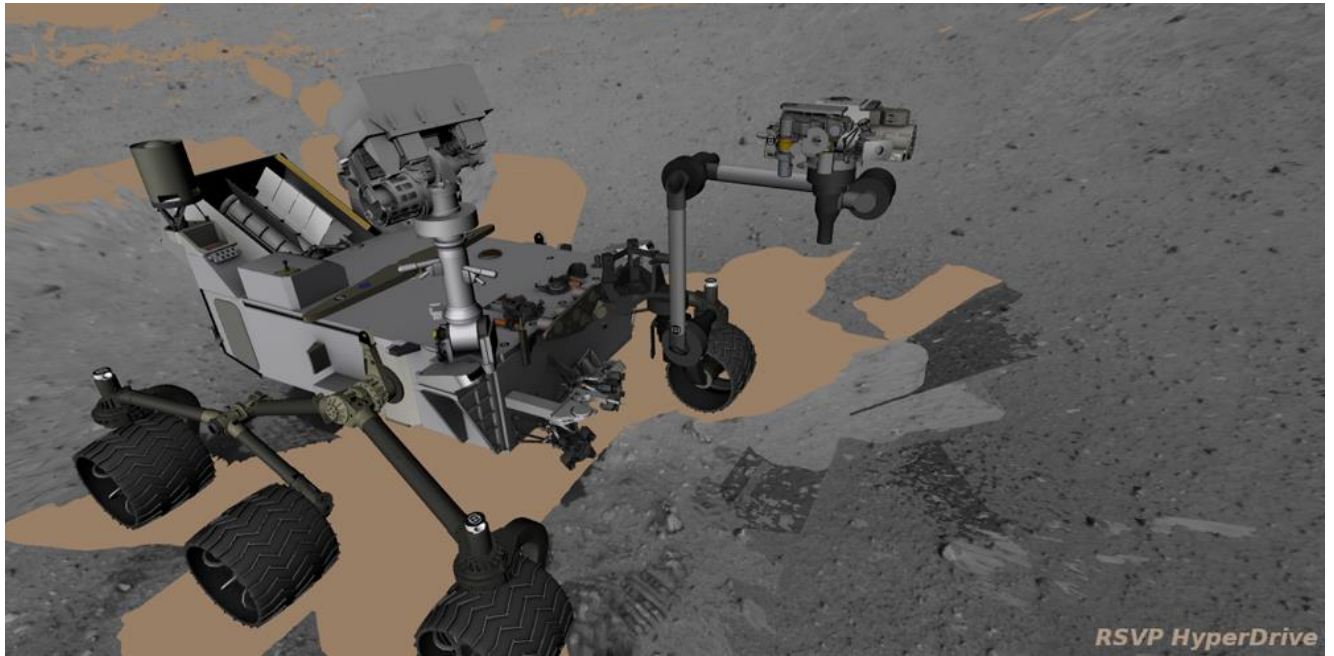


Figure 5.x – Rover location rendered in a unified terrain mesh

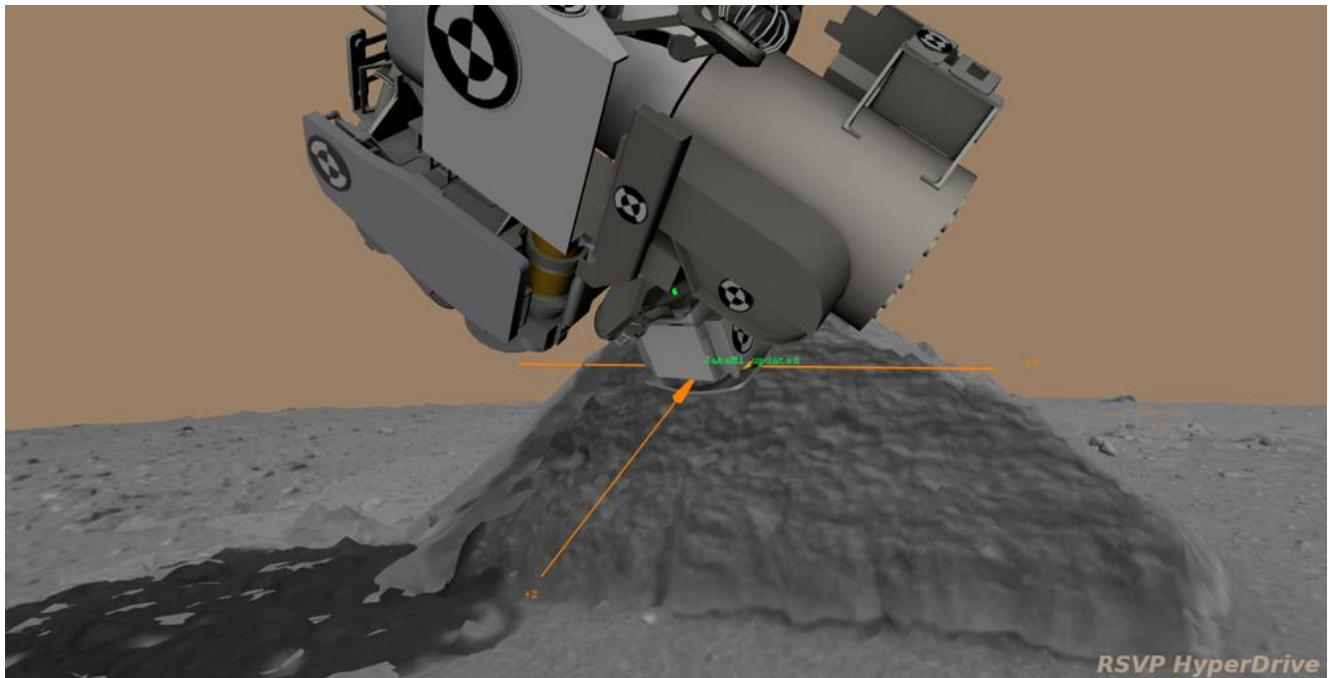


Figure 5.X – Arm payload location rendered in a unified terrain mesh

5.2.1.13 Mosaic RDRs

This section discusses the process of mosaicking multiple frames into a single RDR product using some projection. The text largely reflects the methods applied by MIPL under IDS. It should be noted that governing methods and software can differ between IDS and other operations subsystems or science instrument teams; the algorithms followed by other teams may not be the same as described here.

5.2.1.13.1 Overview of Mosaics in General

Mosaics can embody several important properties, making them very useful products. They assemble small pieces into a larger field of view. Certain projections create a level horizon, removing rover tilt. Mosaics can be calibrated so directions such as north and east can be determined - or they can be made relative to the rover to visualize forward and right. They can provide overhead views (Vertical, Polar, or Orthorectified projections) to help understand the local environment. They can be made from different types of data (such as slope). They can combine different filters to create color. Finally, they are the signature products for public outreach.

Mosaics can be assembled autonomously by tracing a view ray from each mosaic location or pixel into the scene, determining its intersection with a surface model (typically a ground plane), and then querying each input image to determine if that point lies within its field of view. In this fashion mosaics containing perhaps hundreds of images can be assembled for each spectral band. It may be necessary to refine the camera pointing in order to produce accurate mosaics. This requires the determination of the actual azimuth and elevation of each image in order to correct for errors such as gear backlash. One way to do this is to acquire tiepoints between all pairs of overlapping images. Camera pointing parameters are then estimated which cause the camera model to map the tiepoints

to their correct locations. In many cases this can be accomplished automatically, but it often requires human intervention to select tiepoints because of small overlap, parallax, or changing lighting.

5.2.1.13.2 Mosaicking Method

The process used by the MIPL software to create mosaics is described below. It consists of several sub-steps. Conceptually, one can think of the process as adjusting the pointing of the inputs, projecting them down to a surface, and looking at the result from a different point of view (the output projection). In reality, the process is run in reverse for ease of interpolation (this is described below).

- A. **Pointing Correction** - An optional (but important) first step in mosaic production is pointing correction. This is used to minimize geometric seams (discontinuities) between frames. The results of pointing correction are used in mosaics, but they can also be fed back into the RDRs (often at the ILUT stage), resulting in adjusted XYZ and other derived values and corrected meshes.

There are several methods by which improved pointing of the cameras can be determined. The most common method is to pick tiepoints between image pairs, either automatically or with manual assistance, and use those in a global cost function minimization to determine the corrected pointing parameters. Another possibility is to analyze the shape of XYZ data in the overlap region, again using an error minimization process to derive updated pointing parameters. Pointing parameters can also be determined manually.

Regardless of method, the result is encapsulated in a pointing correction or "nav" file. A more detailed description of this file is provided in Section 5.2.1.13.3, but fundamentally, this file contains, for each image being corrected, the original pointing parameters and the revised pointing parameters.

Pointing parameters are simply those numbers which represent how the camera is pointing in the rover frame, reduced to available degrees of freedom. These are used as inputs to the kinematics procedures which derive the camera model. The set of pointing parameters, together with the kinematics algorithm, is referred to as a pointing model. The job of a pointing model is to take a calibration camera model and transform it using the pointing parameters to create a transformed camera model which represents the specific image in question.

A given camera may have multiple available types of pointing models. The mast-mounted cameras have a standard model with two parameters: azimuth and elevation actuator angles. However, another model is available with three: azimuth, elevation, and "twist", which is a rotation around the camera's A axis (A being one of the CAHV camera model parameters). For SHERLOC-WATSON, there are two models: one with six parameters (the XYZ position plus the three Euler angles describing the orientation) and one with seven (XYZ plus the four components of a quaternion). The Hazcam and MARDI models have zero parameters, since they are rigidly attached to the rover body. Other pointing models may exist as well.

The set of available pointing model types and their full descriptions are outside the scope of this SIS; they are defined in a PDS documentation file.

For the mosaic process, the "nav" file is sufficient to describe the pointing parameters. However, if other corrected RDRs need to be produced, such as XYZs or meshes, then the pointing parameters must be stored in the label. This is accomplished via the POINTING_MODEL_NAME and POINTING_MODEL_PARAMS labels. When recomputing a

camera model, if these labels are present they should be used in preference to the normal method of pointing via labels in the ARTICULATION_DEVICE_STATE groups. For most users, however, the GEOMETRIC_CAMERA_MODEL should be used directly; this will be updated properly with respect to the corrected pointing.

- B. **Output Projection Determination** - The output projection is then determined. The parameters describing the projection are listed in Appendix A, and described in detail in Appendix F. The output projection parameters are determined by analysis of the inputs to give the "best" resulting mosaic, but can be overridden by the user. The determination process is outside the scope of this document; the results are what is important and they are in the label.
- C. **Surface Determination** - A surface model is critical for mosaics. This is a mathematical surface, which approximates the actual scene. To the extent that the scene differs from the surface model, distortion and uncorrectable seams due to parallax can result.

Usually the surface model is a flat plane, with normal pointing upwards. This can be adjusted, however, to better match the scene. Regardless, the results are documented in the SURFACE_MODEL_PARMS group.

There are five potential surface models in the MIPL software: PLANE, INFINITY, SPHERE, SPHERE1, and SPHERE2. See SURFACE_MODEL_TYPE in Appendix F for description. Note that an appropriate surface model is often determined automatically as part of the pointing correction process, and the surface model can be stored in the "nav" file. Almost all mosaics are created using the PLANE model.

Note that the parameters (surface normal and ground point, for PLANE) for an appropriate surface model are often determined automatically as part of the pointing correction process, and the surface model can be stored in the "nav" file.

- D. **Computation of Output View Ray** - For each pixel in the output mosaic, a view ray in 3-D space is constructed. How this view ray is constructed depends on the projection type. In this section, the pixel is at location (i,j) in 0-based coordinates, with i corresponding to sample and j to line. (0,0) is in the upper-left-hand corner. Capitalized values represent PDS label items from the SURFACE_PROJECTION_PARMS group. Unit and coordinate system conversions are applied as necessary but are not specified here. The coordinate system used is defined by REFERENCE_COORD_SYSTEM_* in SURFACE_PROJECTION_PARMS.
- E. **Projection from Output to Surface** - Once the view ray is determined, it is projected out until it intersects with the surface model. The resulting point in XYZ space is used in the next step. If the ray does not intersect the surface, the point is assumed to be at infinity in the direction the view ray is pointing. Exception: as mentioned below, the Vertical projection will reverse the direction of its view ray; infinity is assumed only if they both miss.

Note that the INFINITY surface model guarantees the ray will miss the surface at all times.

The difference between the SPHERE1 and SPHERE2 models is that, if the ray intersects the spherical surface more than once, SPHERE1 will take the first intersection, while SPHERE2 will take the second. For normal rover situations, SPHERE1 thus roughly models a convex hill, while SPHERE2 roughly models a concave crater when the rover is outside the sphere.

- F. **Projection from Surface to Input** - The XYZ location (or direction for the infinity case) is then back-projected into each input camera model in turn, using the corresponding input camera model. The first input for which the resulting pixel coordinate is inside the image (excluding border pixels which are thrown away) and non-0 stops the process; that is the image from which the output pixel value is taken. Values of 0 in the input image are ignored, with the effect that they are transparent.

This process results in stacking the images such that the first one in the input list of images "wins". There is no feathering of overlaps; the first image is "on top" of all the others, and an image completely covered by preceding images will not be used at all.

- G. **Interpolation and Storage of the Result** - Finally, a bilinear interpolation is performed on the input image, based on the 4 pixels surrounding the back-projected location. The result of this interpolation is the value of the output pixel.

Bilinear interpolation is optional, but is normally done for image mosaics. Mosaics of other data types such as XYZ or Surface Normal (UVW) generally have interpolation turned off to avoid aliasing from interpolation with invalid pixels.

5.2.1.13.3 Mosaic Ancillary Files

A number of ancillary files are used to support mosaicking, and contain parameters and information describing how the mosaic was produced. With these, it is possible to maintain traceability and provenance for each pixel in a standard mosaic back to the source image. On PDS-released archive volumes, the ancillary files have the same basename as the mosaic to preserve one-to-one matching with the mosaic, albeit with a different extension. On the operational data store (ODS) for operations, they might not necessarily have the same mosaic basename, as several mosaics might share an ancillary file. In such cases, the ancillary file's name may indicate a different product type, projection, eye, filter/color, or geometric or brightness correction than the target mosaic's filename. The other filename fields should always match.

It should be noted that many if not most mosaics are produced at least partially by hand, which explains most of the inconsistencies noted below. The general case is described, but as with any hand work not all conforms exactly.

These files are not described completely here, but we hope the descriptions are sufficient to be able to decipher them:

- a) **List files** - With a ".LIS" extension, list files are simple text files containing the names of the images making up the mosaic, one per line. The first name in the list references the image frame that is "on top" in the mosaic product, covering the image frames that are referenced in the list below it. The list files often contain full pathnames to disks on the operational data store (ODS), directory paths which are not part of a PDS-released archive volume. However, the filenames themselves, minus directory paths, usually will be part of an archive volume. Occasional mosaics may have list files comprised of names for private copies of images (e.g. with "/home" in the pathname) that are the result of different scenarios of special processing: specified and unspecified. In the former case, such image files will be in an archive volume with filenames that carry a character flagging the type of special processing. In the latter case, no special character is present in the filename, though the file's name and/or metadata label will identify the ultimate source image of the unspecified processing. Note that a mosaic's list file content of filenames (minus directory paths) is also referenced in the mosaic's label using keyword INPUT_PRODUCT_ID.

- b) **Nav files** - With a “.NAV” extension, nav files are XML files describing the pointing corrections that have been applied to images in a mosaic, as well as the surface model. The prologue contains identifying information. Note that the “static_parameters” file is listed as “M2020:M2020_pma.point” in some nav files; this should be corrected to “M2020:M2020_mast.point” and refers to the calibration file used to find mast kinematics parameters.

Pointing correction works by applying a set of pointing parameters (e.g. mast azimuth and elevation) to a kinematics algorithm and using that to re-point the camera model. See definition of keyword POINTING_MODEL_NAME in Appendix F for more information.

For each image (<solution> element), the Site and Drive components of the RMC are listed, followed by image identifier information. This identifier information contains the original pointing parameters, which allows the same correction to be applied to e.g. the left and right eyes (irregardless of image ID information).

Following that are the updated pointing parameters, and then the revised (re-pointed) camera model. In rare cases, nav files are edited by hand, which puts the accuracy of the camera model update at risk.

At the end is usually a “surface_model” element describing the surface model determined by the MIPL software program MARSNAV (which creates the nav files). This information is repeated in the SURFACE_MODEL_PARMS group of the mosaic product’s label.

- c) **Tiepoint files** - With a “.TIE” or “.TPT” extension, tiepoint files are XML files containing image tiepoints used as input to program MARSNAV. These can be automatically or manually selected. The prologue relates each image ID to a key, which is used throughout the remainder of the file.

Each tiepoint has a left and right key, and then 1-based coordinates in the corresponding files of the tiepoint. Just <left> and <right> should be used; <projected> has little value. In the flags, “quality” represents the quality of correlation match when the tiepoint was correlated (scale 0-1). The “interactive” flag has little archival value. It does not (as the name suggests) indicate whether the tiepoint was automatically selected or manually tweaked.

Tiepoint type “0” is by far the most common and is a standard image tiepoint. Other types are rarely used and their descriptions are beyond the scope of this document. Full descriptions are in the help documentation for program MARSNAV.

- d) **Brightness Correction files** - With a “.BRT” extension, brightness correction files are XML files containing information used to correct the brightness and contrast of images in a mosaic relative to one another. They are similar in concept and structure to nav files. After the prologue, each image has one <brt_solution>. The most common correction type, LINEAR, specifies an overall additive and multiplicative factor to be applied to each image (MULT is applied first, then ADD). These factors are echoed in the IMAGE_RADIANCE_FACTOR and IMAGE_RADIANCE_OFFSET keywords in the mosaic label. The HSI_LIN type is similar, except the correction (for color images) is done in Hue-Saturation-Intensity (HSI) space, with the correction applied to Intensity only. See also BRIGHTNESS_CORRECTION_TYPE.
- e) **Brightness Overlap files** - With a “.OVR” extension, brightness overlap files are XML files containing information about image statistics in overlap areas, used to create the brightness correction files. They are similar in concept to tiepoint files, except the “tiepoints” are the mean and standard deviation of small areas of overlapping pixels for the mosaic.

They start with a prologue defining the image ID-to-key mapping, as with nav files. Each overlap then has a number of images involved in the overlap, the number of pixels, and a “radius” which is a general description of the maximum size of the overlap. This is followed by the key, mean and standard deviation of the overlapping area in each image. The line and

sample coordinates are provided for an arbitrary point in the overlap, just to help locate where the overlap is. The actual shape of the overlap is not specified.

Overlap type “0” is a standard overlap. Type “1” gives the mean and standard deviation not of



$$\text{azimuth} = i / \text{MAP_RESOLUTION} + \text{START_AZIMUTH}$$

$$\text{elevation} = (\text{ZERO_ELEVATION_LINE} - j) / \text{MAP_RESOLUTION}$$

The view ray emanates from the point PROJECTION_ORIGIN_VECTOR.

Figure 5.2.1.13.4 shows such a mosaic overlaid onto azimuth and elevation grid lines, with individual frame boundaries superimposed and annotated by number. In this case each pixel represents a fixed angle in azimuth and elevation. Rows are of constant elevation in the selected coordinate frame. In this case, a Site frame was used, so the horizon is level, and columns begin clockwise from Mars north.

Figure 5.2.1.13.4 - Cylindrical Projection Mosaic

5.2.1.13.5 “Camera Point Perspective Mosaic” RDR

Figure 5.2.1.13.5 shows a Camera Point Perspective mosaic. It is a perspective projection with horizontal epipolar lines. The mosaic behaves as though the “camera” which acquired the image frames was an instrument with a much larger field of view. For M2020, this type of mosaic is typically in the Rover Frame and thus may have a tilted horizon if the rover is not level.

Point-perspective mosaics give the most natural view of small areas and are suitable for stereo



Figure 5.2.1.13.5 - Camera Point Perspective Mosaic**5.2.1.13.6 “Cylindrical-Perspective Projection Mosaic” RDR**

Cylindrical-Perspective mosaics are used for large stereo panoramas, and work across a full 360 degrees of azimuth. Stereo is preserved because a baseline separation is maintained between the camera eyes at different azimuths.

This projection is the most complicated projection to create. Each column i (counting from 0) in the output mosaic is assigned its own camera model. This is done in several steps:

- 1) Compute initial camera model. This model is a CAHV linearized model derived from the first input to the mosaic, re-pointed to azimuth 0 and elevation PROJECTION_ELEVATION. This model is stored in the GEOMETRIC_CAMERA_MODEL label group.
- 2) The instantaneous field of view of the "central" pixel (at the point where the A vector intersects the image plane) is computed using the formula:

$$\text{ifov} = \text{atan}(1.0 / |(\vec{H} - \vec{A} * (\vec{H} \cdot \vec{A}))|)$$

where the “•” indicates the scalar dot product of the two vectors A and H.

Alternatively, this can be derived from the image size and azimuthal extent (where the azimuths are adjusted by 360 degrees such that the result is minimally positive):

$$\text{ifov} = (\text{STOP_AZIMUTH} - \text{START_AZIMUTH}) / \text{LINE_SAMPLES}$$

- 3) The azimuth of the column is computed:

$$\text{azimuth} = \text{START_AZIMUTH} + i * \text{ifov}$$

- 4) The initial camera model is re-pointed using kinematics as described above under the pointing correction section, using the above azimuth and PROJECTION_ELEVATION. This results in the final camera model for the column.

Step 4 is difficult to duplicate for reconstructing the set of camera models. For that reason, an alternate method is described in this paragraph. The resulting models are exact for mast-mounted cameras with no backlash correction; they are a close approximation for other cases. In general, for mast-mounted cameras, the C points of the column camera models describe a ring in space, whose diameter is approximately the baseline between the cameras. This ring is described by PROJECTION_ORIGIN_VECTOR (center), PROJECTION_AXIS_OFFSET (radius), and CAMERA_ROTATION_AXIS_VECTOR (orientation of the ring axis). These together simulate the kinematics motion of a mast-mounted camera in Rover frame. To compute the camera model for the azimuth defined in Step 3, take the camera model from the label, and rotate the entire camera model around the camera rotation axis by the azimuth amount, using the ring center as the pivot point. The

C point will remain on the ring, while the camera pointing (close to but not identical to the A vector) will remain approximately tangent to the ring at that point. After this, compute the rotation required to transform CAMERA_ROTATION_AXIS_VECTOR into PROJECTION_Z_AXIS_VECTOR (which can be done by taking the cross product to get the rotation axis and the dot product to get the rotation amount). Then rotate the camera model by this amount, again using the ring center as the pivot point. This has the effect of tilting the entire ring so it is perpendicular to the PROJECTION_Z_AXIS_VECTOR. This last rotation is often used to remove the effect of rover tilt, resulting in a flat horizon with the camera model baselines (vector between the left and right eyes) aligned with the horizon (technically, perpendicular to the Z axis in Local Level frame). For this “untilt” case, the PROJECTION_Z_AXIS_VECTOR is the Local Level frame’s Z axis expressed in Rover Nav frame. Note that PROJECTION_ELEVATION and PROJECTION_LINE are measured before this “untilt” rotation takes place, so they end up describing a sinusoid in the final mosaic when untilt is used.

Once the camera models have been defined, the mosaic proceeds through each pixel as with the other projections. The view ray is computed as described below (A, H, and V come from the column’s camera model):

$$\begin{aligned} x_center &= \vec{A} \cdot \vec{H} \\ y_center &= \vec{A} \cdot \vec{V} \\ samp &= x_center \\ line &= y_center + j - PROJECTION_ELEVATION_LINE \end{aligned}$$

where the “•” indicates the scalar dot product of two vectors. This (samp,line) coordinate is then projected into space using the column’s camera model, and this projection becomes the view ray. The origin of the view ray is the column’s C point. See Section 5.4.1 and references [Ref 18] through [Ref 27] for the mathematics of camera models.

Figure 5.2.1.13.6 shows a Cylindrical-Perspective projection in which a 360 degree view can be viewed in stereo. This is a perspective projection similar to Figure 5.2.1.13.5 except that the mosaic acts like a pinhole camera which follows the mosaic in azimuth while maintaining camera baseline separation. If the mosaic is generated with no tilt correction (i.e., CAMERA_ROTATION_AXIS_VECTOR and PROJECTION_Z_AXIS_VECTOR are the same) and the rover is tilted, the horizon will not be level, instead being sinusoidal. This preserves epipolar alignment and allows for better stereo viewing of the panorama. However, for aesthetic reasons,



5.2.1.13.7 “Polar Projection Mosaic” RDR

Polar mosaics create a quasi-overhead view that still allows viewing all the way to the horizon.

MIPL creates the Polar projection by computing the azimuth and elevation of the view ray as follows:

```

x = i - SAMPLE_PROJECTION_OFFSET
y = LINE_PROJECTION_OFFSET - j
range = sqrt(x*x + y*y)
elevation = range / MAP_RESOLUTION - 90 degrees
azimuth = REFERENCE_AZIMUTH + (90 degrees - atan2(y, x))

```

The view ray emanates from the point PROJECTION_ORIGIN_VECTOR.

Figure 5.2.1.13.7 shows a Polar projection. Concentric circles represent constant projected elevation. Mars nadir is at the convergent center and the horizon is corrected for lander tilt. North is up.

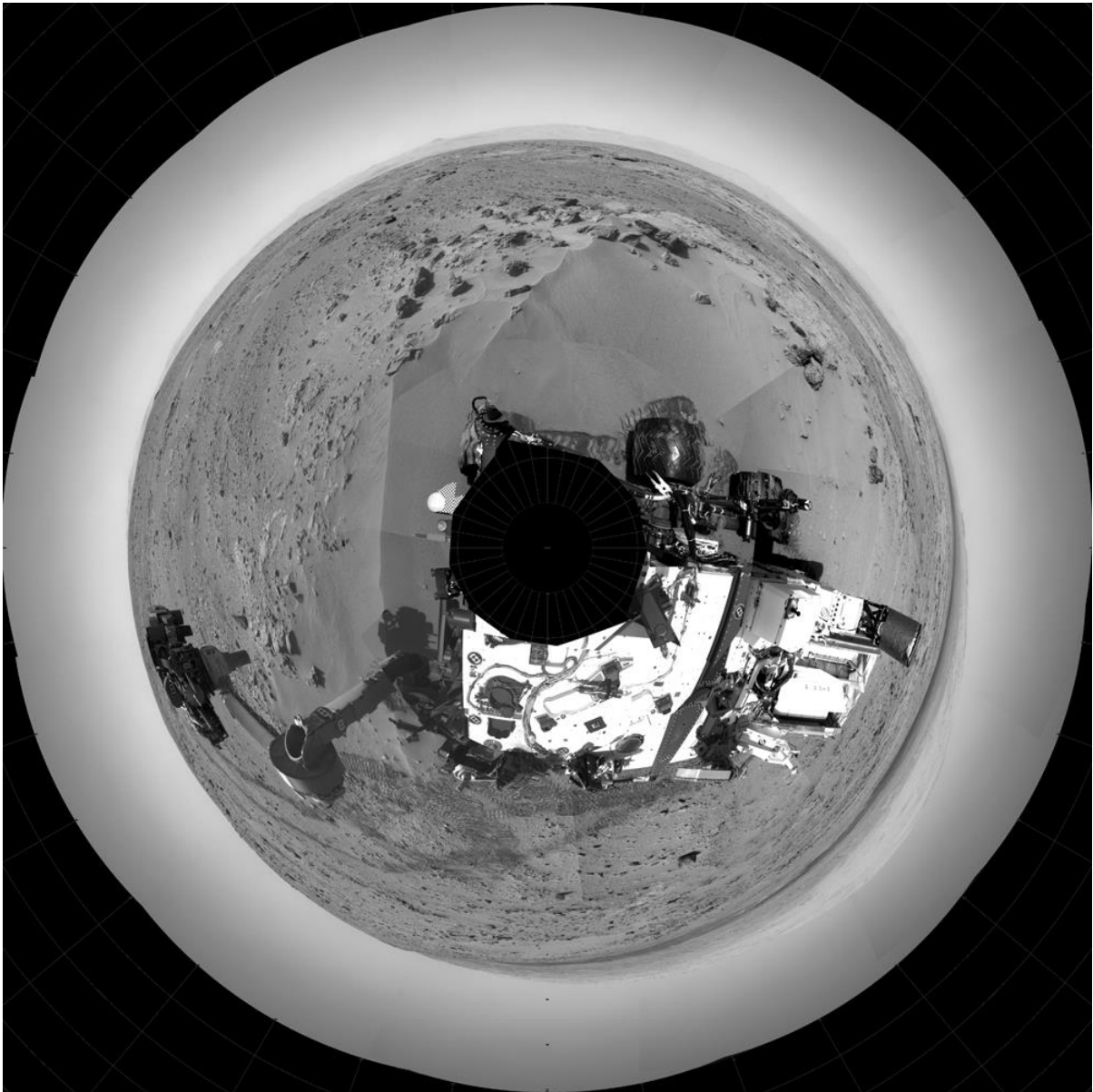


Figure 5.X – Polar projection mosaic

5.2.1.13.8 “Vertical Projection Mosaic” RDR

Vertical mosaics provide a view of the surroundings as if you were looking straight down. They are thus quite useful for establishing the environmental context or comparing with orbital imagery, but suffer from severe distortion with any variance of the scene from the surface model. In particular, rocks are severely elongated, and the terrain is not taken into account.

MIPL creates the Vertical projection as follows:

nl = number of lines in the mosaic (IMAGE object, LINES)

ns = number of samples in the mosaic (IMAGE object, LINE_SAMPLES)

$$x = (nl/2 - j) * MAP_SCALE$$
$$y = (i - ns/2) * MAP_SCALE$$

The view ray emanates from (x, y, 0) and points straight down (0,0,1). If the ray misses the surface in step E of Section 5.3.13.3 above, it is changed to point straight up (0,0,-1).

Figure 5.2.1.13.8 shows a vertical view. It assumes that the field is a plane tangent to the Martian surface with up pointing north. This is not an Orthorectified rendering but is still useful in many situations.

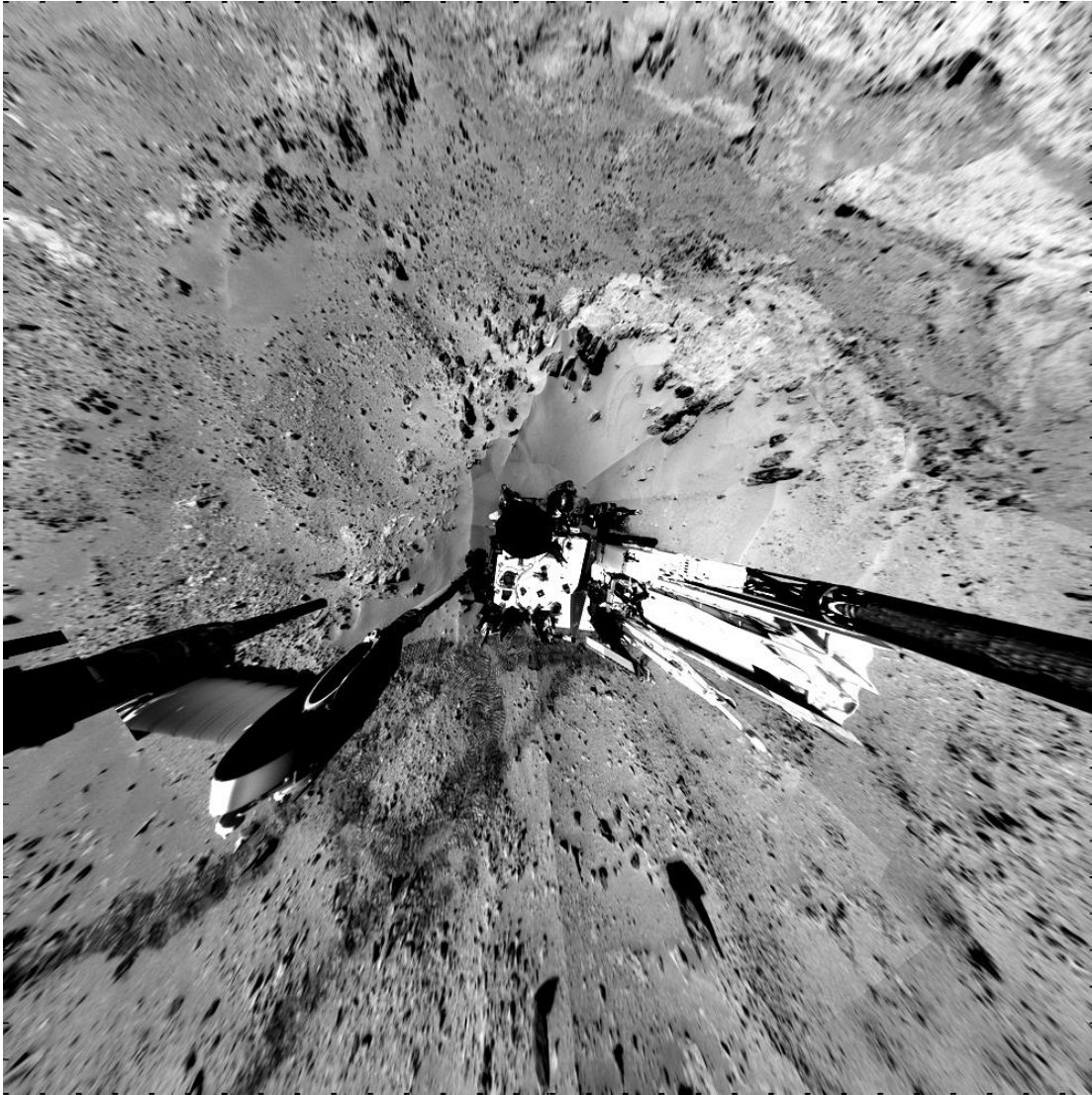


Figure 5.X Vertical projection mosaic

5.2.1.13.9 “Orthographic Projection Mosaic” RDR

The Orthographic projection is a generalization of the Vertical projection intended primarily for use with SHERLOC-WATSON data. It differs from Vertical in that an arbitrary projection plane can be specified.

If O is the point specified by the PROJECTION_ORIGIN_VECTOR and Xhat and Yhat are the unit vectors given by PROJECTION_X_AXIS_VECTOR and PROJECTION_Y_AXIS_VECTOR respectively, then an arbitrary point P will have projection coordinates (X,Y) as follows:

$$\vec{X} = (\vec{P} - \vec{O}) \cdot \hat{X}$$

$$\vec{Y} = (\vec{P} - \vec{O}) \cdot \hat{Y}$$

where the “•” indicates the scalar dot product of two vectors. PROJECTION_Z_AXIS_VECTOR is the direction of projection; the three vectors form a right-handed orthonormal basis.

All of these quantities must be specified with respect to a single frame defined by the REFERENCE_COORD_SYSTEM_NAME and REFERENCE_COORD_SYSTEM_INDEX. Additional relevant parameters for the projection are MAP_SCALE, X_AXIS_MINIMUM, X_AXIS_MAXIMUM, Y_AXIS_MINIMUM, and Y_AXIS_MAXIMUM.

A Vertical projection is the same as Orthographic with PROJECTION_X_AXIS_VECTOR = (1,0,0), PROJECTION_Y_AXIS_VECTOR = (0,1,0), and PROJECTION_Z_AXIS_VECTOR = (0,0,1).

5.2.1.13.10 “Orthorectified Projection Mosaic” RDR

Orthorectified mosaics are used to show a “true” view of the scene from a different point of view, without distortion due to parallax. The point of view is usually overhead, resulting in an image suitable for comparison to satellite imagery. The removal of parallax necessarily leads to holes or gaps in the mosaic, which do not occur with the other projections.

The Orthorectified mosaic is projected to a plane in a similar manner to the Orthographic or Vertical projections. However, unlike any of the other projections, the XYZ location of the pixels are taken into account. This is what allows parallax to be removed.

A simple way to think of this, for the case of an Orthorectified-Vertical projection, is to attach the XYZ coordinate (derived from stereo analysis) to each input image pixel, chop off the Z coordinate, and use the XY coordinates as the position in the output image. The more general (non-Vertical) case is similar in concept, just rotate the XYZ values to the frame defined by the projection plane first.

The most common projection plane is to look straight down, which corresponds to the same point of view as the Vertical projection. For this case, PROJECTION_X_AXIS_VECTOR = (1,0,0), PROJECTION_Y_AXIS_VECTOR = (0,1,0), and PROJECTION_Z_AXIS_VECTOR = (0,0,1).

The specific algorithm must deal with filling holes in the output mosaic and is still to be determined, as the software for this projection remains under development as of this writing.

Figure 5.2.1.13.10 shows an orthorectified rendering.

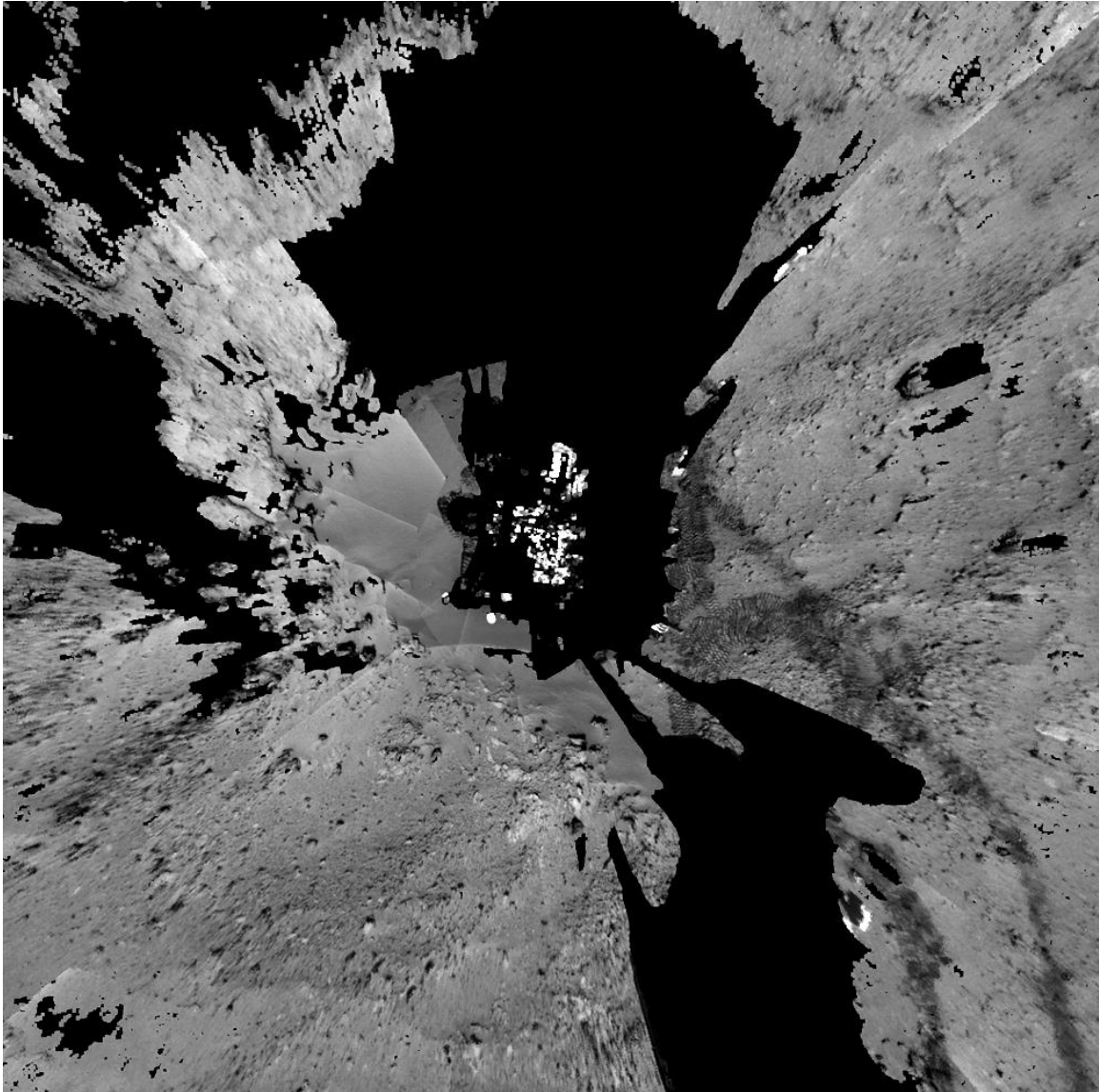


Figure 5.2.1.13.10 - Orthorectified Projection Mosaic

5.2.1.13.11 **Non-image Mosaics**

Normally mosaics are created using imagery, where each pixel is either a raw or radiometrically corrected intensity value. However, mosaics can also be created using other types of pixels. In fact, any of the RDR types using an image format (e.g. not meshes) can be mosaicked. The most useful of these are mosaics of XYZ, surface normal, and the various Slope types.

For example, an XYZ mosaic contains XYZ values for each pixel in the mosaic rather than intensity values. The inputs to the mosaic program are XYZ files (or individual X, Y, or Z components), and the pixels are interpreted in the same way - as the coordinate of the corresponding pixel in Cartesian space. Like XYZ images, the mosaics may consist of a single 3-band file with X, Y, and Z components, or separate 1 band files for each component. A Z-only mosaic of a Vertical or Orthorectified-Vertical projection creates a digital elevation model (DEM) - approximate in the Vertical case, correct for Orthorectified-Vertical.

As another example, Slope mosaics are often created and then overlaid on an image mosaic using the same projection parameters to help with rover navigation.

For M2020, a common product is the "SuperCam Finder" mosaic. This consists of a set of three Polar projection mosaics: the imagery, the range, and the rover exclusion mask. The mask is adjusted based on actual or predicted arm locations. The overlay of the mask mosaic over the image aids the SuperCam team in targeting (to avoid the rover and the arm), while the overlay of range helps determine focus distances for commanding.

Care must be taken while producing these mosaics to ensure that a consistent coordinate system and data type are used for all the input images. No transform is done on the data; the output mosaic may have only one coordinate system in which the values are defined, and one DERIVED_IMAGE_TYPE.

Non-image mosaics are often created without interpolation; the nearest pixel is used instead. This avoids aliasing effects when pixels are interpolated with neighboring invalid pixels.

5.2.1.14 **“Anaglyph” RDR**

A stereo anaglyph is a method of displaying stereo imagery quickly and conveniently using conventional display technology (no special hardware) and red/blue glasses. This is done by displaying the left eye of the stereo pair in the red channel, and displaying the right eye in the green and blue channels. An anaglyph data product simply captures that into a single 3-band color image, which can be displayed using any standard image display program with no knowledge that it is a stereo image. The red (first) band contains the left eye image, while the green and blue (second and third) bands each contain the right eye image (so the right image is duplicated in the file).

The Anaglyph method can also apply to multi-frame mosaic products. MIPL-generated mosaic Anaglyphs occasionally required some subtle pixel-shifting of the right eye mosaic data to improve the stereo effects. Mosaic Anaglyph products are distinguishable in the Mosaic RDR filename convention (see Section 6.1.2).

5.3 RDR Product Format

5.3.1 Image RDRs

The image RDR data products covered by this SIS are listed in Tables 5.3.1.1 and 5.3.1.2 below. Products listed as “1-3” bands are generally 1-band products but could have 3 RGB bands for certain MCZ camera modes.

Table 5.3.1.1 - M2020 Camera Image RDR Binary Formats TBD

Description	Product Identifier	# Bands	Data Type	DERIVED_IMAGE_TYPE Keyword value
CAHV-linearized	LIN	1-3	16-bit signed integer or 8-bit unsigned byte	“IMAGE”
Bayer pattern	BAY	1-3	16-bit signed integer	“IMAGE”
Inverse lookup table (ILUT)	ILT	1-3	16-bit signed integer	“IMAGE”
	ILC	1-3	16-bit signed integer	“IMAGE”
	ILP	1-3	16-bit signed integer	“IMAGE”
Base Image Product	FDR	1-3	16-bit signed integer	“IMAGE”
Radiometrically corrected	RAD	1-3	16-bit signed integer	“IMAGE”
	RAS	1-3	16-bit signed integer	“IMAGE”
	RAF	1-3	Float	“IMAGE”
Radiometrically corrected for Instrument Effects	RIE	1-3	16-bit signed integer	“IMAGE”
	RIF	1-3	Float	“IMAGE”
Radiometrically corrected IOF radiance factor	IOI	1-3	16-bit signed integer	“IMAGE”
	IOF	1-3	16-bit signed integer	“IMAGE”
MCZ-produced	DRX	1-3	16-bit signed integer	“IMAGE”
Radiometrically corrected	DRW	1-3	16-bit signed integer	“IMAGE”
Stereo Disparity	DSP	2	Float	“DISPARITY_MAP”
	DSR	2	Float	“DISPARITY_MAP”
	DSL	1	Float	“DISPARITY_LINE_MAP”
	DSS	1	Float	“DISPARITY_SAMPLE_MAP”
	DSG	1	8-bit unsigned byte	“DISPARITY_MAP”
	DSE	TBD	Float	“DISPARITY_ERROR_MAP”
Stereo Disparity Mask File	MDS	1	8-bit unsigned byte	“MASK”
Stereo Delta Disparity	DDD	2	Float	“DELTA_DISPARITY_MAP”
	DDL	1	Float	“DELTA_DISPARITY_LINE_MAP”
	DDS	1	Float	“DELTA_DISPARITY_SAMPLE_MAP”
Stereo First-stage Disparity	DFF	2	Float	“DISPARITY_MAP”
	DFL	1	Float	“DISPARITY_LINE_MAP”
	DFS	1	Float	“DISPARITY_SAMPLE_MAP”
XYZ	XYZ	3	Float	“XYZ_MAP”
	XYF	3	Float	“XYZ_MAP”
	XYO	3	8-bit unsigned byte	“XYZ_MAP”
	XYE	3	Float	“XYZ_ERROR_MAP”
	XYM	3	Float	“XYZ_MAP”
XYZ Mask File	MXY	1	8-bit unsigned byte	“MASK”
XYZ X-band	XXX	1	Float	“X_MAP”
	XXF	1	Float	“X_MAP”
XYZ Y-band	YYY	1	Float	“Y_MAP”
	YYF	1	Float	“Y_MAP”
	ZZZ	1	Float	“Z_MAP”
XYZ Z-band	ZZF	1	Float	“Z_MAP”
	ZZO	3	8-bit unsigned byte	“Z_MAP”
Surface Normal (UVW)	UVW	3	Float	“UVW_MAP”

Description	Product Identifier	# Bands	Data Type	DERIVED_IMAGE_TYPE Keyword value
	UVS	3	Float	"UVW_MAP"
	UVP	3	Float	"UVW_MAP"
	UVT	1	Float	"ANGLE_MAP"
	UVO	3	8-bit unsigned byte	"UVW_MAP"
	UVF	3	Float	"UVW_MAP"
Surface Normal U-band	UUU	1	Float	"U_MAP"
	UUF	1	Float	"U_MAP"
Surface Normal V-band	VVV	1	Float	"V_MAP"
	VVF	1	Float	"V_MAP"
Surface Normal W-band	WWW	1	Float	"W_MAP"
	WWF	1	Float	"W_MAP"
Surface Roughness	RUD	3	Float	"ROUGHNESS_MAP"
	RUT	2	Float	"ROUGHNESS_MAP"
Range	RNG	1	Float	"RANGE_MAP"
	RNM	1	Float	"RANGE_MAP"
	RNO	3	8-bit unsigned byte	"RANGE_MAP"
	RNF	1	Float	"RANGE_MAP"
	RNE	3	Float	"RANGE_ERROR_MAP"
Arm Reachability	ARM	5	16-bit signed integer	"REACHABILITY_MAP"
	ARO	3	8-bit unsigned byte	"REACHABILITY_MAP"
	ARK	5	16-bit signed integer	"REACHABILITY_MAP"
Arm Reachability Mask File	MAR	1	8-bit unsigned byte	"MASK"
Arm Preload Values	ARP	2	16-bit signed integer	"PRELOAD_MAP"
Slope	SLP	1	Float	"SLOPE_MAP"
	SLO	3	8-bit unsigned byte	"SLOPE_MAP"
	SRD	1	Float	"RADIAL_SLOPE_MAP"
	SRO	3	8-bit unsigned byte	"RADIAL_SLOPE_MAP"
	SHD	1	Float	"SLOPE_HEADING_MAP"
	SHO	3	8-bit unsigned byte	"SLOPE_HEADING_MAP"
	SMG	1	Float	"SLOPE_MAGNITUDE_MAP"
	SMO	3	8-bit unsigned byte	"SLOPE_MAGNITUDE_MAP"
	SNT	1	Float	"NORTHERLY TILT_MAP"
	SNO	3	8-bit unsigned byte	"NORTHERLY TILT_MAP"
Solar Energy	SEN	1	Float	"SOLAR_ENERGY_MAP"
	SEO	3	8-bit unsigned byte	"SOLAR_ENERGY_MAP"
Incidence, Emission, Phase	IEP	3	Float	"IEP_MAP"
	IEF	3	Float	"IEP_MAP"

Table 5.3.1.2 - M2020 Camera Non-image RDR Binary Formats

Description	Product ID	# Bands	Data Type	Data Structure
Terrain Mesh	TBD	N/A	TBD	TBD
JPEG compressed	TBD	TBD	TBD	TBD

5.4 RDR Product Structure

RDR products will have three possible structures. RDRs generated by MIPL will have a VICAR label wrapped by an ODL label, see Figure 5.4, Diagram A). This is the same as the EDR format, with the exception that the binary header data are eliminated. RDR products not generated by MIPL may look the same, or they may omit the VICAR label, containing only an ODL label (Figure 5.4, Diagram B). Or, RDR products conforming to a standard other than PDS, such as JPEG compressed or certain Terrain products (Figure 5.4, Diagram C), may contain no additional labels, instead following the other standard's formatting. In any case, RDRs will also have a detached PDS label. For a description of the ODL labels, see Section 3.2.1, and for a description of the VICAR Label, see Section 3.2.2, and for a mapping between ODL and VICAR, see Section 3.2.3.

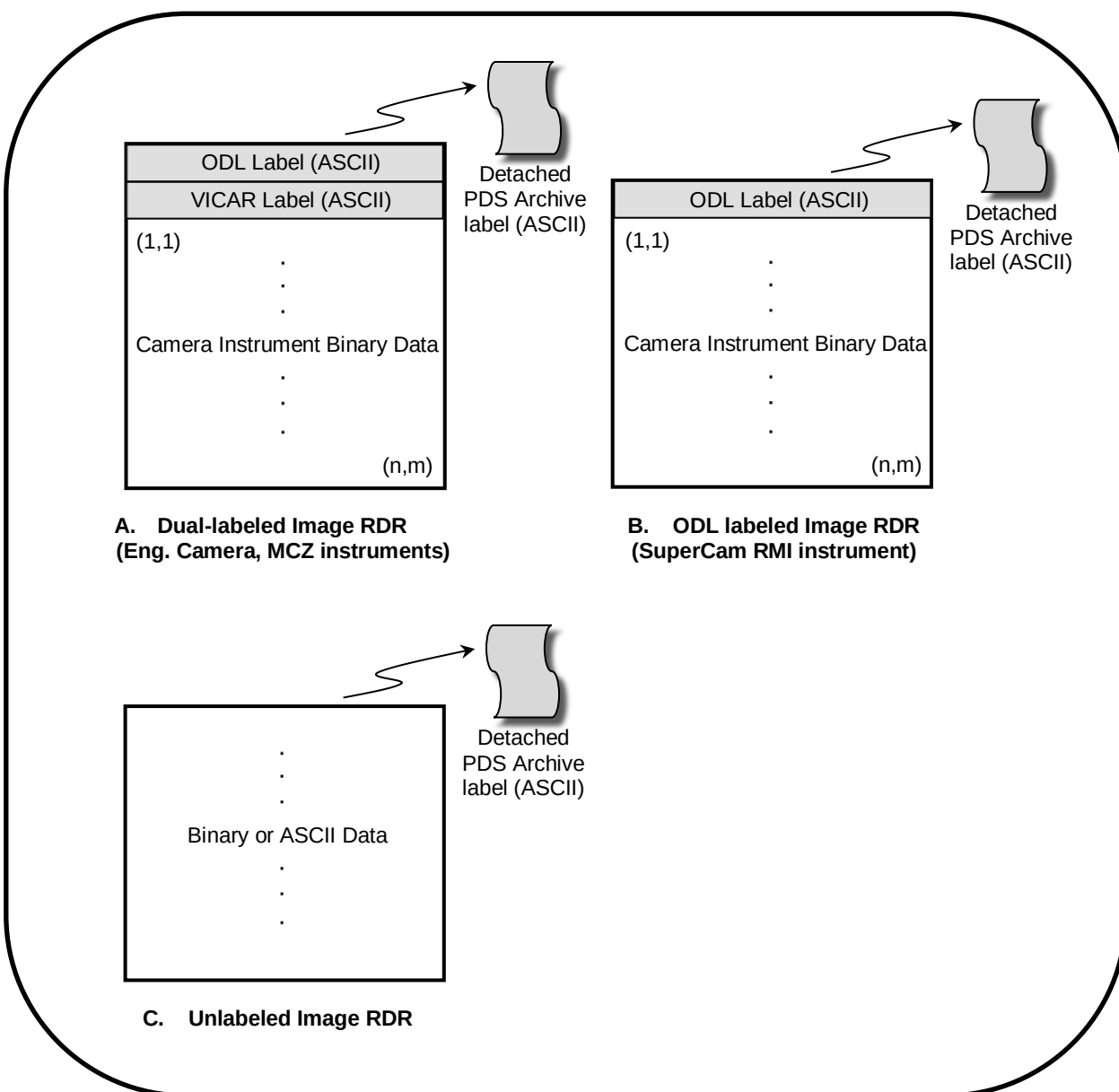


Figure 5.4 - RDR Product Structures **TBD**

The RDR data product is comprised of processed versions of the raw camera data, in both single and multi-frame (mosaic or mesh) form. Most RDR data products will be in “image” format, having detached PDS labels, as well as attached ODL labels or, if generated by MIPL (IDS), dual attached ODL/VICAR labels. Non-labeled RDRs include JPEG compressed products and the Terrain products.

For Mosaic RDRs, the detached PDS label is shared amongst four types of archiveable files: a) the “.IMG” Mosaic image file, b) the “.NAV” navigation file, c), the “.BRT” brightness correction file, and d) the “.LIS” list file indentifying the mosaic component images. For a description of “b” and “c”, see Section 5.2.1.13.3. To support the four types of Mosaic files, the PDS label contains four FILE Objects in the structure. See Appendix B for an example of a Mosaic RDR detached PDS label.

6. STANDARDS USED IN GENERATING PRODUCTS

6.1 File Naming Standards

The file naming scheme has been relieved of the string length constraint imposed by the Level II 36.3 filename standard approved by PDS in 2009. PDS has migrated to a new standard that is PDS-4, and filenames are essentially limited to a character string length of _____. This is a change from the 36.3 convention that M2020 was constrained to using. Use of three- character extensions, such as “.IMG” for image EDRs and RDRs and “.DAT” for spectrum EDRs and state-of-health EDRs, is consistent with the PDS standard.

There are three file naming schemes adapted for the M2020 image and non-image data products. The first applies to the EDR data product and all Single-frame RDR data products. The second applies to all Mosaic RDR data products. The third applies to Terrain products.

The primary attributes of the filename nomenclature are:

- a) **Uniqueness** - It must be unique unto itself without the file system's directory path. This Protects against product overwrite as files are copied/moved within the file system and external to the file system, if managed correctly.
- b) **Metadata** - It should be comprised of metadata fields that keep file bookkeeping and sorting intuitive to the human user. Even though autonomous file processing will be managed via databases, there will always be human-in-the-loop that puts a premium on filename intuition. Secondly, the metadata fields should be smartly selected based on their value to ground processing tools, as it is less CPU-intensive to extract information from the filename than from the label.

NOTE: Metadata information in the filename also resides in the product label.

The metadata fields have been selected based on MER and PHX lessons learned. In general, the metadata fields are arranged to achieve:

- a) **Sortability** - At the beginning of the filename resides a primary time oriented field such as Spacecraft Clock Start Count (SCLK). This allows for sorting of files on the M2020 file system by spacecraft data acquisition time as events occurred on Mars.
- b) **Readability** - An effort is made to alternate Integer fields with ASCII character fields to Optimize differentiation of field boundaries for the human user.

6.1.1 EDR Filename

Each M2020 EDR data product can be uniquely identified by incorporating into the product filename at minimum the Instrument ID, SCLK (or UTC), Product Type identifier, and Version number. The convention is illustrated in Figure 6.1. below.

Table 6.1 EDR/RDR Filename Fields

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	
INSTRUMENT	COLOR/FILTER	SPECIAL FLAG		PRIMARY TIMESTAMP			VENUE		SECONDARY TIMESTAMP											-	TERTIARY TIMESTAMP		PROD TYPE		GEOMETRY	THUMBNAIL		SITE		DRIVE		SEQUENCE/RTT											CAM SPECIFIC		DOWNSAMPLE	COMPRESSION	PRODUCER	VERSION	.		EXT							
Field				Position (size, type)					Description																																																	
Instrument				01 (2, a)					Instrument Identifier: - FL : Front Hazcam Left (RCE-A) - FR : Front Hazcam Right (RCE-A) - FA : Front Hazcam Anaglyph (RCE-A) - FG : Front Hazcam Colorglyph (RCE-A) - BL : Front Hazcam Left (RCE-B) - BR : Front Hazcam Right (RCE-B) - BA : Front Hazcam Anaglyph (RCE-B)																																																	

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58	
INSTRUMENT		COLOR/FILTER		SPECIAL FLAG		PRIMARY TIMESTAMP		VENUE		SECONDARY TIMESTAMP										-		TERTIARY TIMESTAMP		PROD TYPE		GEOMETRY		THUMBNAIL		SITE		DRIVE		SEQUENCE/RTT								CAM SPECIFIC				DOWNSAMPLE		COMPRESSION		PRODUCER		VERSION		.		EXT		
Field						Position (size, type)			Description																																																	
									- BG : Front Hazcam Colorglyph (RCE-B) - CC : Cache Cam - EA : EDL Parachute Uplook Cam A (PUC-A) - EB : EDL Parachute Uplook Cam B (PUC-B) - EC : EDL Parachute Uplook Cam C (PUC-C) - ED : EDL Rover Downlook Cam (RDC) - EL : EDL Lander Vision System (LVS) - ES : EDL Descent Stage Downlook Cam (DSD) - EU : EDL Rover Uplook Cam (RUC) - HN : Mars Helicopter Navigation Cam - HS : Mars Helicopter Scout Cam - ZL : Mastcam-Z Left - ZR : Mastcam-Z Right - ZA : Mastcam Anaglyph - ZG : Mastcam Colorglyph - WS : MEDA Skycam - NL : Navcam Left - NR : Navcam Right - NA : Navcam Anaglyph - NG : Navcam Colorglyph - PC : PIXL Micro Context Cam (MCC) - RL : Rear Hazcam Left - RR : Rear Hazcam Right - RA : Rear Hazcam Anaglyph - RC : Rear Hazcam Colorglyph - SC : SHERLOC Context Imager (ACI) - SI : SHERLOC Imaging (Watson) - SA : SHERLOC Imaging (Watson) Anaglyph - SG : SHERLOC Imaging (Watson) Colorglyph - SL : Watson treated as left-eye - SR : Watson treated as right-eye after G6.3 release - LR : Supercam RMI after G6.3 SR : Supercam RMI for STT																																																	
									Non-imaging instrument identifiers (not described in this SIS):																																																	
									- WE : MEDA Environment - OX : MOXIE																																																	

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INSTRUMENT	COLOR/FILTER	SPECIAL FLAG	PRIMARY TIMESTAMP	VENUE	SECONDARY TIMESTAMP															-	TERTIARY TIMESTAMP	PROD TYPE	GEOMETRY	THUMBNAIL	SITE	DRIVE	SEQUENCE/RTT										CAM SPECIFIC	DOWNSAMPLE	COMPRESSION	PRODUCER	VERSION	.	EXT																																																					
Field			Position (size, type)		Description																																																																																											
					- PE : PIXL Engineering - PS : PIXL Spectrometer - LS : SuperCam Non-Imaging Data after G6.3 release - SF : SuperCam Non-Imaging Data for STT - SS : SHERLOC Spectrometer - XM : RIMFAX Mobile - XS : RIMFAX Stationary																																																																																											
Color/Filter			03 (1, i/a)		Color flag (see Section X.X): - E : Raw Bayer pattern - M : Grayscale image (Monochrome/Panchromatic) - A : Upper green bayer cells (G1 see Section X.X) - D : Lower green bayer cells (G2 see Section X.X) - O : Other _ : N/A For all color cameras, the flag is set based on the color space depending on whether the product is a 3-band product, or an individual band. <table><tr><td>Color Type</td><td>3-Band</td><td>Band 1</td><td>Band 2</td><td>Band 3</td></tr><tr><td>RGB</td><td>F</td><td>R</td><td>G</td><td>B</td></tr><tr><td>XYZ</td><td>T</td><td>X</td><td>Y</td><td>Z</td></tr><tr><td>xyY</td><td>C</td><td>J</td><td>K</td><td>L</td></tr><tr><td>HSI</td><td>P</td><td>H</td><td>S</td><td>I</td></tr></table> For certain instruments, this flag may take on additional values indicating illumination or filter: For PIXL, the flag can be set based on the LEDs used to illuminate the target. <table><tr><td>LED Color</td><td>Flag</td></tr><tr><td>Red</td><td>R</td></tr><tr><td>Green</td><td>G</td></tr><tr><td>Blue</td><td>B</td></tr><tr><td>Multiple</td><td>W</td></tr><tr><td>UV</td><td>U</td></tr></table>																																																							Color Type	3-Band	Band 1	Band 2	Band 3	RGB	F	R	G	B	XYZ	T	X	Y	Z	xyY	C	J	K	L	HSI	P	H	S	I	LED Color	Flag	Red	R	Green	G	Blue	B	Multiple	W	UV	U
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								<table><tr><td rowspan="2">Ground Test SCLK NOT reset</td><td>Year-DOY</td><td>(same as Flight Cruise)</td><td>(same as Flight Cruise)</td><td>(same as Flight Cruise)</td></tr><tr><td>Sol</td><td><nnnn> <aaaa></td><td>"0000", "0001", ... "9999" " _ _ _ _ " (4 underscores)</td><td>0 thru 9999 Value is out of range</td></tr><tr><td rowspan="3">Ground Test SCLK Reset</td><td rowspan="3">DOY-Year</td><td><ddd>A, <ddd>B, . . . <ddd>Z</td><td>001A, 002A, ... 365A, 001B, 002B, ... 365B, . . . 001Z, 002Z, ... 365Z</td><td>DOY 1 – 365 2017, DOY 1 – 365 2018, . . . DOY 1 – 365 2042</td></tr><tr><td><ddd>Z</td><td>001Z, 002Z, ... 365Z</td><td>DOY 1 – 365 2042</td></tr><tr><td><aaaa></td><td>" _ _ _ _ " (4 underscores)</td><td>Value is out of range</td></tr></table>																																																		Ground Test SCLK NOT reset	Year-DOY	(same as Flight Cruise)	(same as Flight Cruise)	(same as Flight Cruise)	Sol	<nnnn> <aaaa>	"0000", "0001", ... "9999" " _ _ _ _ " (4 underscores)	0 thru 9999 Value is out of range	Ground Test SCLK Reset	DOY-Year	<ddd>A, <ddd>B, . . . <ddd>Z	001A, 002A, ... 365A, 001B, 002B, ... 365B, . . . 001Z, 002Z, ... 365Z	DOY 1 – 365 2017, DOY 1 – 365 2018, . . . DOY 1 – 365 2042	<ddd>Z	001Z, 002Z, ... 365Z	DOY 1 – 365 2042	<aaaa>	" _ _ _ _ " (4 underscores)	Value is out of range
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Venue				9 (1, a)				Mission venue identifier: _ : Flight (surface or cruise) - A : AVSTB - F : FSWTB - M : MSTB - S : "Scarecrow" - V : VSTB Other venue identifiers may be defined later.																																																																					
Secondary timestamp				10 (10, i)				Secondary timestamp that is of finer granularity than the Primary timestamp. . Value type is based on either of four scenarios: <u>Flight Cruise</u> SCLK – This field stores the 10-integer SCLK (seconds). Which specific SCLK count (Start or End) is used depends on the instrument, but nominally it is the starting count of the <u>first</u> (i.e., lowest Clock time) acquired instrument data. <u>Flight Surface</u> SCLK – Same as for "Flight Cruise" <u>Ground Test in which SCLK in NOT reset</u> SCLK – Same as for "Flight Cruise" <u>Ground Test in which SCLK is reset</u> ERT - This field stores the ERT time portions Month, Day-of-month, Hour and Seconds as 10 integers in a UTC-like format																																																																					

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Field				Position (size, type)				Description																																																																																
								This field's value type associates with the value type of the Primary timestamp field.																																																																																
								<table><tr><th>Scenario</th><th>Time Type</th><th>Value Format</th><th>Valid Values</th><th>Time Range</th></tr><tr><td rowspan="2">Flight Cruise</td><td rowspan="2">SCLK</td><td><ssssssssss> (Seconds)</td><td>0000000000, 0000000001, • • • 9999999999</td><td>0 thru 9999999999</td></tr><tr><td><aaaaaaaaaa></td><td>" _ _ _ _ _ _ _ _ _ _" (10 underscores)</td><td>Value is out of range</td></tr><tr><td>Flight Surface</td><td>SCLK</td><td>(same as Flight Cruise)</td><td>(same as Flight Cruise)</td><td>(same as Flight Cruise)</td></tr><tr><td>Ground Test SCLK NOT reset</td><td>SCLK</td><td>(same as Flight Cruise)</td><td>(same as Flight Cruise)</td><td>(same as Flight Cruise)</td></tr><tr><td rowspan="2">Ground Test SCLK reset</td><td rowspan="2">ERT</td><td><MMDDHHmmss> (Month, Day-of-month, Hour, Minute, Second)</td><td>"0101010000", "0101010001", • • • "1231235959"</td><td>January 1, 01:00:00 thru December 31, 23:59:59</td></tr><tr><td><aaaaaaaaaa></td><td>" _ _ _ _ _ _ _ _ _ _" (10 underscores)</td><td>Value is out of range</td></tr></table>																																																		Scenario	Time Type	Value Format	Valid Values	Time Range	Flight Cruise	SCLK	<ssssssssss> (Seconds)	0000000000, 0000000001, • • • 9999999999	0 thru 9999999999	<aaaaaaaaaa>	" _ _ _ _ _ _ _ _ _ _" (10 underscores)	Value is out of range	Flight Surface	SCLK	(same as Flight Cruise)	(same as Flight Cruise)	(same as Flight Cruise)	Ground Test SCLK NOT reset	SCLK	(same as Flight Cruise)	(same as Flight Cruise)	(same as Flight Cruise)	Ground Test SCLK reset	ERT	<MMDDHHmmss> (Month, Day-of-month, Hour, Minute, Second)	"0101010000", "0101010001", • • • "1231235959"	January 1, 01:00:00 thru December 31, 23:59:59	<aaaaaaaaaa>	" _ _ _ _ _ _ _ _ _ _" (10 underscores)	Value is out of range
								Scenario	Time Type	Value Format	Valid Values	Time Range																																																																												
								Flight Cruise	SCLK	<ssssssssss> (Seconds)	0000000000, 0000000001, • • • 9999999999	0 thru 9999999999																																																																												
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								Flight Surface	SCLK	(same as Flight Cruise)	(same as Flight Cruise)	(same as Flight Cruise)																																																																												
								Ground Test SCLK NOT reset	SCLK	(same as Flight Cruise)	(same as Flight Cruise)	(same as Flight Cruise)																																																																												
								Ground Test SCLK reset	ERT	<MMDDHHmmss> (Month, Day-of-month, Hour, Minute, Second)	"0101010000", "0101010001", • • • "1231235959"	January 1, 01:00:00 thru December 31, 23:59:59																																																																												
<aaaaaaaaaa>	" _ _ _ _ _ _ _ _ _ _" (10 underscores)	Value is out of range																																																																																						
_				20 (1, a)				Underscore for readability. Always set to "_".																																																																																
Milliseconds				21 (3, i)				Milliseconds of either the SCLK or UTC.																																																																																
Product type				24 (3, a)				Product identifier. EDR: See Table 4.1 for details RDR: See Table 5.3.1.1 for details.																																																																																
Geometry				27 (1, a)				Linearization flag: _ : Non-linearized (raw geometry) - L : Product has been linearized with nominal stereo partner - A : Product has been linearized with an actual stereo partner. Note that for the "A" case, an image can have multiple stereo partners and the linearized images will be different for each partner. A user will need to look in the ODL/VICAR label to determine which partner was used for linearization.																																																																																

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58																																								
INSTRUMENT	COLOR/FILTER	SPECIAL FLAG	PRIMARY TIMESTAMP	VENUE	SECONDARY TIMESTAMP															-	TERTIARY TIMESTAMP	PROD TYPE	GEOMETRY	THUMBNAIL	SITE	DRIVE	SEQUENCE/RTT										CAM SPECIFIC	DOWNSAMPLE	COMPRESSION	PRODUCER	VERSION	.	EXT																																																						
Field				Position (size, type)				Description																																																																																									
								All EDRs are Raw geometry (“_”).																																																																																									
Thumbnail				28 (1, a)				Thumbnail flag: T : Product is a thumbnail N : Product is a non-thumbnail (full-frame, sub-frame, downsample)																																																																																									
Site				29 (3, i/a)				Site identifier (see section X.X on Site frames): Site location count from the RMC where the data was acquired. <table><tr><th>Values</th><th>Range</th></tr><tr><td>000, 001, ..., 999</td><td>0 thru 999</td></tr><tr><td>A00, A01, ..., A99</td><td>1000 thru 1099</td></tr><tr><td>B00, B01, ..., B99</td><td>1100 thru 1199</td></tr><tr><td>...</td><td>...</td></tr><tr><td>Z00, Z01, ... Z99</td><td>3500 thru 3599</td></tr><tr><td>AA0, AA1, ..., AA9</td><td>3600 thru 3609</td></tr><tr><td>AB0, AB1, ..., AB9</td><td>3610 thru 3619</td></tr><tr><td>...</td><td>...</td></tr><tr><td>ZZ0, ZZ1, ..., ZZ9</td><td>10350 thru 10359</td></tr><tr><td>AAA, AAB, ..., AAZ</td><td>10360 thru 10385</td></tr><tr><td>ABA, ABB, ..., ABZ</td><td>10386 thru 10411</td></tr><tr><td>...</td><td>...</td></tr><tr><td>ZZA, ZZB, ..., ZZZ</td><td>27910 thru 27935</td></tr><tr><td>OAA, OAB, ..., OAZ</td><td>27936 thru 27961</td></tr><tr><td>0BA, 0BB, ..., 0BZ</td><td>27962 thru 27987</td></tr><tr><td>...</td><td>...</td></tr><tr><td>7CA, 7CB, ..., 7CZ</td><td>32720 thru 32745</td></tr><tr><td>7DA, 7DB, ..., 7DV</td><td>32746 thru 32767</td></tr><tr><td>---</td><td>Value out of range</td></tr></table>																																																		Values	Range	000, 001, ..., 999	0 thru 999	A00, A01, ..., A99	1000 thru 1099	B00, B01, ..., B99	1100 thru 1199	...	...	Z00, Z01, ... Z99	3500 thru 3599	AA0, AA1, ..., AA9	3600 thru 3609	AB0, AB1, ..., AB9	3610 thru 3619	...	...	ZZ0, ZZ1, ..., ZZ9	10350 thru 10359	AAA, AAB, ..., AAZ	10360 thru 10385	ABA, ABB, ..., ABZ	10386 thru 10411	...	...	ZZA, ZZB, ..., ZZZ	27910 thru 27935	OAA, OAB, ..., OAZ	27936 thru 27961	0BA, 0BB, ..., 0BZ	27962 thru 27987	...	...	7CA, 7CB, ..., 7CZ	32720 thru 32745	7DA, 7DB, ..., 7DV	32746 thru 32767	---	Value out of range
								Values	Range																																																																																								
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7DA, 7DB, ..., 7DV	32746 thru 32767																																																																																																
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Drive				32 (4, i/a)				Drive identifier: Drive count (position within a Site location) from the RMC where the data was acquired. <table><tr><th>Values</th><th>Range</th></tr><tr><td>0000, 0001, ..., 9999</td><td>0 thru 9999</td></tr></table>																																																		Values	Range	0000, 0001, ..., 9999	0 thru 9999																																				
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INSTRUMENT	COLOR/FILTER	SPECIAL FLAG	PRIMARY TIMESTAMP	VENUE	SECONDARY TIMESTAMP															-	TERTIARY TIMESTAMP	PROD TYPE	GEOMETRY THUMBNAI	SITE	DRIVE	SEQUENCE/RTT										CAM SPECIFIC	DOWNSAMPLE	COMPRESSION	PRODUCER	VERSION	.	EXT																															
Field				Position (size, type)				Description																																																																	
								MEDA Skycam EDL Cameras HELI S000 Format key: - D : Single-resolution or multi-resolution downsampling (ECAM Reconstructed images only). Multi-resolution value represents the highest resolution tile(s) of the reconstructed image. <table><tr><td>Valid values</td><td>Description</td></tr><tr><td>A</td><td>Single resolution (Alone)</td></tr><tr><td>L</td><td>Multi-resolution 4x4 (Low-res)</td></tr><tr><td>M</td><td>Multi-resolution 2x2 (Med-res)</td></tr><tr><td>N</td><td>Multi-resolution 1x1 (No downsample)</td></tr></table> - NNN : Observation counter (RMI only). Counter increments by one for each image command used in an RMI sequence (raster). Valid values are 000-999. - PPPP : PIXL motion counter. Valid values TBD. - R : Reconstruction counter (ECAM Reconstructed only). When a new tile is added that changes either the geometry or color of the reconstructed image, the counter increments by one. Valid values are 0-9, then A-Z. - S : Stereo partner counter. Details are TBD. All EDRs are “_”. <table><tr><td>Valid values</td><td>Description</td></tr><tr><td>—</td><td>Nominal (synchronized) stereo or No nominal stereo partner</td></tr><tr><td>A-Z then 0-9</td><td>Details TBD</td></tr></table> - TT: Tile counter (ECAM Tile only). Identifies a unique tile within a reconstructed image. Valid values are 00-99. - ZZZ : Focus value in millimeters (MCZ only). Valid values are 000-999. _ : Literal underscore. As last character of camera specific field, identifies an ECAM Tile. 0 : Literal zero. Undefined field placeholder.																																																		Valid values	Description	A	Single resolution (Alone)	L	Multi-resolution 4x4 (Low-res)	M	Multi-resolution 2x2 (Med-res)	N	Multi-resolution 1x1 (No downsample)	Valid values	Description	—	Nominal (synchronized) stereo or No nominal stereo partner	A-Z then 0-9	Details TBD
																																																										Valid values	Description														
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[illegible]

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	50	51	52	53	54	55	56	57	58
INSTRUMENT	COLOR/FILTER	SPECIAL FLAG	PRIMARY TIMESTAMP				VENUE						SECONDARY TIMESTAMP							TERTIARY TIMESTAMP			PROD TYPE		GEOMETRY THUMBNAIL		SITE			DRIVE						SEQUENCE/RTT							CAM SPECIFIC		DOWNSAMPLE	COMPRESSION	PRODUCER	VERSION	.	EXT							
Field				Position (size, type)				Description																																																	
								Every version need not exist. E.g. version 01, 02, and 04 may exist but not 03. In general, the highest-numbered version represents the best version of that product. This field increments independently of all fields.																																																	
.				55 (1, a)				Separator for filename and extension. Always set to “.”																																																	
Extension				56 (3, a)				File extension: • VIC : Used for VICAR files. • IMG : Same as .VIC with ODL label. • TIF : TIFF formatted image file. (no label) • JPG : JPEG formatted image file. (no label) • PNG : PNG formatted image file. (no label) • TXT : ASCII text file. • iv : Inventor-format file • ht : Height-map file (VICAR format) • rgb : Skin file in SGI RGB format • obj: Mesh file in Wavefront OBJ format • mtl : OBJ Material file • png : Lower case for meshes • mlp : MeshLab project file • xml : PDS4 label file Note: other file extensions to be added in the future																																																	

6.1.2 Mosaic RDR Filename

The filename convention follows:

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	36	38	39	40	41	42	43	44
INSTRUMENT 1	INSTRUMENT 2	EYE	COLOR/FILTER			SPECIAL FLAG	SOL				MULTI SOL	PRODUCT TYPE	VENUE	SITE			DRIVE			MULTI DRIVE	PROJECTION	GEOM CORR	COORD FRAME	BRT CORR	DESCRIPTION						PRODUCER	VERSION	.	EXT									

Field	Position (size, type)	Description
Instrument 1	1 (1,a)	First Instrument Identifier: - N : Navcam - Z : Mastcam-Z - S : Supercam RMI - P : PIXL Micro Context Cam (MCC) - I : SHERLOC Imaging (WATSON) - C : SHERLOC ACI - F : Front Hazcam (RCE-A) - B : Front Hazcam (RCE-B) - R : Rear Hazcam - E : EDL Camera - V : Mars Helicopter Navigation Cam
Instrument 2	2 (1,a)	Second Instrument Identifier: - N : Navcam - Z : Mastcam-Z - S : Supercam RMI - P : PIXL Micro Context Cam (MCC) - I : SHERLOC Imaging (WATSON) - C : SHERLOC ACI - F : Front Hazcam (RCE-A) - B : Front Hazcam (RCE-B) - R : Rear Hazcam - E : EDL Camera - V : Mars Helicopter Navigation Cam - X : Use when there are more than two instruments

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	36	38	39	40	41	42	43	44
INSTRUMENT 1	INSTRUMENT 2	EYE	COLOR/FILTER	SPECIAL FLAG	SOL	MULTI SOL	PRODUCT TYPE	VENUE	SITE	DRIVE	MULTI DRIVE	PROJECTION	GEOM CORR	COORD FRAME	BRT CORR	DESCRIPTION	PRODUCER	VERSION	.	EXT																							

Field	Position (size, type)	Description																									
		_ : Use when there is only one instrument																									
Eye	3 (1,a)	Identifies the eye of the data used to make this mosaic. For stereo mosaics, indicates which eye should view this product (in the unlikely event these two definitions are in conflict, the second use prevails for stereo mosaics). - A : Anaglyph - C : Colorglyph - X : Mixed, used for by products which have more than one image. - L : Left eye - M : Mono - R : Right eye - S : Stereo 2-band																									
Color/Filter	4 (3,a)	Indicates the filter or color used for each band of the mosaic. Single-band mosaics should repeat the same character 3 times. Three-band color spaces may either list the three one-band components (e.g. "RGB") or repeat the 3-band code 3 times (e.g. "FFF"). Valid values: 0 or _ : Filter not applicable - M : Grayscale image (Monochrome/Panchromatic) - U : UV (PIXL only) <table><tr><th>Color Type</th><th>3-Band</th><th>Band 1</th><th>Band 2</th><th>Band 3</th></tr><tr><td>RGB</td><td>F</td><td>R</td><td>G</td><td>B</td></tr><tr><td>XYZ</td><td>T</td><td>X</td><td>Y</td><td>Z</td></tr><tr><td>xyY</td><td>C</td><td>J</td><td>K</td><td>L</td></tr><tr><td>HSI</td><td>P</td><td>H</td><td>S</td><td>I</td></tr></table> Other color flags may be defined in the future.	Color Type	3-Band	Band 1	Band 2	Band 3	RGB	F	R	G	B	XYZ	T	X	Y	Z	xyY	C	J	K	L	HSI	P	H	S	I
Color Type	3-Band	Band 1	Band 2	Band 3																							
RGB	F	R	G	B																							
XYZ	T	X	Y	Z																							
xyY	C	J	K	L																							
HSI	P	H	S	I																							
Special	7 (1,a)	Special Processing flag. The special processing character is used to indicate off-nominal or special processing of the image. Examples include use of different correlation parameters, special stretches to eliminate shadows, reprocessing with different camera pointing, etc. The meaning of any individual character in this field (other than "_" which means nominal processing) will be defined on an ad-hoc basis as needed during the mission. Within one Sol or a range of Sols, the character will be used consistently. So, this field can be used to group together all derived products resulting from one kind of special processing. An attempt will be made to maintain consistency across different Sols as well, but this may not always be																									

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	36	38	39	40	41	42	43	44			
INSTRUMENT 1	INSTRUMENT 2	EYE	COLOR/FILTER	SPECIAL FLAG	SOL	MULTI SOL	PRODUCT TYPE	VENUE	SITE	DRIVE	MULTI DRIVE	PROJECTION	GEOM CORR	COORD FRAME	BRT CORR	DESCRIPTION	PRODUCER	VERSION	.	EXT																										

Field	Position (size, type)	Description																						
		possible; thus the meaning of characters may change across different individual or ranges of Sols. An ASCII text file will be maintained containing all special processing designators that are used, the Sols they relate to, and a description of the special processing that was done. This file will be included in the PDS archive.																						
Sol	8 (4,i)	- Sol : for flight (Cruise & Surface) or flight-like ground tests when the Secondary timestamp continuously increments, this field stores the 4-integer Sol (Mars solar day) of the first (i.e lowest) clock time acquired instrument data. - Year : for ground tests when the Secondary timestamp resets and repeats, the uniqueness as a timestamp is lost. Wall Clock is used to restore uniqueness, represented in a UTC-like format across Primary, Secondary, and Tertiary time fields. This field is used to store the Earth Year portion of the UTC-like time value, designated by a “Y” followed by the last three digits of the year. <table><tr><th colspan="2">Time Type</th><th>Values</th><th>Valid Time Range</th></tr><tr><td rowspan="3">Sol</td><td>Cruise</td><td>0000</td><td>NULL</td></tr><tr><td rowspan="2">Surface</td><td>0001 – 9999</td><td>1 thru 9999</td></tr><tr><td>----</td><td>Value is out of range</td></tr><tr><td rowspan="3">Year of UTC</td><td>Cruise Test</td><td>Y000</td><td>NULL</td></tr><tr><td rowspan="2">Surface Test</td><td>Y017 – Y099</td><td>2017 thru 2099</td></tr><tr><td>----</td><td>Value is out of range</td></tr></table>	Time Type		Values	Valid Time Range	Sol	Cruise	0000	NULL	Surface	0001 – 9999	1 thru 9999	----	Value is out of range	Year of UTC	Cruise Test	Y000	NULL	Surface Test	Y017 – Y099	2017 thru 2099	----	Value is out of range
Time Type		Values	Valid Time Range																					
Sol	Cruise	0000	NULL																					
	Surface	0001 – 9999	1 thru 9999																					
		----	Value is out of range																					
Year of UTC	Cruise Test	Y000	NULL																					
	Surface Test	Y017 – Y099	2017 thru 2099																					
		----	Value is out of range																					
Multisol	12 (1,a)	Flag indicating that a significant percentage of the data content was acquired on more than a single Sol, i.e. across multiple Sols. Specification of this flag is at the discretion of the mosaic product's producer, who determines what percentage is "significant". Valid values: _ : Flags mosaic as single sol (insignificant multiple sol contribution) - X : Flags mosaic as containing significant content from multiple sols																						
Product Type	13 (3,a)	Product Identifier. See Table 5.3.1.1 for details.																						
Phase	16 (1,a)	Mission phase identifier:																						

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	36	38	39	40	41	42	43	44																																									
INSTRUMENT 1	INSTRUMENT 2	EYE	COLOR/FILTER		SPECIAL FLAG	SOL			MULTI SOL			PRODUCT TYPE		VENUE	SITE			DRIVE			MULTI DRIVE		PROJECTION		GEOM CORR	COORD FRAME	BRT CORR	DESCRIPTION					PRODUCER	VERSION	.	EXT																																																
Field						Position (size, type)			Description																																																																											
									_ : Flight (surface only) - A : ATLO - C : Cruise - T : Test																																																																											
Site						17 (3, i/a)			Site identifier (see section X.X on Site frames): Site location count from the RMC where the data was acquired. <table><tr><th>Values</th><th>Range</th></tr><tr><td>000, 001, ..., 999</td><td>0 thru 999</td></tr><tr><td>A00, A01, ..., A99</td><td>1000 thru 1099</td></tr><tr><td>B00, B01, ..., B99</td><td>1100 thru 1199</td></tr><tr><td>...</td><td>...</td></tr><tr><td>Z00, Z01, ... Z99</td><td>3500 thru 3599</td></tr><tr><td>AA0, AA1, ..., AA9</td><td>3600 thru 3609</td></tr><tr><td>AB0, AB1, ..., AB9</td><td>3610 thru 3619</td></tr><tr><td>...</td><td>...</td></tr><tr><td>ZZ0, ZZ1, ..., ZZ9</td><td>10350 thru 10359</td></tr><tr><td>AAA, AAB, ..., AAZ</td><td>10360 thru 10385</td></tr><tr><td>ABA, ABB, ..., ABZ</td><td>10386 thru 10411</td></tr><tr><td>...</td><td>...</td></tr><tr><td>ZZA, ZZB, ..., ZZZ</td><td>27910 thru 27935</td></tr><tr><td>0AA, 0AB, ..., 0AZ</td><td>27936 thru 27961</td></tr><tr><td>0BA, 0BB, ..., 0BZ</td><td>27962 thru 27987</td></tr><tr><td>...</td><td>...</td></tr><tr><td>7CA, 7CB, ..., 7CZ</td><td>32720 thru 32745</td></tr><tr><td>7DA, 7DB, ..., 7DV</td><td>32746 thru 32767</td></tr><tr><td>— — —</td><td>Value out of range</td></tr></table>																																				Values	Range	000, 001, ..., 999	0 thru 999	A00, A01, ..., A99	1000 thru 1099	B00, B01, ..., B99	1100 thru 1199	...	...	Z00, Z01, ... Z99	3500 thru 3599	AA0, AA1, ..., AA9	3600 thru 3609	AB0, AB1, ..., AB9	3610 thru 3619	...	...	ZZ0, ZZ1, ..., ZZ9	10350 thru 10359	AAA, AAB, ..., AAZ	10360 thru 10385	ABA, ABB, ..., ABZ	10386 thru 10411	...	...	ZZA, ZZB, ..., ZZZ	27910 thru 27935	0AA, 0AB, ..., 0AZ	27936 thru 27961	0BA, 0BB, ..., 0BZ	27962 thru 27987	...	...	7CA, 7CB, ..., 7CZ	32720 thru 32745	7DA, 7DB, ..., 7DV	32746 thru 32767	— — —	Value out of range
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— — —	Value out of range																																																																																			
Drive						20 (4, i/a)			Drive identifier: Drive count (position within a Site location) from the RMC where the data was acquired.																																																																											

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INSTRUMENT 1	INSTRUMENT 2	EYE	COLOR/FILTER		SPECIAL FLAG	SOL			MULTI SOL			PRODUCT TYPE		VENUE	SITE		DRIVE			MULTI DRIVE		PROJECTION		GEOM CORR	COORD FRAME	BRT CORR	DESCRIPTION					PRODUCER	VERSION		.	EXT							

Field	Position (size, type)	Description																															
			<table><tr><th>Values</th><th>Range</th></tr><tr><td>0000, 0001, ..., 9999</td><td>0 thru 9999</td></tr><tr><td>A000, A001, ..., A999</td><td>10000 thru 10999</td></tr><tr><td>B000, B001, ..., B999</td><td>11000 thru 11999</td></tr><tr><td>...</td><td>...</td></tr><tr><td>Z000, Z001, ... Z999</td><td>35000 thru 35999</td></tr><tr><td>AA00, AA01, ..., AA99</td><td>36000 thru 36099</td></tr><tr><td>AB00, AB01, ..., AB99</td><td>36100 thru 36199</td></tr><tr><td>...</td><td>...</td></tr><tr><td>AZ00, AZ01, ..., AZ99</td><td>38500 thru 38599</td></tr><tr><td>BA00, BA01, ..., BA99</td><td>38600 thru 38699</td></tr><tr><td>BB00, BB01, ..., BB99</td><td>38700 thru 38799</td></tr><tr><td>...</td><td>...</td></tr><tr><td>LJ00, LJ01, ..., LJ35</td><td>65500 thru 65535</td></tr><tr><td>-----</td><td>Value is out of range</td></tr></table>	Values	Range	0000, 0001, ..., 9999	0 thru 9999	A000, A001, ..., A999	10000 thru 10999	B000, B001, ..., B999	11000 thru 11999	...	...	Z000, Z001, ... Z999	35000 thru 35999	AA00, AA01, ..., AA99	36000 thru 36099	AB00, AB01, ..., AB99	36100 thru 36199	...	...	AZ00, AZ01, ..., AZ99	38500 thru 38599	BA00, BA01, ..., BA99	38600 thru 38699	BB00, BB01, ..., BB99	38700 thru 38799	...	...	LJ00, LJ01, ..., LJ35	65500 thru 65535	-----	Value is out of range
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		B000, B001, ..., B999	11000 thru 11999																														
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		AA00, AA01, ..., AA99	36000 thru 36099																														
		AB00, AB01, ..., AB99	36100 thru 36199																														
		...	...																														
		AZ00, AZ01, ..., AZ99	38500 thru 38599																														
		BA00, BA01, ..., BA99	38600 thru 38699																														
		BB00, BB01, ..., BB99	38700 thru 38799																														
		...	...																														
		LJ00, LJ01, ..., LJ35	65500 thru 65535																														
-----	Value is out of range																																
Multidrive	24 (1,a)	Flag indicating that a significant percentage of the data content was acquired on more than a single Drive position, i.e. across multiple drives. Valid values: _ : Flags mosaic as single drive position - X : Flags mosaic as containing content from multiple Drive positions																															
Projection	25 (3,a)	Projection type. Valid values: - CYL - Cylindrical - CYP - Cylindrical-Perspective - ORT - Orthographic - ORR - Orthorectified - PER - Perspective - POL - Polar - VRT - Vertical (special case of Orthographic)																															
Geometric Correction	28 (1,a)	Geometric correction type indicator. Specifies the correction type that was applied to the largest percentage of data content. Valid values are: _ : No correction (raw pointing) - A : Auto-correction via tiepointing																															

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	36	38	39	40	41	42	43	44																
INSTRUMENT 1	INSTRUMENT 2	EYE		COLOR/FILTER		SPECIAL FLAG		SOL				MULTI SOL		PRODUCT TYPE		VENUE		SITE		DRIVE				MULTI DRIVE		PROJECTION		GEOM CORR	COORD FRAME	BRT CORR		DESCRIPTION								PRODUCER	VERSION	.		EXT															
Field						Position (size, type)						Description																																															
												- F : Auto-correction via tiepointing & auto-registration with fiducials - G : Auto-correction via tiepointing & manual registration with fiducials - T : Manual tiepointing - R : Manual tiepointing & auto-registration with fiducials - M : Manual tiepointing & manual registration with fiducials - O : Other correction not listed above																																															
Coordinate Frame						29 (1,a)						Coordinate system (frame) type. Valid values: - S : Site frame - L : Local Level frame - R : Rover Frame - U : Untilt (CYP only, is Rover frame with rotation) - O : Other																																															
Brt Corr.						30 (1, a)						Brightness correction type indicator. Specifies the correction type that was applied to the largest percentage of data content. Valid values are: _ : No correction - B : Automatic brightness adjustment (multiplicative and/or additive factor applied to each frame) - M : Manual brightness adjustment (same factors as “B”) - V : Anti-vignetting adjustment applied to some or all frames - G : General brightness correction - A : General brightness correction that can vary across the frame (automatic) - O : Other correction not listed above																																															
Description						31 (7,a)						General purpose mosaic identifier. Can be set to anything to help identify the mosaic, such as target name, panorama name, theme name, method of production, etc. Valid values include A-Z, 0-9, and underscore. Must always pad to 7 characters, using underscores as necessary.																																															
Producer						38 (1,a)						Identifier for the institution/team that created this product: - J : JPL (IDS/MIPL) - P : Principal investigator of instrument. <table><tr><td>Instrument</td><td>PI</td></tr><tr><td>ECAM</td><td>JPL</td></tr><tr><td>MCZ</td><td>ASU (Tempe, AZ)</td></tr><tr><td>SCAM RMI</td><td>IRAP (France)</td></tr><tr><td>PIXL MCC</td><td>TBD</td></tr></table>																																						Instrument	PI	ECAM	JPL	MCZ	ASU (Tempe, AZ)	SCAM RMI	IRAP (France)	PIXL MCC	TBD
Instrument	PI																																																										
ECAM	JPL																																																										
MCZ	ASU (Tempe, AZ)																																																										
SCAM RMI	IRAP (France)																																																										
PIXL MCC	TBD																																																										

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	36	38	39	40	41	42	43	44
INSTRUMENT 1	INSTRUMENT 2	EYE		COLOR/FILTER	SPECIAL FLAG		SOL				MULTI SOL	PRODUCT TYPE		VENUE		SITE		DRIVE		MULTI DRIVE	PROJECTION		GEOM CORR	COORD FRAME	BRT CORR		DESCRIPTION						PRODUCER	VERSION	.		EXT						

Field	Position (size, type)	Description																				
		<table><tr><td>SHERLOC</td><td>JPL</td></tr><tr><td>MEDA Skycam</td><td>Ministry of Education and Science (Spain)</td></tr><tr><td>EDL Cameras</td><td>JPL</td></tr><tr><td>HELI RTE/NAV</td><td>JPL</td></tr></table> - A – I, K – O, Q – Z : Co-I to be identified per instrument at the discretion of the instrument PI. _ : undefined/other Other producer codes will be added in the future.	SHERLOC	JPL	MEDA Skycam	Ministry of Education and Science (Spain)	EDL Cameras	JPL	HELI RTE/NAV	JPL												
SHERLOC	JPL																					
MEDA Skycam	Ministry of Education and Science (Spain)																					
EDL Cameras	JPL																					
HELI RTE/NAV	JPL																					
Version	39 (2,a)	Product version number. Increments by one whenever a previously generated file with an otherwise identical filename exists. <table><tr><th>Values</th><th>Range</th></tr><tr><td>00, 01, 02 ..., 99</td><td>0 thru 99</td></tr><tr><td>A0, A1, ..., A9</td><td>100 thru 109</td></tr><tr><td>AA, AB, ..., AZ</td><td>110 thru 135</td></tr><tr><td>B0, B1, B2 ..., B9</td><td>136 thru 145</td></tr><tr><td>BA, BB, ..., BZ</td><td>146 thru 171</td></tr><tr><td>...</td><td>...</td></tr><tr><td>Z0, Z1, ..., Z9</td><td>1000 thru 1009</td></tr><tr><td>ZA, ZB, ..., ZZ</td><td>1010 thru 1035</td></tr><tr><td>--</td><td>Value is out of range</td></tr></table> Every version need not exist. E.g. version 01, 02, and 04 may exist but not 03. In general, the highest-numbered version represents the best version of that product. This field increments independently of all fields.	Values	Range	00, 01, 02 ..., 99	0 thru 99	A0, A1, ..., A9	100 thru 109	AA, AB, ..., AZ	110 thru 135	B0, B1, B2 ..., B9	136 thru 145	BA, BB, ..., BZ	146 thru 171	...	...	Z0, Z1, ..., Z9	1000 thru 1009	ZA, ZB, ..., ZZ	1010 thru 1035	--	Value is out of range
Values	Range																					
00, 01, 02 ..., 99	0 thru 99																					
A0, A1, ..., A9	100 thru 109																					
AA, AB, ..., AZ	110 thru 135																					
B0, B1, B2 ..., B9	136 thru 145																					
BA, BB, ..., BZ	146 thru 171																					
...	...																					
Z0, Z1, ..., Z9	1000 thru 1009																					
ZA, ZB, ..., ZZ	1010 thru 1035																					
--	Value is out of range																					
.	41 (1,a)	Separator for filename and extension. Always set to “.”																				
Extension	42 (3,a)	File extension: - VIC : Used for VICAR files. - IMG : Same as .VIC with ODL label.																				

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	36	38	39	40	41	42	43	44
INSTRUMENT 1	INSTRUMENT 2	EYE	COLOR/FILTER			SPECIAL FLAG	SOL				MULTI SOL	PRODUCT TYPE			VENUE	SITE			DRIVE			MULTI DRIVE	PROJECTION			GEOM CORR	COORD FRAME	BRT CORR	DESCRIPTION								PRODUCER	VERSION	.	EXT			

Field	Position (size, type)	Description
		• TIF : TIFF formatted image file. (no label) • JPG : JPEG formatted image file. (no label) • PNG : PNG formatted image file. (no label) • TXT : ASCII text file. • xml : PDS4 label file

Figure 6.1.2 – Mosaic RDR Filename Convention

6.1.3 Unified Terrain Mesh RDR Filename **TBD**

INSERT FILENAME GRAPHIC HERE

6.2 PDS Standards

The M2020 camera instrument EDR data product complies with Planetary Data System standards for file formats and labels, as specified in the PDS Standards Reference [Ref 14]. See Section 4.2 for a description of the PDS Label and the specific conventions adopted by M2020.

6.3 Time Standards

The EDR ODL label use keywords containing time values. Each time value standard is defined according to the keyword description. See Appendix B.

6.4 Coordinate Frame Standards

The M2020 Frame Manager defines several dozen coordinate frames, which can be used for commanding pointing among other things. Refer to the Pointing, Positioning, Phasing and Coordinate Systems (PPPCS) document [Ref 1] or the Surface Attitude, Positioning and Pointing (SAPP) Functional Design Description (FDD) [Ref 2] for more details on all these coordinate frames. Only a few of them are used by the products and processes described by this SIS. This subset is described in detail in this section. The only place in this SIS where the full set of frames can appear is in the INSTRUMENT_COORD_FRAME_ID label, which is a command echo.

A subset of these frames needed for a specific image or data set are defined by the *_COORDINATE_SYSTEM groups.

Note that the PLACES database [Ref X] maintains both telemetered and re-localized versions of the Site and Rover Nav frames at every available index.

Table 6.4 - Coordinate Frames Used for M2020 Surface Operations

Frame Name (Label Keyword Value)	Short Name (SAPP FDD)	Reference Frame (Used to Define)	Coordinate Frame	
			Origin	Orientation
ROVER_NAV_FRAME	RNAV	Enclosing SITE_FRAME	Attached to rover	Aligned with rover
ROVER_MECH_FRAME	RMECH	Enclosing SITE_FRAME	Attached to rover	Aligned with rover
LOCAL_LEVEL_FRAME	LL	Enclosing SITE_FRAME	Attached to rover (coincident with Rover Nav Frame)	North/East/Nadir
SITE_FRAME	SITE(n)	Previous SITE_FRAME	Attached to surface	North/East/Nadir
RSM_HEAD_FRAME	RSM_HEAD	ROVER_NAV_FRAME	Attached to mast head	Aligned with pointing of mast head. This corresponds to RSM_HEAD in the Frame Manager
Arm Frames: TBD ARM_BASE_FRAME ARM_TURRET_FRAME ARM_DRILL_FRAME ARM_GDST_FRAME ARM_TOOL_FRAME ARM_WATSON_FRAME ARM_SHERLOC_FRAME ARM_PIXL_FRAME	Arm Frames: TBD RA_BASE TURRET DRILL GDRT TOOL WATSON SHERLOC PIXL	ROVER_NAV_FRAME	Attached to the tool; see PPPCS for the specific tool frame.	Aligned with tool in some way; see PPPCS [Ref X] for the specific tool Frame.

6.4.1 Rover Navigation (Rover Nav) Frame

The Rover Nav frame (RNAV) is the one used for surface navigation and mobility. By definition, the frame is attached to the rover, and moves with it when the rover moves while on the surface. Its Y origin is centered on the rover and the X origin is aligned with the middle wheels' rotation axis for the deployed rover and suspension system on a flat plane. The Z origin is defined to be at the nominal surface, which is a fixed position with respect to the rover body. The actual surface will likely not be at exactly $Z=0$ due to the effects of suspension sag, rover tilt, rocker bogie angles, etc. The +X axis points to the front of the rover, +Y to the right side, and +Z down (perpendicular to the chassis deck). See Figure 6.4.1.1.

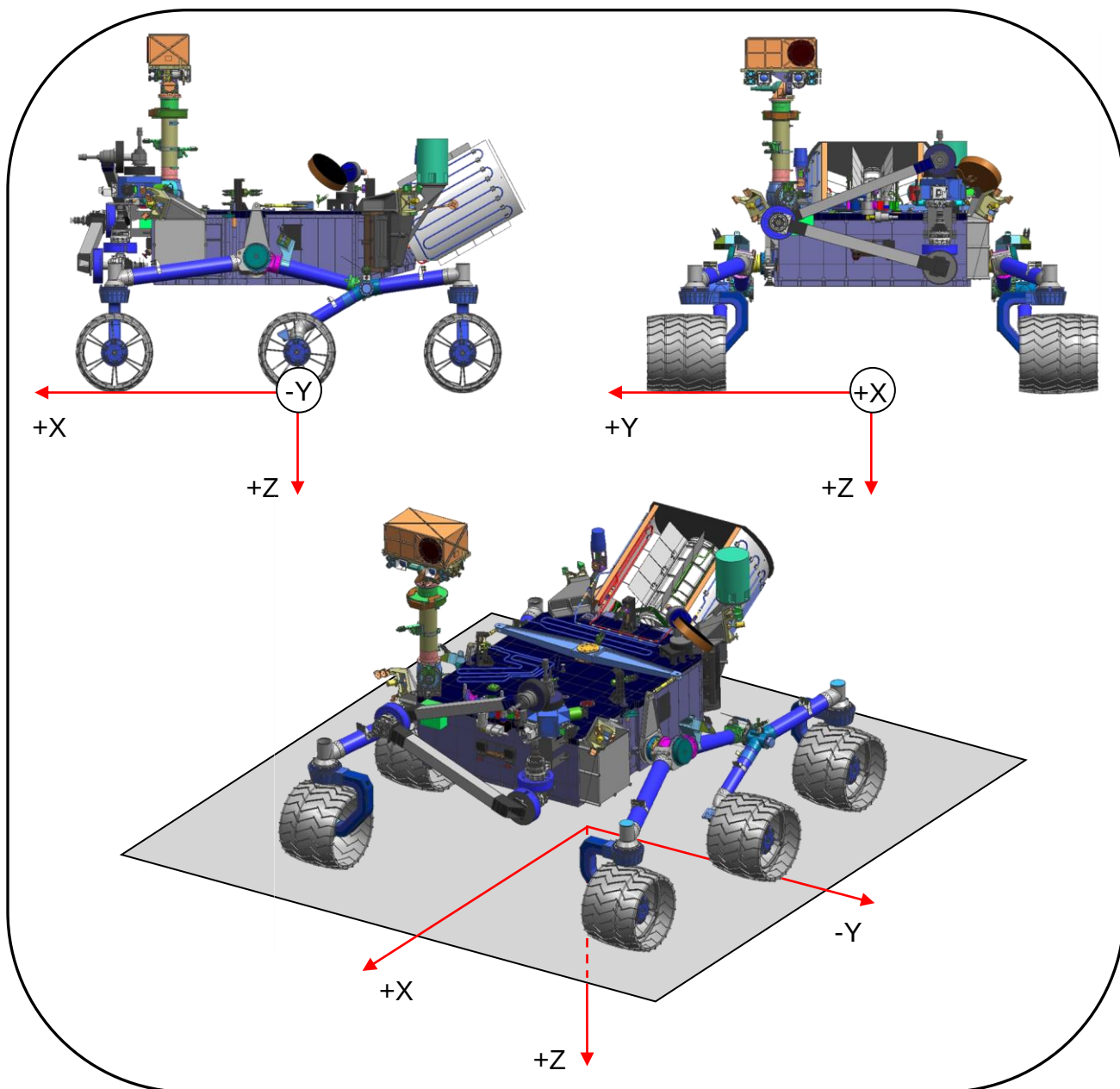


Figure 6.4.1.1 – Rover Navigation (RNAV) Coordinate Frame

The Rover Nav frame is specified via an offset from the current Site frame, and a quaternion that represents the rotation between the two. A new instance of the Rover Nav frame, with a potentially unique offset/quaternion, is created every time the ROVER_MOTION_COUNTER increments.

Orientation of the rover (and thus Rover Nav) with respect to Local Level or Site is also sometimes described by Euler angles as shown in Figure 6.4.1.2, where ψ is heading, θ is attitude or pitch, and ϕ is bank or roll.

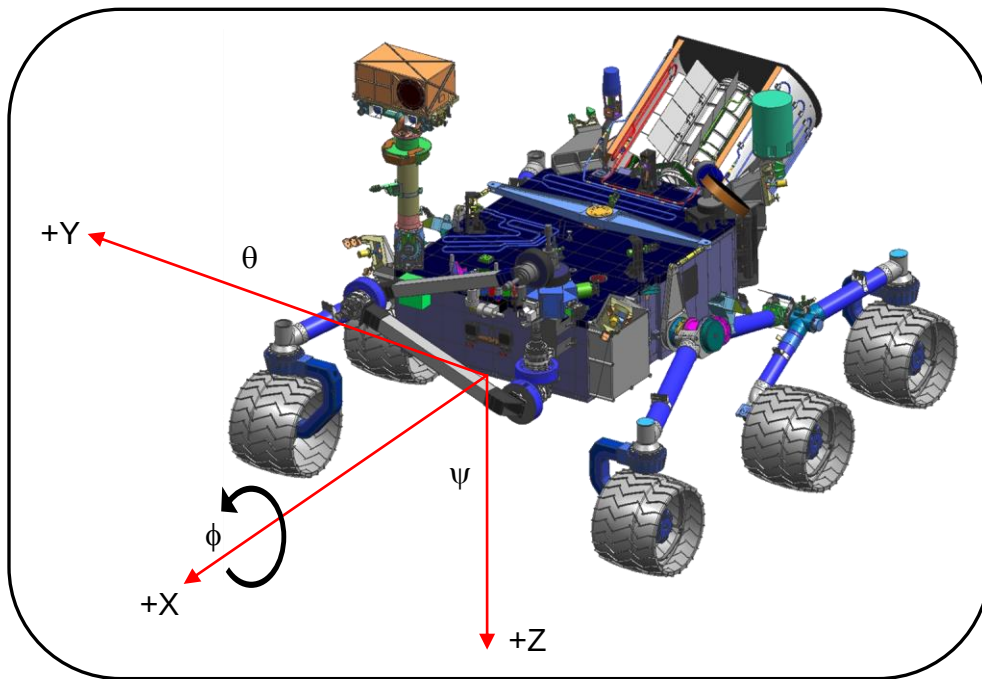


Figure 6.4.1.2 – Yaw, Pitch and Roll Definitions

6.4.2 Rover Mechanical (Rover Mech) Frame

The Rover Mechanical (RMECH) frame is oriented identically to the Rover Nav frame. The origin is forward of Rover Nav by $x=0.09002$ meters. In other words, given a point expressed in Rover Mech, if you add $(0.09002, 0.0, 0.0)$ you will get the same point expressed in Rover Nav. Rover Mech is not used by any nominal products (EDR or RDR) but could appear in certain special products, generally having to do with arm kinematics.

6.4.3 Local Level Frame

The Local Level frame is coincident with the Rover Nav frame, i.e. they share the same origin at all times. The orientation is different, however. The +X axis points North, +Z points down to nadir along

the local gravity vector, and +Y completes the right-handed system. Thus the orientation matches the orientation of Site frames.

Local Level frames are defined by an offset from the current Site frame, with an identity quaternion.

6.4.4 Site Frame

Site frames are used to reduce accumulation of rover localization error. They are used to provide a common reference point for all operations within a local area. Rover Nav and Local Level frames are specified using an offset from this origin. When a new Site is declared, that becomes the new reference, and the offset is zeroed. In this way, long-term localization error is relegated to the offset between Sites, becoming irrelevant to local operations, because the positions are reset with each new Site.

When a Site frame is declared, it is identical to the Local Level frame, sharing both orientation and position. However, the Site frame is fixed to the Mars surface; when the rover moves, Local Level moves with it but Site stays put. Therefore, for the Site frame, +X points North, +Z points down to nadir along the local gravity vector, and +Y completes the right-handed system.

Sites are indexed, meaning there are multiple instances. Site 1 by definition represents the landing location. New Sites are declared as needed during operations, as the rover moves away from the local area. See Figure 6.4.4.

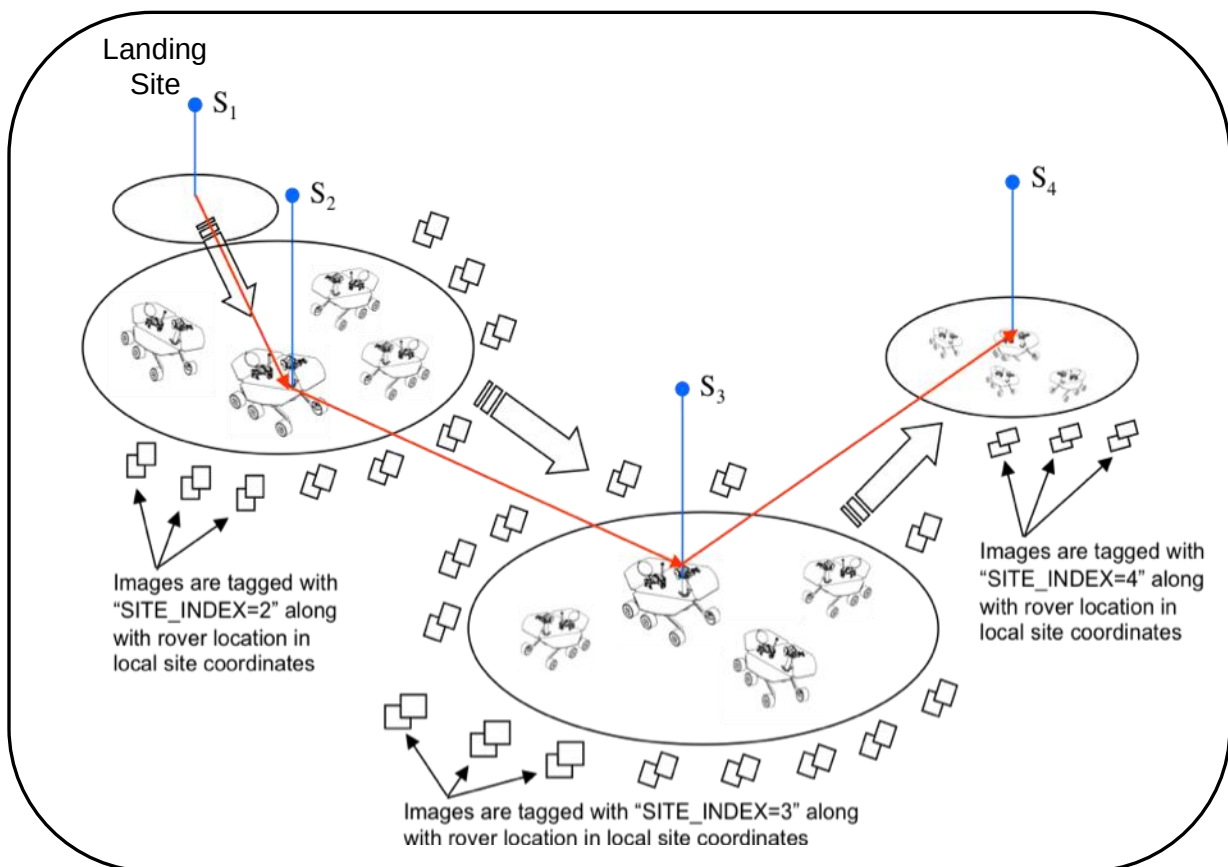


Figure 6.4.4 – Site and Rover Frames

The PLACES database [Ref 10] stores the set of all site-to-site offsets; such offsets are not in every image label.

6.4.5 RSM Frame

The RSM frame is attached to the Remote Sensing Mast (RSM) camera head, and moves with it. See the PPPCS for specific definition. It is expressed as an offset and quaternion from the Rover Nav frame.

6.4.6 Arm Frames

The frame representing the currently selected arm tool is reported in the arm coordinate system group. The selected tool, given by ARTICULATION_DEV_INSTRUMENT_ID, is arbitrary for any given image and may be surprising; for example SHERLOC-WATSON may not be the selected tool for a SHERLOC-WATSON image. The various tool frames are attached to and aligned with the tool in some manner specific to that tool. See the PPPCS [Ref 1] for actual frame definitions.

7. APPLICABLE SOFTWARE

The instrument data downlink processing software is focused on rapid reduction, calibration, and visualization (in the case of images) of products in order to make discoveries, to accurately and expeditiously characterize the geologic environment around the rover, and to provide timely input for operational decisions concerning rover navigation and Arm target selection. Key software tools have

been developed at JPL as part of the IDS and APSS subsystems, and at LANL/IRAP/CNES by the SuperCam team. These toolsets can be used to process data to yield substantial scientific potential in addition to their operational importance.

7.1 Utility Programs

Table 7.1 lists (in no particular order) the primary software tools that will be used to process and manipulate downlinked M2020 instrument payload data. Instrument data processing software executed by teams working the IDS and APSS subsystems at JPL will be capable of reading and writing image and spectra data in PDS format. Within IDS, the “M2020edrgen” program will generate EDRs and the Mars Program Suite of VICAR programs will generate RDRs in PDS format. An IDS pipeline system will deliver the products to the FEI server for transfer to M2020’s ODS as rapidly as possible after receipt of telemetry.

Table 7.1 - Key Software Toolsets

Name	Description	Primary Development Responsibility
SuperCam Ops Software	LIBS software development is primarily performed in IDL code, with LANL (New Mexico) as lead institute.	Dot Delapp (LANL)
	RMI software development is primarily performed in IDL code, with IRAP (France) as lead institute.	David Baratoux (IRAP)
M2020edrgen	Fetches the image Data Product Object (DPO) records from M2020 Data Product (DP) files, reconstructing the image file from the telemetry data into a PDS-labelled image EDR data product. VICAR code.	Alice Stanboli, Costin Radulescu(JPL / MIPL)
Mars Program Suite	Stereo image processing software using EDRs or calibrated images (RDRs), image mosaicking software, 3-D terrain building software. VICAR code: • MARSCAHV – Generates a geometrically corrected version of the EDR, applying the C, A, H and V camera model vectors. • MARSRAD – Generates a radiometrically corrected image from a single input EDR. • MARSJPLSTEREO – Generates a disparity map from a stereo pair of input EDRs, applying a 1-D correlator (fast). • MARSCOR3 – Generates a disparity map from a stereo pair of input EDRs, applying a 2-D correlator (more robust). • MARSXYZ – Generates an XYZ image from an input disparity map. • MARSRANGE – Generates a range image from an input XYZ map. • M2020REACH – Generates an arm reachability map from an input XYZ map.	Bob Deen (JPL / MIPL)

Name	Description	Primary Development Responsibility
	• MARSINVERTER – Generates inverse lookup table (ILUT) products. • MARSDEBAYER – Generates de-Bayered images. • M2020ROUGH – Generates roughness maps. • MARSERROR – Generates XYZ and range error maps. • MARSSLOPE – Generates slope maps. • MARSBRT – Generates brightness/contrast correction file for mosaic processing. • M2020FILTER – Generates XML file for image mask files. • MARSFILTER – Generates image mask files. • MARSMASK – Applies image mask files to image files. • MARSDISPCOMPARE – Checks consistency for left-to-right and right-to-left stereo image correlations. • MARSUVW - Generates a surface normal image, wherein XYZ is computed normal to the surface. • MARSMAP – Generates a Cylindrical, Polar or Vertical projection mosaic from a list of input EDRs. • MARSMOS – Produces pinhole camera mosaics using uncorrected input images and CAHVOR camera model. • MARSMCAULEY – Generates a combination Cylindrical-Perspective projection mosaic from a list of input EDRs. • MARSTIE – Generates pointing corrections (tiepoint file) from an overlapping set of input EDRs. • MARSNAV – Generates an updated azimuth and elevation file based on comparison with existing image data that can be directly compared. • XVD – De facto image reader software capable of displaying VICAR-labeled image files.	
	• CRUMBS – 3-D terrain building software	Oleg Pariser (JPL / MIPL)
APSS / RSVP	Visualization, planning, and sequence generation software for use by Sequence Team to create Sol sequences based on activity lists generated by PSI during planning meetings. Java, C and C++ code.	Brian Cooper (JPL)

7.2 Applicable PDS Software Tools

PDS-labeled images and tables can be viewed with the program NASAView, developed by the PDS and available for a variety of computer platforms from the PDS web site http://pds.jpl.nasa.gov/tools/software_download.cfm. There is no charge for NASAView.

7.3 Software Distribution and Update Procedures

The FEI distribution tool and Mars Image Processing Program Suite are available to researchers and academic institutions. Refer to the MIPL Web site at <http://www-mipl.jpl.nasa.gov> for contact information. FEI is described in detail at <http://www-mipl.jpl.nasa.gov/MDMS.html>

APPENDIX A – Data Product Names ****DRAFT**** (Red = TBD)

Instrument	APID #	APID Names C=Cruise, S=Surface, B=Both	DPOs Used as Sources for Metadata		
			IDPH	Ancillary	Supplementary
Hazcams		Fullframe, Subframe, Downsampled Images:	ImgIdph (B)	n/a	n/a
	317	ImgImageFhl (B)			
	318	ImgImageFhr (B)			
	333	ImgImageRhl (B)			
	334	ImgImageRhr (B)			
		Fullframe, Subframe, Downsampled Images (ICER):			
	319	ImgImageIcerFhl (B)			
	320	ImgImageIcerFhr (B)			
	323	ImgImageIcerRhl (B)			
	324	ImgImageIcerRhr (B)			
		Fullframe, Subframe, Downsampled Images (LOCO):			
	325	ImgImageLocoFhl (B)			
	326	ImgImageLocoFhr (B)			
	329	ImgImageLocoRhl (B)			
	330	ImgImageLocoRhr (B)			
		Fullframe, Subframe, Downsampled Images (JPEG):			
	50	ImgImageJpegFhl (S)			
	51	ImgImageJpegFhr (S)			
	52	ImgImageJpegRhl (S)			
	53	ImgImageJpegRhr (S)			
		Thumbnail Images:	ImgIdph (B)	n/a	n/a
	362	ImgThumbFhl (B)			
	363	ImgThumbFhr (B)			
	378	ImgThumbRhl (B)			
	379	ImgThumbRhr (B)			
		Thumbnail Images (ICER):			
	364	ImgThumbIcerFhl (B)			
	365	ImgThumbIcerFhr (B)			
	368	ImgThumbIcerRhl (B)			
	369	ImgThumbIcerRhr (B)			
		Thumbnail Images (LOCO):			
	370	ImgThumbLocoFhl (B)			
	371	ImgThumbLocoFhr (B)			
	374	ImgThumbLocoRhl (B)			
	375	ImgThumbLocoRhr (B)			
		Thumbnail Images (JPEG):			
	57	ImgThumbJpegFhl (S)			
	58	ImgThumbJpegFhr (S)			
	59	ImgThumbJpegRhl (S)			
	60	ImgThumbJpegRhr (S)			
		Histogram Products:	ImgIdph (B)	n/a	n/a
	311	ImgHistogramFhl (B)			
	312	ImgHistogramFhr (B)			
	315	ImgHistogramRhl (B)			
	316	ImgHistogramRhr (B)			
		Row-summed Products:	ImgIdph (B)	n/a	n/a
	355	ImgRowsumsFhl (B)			
	356	ImgRowsumsFhr (B)			
	359	ImgRowsumsRhl (B)			
	360	ImgRowsumsRhr (B)			

Instrument	APID #	APID Names C=Cruise, S=Surface, B=Both	DPOs Used as Sources for Metadata		
			IDPH	Ancillary	Supplementary
	303 304 307 308	Column-summed Products: ImgColsumsFhl (B) ImgColsumsFhr (B) ImgColsumsRhl (B) ImgColsumsRhr (B)	ImgIdph (B)	n/a	n/a
Navcam	331 332	Fullframe, Subframe, Downsampled Images: ImgImageNI (B) ImgImageNr (B)	ImgIdph (B)	n/a	n/a
	321 322	Fullframe, Subframe, Downsampled Images (ICER): ImgImageIcerNI (B) ImgImageIcerNr (B)			
	327 328	Fullframe, Subframe, Downsampled Images (LOCO): ImgImageLocoNI (B) ImgImageLocoNr (B)			
	54 55	Fullframe, Subframe, Downsampled Images (JPEG): ImgImageJpegNI (S) ImgImageJpegNr (S)			
	376 377	Thumbnail Images: ImgThumbNI (B) ImgThumbNr (B)	ImgIdph (B)	n/a	n/a
	366 367	Thumbnail Images (ICER): ImgThumbIcerNI (B) ImgThumbIcerNr (B)			
	372 373	Thumbnail Images (LOCO): ImgThumbLocoNI (B) ImgThumbLocoNr (B)			
	47 48	Thumbnail Images (JPEG): ImgThumbJpegNI (S) ImgThumbJpegNr (S)			
	313 314	Histogram Products: ImgHistogramNI (B) ImgHistogramNr (B)	ImgIdph (B)	n/a	n/a
	357 358	Row-summed Products: ImgRowsumsNI (B) ImgRowsumsNr (B)			
	305 306	Column-summed Products: ImgColsumsNI (B) ImgColsumsNr (B)	ImgIdph (B)	n/a	n/a
CacheCam	13	Fullframe, Subframe, Downsampled Images: ImgImageCc (B)	ImgIdph (B)	n/a	n/a
	14	Fullframe, Subframe, Downsampled Images (ICER): ImgImageIcerCc (B)			
	15	Fullframe, Subframe, Downsampled Images (LOCO): ImgImageLocoCc (B)			
	56	Fullframe, Subframe, Downsampled Images (JPEG): ImgImageJpegCc (S)			

Instrument	APID #	APID Names C=Cruise, S=Surface, B=Both	DPOs Used as Sources for Metadata		
			IDPH	Ancillary	Supplementary
	16	Thumbnail Images: ImgThumbCc (B)	Imgldph (B)	n/a	n/a
	17	Thumbnail Images (ICER): ImgThumbIcerCc (B)			
	18	Thumbnail Images (LOCO): ImgThumbLocoCc (B)			
	49	Thumbnail Images (JPEG): ImgThumbJpegCc (S)			
	22	Histogram Products: ImgHistogramCc (B)	Imgldph (B)	n/a	n/a
	23	Row-summed Products: ImgRowsumsCc (B)	Imgldph (B)	n/a	n/a
	24	Column-summed Products: ImgColsumsCc (B)	Imgldph (B)	n/a	n/a
SuperCam RMI	426 640	Fullframe, Subframe Images: ScamScidata (C) ScamRmiImage (S)	ImfldphSubset6 (S)	ScamAncillaryTemperatures (S) ScamAncillaryArgs (S) ScamAncillaryParams (S)	n/a
	415	Fullframe, Subframe Images (ICER): ScamRmiImageIcer (S)			
	642	Fullframe, Subframe Images (JPEG): ScamRmiImageJpeg (S)			
	643	Zstack: ScamRmiZstackImage (S)	ImfldphSubset6 (S)	ScamAncillaryTemperatures (S) ScamAncillaryArgs (S) ScamAncillaryParams (S)	n/a
	644	Zstack (ICER): ScamRmiZstackImageIcer (S)			
	645	Zstack (JPEG): ScamRmiZstackImageJpeg (S)			
Mastcam-Z	418 419 677 689	Images: MczLScidata (C4.0.1) MczRScidata (C4.0.1) MczLImage (B) MczRImage(B)	DpoCidph (C) ImfldphSubset6 (S)	MczImageAncillaryData (C) MmmImageAncillaryData (S)	MczSupplementaryCommand Arguments (S)
	682 694	Thumbnail Images: MczLThumbnail (S) MczRThumbnail (S)	DpoCidph (C) ImfldphSubset6 (S)	MmmImageAncillaryData (S)	
	684 696	Video: MczLVideo (S) MczRVideo (S)	DpoCidph (C) ImfldphSubset6 (S)	MmmVideoAncillaryData (S)	
	679 691	Recovered Product: MczLRecoveredProduct (S) MczRRecoveredProduct (S)	DpoCidph (C) ImfldphSubset6 (S)	MmmRecoveredProductAncillaryData (S)	
	680 692	Recovered Thumbnail: MczLRecoveredThumbnail (S) MczRRecoveredThumbnail (S)	DpoCidph (C) ImfldphSubset6 (S)	MmmRecoveredThumbnailAncillaryData (S)	
	685 697	Zstack: MczLZstack (S) MczRZstack (S)	DpoCidph (C) ImfldphSubset6 (S)	MmmZstackAncillaryData (S)	
	678 690	Range Map: MczLRangemap (S) MczRRangemap (S)	DpoCidph (C) ImfldphSubset6 (S)	TBD	TBD

Instrument	APID #	APID Names C=Cruise, S=Surface, B=Both	DPOs Used as Sources for Metadata		
			IDPH	Ancillary	Supplementary
SHERLOC WATSON ACI	425 719	Images: SrcImgScidata (C) SrcImgImage (S)	DpoCidph (C) ImfldphSubset6 (S)	MmmImageAncillaryData (S)	SrclmgSupplementaryCmdAr guments (S)
	720	Thumbnail Images: SrcImgThumbnail (S)	DpoCidph (C) ImfldphSubset6 (S)	MmmImageAncillaryData (S)	SrclmgSupplementaryCmdAr guments (S)
	721	Video: SrcImgVideo (S)	DpoCidph (C) ImfldphSubset6 (S)	MmmVideoAncillaryData (S)	SrclmgSupplementaryCmdAr guments (S)
	724	Recovered Product: SrcImgRecoveredProduct (S)	DpoCidph (C) ImfldphSubset6 (S)	MmmReoveredProductAncillar yData (S)	SrclmgSupplementaryCmdAr guments (S)
	725	Recovered Thumbnail: SrcImgRecoveredThumbnail (S)	DpoCidph (C) ImfldphSubset6 (S)	MmmReoveredThumbnailAncil laryData (S)	SrclmgSupplementaryCmdAr guments (S)
	722	Zstack: SrcImgZstack (S)	DpoCidph (C) ImfldphSubset6 (S)	MmmZstackAncillaryData (S)	SrclmgSupplementaryCmdAr guments (S)
	723	Range Map: SrcImgRangemap (S)	DpoCidph (C) ImfldphSubset6 (S)	TBD	TBD
PIXL	422 35 36	Images: PixlSciEng (C) PixlSci (S) PixlSciDec (S)	TBD	TBD	TBD
MEDA Skycam	420 43 46	Fullframe, Subframe, Downsamled Images: MedaSciData (C) MedaUnprocessedImage (S) Fullframe, Subframe, Downsmpled Images (compressed): MedaCompressedImage (S)	TBD	MedalmgHdrDpo (S)	TBD
	45	Thumbnail Images: MedaThumbnailImage (S)	TBD	TBD	TBD
EDL Cam PUCAM1 PUCAM2 PUCAM3 DDCAM RDCAM RUCAM	416	Images: ImgEdlcamScidata (B)	n/a	n/a	n/a
LCAM	630 631 631	Images: VcemgrScidata (B) VcemgrLvsPackets (C) VcemgrVcePackets (S)	n/a	VcemgrMeasurementPacket (C)	TBD
Heli	417 417	Images: HeloScidata (C) LeoSciData (S)	n/a	n/a	n/a

APPENDIX B – Product Label Keyword Definitions, Values, Sources

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
Ops Keyword ACTIVE_FLIGHT_STRING_ID Archive Attribute TBD Definition Indicates which flight computer “string” (separate sets of electronics) was active when this product was acquired. For M2020 there are two redundant flight computers (called “strings”), also known as Rover Compute Elements (RCE’s). Either string (A or B) may be active at any given time.	Valid Values All Cameras “A”, “B” Type string Units n/a Location IDENTIFICATION (Class) X-path TBD Reference TBD	Mode EMD in XML format Field as “<xml name>:[<element>]:[<element>]:<field>” “M20EarthProductMetadata:M20ProductMetadata:CreationStringId” Type U16
Ops Keyword ACTIVITY_ID Archive Attribute TBD Definition Indicates the requesting activity id.	Valid Values Eng. Cameras (S) “N/A” Type string Units n/a Location IDENTIFICATION (Class) X-path TBD Reference TBD	Mode DPO in XML format (referenced to APID Name in Appendix A) Field as “<xml name>:[<element>]:[<element>]:<field>” “<IDPH DPO>:idph:activity_id” TBD Type U32
Ops Keyword APPLICATION_PROCESS_ID Archive Attribute TBD Definition Specifies the name associated with the source or process which created the data. For M2020, the Application Process Identifier (APID) identifies the data type encapsulated in the packet, including whether the packet is a data product packet or a non-data product packet. M2020 shall implement the following APID assignments (all numbers below are in decimal, all ranges are inclusive): a) APID 2047 is used for Idle Packets.	Valid Values “0” to “2039” (B) Type integer Units n/a Location TELEMETRY (Class) X-path TBD Reference TBD	Mode EMD in XML format Field as “<xml name>:[<element>]:[<element>]:<field>” “M20EarthProductMetadata:M20ProductMetadata:Apid” Type n/a

Output Metadata (Product Label)		Input Metadata (Source)
• Ops Keyword • Archive Label Attribute • Definition	• Mission Phase C=Cruise, S=Surface, B=Both • Valid Values (quoted) • Type • Units • Keyword Location in Ops Label • Attribute X-path in Archive label • Attribute Reference in Archive Label (bracketed)	• Mission Phase C=Cruise, S=Surface, B=Both • Mode • Metadata Field • Type
b) APID 2040-2046 will not be produced by M2020 as these APIDs are reserved by the CCSDS standard. c) APID 0 will not be produced by M2020 . d) APID 1 is reserved for X-band time correlation packets. e) APIDs 2-99 are assigned to non-data product packets. Individual values will be assigned to particular packet types as they are identified. Definitions of these packet formats are defined in this document. Definition of APID assignments is included in the Rover Flight Software APID XML. f) APIDs 100-2039 are assigned to RCE Flight Software data product packets. Definition of APID assignments is included in the Rover Flight Software APID XML. g) APIDs 1500-2039 are reserved for SSE (Simulation and Support Equipment) product types. (Above text from M2020 FGICD Volume 1, Rev A) For M2020, only APID Names uniquely identify Data Product types across all FSW versions. For this reason, the integer APIDs are not documented here. See also APPLICATION_PROCESS_NAME and Appendix A.		
Ops Keyword APPLICATION_PROCESS_NAME Archive Attribute TBD Definition Specifies the name associated with the source or process which created the data. For M2020, only APID Names uniquely identify Data Product types across all FSW versions. See also APPLICATION_PROCESS_ID and Appendix A.	Valid Values Surface See Appendix A. Cruise See Appendix A. Type String (256) Units n/a Location TELEMETRY (Class) X-path TBD Reference TBD	Mode EMD in XML format Field as “<xml name>:[<element>]:<field>” “M20EarthProductMetadata:M20ProductMetadata:ProductName” Type n/a
Ops Keyword ARTICULATION_DEVICE_ANGLE Archive Attribute TBD Definition Specifies the value of an angle between two parts or segments of an articulated device.	Valid Value Surface • Eng. Cameras (S) • SCAM RMI (S) • MCZ (S) • SHERLOC (S) • PIXL (S) • MEDA Skycam (S) 1) <u>CHASSIS</u>	Mode DPO in XML format (referenced to APID Name in Appendix A) Field as “<xml name>:[<element>]:<field>” Surface • Eng. Cameras (S) 1) “<IDPH DPO>.idph:steer_fl”, “<IDPH DPO>.idph:steer_fr”,

Output Metadata (Product Label)	Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)
NOTE: M2020 uses radians. The PDS default unit for this keyword is degrees, so the <rad> tag is required. For the RSM, the "MEASURED" values (see ARTICULATION_DEVICE_ANGLE_NAME) represent the value of the resolver (attached to the output side of the joint), while the "FINAL" values represent the encoder (attached to the motor). The resolver should be preferentially used if available, as it measures the angle after joint backlash. A value of 1e+30 indicates the angle is not available, in which case the encoder should be used instead. Note that the "INITIAL" and "REQUESTED" values are also encoder measurements, and could be used to determine the joint's direction of motion for backlash determination.	n/a 2) <u>HGA</u> n/a 3) <u>RSM</u> n/a 4) <u>ARM</u> n/a 5) <u>SHA</u> n/a 6) <u>SCS</u> n/a 7) <u>DRILL</u> n/a Cruise - Eng. Cameras (C) <u>CHASSIS</u> n/a <u>HGA</u> n/a <u>RSM</u> n/a <u>ARM</u> n/a - MCZ (C) <u>CHASSIS</u> n/a <u>HGA</u> n/a <u>RSM</u> n/a Type Surface - Eng. Cameras (S) - SCAM RMI (S) - MCZ (S) - SHERLOC (S) - PIXL (S) - MEDA Skycam (S) - float array[14] - float array[2] - float array[8] - float array[11] - float array[6] - float array[2] - float array[4] Cruise - Eng. Cameras (C) - float array[8] - float array[2] - float array[8] - float array[10] - MCZ (C) - float array[8]
	"<IDPH DPO>:idph:steer_rl", "<IDPH DPO>:idph:steer_rr", "<IDPH DPO>:idph:resolver[0]", "<IDPH DPO>:idph:resolver[1]", "<IDPH DPO>:idph:resolver[2]", "<IDPH DPO>:idph:resolver[3]" "<IDPH DPO>:idph:drive_fl", "<IDPH DPO>:idph:drive_fr", "<IDPH DPO>:idph:drive_ml", "<IDPH DPO>:idph:drive_mr", "<IDPH DPO>:idph:drive_rl", "<IDPH DPO>:idph:drive_rr" 2) "<IDPH DPO>:idph:hga_azimuth", "<IDPH DPO>:idph:hga_elevation" 3) "<IDPH DPO>:idph:rsm_final_azimuth_r", "<IDPH DPO>:idph:rsm_final_elevation_r", "<IDPH DPO>:idph:rsm_initial_azimuth_r", "<IDPH DPO>:idph:rsm_initial_elevation_r", "<IDPH DPO>:idph:rsm_target_azimuth", "<IDPH DPO>:idph:rsm_target_elevation", "<IDPH DPO>:idph:rsm_final_azimuth", "<IDPH DPO>:idph:rsm_final_elevation", "<IDPH DPO>:idph:rsm_initial_azimuth", "<IDPH DPO>:idph:rsm_initial_elevation" 4) "<IDPH DPO>:idph:arm:q_hall[0]", "<IDPH DPO>:idph:arm:q_hall[1]", "<IDPH DPO>:idph:arm:q_hall[2]", "<IDPH DPO>:idph:arm:q_hall[3]", "<IDPH DPO>:idph:arm:q_hall[4]", "<IDPH DPO>:idph:arm:q_res[0]", "<IDPH DPO>:idph:arm:q_res[1]", "<IDPH DPO>:idph:arm:q_res[2]", "<IDPH DPO>:idph:arm:q_res[3]", "<IDPH DPO>:idph:arm:q_res[4]", "<IDPH DPO>:idph:arm:docking_clock_angle" 5) "<IDPH DPO>:idph:aca:scs_sha_joint_angles_hall[0]", "<IDPH DPO>:idph:aca:scs_sha_joint_angles_hall[1]", "<IDPH DPO>:idph:aca:scs_sha_joint_angles_hall[2]", "<IDPH DPO>:idph:aca:scs_sha_joint_angles_res[0]", "<IDPH DPO>:idph:aca:scs_sha_joint_angles_res[1]", "<IDPH DPO>:idph:aca:scs_sha_joint_angles_res[2]" 6) "<IDPH DPO>:idph:aca:scs_bit_carousel_angle_hall", "<IDPH DPO>:idph:aca:scs_bit_carousel_angle_res" 7) "<IDPH DPO>:idph:drill:scs_spindle", "<IDPH DPO>:idph:drill:scs_chuck_mech", "<IDPH DPO>:idph:drill:scs_percuss_mech"

Output Metadata (Product Label)	Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)
2) float array[2] 3) float array[8] Units radians (<rad> unit tag required) Location 1) CHASSIS ARTICULATION_STATE (Group) 2) HGA ARTICULATION_STATE (Group) 3) RSM ARTICULATION_STATE (Group) 4) ARM ARTICULATION_STATE (Group) 5) SHA ARTICULATION_STATE (Group) 6) SCS ARTICULATION_STATE (Group) 7) DRILL ARTICULATION_STATE (Group) X-path TBD Reference TBD	- SuperCam RMI (S5.2) - MCZ Cameras (S6) - SHERLOC (S6) - PIXL (S6) "<IDPH DPO>:idph:mob:steer_lf", "<IDPH DPO>:idph:mob:steer_rf", "<IDPH DPO>:idph:mob:steer_lr", "<IDPH DPO>:idph:mob:steer_rr", "<IDPH DPO>:idph:mob:left_bogie", "<IDPH DPO>:idph:mob:right_bogie", "<IDPH DPO>:idph:mob:left_differential", "<IDPH DPO>:idph:mob:right_differential", "<IDPH DPO>:idph:rvr:drive_fl", "<IDPH DPO>:idph:rvr:drive_fr", "<IDPH DPO>:idph:rvr:drive_lm", "<IDPH DPO>:idph:rvr:drive_rm", "<IDPH DPO>:idph:rvr:drive_rl", "<IDPH DPO>:idph:rvr:drive_rr" "<IDPH DPO>:idph:rvr:hga_azimuth", "<IDPH DPO>:idph:rvr:hga_elevation" "<IDPH DPO>:idph:rsm:resolver_azimuth", "<IDPH DPO>:idph:rsm:resolver_elevation", "<IDPH DPO>:idph:rsm:initial_resolver_azimuth", "<IDPH DPO>:idph:rsm:initial_resolver_elevation", "<IDPH DPO>:idph:rsm:target_azimuth", "<IDPH DPO>:idph:rsm:target_elevation", "<IDPH DPO>:idph:rsm:encoder_azimuth", "<IDPH DPO>:idph:rsm:encoder_elevation", "<IDPH DPO>:idph:rsm:initial_encoder_azimuth", "<IDPH DPO>:idph:rsm:initial_encoder_elevation" "<IDPH DPO>:idph:arm:q_hall[0]", "<IDPH DPO>:idph:arm:q_hall[1]", "<IDPH DPO>:idph:arm:q_hall[2]", "<IDPH DPO>:idph:arm:q_hall[3]", "<IDPH DPO>:idph:arm:q_hall[4]", "<IDPH DPO>:idph:arm:q_res[0]", "<IDPH DPO>:idph:arm:q_res[1]", "<IDPH DPO>:idph:arm:q_res[2]", "<IDPH DPO>:idph:arm:q_res[3]", "<IDPH DPO>:idph:arm:q_res[4]", "<IDPH DPO>:idph:arm:docking_clock_angle" "<IDPH DPO>:idph:scs:sha_joint_angles_hall[0]", "<IDPH DPO>:idph:scs:sha_joint_angles_hall[1]", "<IDPH DPO>:idph:scs:sha_joint_angles_hall[2]", "<IDPH DPO>:idph:scs:sha_joint_angles_res[0]", "<IDPH DPO>:idph:scs:sha_joint_angles_res[1]", "<IDPH DPO>:idph:scs:sha_joint_angles_res[2]" "<IDPH DPO>:idph:scs:bit_carousel_angle_hall", "<IDPH DPO>:idph:scs:bit_carousel_angle_res" "<IDPH DPO>:idph:scs:spindle", "<IDPH DPO>:idph:scs:chuck_mech", "<IDPH DPO>:idph:scs:percuss_mech", Cruise and S5.2 - Eng. Cameras (C)

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
		1) – 4) Same as MSL Surface 5) – 7) Not in Cruise FSW <u>MCZ Cameras (C, S5.2)</u> 1) "<IDPH DPO>:cidph:steering:front_left", "<IDPH DPO>:cidph:steering:front_right", "<IDPH DPO>:cidph:steering:rear_left", "<IDPH DPO>:cidph:steering:rear_right", "<IDPH DPO>:cidph:resolvers:bogie_left", "<IDPH DPO>:cidph:resolvers:bogie_right", "<IDPH DPO>:cidph:resolvers:differential_left", "<IDPH DPO>:cidph:resolvers:differential_right" 2) "<IDPH DPO>:cidph:hga:azimuth", "<IDPH DPO>:cidph:hga:elevation" 3) "<IDPH DPO>:cidph:rsm_state:res:azimuth", "<IDPH DPO>:cidph:rsm_state:res:elevation", "<IDPH DPO>:cidph:rsm_state:target:azimuth", "<IDPH DPO>:cidph:rsm_state:target:elevation", "<IDPH DPO>:cidph:rsm_state:initial:azimuth", "<IDPH DPO>:cidph:rsm_state:initial:elevation", "<IDPH DPO>:cidph:rsm_state:final:azimuth", "<IDPH DPO>:cidph:rsm_state:final:elevation" 4) – 7) Not in Cruise FSW Type <u>Eng. Cameras</u> 1) – 6) F32 7) F64 <u>SCAM RMI, MCZ, SHERLOC, PIXL MCC, MEDA SkyCam</u> 1) – 7) F32
Ops Keyword ARTICULATION_DEVICE_ANGLE_NAME Archive Attribute TBD Definition Specifies the formal name which identifies each of the values used in ARTICULATION_DEVICE_ANGLE.	Valid Values Surface - Eng. Cameras (S) - SCAM RMI (S) - MCZ (S) - SHERLOC (S) - PIXL (S) - MEDA Skycam (S) 1) <u>CHASSIS</u> ("LEFT FRONT WHEEL STEER MOTOR", "RIGHT FRONT WHEEL STEER MOTOR", "LEFT REAR WHEEL STEER MOTOR", "RIGHT REAR WHEEL STEER MOTOR", "LEFT BOGIE", "RIGHT BOGIE", "LEFT DIFFERENTIAL", "RIGHT DIFFERENTIAL", "LEFT FRONT WHEEL DRIVE MOTOR", "RIGHT FRONT WHEEL DRIVE MOTOR", "MIDDLE LEFT WHEEL DRIVE MOTOR", "MIDDLE RIGHT WHEEL DRIVE MOTOR", "LEFT REAR WHEEL DRIVE MOTOR", "RIGHT REAR WHEEL DRIVE MOTOR") 2) <u>HGA</u>	Mode Static values Field as "<xml name>:[<element>]:[<element>]:<field>" n/a Type n/a

Output Metadata (Product Label)	Input Metadata (Source)
• Ops Keyword • Archive Label Attribute • Definition	• Mission Phase C=Cruise, S=Surface, B=Both • Valid Values (quoted) • Type • Units • Keyword Location in Ops Label • Attribute X-path in Archive label • Attribute Reference in Archive Label (bracketed)
	(“AZIMUTH”, “ELEVATION”) 3) <u>RSM</u> (“AZIMUTH FINAL-RESOLVER”, “ELEVATION FINAL-RESOLVER”, “AZIMUTH INITIAL-RESOLVER”, “ELEVATION INITIAL-RESOLVER”, “AZIMUTH REQUESTED”, “ELEVATION REQUESTED”, “AZIMUTH FINAL-ENCODER”, “ELEVATION FINAL-ENCODER”, “AZIMUTH INITIAL-ENCODER”, “ELEVATION INITIAL-ENCODER” 4) <u>ARM</u> (“JOINT 1 AZIMUTH-HALL SENSOR”, “JOINT 2 ELEVATION-HALL SENSOR”, “JOINT 3 ELBOW-HALL SENSOR”, “JOINT 4 WRIST-HALL SENSOR”, “JOINT 5 TURRET-HALL SENSOR”, “JOINT 1 AZIMUTH-RESOLVER”, “JOINT 2 ELEVATION-RESOLVER”, “JOINT 3 ELBOW-RESOLVER”, “JOINT 4 WRIST-RESOLVER”, “JOINT 5 TURRET-RESOLVER”, “SCS DOCKING CLOCK ANGLE”) 5) <u>SHA</u> (“SHA JOINT 1 Z- HALL SENSOR”, “SHA JOINT 2 SHOULDER-HALL SENSOR”, “SHA JOINT 3 ELBOW-HALL SENOSR”, “SHA JOINT 1 Z-RESOLVER”, “SHA JOINT 2 SHOULDER-RESOLVER”, “SHA JOINT 3 ELBOW-RESOLVER”) 6) <u>SCS</u> (“SCS BIT CAROUSEL-HALL SENSOR”, “SCS BIT CAROUSEL-RESOLVER”) 7) <u>DRILL</u> (“SPINDLE OUTPUT POSITION-RESOLVER”, “CHUCK POSITION-RESOLVER”, “PERCUSS POSITION-RESOLVER”) Cruise • Eng. Cameras (C) 1) <u>CHASSIS</u> (“LEFT FRONT WHEEL STEER MOTOR”, “RIGHT FRONT WHEEL STEER MOTOR”, “LEFT REAR WHEEL STEER MOTOR”, “RIGHT REAR WHEEL STEER MOTOR”, “LEFT BOGIE”, “RIGHT BOGIE”, “LEFT DIFFERENTIAL”, “RIGHT DIFFERENTIAL”) 2) <u>HGA</u>

Output Metadata (Product Label)	Input Metadata (Source)
• Ops Keyword • Archive Label Attribute • Definition	• Mission Phase C=Cruise, S=Surface, B=Both • Valid Values (quoted) • Type • Units • Keyword Location in Ops Label • Attribute X-path in Archive label • Attribute Reference in Archive Label (bracketed)
	(“AZIMUTH”, “ELEVATION”) 3) <u>RSM</u> (“AZIMUTH FINAL-RESOLVER”, “ELEVATION FINAL-RESOLVER”, “AZIMUTH REQUESTED”, “ELEVATION REQUESTED”, “AZIMUTH INITIAL-ENCODER”, “ELEVATION INITIAL-ENCODER”, “AZIMUTH FINAL-ENCODER”, “ELEVATION FINAL-ENCODER”) 4) <u>ARM</u> (“JOINT 1 AZIMUTH-HALL SENSOR”, “JOINT 2 ELEVATION-HALL SENSOR”, “JOINT 3 ELBOW-HALL SENSOR”, “JOINT 4 WRIST-HALL SENSOR”, “JOINT 5 TURRET-HALL SENSOR”, “JOINT 1 AZIMUTH-RESOLVER”, “JOINT 2 ELEVATION-RESOLVER ”, “JOINT 3 ELBOW-RESOLVER ”, “JOINT 4 WRIST-RESOLVER ”, “JOINT 5 TURRET-RESOLVER ”) • MCZ (C) 1) <u>CHASSIS</u> (“LEFT FRONT WHEEL STEER MOTOR”, “RIGHT FRONT WHEEL STEER MOTOR”, “LEFT REAR WHEEL STEER MOTOR”, “RIGHT REAR WHEEL STEER MOTOR”, “LEFT BOGIE”, “RIGHT BOGIE”, “LEFT DIFFERENTIAL”, “RIGHT DIFFERENTIAL”) 2) <u>HGA</u> (“AZIMUTH”, “ELEVATION”) 3) <u>RSM</u> (“AZIMUTH FINAL-RESOLVER”, “ELEVATION FINAL-RESOLVER”, “AZIMUTH REQUESTED”, “ELEVATION REQUESTED”, “AZIMUTH INITIAL-ENCODER”, “ELEVATION INITIAL-ENCODER”, “AZIMUTH FINAL-ENCODER”, “ELEVATION FINAL-ENCODER”) Type Cruise 1) string array[8] 2) string array[2] 3) string array[8] 4) string array[10] Surface 1) string array[14]

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
	2) string array[2] 3) string array[8] 4) string array[10] 5) string array[6] 6) string array[2] 7) string array[4] Units n/a Location 1) CHASSIS ARTICULATION_STATE (Group) 2) HGA ARTICULATION_STATE (Group) 3) RSM ARTICULATION_STATE (Group) 4) ARM ARTICULATION_STATE (Group) 5) SHA ARTICULATION_STATE (Group) 6) SCS ARTICULATION_STATE (Group) 7) DRILL ARTICULATION_STATE (Group) X-path TBD Reference TBD	
Ops Keyword ART_DEV_COMPONENT_STATE Archive Attribute TBD Definition	Valid Values Surface - Eng. Cameras - SuperCam - MCZ - SHERLOC Preload Value[0]: n/a Preload Tool[1]: 0 = "TURRET" 1 = "DRILL" 2 = "DOCKING POST" 3 = "PIXL" 4 = "GDRT" 5 = "FCS" 6 = "WATSON" 7 = "SHERLOC" 8 = "CUSTOM TCP" Type string[2] Units Newtons Location ARM ARTICULATION_STATE X-path TBD Reference TBD	Mode DPO in XML format (referenced to APID Name in Appendix A) Field as "<xml name>:[<element>]:[<element>]:<field>" Surface - Eng. Cameras (S5.2,S6) - SuperCam RMI (S5.2,S6) - MCZ (S6) - SHERLOC (S6) "<IDPH DPO>:idph:arm:preload", "<IDPH DPO>:idph:arm:preload_tool" - MCZ (S5.2) N/A - SHERLOC (S5.2) N/A Type n/a

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
Ops Keyword ART_DEV_COMPONENT_STATE_NAME Archive Attribute TBD Definition	Valid Values Surface All Cameras (S) ("PRELOAD_VALUE", "PRELOAD_TOOL") Type string Units n/a Location ARM_ARTICULATION_STATE X-path TBD Reference TBD	Mode Static values Field as "<xml name>:[<element>]:<field>" n/a Type n/a
Ops Keyword ARTICULATION_DEVICE_ID Archive Attribute TBD Definition Specifies the unique abbreviated identification of an articulation device. An articulation device is anything that can move independently of the spacecraft to which it is attached, (e.g., mast heads, wheel bogies, arms, etc.). NOTE: The ARTICULATION_DEVICE_ID is not a unique identifier for a given articulated device. Note also that the associated ARTICULATION_DEVICE_NAME element provides the full name of the articulated device.	Valid Values Surface - Eng. Cameras (B) - SCAM RMI (S) - MCZ (B) - SHERLOC (S) - PIXL (S) - MEDA Skycam (S) "CHASSIS" "HGA" "RSM" "ARM" "SHA" "SCS" "DRILL" Cruise - Eng. Cameras (C) "CHASSIS" "HGA" "RSM" "ARM" - MCZ (C) "CHASSIS" "HGA" "RSM" Type string Units n/a Location - CHASSIS_ARTICULATION_STATE (Group) - HGA_ARTICULATION_STATE (Group) - RSM_ARTICULATION_STATE (Group) - ARM_ARTICULATION_STATE (Group) - SHA_ARTICULATION_STATE (Group) - SCS_ARTICULATION_STATE (Group)	Mode Static values Field as "<xml name>:[<element>]:<field>" n/a Type n/a

Output Metadata (Product Label)	Input Metadata (Source)
• Ops Keyword • Archive Label Attribute • Definition	• Mission Phase C=Cruise, S=Surface, B=Both • Valid Values (quoted) • Type • Units • Keyword Location in Ops Label • Attribute X-path in Archive label • Attribute Reference in Archive Label (bracketed)
	7) DRILL_ARTICULATION_STATE (Group) <u>X-path</u> TBD <u>Reference</u> TBD
Ops Keyword ARTICULATION_DEVICE_MODE Archive Attribute TBD Definition Specifies the deployment state (i.e., physical configuration) of an articulation device at the time of data acquisition. The SCS tube states are defined as followed: "EMPTY" – tube has not been used to collect a sample. "FILLED" – tube has been used to collect a sample (regardless of volume). "SEAL DISPENSED" – seal has been inserted into the tube, but not sealed. "SEAL ACTIVATED" – tube is sealed. The drill core break lockout rod (CBLO) states are defined as follows: "INVALID POSE" – CBLO is outside valid range of motion. "INDETERMINANT" – CBLO is not near one of the predefined positions, but is within valid range of motion. "RETRACTED" – CBLO is at the retracted (spindle homing) hardstop. "DRILL" – CBLO is at the drilling position. "STUCK RECOVERY" – CBLO is at the stuck recovery position. "EXTENDED" – CBLO is at the extended (core break) hardstop.	Valid Values <u>Surface</u> • Eng. Cameras (S) • SCAM RMI (S) • MCZ (S) • SHERLOC (S) • PIXL (S) • MEDA Skycam (S) <u>CHASSIS</u> "DEPLOYED" <u>HGA</u> "DEPLOYED" <u>RSM</u> 0 = "STOWED" 1 = "DEPLOYED" <u>SCS</u> 0 = "EMPTY" 1 = "FILLED" 2 = "SEAL DISPENSED" 3 = "SEAL ACTIVATED" <u>DRILL</u> 0 = "INVALID POSE" 1 = "INDETERMENT" 2 = "RETRACTED" 3 = "DRILL" 4 = "STUCK RECOVERY" 5 = "EXTENDED" Cruise • Eng. Cameras (C) • MCZ (C) 1) <u>CHASSIS</u> "DEPLOYED" 2) <u>HGA</u> "DEPLOYED" 3) <u>RSM</u> 0 = "STOWED" 1 = "DEPLOYED" Type string Units n/a Location
	Mode • Static value: 1) for Chassis 2) for HGA • DPO in XML format (referenced to APID Name in Appendix A) Field as "<xml name>:[<element>]:[<element>]:[<field>]" <u>Surface</u> • <u>Eng. Cameras</u> (S5.2, S6) 3) "<IDPH DPO>:idph:rsm_deployed" 4) "<IDPH DPO>:idph:aca:sca_active_tube_state" 5) "<IDPH DPO>:idph:drill:scs_core_break_lock" • <u>SCAM RMI</u> (S5.2, S6) • <u>MCZ Cameras</u> (S6) • <u>SHERLOC</u> (S6) • <u>PIXL</u> (S5.2?, S6) • <u>MEDA Skycam</u> (S5.2?, S6) 3) "<IDPH DPO>:idph:rsm:deployed" 4) "<IDPH DPO>:idph:scs:active_tube_state" 5) "<IDPH DPO>:idph:scs:core_break_lock" Cruise & S5.2 • <u>Eng. Cameras</u> (C) 3) "<IDPH DPO>:idph:rsm_deployed" • <u>MCZ Cameras</u> (C, S5.2) 3) "<IDPH DPO>:cidph:rsm_state:deployed" Type 1) n/a 2) n/a 3) Eng. Bool All other instruments U8 4) U8 5) U8

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
	1) CHASSIS ARTICULATION_STATE (Group) 2) HGA ARTICULATION_STATE (Group) 3) RSM ARTICULATION_STATE (Group) 4) SCS ARTICULATION_STATE (Group) 5) DRILL ARTICULATION_STATE (Group) <u>X-path</u> TBD <u>Reference</u> TBD	
<u>Ops Keyword</u> ARTICULATION_DEVICE_NAME <u>Archive Attribute</u> TBD <u>Definition</u> Specifies the common name of an articulation device. An articulation device is anything that can move independently of the spacecraft to which it is attached, (e.g. mast heads, wheel bogies, arms, etc.) NOTE: The associated ARTICULATION_DEVICE_ID element provides an abbreviated name or acronym for the articulated device.	<u>Valid Values</u> <u>Surface</u> - Eng. Cameras (S) - SCAM RMI (S) - MCZ (S) - SHERLOC (S) - PIXL (S) - MEDA Skycam (S) "MOBILITY CHASSIS" "HIGH GAIN ANTENNA" "REMOTE SENSING MAST" "SAMPLE ARM" "SAMPLE HANDLING ARM" "SAMPLE CACHE SYSTEM" "DRILL" <u>Cruise</u> - Eng. Cameras (C) "MOBILITY CHASSIS" "HIGH GAIN ANTENNA" "REMOTE SENSING MAST" "SAMPLE ARM" - MCZ (C) "MOBILITY CHASSIS" "HIGH GAIN ANTENNA" "REMOTE SENSING MAST" <u>Type</u> string <u>Units</u> n/a <u>Location</u> - CHASSIS ARTICULATION_STATE (Group) - HGA ARTICULATION_STATE (Group) - RSM ARTICULATION_STATE (Group) - ARM ARTICULATION_STATE (Group) - SHA ARTICULATION_STATE (Group) - SCS ARTICULATION_STATE (Group) - DRILL ARTICULATION_STATE (Group) <u>X-path</u> TBD <u>Reference</u> TBD	<u>Mode</u> Static values <u>Field as "<xml name>:[<element>]:[<element>]:[<field>]"</u> n/a <u>Type</u> n/a
<u>Ops Keyword</u>	<u>Valid Values</u>	<u>Mode</u>

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
ARTICULATION_DEVICE_TEMP Archive Attribute TBD Definition Specifies the temperature, in degrees Celsius, of an articulated device or some part of an articulated device.	All Cameras (S) "-3.4e38" to "3.4e38" Type Surface a. float array[7] b. float array[2] Cruise a. float array[5] b. float array[2] Units deg C (<degC> unit tag required) Location a. ARM_ARTICULATION_STATE (Group) b. RSM_ARTICULATION_STATE (Group) X-path TBD Reference TBD	DPO in XML format (referenced to APID Name in Appendix A) Field as "<xml name>:[<element>]:<field>" Surface - Eng. Cameras (S5.2, S6) a. "<IDPH DPO>:idph:arm:temp_actuator[0]" "<IDPH DPO>:idph:arm:temp_actuator[1]" "<IDPH DPO>:idph:arm:temp_actuator[2]" "<IDPH DPO>:idph:arm:temp_actuator[3]" "<IDPH DPO>:idph:arm:temp_actuator[4]" "<IDPH DPO>:idph:arm:temp_deflection[0]" "<IDPH DPO>:idph:arm:temp_deflection[1]" b. "<IDPH DPO>:idph:temp[20]", "<IDPH DPO>:idph:temp[21]" - SuperCam (S5.2, S6) - MCZ (S6) - SHERLOC (S6) a. "<IDPH DPO>:idph:arm:temp_actuator[0]" "<IDPH DPO>:idph:arm:temp_actuator[1]" "<IDPH DPO>:idph:arm:temp_actuator[2]" "<IDPH DPO>:idph:arm:temp_actuator[3]" "<IDPH DPO>:idph:arm:temp_actuator[4]" "<IDPH DPO>:idph:arm:temp_deflection[0]" "<IDPH DPO>:idph:arm:temp_deflection[1]" b. "<IDPH DPO>:idph:rsm:camera_plate_temp2", "<IDPH DPO>:idph:rsm:camera_plate_temp1" Cruise and S5.2 - Eng. Cameras (C) a. "<IDPH DPO>:idph:arm:temp_actuator[0]" "<IDPH DPO>:idph:arm:temp_actuator[1]" "<IDPH DPO>:idph:arm:temp_actuator[2]" "<IDPH DPO>:idph:arm:temp_actuator[3]" "<IDPH DPO>:idph:arm:temp_actuator[4]" b. "<IDPH DPO>:idph:temp[20]", "<IDPH DPO>:idph:temp[21]" - MCZ (C, S5.2) a. "<IDPH DPO>:cidph:arm_state:temp[0]" "<IDPH DPO>:cidph:arm_state:temp[1]" "<IDPH DPO>:cidph:arm_state:temp[2]" "<IDPH DPO>:cidph:arm_state:temp[3]" "<IDPH DPO>:cidph:arm_state:temp[4]" Type Surface F32[7] Cruise F32[5]
Ops Keyword ARTICULATION_DEVICE_TEMP_NAME Archive Attribute TBD Definition	Valid Values Surface a. ("AZIMUTH JOINT", "ELEVATION JOINT", "ELBOW JOINT", "WRIST JOINT", "TURRET JOINT", "DEFLECTION BODY DELTA TEMPERATURE", "DEFLECTION ARM DELTA TEMPERATURE")	Mode Static value Field as "<xml name>:[<element>]:<field>" n/a Type

Output Metadata (Product Label)		Input Metadata (Source)
• Ops Keyword • Archive Label Attribute • Definition	• Mission Phase C=Cruise, S=Surface, B=Both • Valid Values (quoted) • Type • Units • Keyword Location in Ops Label • Attribute X-path in Archive label • Attribute Reference in Archive Label (bracketed)	• Mission Phase C=Cruise, S=Surface, B=Both • Mode • Metadata Field • Type
Specifies the array of formal names identifying each of the values used in ARTICULATION_DEVICE_TEMP.	b. (“NAVCAM LEFT CAMERA PLATE”, “NAVCAM RIGHT CAMERA PLATE”) Cruise a. (“AZIMUTH JOINT”, “ELEVATION JOINT”, “ELBOW JOINT”, “WRIST JOINT”, “TURRET JOINT”) b. (“NAVCAM LEFT CAMERA PLATE”, “NAVCAM RIGHT CAMERA PLATE”) Type Surface a. string array[7] b. string array[2] Cruise a. string array[5] b. string array[2] Units n/a Location a. ARM_ARTICULATION_STATE (Group) b. RSM_ARTICULATION_STATE (Group) X-path TBD Reference TBD	n/a
Ops Keyword ARTICULATION_DEV_INSTRUMENT_ID Archive Attribute TBD Definition Specifies an abbreviated name or acronym which identifies the instrument mounted on the articulation device.	Valid Values Surface 0 = “TURRET” 1 = “DRILL” 2 = “DOCKING POST” 3 = “PIXL” 4 = “GDRT” 5 = “FCS” 6 = “WATSON” 7 = “SHERLOC” 8 = “CUSTOM TCP” Cruise 0 = “TURRET” 1 = “DRILL” 2 = “DOCKING POST” 3 = “PIXL” 4 = “GDRT” 5 = “FCS” 6 = “WATSON” 7 = “SHERLOC” 8 = “CUSTOM TCP” Type string Units n/a	Mode DPO in XML format (referenced to APID Name in Appendix A) Field as “<xml name>:[<element>]:<field>” Surface • <u>Eng. Cameras</u> (S5.2, S6) • <u>SuperCam</u> (S5.2, S6) • <u>MCZ</u> (S6) • <u>SHERLOC</u> (S6) • <u>MEDA</u> (S5.2?, S6) • <u>PIXL</u> (S5.2?, S6) “<IDPH DPO>:idph:arm:tool” Cruise and S5.2 • <u>Eng. Cameras</u> (C) “<IDPH DPO>:idph:arm_instrument” • <u>MCZ</u> (C, S5.2) “<IDPH DPO>:cidph:arm_state:instrument” Type U8

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
	<u>Location</u> ARM_ARTICULATION_STATE (Group) <u>X-path</u> TBD <u>Reference</u> TBD	
<u>Ops Keyword</u> ARTICULATION_DEV_POSITION <u>Archive Attribute</u> TBD <u>Definition</u> Specifies the set of indices for articulation devices that contain moving parts with discrete positions. The associated ARTICULATION_DEV_POSITION_NAME names each moving device. In order to get the actual filter used for this specific image, the FILTER_NAME/FILTER_NUMBER keywords in the INSTRUMENT_DATA group should be used.	<u>Valid Values</u> n/a <u>Type</u> int array[3] <u>Units</u> n/a <u>Location</u> MINI_HEADER (Group)	<u>Mode</u> DPO in XML format (referenced to APID Name in Appendix A) Calculation: Filter = filter index * 294 <u>Field as “<xml name>:[<element>]:<field>”</u> - MCZ - SHERLOC MMM_Image_Mini_Header:mech[1] MMM_Image_Mini_Header:mech[2] MMM_Image_Mini_Header:mech[3] <u>Type</u> Integer <u>Note:</u> For MCZ and SHERLOC products, mech values “focus”, “zoom”, and “filter” are comprised of 8 bytes starting at byte offset 44 (words 11 and 12): Focus = byte offset [44] – [47] Zoom = byte offset [48] – [49] Filter = byte offset [50] – [51]
<u>Ops Keyword</u> ARTICULATION_DEV_POSITION_NAME <u>Archive Attribute</u> TBD <u>Definition</u> Specifies an array of values that provides the formal names for each entry in ARTICULATION_DEV_POSITION. This element names the actual device doing the moving, (e.g., a filter wheel), not the name of a position (e.g., the filter itself).	<u>Valid Values</u> ("FOCUS", "ZOOM", "FILTER") <u>Type</u> string array[3] <u>Units</u> n/a <u>Location</u> MINI_HEADER (Group)	<u>Mode</u> Static value <u>Field as “<xml name>:[<element>]:<field>”</u> - MCZ - SHERLOC n/a <u>Type</u> TBD
<u>Ops Keyword</u> ARTICULATION_DEV_VECTOR <u>Archive Attribute</u> TBD <u>Definition</u> Specifies the direction and magnitude of an external force acting on the articulation device, in the rover's coordinate system, at the time the pose was computed.	<u>Valid Values</u> “-1.0” to “1.0” <u>Type</u> float array[3] <u>Units</u> n/a <u>Location</u> 1) ARM_ARTICULATION_STATE (Group) 2) CHASSIS_ARTICULATION_STATE (Group)	<u>Mode</u> DPO in XML format (referenced to APID Name in Appendix A) <u>Field as “<xml name>:[<element>]:<field>”</u> <u>Surface</u> Eng_Cameras 1) “<IDPH DPO>:idph:arm.att_ref[0]” “<IDPH DPO>:idph:arm.att_ref[1]” “<IDPH DPO>:idph:arm.att_ref[2]” “<IDPH DPO>:idph:arm.att_ref[3]”

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
	<u>X-path</u> TBD <u>Reference</u> TBD	NEED TO CONVERT to GRAVITY 2) "<IDPH DPO>:idph:raw_gravity[0]" "<IDPH DPO>:idph:raw_gravity[1]" "<IDPH DPO>:idph:raw_gravity[2]" - SuperCam - MCZ - SHERLOC - PIXL 1) "<IDPH DPO>:idph:arm.att_ref[0]" "<IDPH DPO>:idph:arm.att_ref[1]" "<IDPH DPO>:idph:arm.att_ref[2]" "<IDPH DPO>:idph:arm.att_ref[3]" NEED TO CONVERT to GRAVITY 2) "<IDPH DPO>:idph:rvr:sapp_raw_gravity[0]" "<IDPH DPO>:idph:rvr:sapp_raw_gravity[1]" "<IDPH DPO>:idph:rvr:sapp_raw_gravity[2]" <u>Type</u> 1) F32[3] 2) F32[3]
<u>Ops Keyword</u> ARTICULATION_DEV_VECTOR_NAME <u>Archive Attribute</u> TBD <u>Definition</u> Specifies the formal name of the vector type acting on the articulation device.	<u>Valid Values</u> "GRAVITY" <u>Type</u> string <u>Units</u> n/a <u>Location</u> ARM ARTICULATION_STATE (Group) CHASSIS ARTICULATION_STATE (Group) <u>X-path</u> TBD <u>Reference</u> TBD	<u>Mode</u> Static value <u>Field as "<xml name>:[<element>]:[<element>]:<field>"</u> n/a <u>Type</u> n/a
<u>Ops Keyword</u> ATTITUDE_PROPAGATION_COUNTER <u>Archive Attribute</u> TBD <u>Definition</u> Specifies the total number of RTIs spent propagating since the last gyroscopic propagation counter reset.	<u>Valid Values</u> "N/A" <u>Type</u> float <u>Units</u> n/a <u>Location</u> ROVER_COORDINATE_SYSTEM <u>X-path</u> TBD <u>Reference</u> TBD	<u>Mode</u> DPO in XML format (referenced to APID Name in Appendix A) <u>Field as "<xml name>:[<element>]:[<element>]:<field>"</u> Eng. Cameras (S) "<IDPH DPO>idph:sapp_counter" <u>Type</u> F64
<u>Ops Keyword</u> ATTITUDE_PROPAGATION_DURATION <u>Archive Attribute</u>	<u>Valid Values</u> "N/A" <u>Type</u>	<u>Mode</u> DPO in XML format (referenced to APID Name in Appendix A)

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
TBD Definition Specifies the total time spent propagating on the gyrosince the last gyroscopic propagation counter reset.	float <u>Units</u> Seconds <u>Location</u> ROVER_COORDINATE_SYSTEM <u>X-path</u> TBD <u>Reference</u> TBD	Calculation: 1) ECAM = sapp_counter / 8 Field as "<xml name>:[<element>]:[<element>]:<field>" Eng. Cameras (S) "<IDPH DPO>idph:sapp_counter" <u>Type</u> F64
Ops Keyword AUTO_DELETE_FLAG Archive Attribute TBD Definition Indicates if the product will be automatically deleted onboard after it is transmitted.	Valid Values 0 = "FALSE" 1 = "TRUE" <u>Type</u> string <u>Units</u> n/a <u>Location</u> TELEMETRY (Class) <u>X-path</u> TBD <u>Reference</u> TBD	Mode EMD in XML format Field as "<xml name>:[<element>]:[<element>]:<field>" "M20EarthProductMetadata:M20ProductMetadata:DeleteOnSend" <u>Type</u> n/a
Ops Keyword AUTO_EXPOSURE_LOWER_LIMIT Archive Attribute TBD Definition Specifies the maximum number of pixels allowed that are below the lower threshold DN value, as defined by the AUTO_EXPOSURE_LOWER_THRESHOLD keyword.	Valid Values "0" to "1048575" (B) <u>Type</u> integer <u>Units</u> n/a <u>Location</u> OBSERVATION_REQUEST_PARMS (Group) MINI_HEADER (Group) <u>X-path</u> TBD <u>Reference</u> TBD	Mode SCAM Image DPO mini-header Field as "<xml name>:[<element>]:[<element>]:<field>" "SCAM_Image_Mini_Header:autoexp_num_pixels_under_lower_threshold" <u>Type</u> U8 Notes: For SCAM RMI products, "auto exposure lower limit" is comprised of three bytes coming from the Image DPO mini-header at offset 17.
Ops Keyword AUTO_EXPOSURE_ROI_SIZE Archive Attribute TBD Definition Specifies the autoexposure ROI size as red pixel square side.	Valid Values "0" to "2046" (B) <u>Type</u> integer <u>Units</u> n/a <u>Location</u>	Mode SCAM Image DPO mini-header Field as "<xml name>:[<element>]:[<element>]:<field>" "SCAM_Image_Mini_Header:autoexp_roi_size" <u>Type</u> U16

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
Note: Must be even.	OBSERVATION_REQUEST_PARMS (Group) MINI_HEADER (Group) <u>X-path</u> TBD <u>Reference</u> TBD	Notes: For SCAM RMI products, "autoexposure roi size" is comprised of two bytes coming from the Image DPO mini-header at offset 23.
<u>Ops Keyword</u> AUTO_EXPOSURE_ROI_Y_POSITION <u>Archive Attribute</u> TBD <u>Definition</u> Specifies the autoexposure ROI top left coordinate in y. Note: Red pixels only.	<u>Valid Values</u> "2" to "2048" (B) <u>Type</u> integer <u>Units</u> n/a <u>Location</u> OBSERVATION_REQUEST_PARMS (Group) MINI_HEADER (Group) <u>X-path</u> TBD <u>Reference</u> TBD	<u>Mode</u> SCAM Image DPO mini-header <u>Field as "<xml name>:[<element>]:[<element>]:<field>"</u> "SCAM_Image_Mini_Header:autoexp_roi_y_size" <u>Type</u> U16 Notes: For SCAM RMI products, "autoexposure roi y position" is comprised of two bytes coming from the Image DPO mini-header at offset 25.
<u>Ops Keyword</u> AUTO_EXPOSURE_LOWER_THRESHOLD <u>Archive Attribute</u> TBD <u>Definition</u> Specifies the lower threshold DN value in which a specified number of pixels is permitted to exceed. AUTO_EXPOSURE_LOWER_LIMIT keyword defines the number of pixels allowed to exceed the upper threshold.	<u>Valid Values</u> "0" to "1024" (B) <u>Type</u> integer <u>Units</u> n/a <u>Location</u> 1) a. OBSERVATION_REQUEST_PARMS (Group) b. MINI_HEADER (Group) <u>X-path</u> TBD <u>Reference</u> TBD	<u>Mode</u> SCAM Image DPO mini-header <u>Field as "<xml name>:[<element>]:[<element>]:<field>"</u> "SCAM_Image_Mini_Header:autoexp_lower_threshold" <u>Type</u> U16 Notes: For SCAM RMI products, "autoexposure lower threshold" is comprised of two bytes coming from the Image DPO mini-header at offset 13.
<u>Ops Keyword</u> AUTO_EXPOSURE_UPPER_LIMIT <u>Archive Attribute</u> TBD <u>Definition</u> Specifies the lower threshold DN value in which a specified number of pixels is permitted to exceed. AUTO_EXPOSURE_LOWER_LIMIT keyword defines the number of pixels allowed to exceed the upper threshold.	<u>Valid Values</u> "0" to "1048576" (B) <u>Type</u> integer <u>Units</u> n/a <u>Location</u> OBSERVATION_REQUEST_PARMS (Group) MINI_HEADER (Group) <u>X-path</u> TBD	<u>Mode</u> SCAM Image DPO mini-header <u>Field as "<xml name>:[<element>]:[<element>]:<field>"</u> "SCAM_Image_Mini_Header:autoexp_max_num_pixels_above_upper_threshold" <u>Type</u> U8[] Notes:

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
	<u>Reference</u> TBD	For SCAM RMI products, "auto exposure upper limit" is comprised of three bytes coming from the Image DPO mini-header at offset 20.
<u>Ops Keyword</u> AUTO_EXPOSURE_UPPER_THRESHOLD <u>Archive Attribute</u> TBD <u>Definition</u> Specifies the upper threshold DN value in which a specified number of pixels is permitted to exceed. AUTO_EXPOSURE_UPPER_LIMIT keyword defines the number of pixels allowed to exceed the upper threshold.	<u>Valid Values</u> "0" to "1023" <u>Type</u> integer <u>Units</u> n/a <u>Location</u> 1) a. OBSERVATION_REQUEST_PARMS (Group) b. MINI_HEADER (Group) <u>X-path</u> TBD <u>Reference</u> TBD	<u>Mode</u> SCAM Image DPO mini-header <u>Field as "<xml name>:[<element>]:<field>"</u> "SCAM_Image_Mini_Header:autoexp_upper_threshold" <u>Type</u> U8[] Notes: For SCAM RMI products, "autoexposure upper threshold" is comprised of two bytes coming from the Image DPO mini-header at offset 15.
<u>Ops Keyword</u> AUTO_EXPOSURE_DATA_CUT <u>Archive Attribute</u> TBD <u>Definition</u> Specifies the DN value which a specified fraction of pixels is permitted to exceed. The fraction is specified using the keyword AUTO_EXPOSURE_DATA_FRACTION.	<u>Valid Values</u> - Eng. Cameras - MCZ "0" to n <u>Type</u> integer <u>Units</u> n/a <u>Location</u> - Eng. Cameras OBSERVATION_REQUEST_PARMS (Group) - MCZ a. OBSERVATION_REQUEST_PARMS (Group) b. MINI_HEADER (Group) <u>X-path</u> TBD <u>Reference</u> TBD	<u>Mode</u> DPO in XML format (referenced to APID Name in Appendix A) <u>Field as "<xml name>:[<element>]:<field>"</u> - Eng. Cameras "<IDPH DPO>:idph:params:exp_auto_dn" - MCZ a. "<Ancillary DPO>:cmd_arguments_image:exp_target_DN" b. "MMM_Image_Mini_Header:Image:cmd[4]" <u>Type</u> - Eng. Cameras U16 - MCZ U8 Note: For MCZ products, "auto exposure data cut" is comprised of one byte coming from the MMM mini-header at offset 28.
<u>Ops Keyword</u> AUTO_EXPOSURE_PERCENT <u>Archive Attribute</u> TBD <u>Definition</u> Specifies the auto-exposure early-termination percent. If the calculated exposure time has written this value, then terminate auto exposure early.	<u>Valid Values</u> - Eng. Cameras (B) - MCZ (B) "0.0" to "100.0" <u>Type</u> float <u>Units</u> n/a <u>Location</u> - Eng. Cameras OBSERVATION_REQUEST_PARMS (Group)	<u>Mode</u> DPO in XML format (referenced to APID Name in Appendix A) <u>Field as "<xml name>:[<element>]:<field>"</u> - Eng. Cameras "<IDPH DPO>:idph:params:exp_auto_percent" - MCZ a. "<AncillaryDPO>:cmd_arguments_image:exp_early_termination" b. "MCZ_Image_Mini_Header:Image:cmd[4]" <u>Type</u>

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
	<u>MCZ</u> - OBSERVATION_REQUEST_PARMS (Group) - MINI_HEADER (Group) <u>X-path</u> TBD <u>Reference</u> TBD	U8 Note: For MCZ products, "auto exposure percent" is comprised of two bytes coming from the MMM mini-header at offset 30.
Ops Keyword AUTO_EXPOSURE_PIXEL_FRACTION Archive Attribute TBD Definition Specifies the percentage of pixels whose targeted value is higher than the AUTO_EXPOSURE_DATA_CUT keyword.	Valid Values <u>Eng. Cameras</u> (B) <u>MCZ</u> (B) "0.0" to "100.0" Type <u>Eng. Cameras</u> float <u>MCZ</u> unsigned int Units n/a Location - OBSERVATION_REQUEST_PARMS (Group) - OBSERVATION_REQUEST_PARMS (Group) - MINI_HEADER (Group) <u>X-path</u> TBD <u>Reference</u> TBD	Mode DPO in XML format (referenced to APID Name in Appendix A) Field as "<xml name>:[<element>]:[<element>]:<field>" <u>Eng. Cameras</u> "<IDPH DPO>:idph:params:exp_auto_frac" <u>MCZ</u> "<Ancillary DPO>:cmd_arguments_image:exp_pixel_fraction" "MCZ_Image_Mini_Header:Image:cmd[4]" Type <u>Eng. Cameras</u> F32 <u>MCZ</u> U8 Note: For MCZ products, "auto pixel fraction" is comprised of two bytes coming from the MMM mini-header at offset 29.
Ops Keyword AZIMUTH_FOV Archive Attribute TBD Definition Specifies the angular measure of the horizontal field of view of an imaged scene. NOTE: For MER, this was computed as IFOV times the number of horizontal pixels. For M2020, it is computed by projecting rays from the left and right edges of the image at the center through the camera model, and computing the angle subtended by those rays.	Valid Values "0.0" to "360.0" Type float Units deg (<deg> unit tag required) Location INSTRUMENT_STATE_PARMS (Group) <u>X-path</u> TBD <u>Reference</u> TBD	Mode 1) DPO in XML format (referenced to APID Name in Appendix A) 2) DPO in XML format (referenced to APID Name in Appendix A) or Image DPO mini-header - Calculation: - Line/sample is translated into an origin and "look direction" vectors (left and right). The vectors are then used to calculate Azimuth field of view. Field as "<xml name>:[<element>]:[<element>]:<field>" <u>Eng. Cameras</u> "<IDPH DPO>:idph:cols" "<IDPH DPO>:idph:rows" "<IDPH DPO>:idph:res_cols" "<IDPH DPO>:idph:res_rows" <u>MCZ Cameras</u> "<Ancillary DPO>:cmd_arguments_image>window_width" "<Ancillary DPO>:cmd_arguments_image>window_height" "MCZ_Image_Mini_Header:cmd[2]"

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
		d. "MCZ_Image_Mini_Header:cmd[2]" Type 1) U16 2) U8 NOTES: - For MCZ non-recovered data products (cases "a" and "b"), float values come from the Ancillary DPO. - For MCZ recovered data products (cases "c" and "d"), values analogous to cases "a" (window_width) and "b" (window_height) are each comprised of one byte coming from Image DPO mini-header at byte offsets 22 and 23, respectively. - For MCZ, parm "window_width" is number of image samples. - For MCZ, parm "window_height" is number of image lines. - For MCZ non-Thumbnail data products, multiply by factor of 8 to convert to correct line/sample value.
Ops Keyword BAD_PIXEL_REPLACEMENT_FLAG Archive Attribute TBD Definition Specifies whether or not bad pixel replacement processing was requested or completed. If set to TRUE, certain pixels in the image were replaced based on a bad pixel table. See BAD_PIXEL_REPLACEMENT_ID.	Valid Values Eng. Cameras 0 = "FALSE" 1 = "TRUE" Type string(5) Units n/a Location OBSERVATION_REQUEST_PARMS (Group) X-path TBD Reference TBD	Mode DPO in XML format (referenced to APID Name in Appendix A) Field as "<xml name>:[<element>]:<field>" Eng. Cameras "<IDPH DPO>:idph:params:bad" Type boolean
Ops Keyword BAD_PIXEL_REPLACEMENT_ID Archive Attribute TBD Definition Specifies the ID of the bad pixel table used in the bad pixel replacement process. The bad pixel table ID is incremented every time an update to the bad pixel table is made. See BAD_PIXEL_REPLACEMENT_FLAG.	Valid Values Eng. Cameras 0 = "N/A" "1" to "65535" Type string Units n/a Location INSTRUMENT_STATE_PARMS (Group) X-path	Mode DPO in XML format (referenced to APID Name in Appendix A) Field as "<xml name>:[<element>]:<field>" Eng. Cameras "<IDPH DPO>:idph:bad" Type U16

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
	TBD <u>Reference</u> TBD	
Ops Keyword BANDS Archive Attribute TBD Definition Specifies the number of spectral bands in image or other object.	Valid Values "1" to "16" Type integer Units n/a Location IMAGE (Object) X-path TBD Reference TBD	Mode Static Value Field as "<xml name>:[<element>]:[<element>]:<field>" n/a Type n/a
Ops Keyword BAND_STORAGE_TYPE Archive Attribute TBD Definition Specifies the number of spectral bands in image or other object.	Valid Values "BAND_SEQUENTIAL", "SAMPLE_INTERLEAVED", "LINE_INTERLEAVED" Type string(20) Units n/a Location IMAGE (Object) X-path TBD Reference TBD	Mode VICAR label Field as "<xml name>:[<element>]:[<element>]:<field>" n/a Type n/a
Ops Keyword BAYER_METHOD Archive Attribute TBD Definition Specifies whether or not the data are encoded in a Bayer pattern, and if not, how the pattern was removed. Valid value definitions; other teams may add their own algorithm names as needed: a) "NONE" – Imager is not a Bayer-pattern imager. Default if keyword is missing. b) "RAW_BAYER" – Image has the raw Bayer pattern; no	Valid Values Surface EDRs - Eng. Cameras For EDR: 0 = "RAW_BAYER" 1 = "PANCHROMATIC" 2 = "RED_AVERAGING" 3 = "GREEN_AVERAGING" 4 = "BLUE_AVERAGING" - SuperCam (S) - MCZ (S) - SHERLOC-WATSON (S) TBD Surface RDRs: "NONE", "RAW_BAYER", "MALVAR", "ZHANG_WU", "RED_AVERAGED", "GREEN_AVERAGED", "BLUE_AVERAGED", "RED_BILINEAR",	Field as "<xml name>:[<element>]:[<element>]:<field>" Surface - Eng. Cameras a. "<IDPH DPO:idph:color>" b. "<IDPH DPO:idph:params:color>" Cruise - Eng. Cameras a. "<IDPH DPO>:idph:voff" Type Surface - Eng. Cameras I32 Cruise - Eng. Cameras (C) U16

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
de-Bayering has been done. c) "MALVAR" – Image has been converted to RGB using the Malvar algorithm. d) "ZHANG_WU" – Image has been converted to RGB using the Zhang-Wu algorithm. e) "RED_AVERAGED" – Pixels are created by averaging 1 (or more) pixels from the red band exclusively. f) "GREEN_AVERAGED" – Pixels are created by averaging 1 (or more) pixels from the green band exclusively. g) "BLUE_AVERAGED" – Pixels are created by averaging 1 (or more) pixels from the blue band exclusively. h) "RED_BILINEAR" – Pixels are created by bilinear interpolation of surrounding pixels taken exclusively from the red band. i) "GREEN_BILINEAR" – Pixels are created by bilinear interpolation of surrounding pixels taken exclusively from the green band. j) "BLUE_BILINEAR" – Pixels are created by bilinear interpolation of surrounding pixels taken exclusively from the blue band. k) "AVERAGED" – RGB image created by averaging individual cells. Generally a simple composite of RED GREEN BLUE_AVERAGED elements. l) "BILINEAR" – Bilinear interpolation of each color treated independently. m) "PANCHROMATIC" – Pixels are created by averaging one or more complete Bayer cells, creating a panchromatic image. n) "IDENTITY" – Sensor has a Bayer pattern but the image can be used as-is due to the presence of other filters.	"GREEN_BILINEAR", "BLUE_BILINEAR", "AVERAGED", "BILINEAR", "PANCHROMATIC", "IDENTITY" Cruise - Eng. Cameras (C) 0 - 15 = "RAW_BAYER" 16 - 19 = "RED_AVERAGED" 20 - 23 = "GREEN_AVERAGED" 24 - 27 = "BLUE_AVERAGED" 28 - 31 = "PANCHROMATIC" 32 = "RED_AVERAGED" 33 = "GREEN_AVERAGED" 34 = "BLUE_AVERAGED" 35 = "PANCHROMATIC" - SuperCam (C) "RAW_BAYER" Type string Units n/a Location a. INSTRUMENT_STATE_PARS (Group) b. OBSERVATION_REQUEST_PARS (Group) X-path TBD Reference TBD	
Ops Keyword BIAS_COEFFS_FILE_DESC Archive Attribute TBD Definition Specifies a description of bias coefficients file named in BIAS_COEFFS_FILE_NAME.	Valid Values n/a Type string Units n/a Location DERIVED_IMAGE_PARS (Group) X-path TBD Reference TBD	Mode RDR-generating software Field as "<xml name>:[<element>]:[<element>]:<field>" n/a Type n/a
Ops Keyword	Valid Values	Mode

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
BIAS_COEFFS_FILE_NAME <u>Archive Attribute</u> TBD <u>Definition</u> Specifies the name of bias coefficients file used in generating the RDR.	n/a <u>Type</u> string <u>Units</u> n/a <u>Location</u> DERIVED_IMAGE_PARMS (Group) <u>X-path</u> TBD <u>Reference</u> TBD	RDR-generating software <u>Field as “<xml name>:[<element>]:[<element>]:<field>”</u> n/a <u>Type</u> n/a
<u>Ops Keyword</u> BOOT_COUNT <u>Archive Attribute</u> TBD <u>Definition</u> Unsigned integer count incremented each time the RCE boots.	<u>Valid Values</u> n/a <u>Type</u> string <u>Units</u> n/a <u>Location</u> TELEMETRY (Class) <u>X-path</u> TBD <u>Reference</u> TBD	<u>Mode</u> EMD in XML format <u>Field as “<xml name>:[<element>]:[<element>]:<field>”</u> "M20EarthProductMetadata:M20ProductMetadata:Boot Counter" <u>Type</u> Unsigned integer
<u>Ops Keyword</u> BRIGHTNESS_CORRECTION_FILE <u>Archive Attribute</u> TBD <u>Definition</u> Specifies the name of the file containing brightness correction parameters for each input file of a mosaic. The file is in XML format.	<u>Valid Values</u> n/a <u>Type</u> string <u>Units</u> n/a <u>Location</u> DERIVED_IMAGE_PARMS (Group) <u>X-path</u> TBD <u>Reference</u> TBD	<u>Mode</u> RDR-generating software <u>Field as “<xml name>:[<element>]:[<element>]:<field>”</u> n/a <u>Type</u> n/a
<u>Ops Keyword</u> BRIGHTNESS_CORRECTION_TYPE <u>Archive Attribute</u> TBD <u>Definition</u> Identifies the type of brightness correction that was applied to	<u>Valid Values</u> "NONE", "LINEAR", "MIXED" <u>Type</u> string <u>Units</u> n/a	<u>Mode</u> RDR-generating software <u>Field as “<xml name>:[<element>]:[<element>]:<field>”</u> n/a <u>Type</u> n/a

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
a mosaic. Brightness_correction is defined as a mosaic radiometric_seam matching process that is done on top of ordinary radiometric correction. Its intent is to reduce visual seams at the expense of radiometric accuracy. Applies to RDRs only, not EDRs. Types may be added over time but currently they consist of: a) "NONE" - No correction made. b) "LINEAR" - Each input image gets a single additive and a single multiplicative factor, which are applied to every pixel in the image as follows: (new_image = image * mult + add). These factors are defined via the IMAGE_RADIANCE_FACTOR and IMAGE_RADIANCE_OFFSET keywords. c) "MIXED" - Multiple types of correction were applied to different input images. This should be avoided but is included as a placeholder.	Location DERIVED_IMAGE_PARMS (Group) X-path TBD Reference TBD	
Ops Keyword BYTES Archive Attribute TBD Definition Specifies the number of bytes allocated for a particular data representation.	Valid Values "0" to n Type integer Units n/a Location IMAGE_HEADER (Object) X-path TBD Reference TBD	Mode Calculation based on size of VICAR label Field as "<xml name>:[<element>]:[<element>]:[<field>]" n/a Type n/a
Ops Keyword CALIBRATION_SOURCE_ID Archive Attribute TBD Definition Specifies a unique identifier (within a data set) indicating the source of the calibration data used in generating the entity described by the enclosing group (often, a camera model). The construction of this identifier is mission-specific, but should indicate which specific calibration data set was used (via date or other means) and may also indicate the calibration method.	Valid Values n/a Type string(47) Units n/a Location GEOMETRIC_CAMERA_MODEL (Group) X-path TBD Reference TBD	Mode DPO in XML format (referenced to APID Name in Appendix A) Field as "<xml name>:[<element>]:[<element>]:[<field>]" - Eng. Cameras "<IDPH DPO>:idph:cmmod_model_id" - MCZ Cameras "<IDPH DPO>:cidph:camera_model:mid" Type U16
Ops Keyword CAMERA_ROTATION_AXIS_VECTOR Archive Attribute TBD	Valid Values n/a Type float array[3]	Mode RDR-generating software Field as "<xml name>:[<element>]:[<element>]:[<field>]"

Output Metadata (Product Label)			Input Metadata (Source)																																																																																																											
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)		- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type																																																																																																											
Definition Specifies the axis around which the camera rotates. For the Cylindrical-Perspective projection, this is the axis around which one rotates the camera model in azimuth, before (optionally) correcting for rover tilt using PROJECTION_Z_AXIS_VECTOR. It corresponds to the physical azimuth rotation axis for mast-mounted cameras.	Units n/a Location DERIVED_IMAGE_PARMS (Group) X-path TBD Reference TBD		n/a Type n/a																																																																																																											
Ops Keyword CAMERA_SERIAL_NUMBER Archive Attribute TBD Definition Specifies the manufacturer’s serial number assigned to a camera instrument. This number may be used to uniquely identify a particular camera instrument for tracing its components or determining its calibration history, for example. For M2020, the value might not be identical to that of INSTRUMENT_SERIAL_NUMBER.	Valid Values <table><tr><td></td><td colspan="2">EM Strings (RCE)</td><td colspan="2">FM Strings (</td></tr><tr><td><u>Instrument</u></td><td><u>A</u></td><td><u>B</u></td><td><u>A</u></td><td></td></tr><tr><td>Fhaz Left</td><td>TBD</td><td>TBD</td><td>TBD</td><td></td></tr><tr><td>Fhaz Right</td><td>TBD</td><td>TBD</td><td>TBD</td><td></td></tr><tr><td>Rhaz Left</td><td></td><td>TBD</td><td></td><td>TBD</td></tr><tr><td>Rhaz Right</td><td></td><td>TBD</td><td></td><td>TBD</td></tr><tr><td>Navcam Left</td><td></td><td>TBD</td><td></td><td>TBD</td></tr><tr><td>Navcam Right</td><td></td><td>TBD</td><td></td><td>TBD</td></tr><tr><td>Cache</td><td></td><td>TBD</td><td></td><td>TBD</td></tr></table> - SuperCam<table><tr><td><u>Instrument</u></td><td><u>EM</u></td><td><u>FM</u></td></tr><tr><td>RMI</td><td>TBD</td><td>TBD</td></tr></table> - Mastcam-Z<table><tr><td><u>Instrument</u></td><td><u>EM</u></td><td><u>FM</u></td></tr><tr><td>Mastcam-Z Left</td><td>TBD</td><td>TBD</td></tr><tr><td>Mastcam-Z Right</td><td>TBD</td><td>TBD</td></tr></table> - SHERLOC<table><tr><td><u>Instrument</u></td><td><u>EM</u></td><td><u>FM</u></td></tr><tr><td>WATSON</td><td>TBD</td><td>TBD</td></tr><tr><td>ACI</td><td>TBD</td><td>TBD</td></tr></table> - PIXL<table><tr><td><u>Instrument</u></td><td><u>EM</u></td><td><u>FM</u></td></tr><tr><td>MCC</td><td>TBD</td><td>TBD</td></tr></table> - MEDA<table><tr><td><u>Instrument</u></td><td><u>EM</u></td><td><u>FM</u></td></tr><tr><td>SkyCam</td><td>TBD</td><td>TBD</td></tr></table> - EDL<table><tr><td><u>Instrument</u></td><td><u>EM</u></td><td><u>FM</u></td></tr><tr><td>LVS</td><td>TBD</td><td>TBD</td></tr><tr><td>PUC A</td><td>TBD</td><td>TBD</td></tr><tr><td>PUC B</td><td>TBD</td><td>TBD</td></tr><tr><td>PUC C</td><td>TBD</td><td>TBD</td></tr><tr><td>DDC</td><td>TBD</td><td>TBD</td></tr><tr><td>RDC</td><td>TBD</td><td>TBD</td></tr><tr><td>RUC</td><td>TBD</td><td>TBD</td></tr></table> Type string			EM Strings (RCE)		FM Strings (		<u>Instrument</u>	<u>A</u>	<u>B</u>	<u>A</u>		Fhaz Left	TBD	TBD	TBD		Fhaz Right	TBD	TBD	TBD		Rhaz Left		TBD		TBD	Rhaz Right		TBD		TBD	Navcam Left		TBD		TBD	Navcam Right		TBD		TBD	Cache		TBD		TBD	<u>Instrument</u>	<u>EM</u>	<u>FM</u>	RMI	TBD	TBD	<u>Instrument</u>	<u>EM</u>	<u>FM</u>	Mastcam-Z Left	TBD	TBD	Mastcam-Z Right	TBD	TBD	<u>Instrument</u>	<u>EM</u>	<u>FM</u>	WATSON	TBD	TBD	ACI	TBD	TBD	<u>Instrument</u>	<u>EM</u>	<u>FM</u>	MCC	TBD	TBD	<u>Instrument</u>	<u>EM</u>	<u>FM</u>	SkyCam	TBD	TBD	<u>Instrument</u>	<u>EM</u>	<u>FM</u>	LVS	TBD	TBD	PUC A	TBD	TBD	PUC B	TBD	TBD	PUC C	TBD	TBD	DDC	TBD	TBD	RDC	TBD	TBD	RUC	TBD	TBD	Mode DPO in XML format (referenced to APID Name in Appendix A) Field as “<xml name>:[<element>]:[<element>]:<field>” “<IDPH DPO>:idph:cmmod_serial_no” Type U8		
	EM Strings (RCE)		FM Strings (																																																																																																											
<u>Instrument</u>	<u>A</u>	<u>B</u>	<u>A</u>																																																																																																											
Fhaz Left	TBD	TBD	TBD																																																																																																											
Fhaz Right	TBD	TBD	TBD																																																																																																											
Rhaz Left		TBD		TBD																																																																																																										
Rhaz Right		TBD		TBD																																																																																																										
Navcam Left		TBD		TBD																																																																																																										
Navcam Right		TBD		TBD																																																																																																										
Cache		TBD		TBD																																																																																																										
<u>Instrument</u>	<u>EM</u>	<u>FM</u>																																																																																																												
RMI	TBD	TBD																																																																																																												
<u>Instrument</u>	<u>EM</u>	<u>FM</u>																																																																																																												
Mastcam-Z Left	TBD	TBD																																																																																																												
Mastcam-Z Right	TBD	TBD																																																																																																												
<u>Instrument</u>	<u>EM</u>	<u>FM</u>																																																																																																												
WATSON	TBD	TBD																																																																																																												
ACI	TBD	TBD																																																																																																												
<u>Instrument</u>	<u>EM</u>	<u>FM</u>																																																																																																												
MCC	TBD	TBD																																																																																																												
<u>Instrument</u>	<u>EM</u>	<u>FM</u>																																																																																																												
SkyCam	TBD	TBD																																																																																																												
<u>Instrument</u>	<u>EM</u>	<u>FM</u>																																																																																																												
LVS	TBD	TBD																																																																																																												
PUC A	TBD	TBD																																																																																																												
PUC B	TBD	TBD																																																																																																												
PUC C	TBD	TBD																																																																																																												
DDC	TBD	TBD																																																																																																												
RDC	TBD	TBD																																																																																																												
RUC	TBD	TBD																																																																																																												

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
	<u>Units</u> n/a <u>Location</u> GEOMETRIC_CAMERA_MODEL (Group) <u>X-path</u> TBD <u>Reference</u> TBD	
<u>Ops Keyword</u> CFA_TYPE <u>Definition</u> Defines the type of Color Filter Array (CFA) used to encode multiple colors in a single exposure. It is optional if there is no CFA. Valid value definitions: a) "NONE" – no CFA b) "BAYER_RGGB" – repated 2x2 square of pixels.	<u>Valid Values</u> - MEDA SkyCam, SHERLOC-ACI, PIXL MCC: "NONE" - Eng. Cameras, SuperCam RMI, MCZ, SHERLOC Imaging, EDL cameras: "BAYER_RGGB" <u>Type</u> string <u>Units</u> n/a <u>Location</u> INSTRUMENT_STATE_PARMS (Group) <u>X-path</u> TBD <u>Reference</u> TBD	<u>Mode</u> Static value <u>Field as "<xml name>:[<element>]:[<element>]:[<field>]"</u> n/a <u>Type</u> n/a
<u>Ops Keyword</u> CFA_VENUE <u>Definition</u> For instruments using a Color Filter Array (CFA), this keyword defines where the CFA pattern was removed from the data. It is optional if there no CFA.	<u>Valid Values</u> "NONE", "ONBOARD", "GROUND" SuperCam RMI Cruise: "NONE" <u>Type</u> string <u>Units</u> n/a <u>Location</u> INSTRUMENT_STATE_PARMS (Group) <u>X-path</u> TBD <u>Reference</u> TBD	<u>Mode</u> DPO in XML format <u>Field as "<xml name>:[<element>]:[<element>]:[<field>]"</u> n/a <u>Type</u> n/a
<u>Ops Keyword</u> COMMAND_INSTRUMENT_ID <u>Archive Attribute</u>	<u>Valid Values</u> 0 = "NONE" 1 = "FRONT_HAZCAM_LEFT" 2 = "FRONT_HAZCAM_RIGHT"	<u>Mode</u> DPO in XML format (referenced to APID Name in Appendix A)

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
TBD Definition Specifies an abbreviated name or acronym which identifies the instrument that was commanded. NOTE: INSTRUMENT_ID is not a unique identifier for a given instrument. Note also that the associated INSTRUMENT_NAME element provides the full name of the instrument. Example values: IRTM (for Viking Infrared Thermal Mapper), PWS (for plasma wave spectrometer).	3 = "REAR_HAZCAM_LEFT" 4 = "REAR_HAZCAM_RIGHT" 5 = "NAVCAM_LEFT" 6 = "NAVCAM_RIGHT" 7 = "CACHECAM" 8 = "CAMERA_IDS" Type string(20) Units n/a Location OBSERVATION_REQUEST_PARMS (Group) X-path TBD Reference TBD	Field as "<xml name>:[<element>]:<field>" Surface - Eng. Cameras "<IDPH DPO>:idph:params:camera" - MCZ Cameras "<Ancillary DPO>:cmd_arguments_image:camera" Cruise - Eng. Cameras "<IDPH DPO>:idph:params:camera" - MCZ Cameras "<Ancillary DPO>:cmd_arguments_image:camera" Type enum
Ops Keyword COMMAND_SEQUENCE_NUMBER Archive Attribute TBD Definition Specifies a numeric identifier for a sequence of commands sent to a spacecraft or instrument. NOTE: For M2020, this is the command number which identifies the specific generating command within the specified sequence.	Valid Values n/a Type integer Units n/a Location IDENTIFICATION (Class) X-path TBD Reference TBD	Mode EMD in XML format Field as "<xml name>:[<element>]:<field>" "M20EarthProductMetadata:M20ProductMetadata:CommandNumber" Type n/a
Ops Keyword COMMUNICATION_SESSION_ID Archive Attribute TBD Definition Active Communication Session ID at time of MPDU (Metadata Protocol Data Unit) creation. For context, the MPDU is the first PDU (Protocol Data Unit) produced for a data product, and contains general and M2020 specific "metadata". It is wholly contained in a single packet. See also EXPECTED_TRANSMISSION_PATH.	Valid Values n/a Type string Units n/a Location TELEMETRY (Class) X-path TBD Reference TBD	Mode EMD in XML format Field as "<xml name>:[<element>]:<field>" "M20EarthProductMetadata:M20ProductMetadata:CommunicationSessionId" Type n/a
Ops Keyword CONFIGURATION_BIT_ID Archive Attribute	Valid Values "ARM_SO_EU_WU", "ARM_SO_EU_WD", "ARM_SO_ED_WU", "ARM_SO_ED_WD", "ARM_SI_EU_WU", "ARM_SI_EU_WD",	Mode RDR-generating software, Static value

Output Metadata (Product Label)		Input Metadata (Source)
• Ops Keyword • Archive Label Attribute • Definition	• Mission Phase C=Cruise, S=Surface, B=Both • Valid Values (quoted) • Type • Units • Keyword Location in Ops Label • Attribute X-path in Archive label • Attribute Reference in Archive Label (bracketed)	• Mission Phase C=Cruise, S=Surface, B=Both • Mode • Metadata Field • Type
TBD Definition Specifies an array of strings identifying the configuration of the arm represented by the bits in the reachability product. Each configuration requires 2 bits to describe. The first entry in the array contains the configuration for the 2 most significant bits, while the last entry contains the configuration for the 2 least significant bits. For M2020, the instrument/tools that are useful in a reachability map are DRILL, BITCAR, SHERLOC-WATSON, APXS and SCOOP_TIP. For M2020, the 8 configuration values are interpreted as follows: a) "ARM_SO_EU_WU" - Shoulder out, Elbow up, Wrist up. b) "ARM_SO_EU_WD" - Shoulder out, Elbow up, Wrist down. c) "ARM_SO_ED_WU" - Shoulder out, Elbow down, Wrist up. d) "ARM_SO_ED_WD" - Shoulder out, Elbow down, Wrist down. e) "ARM_SI_EU_WU" - Shoulder in, Elbow up, Wrist up. f) "ARM_SI_EU_WD" - Shoulder in, Elbow up, Wrist down. g) "ARM_SI_ED_WU" - Shoulder in, Elbow down, Wrist up. h) "ARM_SI_ED_WD" - Shoulder in, Elbow down, Wrist down. See also INSTRUMENT_BAND_ID.	"ARM_SI_ED_WU", "ARM_SI_ED_WD" Type string array[8] Units n/a Location DERIVED_IMAGE_PARMS (Group) X-path TBD Reference TBD	Field as "<xml name>:[<element>]:<field>" n/a Type n/a
Ops Keyword CONTACT_SENSOR_STATE Archive Attribute TBD Definition Specifies an array of identifiers for the state of an instrument or an instrument host's contact sensors at a specified time.	Valid Values Surface • Bit 0: FCS A csw triggered 0 = "NO CONTACT" 1 = "CONTACT" • Bit 1: FCS B csw triggered 0 = "NO CONTACT" 1 = "CONTACT" • Bit 2: DRILL STABILIZER 1 csw triggered 0 = "NO CONTACT" 1 = "CONTACT" • Bit 3: DRILL STABILIZER 2 csw triggered 0 = "NO CONTACT" 1 = "CONTACT" • Bit 4: DOCK 1 csw triggered 0 = "NO CONTACT" 1 = "CONTACT" • Bit 5: DOCK 2 csw triggered 0 = "NO CONTACT" 1 = "CONTACT"	Mode DPO in XML format (referenced to APID Name in Appendix A) Field as "<xml name>:[<element>]:<field>" Surface • All Cameras "<IDPH DPO>:idph:arm:contact" Type U8

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
	- Bit 6: DOCK 3 csw triggered 0 = "NO CONTACT" 1 = "CONTACT" - Bit 7: DOCK 4 csw triggered 0 = "NO CONTACT" 1 = "CONTACT" Type string array[8] Units n/a Location ARM_ARTICULATION_STATE (Group) X-path TBD Reference TBD	
Ops Keyword CONTACT_SENSOR_STATE_NAME Archive Attribute TBD Definition Specifies the possible value that can be contained in the CONTACT_SENSOR_STATE array. FCS – FACILITY CONTACT SENSOR	Valid Values TBD ("FACILITY CONTACT SENSOR A", "FACILITY CONTACT SENSOR B", "DRILL STABILIZER 1", "DRILL STABILIZER 2", "DOCK 1", "DOCK 2", "DOCK 3", "DOCK 4") Type string array[8] Units n/a Location ARM_ARTICULATION_STATE (Group) X-path TBD Reference TBD	Mode Static value: - Single value representing array of Bits 0 thru 7 from CONTACT_SENSOR_STATE. Field as "<xml name>:[<element>]:<field>" n/a Type n/a
Ops Keyword COORDINATE_SYSTEM_INDEX Archive Attribute TBD Definition Specifies an integer array used to record and track the movement of a rover or lander during surface operations. When in a COORDINATE_SYSTEM_STATE group, this keyword identifies which instance of the coordinate frame, named by COORDINATE_SYSTEM_NAME, is being defined by the group. For M2020, these indices are based on the ROVER_MOTION_COUNTER and are in the same order as specified by	Valid Values n/a Type integer array[10] Units n/a Location - For EDRs ... - ROVER_COORDINATE_SYSTEM (Group) - RSM_COORDINATE_SYSTEM (Group) - ARM_COORDINATE_SYSTEM (Group) - TURRET_COORDINATE_SYSTEM (Group) - TOOL_COORDINATE_SYSTEM (Group) - SITE_COORDINATE_SYSTEM (Group)	Mode DPO in XML format (referenced to APID Name in Appendix A), Calculation Field as "<xml name>:[<element>]:<field>" Surface - Eng. Cameras (B) "<IDPH DPO>:idph:rmc_site", "<IDPH DPO>:idph:rmc_drive", "<IDPH DPO>:idph:rmc_pose", "<IDPH DPO>:idph:rmc_arm", "<IDPH DPO>:idph:rmc_sha", "<IDPH DPO>:idph:rmc_drill", "<IDPH DPO>:idph:rmc_rsm", "<IDPH DPO>:idph:rmc_hga", "<IDPH DPO>:idph:rmc_bitcar",

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
ROVER_MOTION_COUNTER_NAME. - EDRs will always contain all 10 elements for this keyword. - For RDRs, the number of indices can be anything from 1 (used for SITE_FRAME) up to 10; however only 1, 2, 3, and 10 indices are common in RDRs. Example: COORDINATE_SYSTEM_INDEX=(1,10)	<u>X-path</u> TBD <u>Reference</u> TBD	"<IDPH DPO>:idph:rmc_seal" 2) same as #1 above 3) same as #1 above 4) same as #1 above 5) same as #1 above 6) "<IDPH DPO>:idph:rmc_site" <u>SuperCam</u> (S) <u>MCZ</u> (S) <u>SHERLOC</u> (S) <u>MEDA SkyCam</u> (S) <u>PIXL MCC</u> (S) 1) "<IDPH DPO>:idph:system:rmc:site", "<IDPH DPO>:idph:system:rmc:drive", "<IDPH DPO>:idph:system:rmc:pose", "<IDPH DPO>:idph:system:rmc:arm", "<IDPH DPO>:idph:system:rmc:sha", "<IDPH DPO>:idph:system:rmc:drill", "<IDPH DPO>:idph:system:rmc:rsm", "<IDPH DPO>:idph:system:rmc:hga", "<IDPH DPO>:idph:system:rmc:bitcar", "<IDPH DPO>:idph:system:rmc:seal" 2) same as #1 above 3) same as #1 above 4) same as #1 above 5) same as #1 above <u>Cruise</u> <u>MCZ</u> (C) 1) "<IDPH DPO>:cidph:rmc:site", "<IDPH DPO>:cidph:rmc:drive", "<IDPH DPO>:cidph:rmc:pose", "<IDPH DPO>:cidph:rmc:arm", "<IDPH DPO>:cidph:rmc:sha", "<IDPH DPO>:cidph:rmc:drill", "<IDPH DPO>:cidph:rmc:rsm", "<IDPH DPO>:cidph:rmc:hga", "<IDPH DPO>:cidph:rmc:bitcar", "<IDPH DPO>:cidph:rmc:seal" 2) same as #1 above 3) same as #1 above 4) same as #1 above 5) same as #1 above <u>Type</u> U16
<u>Ops Keyword</u> COORDINATE_SYSTEM_INDEX_NAME <u>Archive Attribute</u> TBD <u>Definition</u> Specifies an array of the formal names identifying each integer specified in COORDINATE_SYSTEM_INDEX. NOTE: Should match the number of values in COORDINATE_SYSTEM_INDEX.	<u>Valid Values</u> 1) "SITE" 2) "DRIVE" 3) "POSE" 4) "ARM" 5) "SHA" 6) "DRILL" 7) "RSM" 8) "HGA" 9) "BITCAR" 10) "SEAL" <u>Type</u> string array[10]	<u>Mode</u> Static values: 1) for RMC element Site 2) for RMC element Drive 3) for RMC element Pose 4) for RMC element Arm 5) for RMC element SHA 6) for RMC element Drill 7) for RMC element RSM 8) for RMC element HGA 9) for RMC element BITCAR 10) for RMC element SEAL

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
	<u>Units</u> n/a <u>Location</u> - For EDRs ... - ROVER_COORDINATE_SYSTEM (Group) - RSM_COORDINATE_SYSTEM (Group) - ARM_COORDINATE_SYSTEM (Group) - TURRET_COORDINATE_SYSTEM (Group) - TOOL_COORDINATE_SYSTEM (Group) - SITE_COORDINATE_SYSTEM (Group) <u>X-path</u> TBD <u>Reference</u> TBD	<u>Field as "<xml name>:[<element>]:<field>"</u> n/a <u>Type</u> n/a
<u>Ops Keyword</u> COORDINATE_SYSTEM_NAME <u>Archive Attribute</u> TBD <u>Definition</u> Specifies the full name of the coordinate system to which the state vectors are referenced. When in a COORDINATE_SYSTEM group, this keyword provides the full name of the coordinate system being defined by the group. The rest of the keywords in the group describe how this coordinate system is related to some other (the "reference"). Non-unique coordinate systems (such as "SITE" for rover or lander missions), which have multiple instances using the same name, also require COORDINATE_SYSTEM_INDEX to completely identify the coordinate system. NOTE: A CS is named by three things: 1) the CS name (e.g. site, rover), 2) the set of indices, and 3) the solution ID (see SOLUTION_ID). A set of index-names is sometimes included for documentation purposes only. The solution ID is often omitted to indicate the default, but is logically part of the name. Any time there's a location (XYZ) or an orientation (in quaternion or angle form), the label group containing that location/orientation should contain the name of the CS in which it is expressed. This means it should contain the three "REFERENCE_*" keywords (or two if the solution ID is omitted). This includes CS definition groups, which have to define in what frame the origin/rotation numbers are expressed (i.e. what is the reference frame).	<u>Valid Values</u> - For EDRs ... "ROVER_NAV_FRAME" "RSM_HEAD_FRAME" 0 = "ARM_TURRET_FRAME" "ARM_DRILL_FRAME" "ARM_DOCKING_POST_FRAME" "ARM_PIXL_FRAME" "ARM_GDRT_FRAME" "ARM_FCS_FRAME" "ARM_WATSON_FRAME" "ARM_SHERLOC_FRAME" "ARM_CUSTOM_TCP_FRAME" "ARM_TURRET_FRAME" - Same as #3 above <u>Type</u> string <u>Units</u> n/a <u>Location</u> - For EDRs ... - ROVER_COORDINATE_SYSTEM (Group) - RSM_COORDINATE_SYSTEM (Group) - ARM_COORDINATE_SYSTEM (Group) - TURRET_COORDINATE_SYSTEM (Group) - TOOL_COORDINATE_SYSTEM (Group) <u>X-path</u> TBD <u>Reference</u> TBD	<u>Mode</u> - Static value: - for Rover Navigation Frame - for RSM Head Frame - for Turret Frame - DPO in XML format (referenced to APID Name in Appendix A), <u>Field as "<xml name>:[<element>]:<field>"</u> "<IDPH DPO>:idph:arm:tool" - same as #3 above <u>Type</u> n/a
<u>Ops Keyword</u> CORRELATION_AVERAGE_SCALE <u>Archive Attribute</u> TBD	<u>Valid Values</u> n/a <u>Type</u> float	<u>Mode</u> RDR-generating software <u>Type</u>

Output Metadata (Product Label)		Input Metadata (Source)
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Definition Indicates the average scale factor between the left and right image.	Units n/a Location DERIVED_IMAGE_PARMS (Group) X-path TBD Reference TBD	n/a
Ops Keyword CORRELATION_PIXEL_COUNT Archive Attribute TBD Definition Contains the number of pixels that successfully correlated.	Valid Values n/a Type integer Units n/a Location DERIVED_IMAGE_PARMS (Group) X-path TBD Reference TBD	Mode RDR-generating software Type n/a
Ops Keyword DARK_CURRENT_FILE_NAME Archive Attribute TBD Definition Specifies the name, or array of names, of the dark current image file(s) (image taken without opening the camera shutter) which should be used to perform radiometric calibration of the image. The dark current image provides a reference label of the build-up of any charges on the sensor that need to be subtracted from a shuttered image during calibration. Selection of the appropriate dark current image may be based on time, camera, temperature, readout conditions, light flood, gain and offset.	Valid Values n/a Type string array Units n/a Location DERIVED_IMAGE_PARMS (Group) X-path TBD Reference TBD	Mode RDR-generating software Field as “<xml name>:[<element>]:[<element>]:<field>” n/a Type n/a
Ops Keyword DARK_CURRENT_FILE_DESC Archive Attribute TBD Definition Specifies a description of the corresponding dark current files listed in DARK_CURRENT_FILE_NAME.	Valid Values n/a Type string Units n/a Location DERIVED_IMAGE_PARMS (Group) X-path	Mode RDR-generating software Field as “<xml name>:[<element>]:[<element>]:<field>” n/a Type n/a

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
	TBD <u>Reference</u> TBD	
Ops Keyword DATA_LENGTH Archive Attribute TBD Definition Size of image in bytes.	Valid Values "0" to "8388608" (B) Type integer Units bytes Location MINI_HEADER (Group) X-path TBD Reference TBD	Mode DPO in XML format (referenced to APID Name in Appendix A) Field as "<xml name>:[<element>]:<field>" "SCAM_Image_Mini_Header:data_length" Type U32 Notes: For SCAM RMI products, "rmi_header_size" is comprised of four bytes coming from the Image DPO mini-header at offset 197.
Ops Keyword DERIVED_IMAGE_TYPE Archive Attribute TBD Definition Specifies how to interpret the pixel values in a derived image RDR (or colloquially, the type of the derived image itself). Values are defined as: "IMAGE" - Standard image, where pixels represent intensity. Note that this implies nothing about radiometric, geometric, or other corrections that may have been applied. "DISPARITY_MAP" - Pixels represent line and sample disparity coordinates with respect to another image (2 bands). "DISPARITY_LINE_MAP" - Pixels represent line disparity only. "DISPARITY_SAMPLE_MAP" - Pixels represent sample disparity only. "DISPARITY_ERROR_MAP" - Pixels represent a disparity error metric. "DELTA_DISPARITY_MAP" - Pixels represent line and sample as an offset with respect to another image (2 bands). "DELTA_DISPARITY_LINE_MAP" - Pixels represent the line delta disparity only.	Valid Values "IMAGE", "DISPARITY_MAP", "DISPARITY_LINE_MAP", "DISPARITY_SAMPLE_MAP", "DISPARITY_ERROR_MAP", "DELTA_DISPARITY_MAP", "DELTA_DISPARITY_LINE_MAP", "DELTA_DISPARITY_SAMPLE_MAP", "MASK", "XYZ_MAP", "XYZ_ERROR_MAP", "X_MAP", "Y_MAP", "Z_MAP", "ANGLE_MAP", "RANGE_MAP", "RANGE_ERROR_MAP", "UVW_MAP", "U_MAP", "V_MAP", "W_MAP", "ROUGHNESS_MAP", "REACHABILITY_MAP", "PRELOAD_MAP", "SLOPE_MAP", "RADIAL_SLOPE_MAP", "SLOPE_HEADING_MAP", "SLOPE_MAGNITUDE", "NORTHERLY_TILT_MAP", "SOLAR_ENERGY_MAP", "IEP_MAP" Type string Units n/a Location DERIVED_IMAGE_PARMS (Group) X-path TBD Reference TBD	Mode RDR-generating software Field as "<xml name>:[<element>]:<field>" n/a Type n/a

Output Metadata (Product Label)		Input Metadata (Source)
• Ops Keyword • Archive Label Attribute • Definition	• Mission Phase C=Cruise, S=Surface, B=Both • Valid Values (quoted) • Type • Units • Keyword Location in Ops Label • Attribute X-path in Archive label • Attribute Reference in Archive Label (bracketed)	• Mission Phase C=Cruise, S=Surface, B=Both • Mode • Metadata Field • Type
• "DELTA_DISPARITY_SAMPLE_MAP" - Pixels represent the sample delta disparity only. • "MASK" - Pixels represent a mask to apply to another image. • "XYZ_MAP" - Pixels represent XYZ values (3 bands). • "XYZ_ERROR_MAP" - Pixels represent an XYZ error metric. • "X_MAP" - Pixels represent the X component of an XYZ image. • "Y_MAP" - Pixels represent the Y component of an XYZ image. • "Z_MAP" - Pixels represent the Z component of an XYZ image. • "ANGLE_MAP" - Pixels represent an angle between 2 vectors. • "RANGE_MAP" - Pixels represent a distance from the camera center. • "RANGE_ERROR_MAP" - Pixels represent a range error metric. • "UVW_MAP" - Pixels represent Surface Normal values (3 bands associating to X,Y,Z). • "U_MAP" - Pixels represent the U (X) component of a Surface Normal image. • "V_MAP" - Pixels represent the V (Y) component of a Surface Normal image. • "W_MAP" - Pixels represent the W (Z) component of a Surface Normal image. • "ROUGHNESS_MAP" - Pixels represent a measure of surface roughness. • "REACHABILITY_MAP" - Pixels flag what is reachable on the target feature by the respective IDD instrument. • "PRELOAD_MAP" - Pixels represent an arm preload value. • "SLOPE_MAP" - Pixels represent a slope angle. • "RADIAL_SLOPE_MAP" - Pixels represent the component of slope radial to the rover. • "SLOPE_HEADING_MAP" - Pixels represent a slope heading. • "SLOPE_MAGNITUDE" - Pixels represent a slope magnitude. • "NORTHERLY_TILT_MAP" - Pixels represent the north-		

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- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
facing component of slope. "SOLAR_ENERGY_MAP" - Pixels represent the available fraction of maximum solar energy. "IEP_MAP" - Pixels represent Incidence, Emission and Phase angles.		
Ops Keyword ^DESCRIPTION Archive Attribute TBD Definition Specifies a pointer that provides a free-form, unlimited-length character string that represents or gives an account of something.	Valid Values "VICAR2.TXT" Type string Units n/a Location IMAGE_HEADER (Object) X-path TBD Reference TBD	Mode Static Value Field as "<xml name>:[<element>]:<field>" n/a Type n/a
Ops Keyword DETECTOR_ERASE_COUNT Archive Attribute TBD Definition Specifies the number of times a detector has been flushed of data in raw counts. For SuperCam on M2020, specifies the commanded number of images before transferring the current image to the FPGA. For MCZ cameras on M2020, this reports the value of the camera head vertical register (vflush) parameter.	Valid Values "0" to "15" Type integer Units n/a Location - Eng. Cameras - SuperCam - Skycam OBSERVATION_REQUEST_PARMS (Group) - MCZ Cameras OBSERVATION_REQUEST_PARMS (Group) MINI_HEADER (Group) X-path TBD Reference TBD	Mode DPO in XML format (referenced to APID Name in Appendix A) Field as "<xml name>:[<element>]:<field>" - Eng. Cameras "IDPH DPO>:idph:params:flush" - SuperCam "Ancillary DPO>:cmd_parameters:CCD_cleanCount" TBD - MCZ Cameras "MCZ_Image_Mini_Header:cmd[0]" - Skycam "MEDA Image Header:skycam_fast_flush_cntr_data" Type U8 Note: For MCZ products, "detector erase count" is comprised of two bytes coming from the MMM mini-header starting at offset 12.
Ops Keyword DETECTOR_FIRST_LINE Archive Attribute TBD Definition Specifies the starting row from the hardware, such as a charge-coupled device (CCD), that contains data.	Valid Values - Eng. Cameras "1" to "960" - SuperCam RMI TBD - MCZ Cameras "TBD" Type integer	Mode DPO in XML format (referenced to APID Name in Appendix A) Field as "<xml name>:[<element>]:<field>" - Eng. Cameras "IDPH DPO>:idph:hw_row0" - SuperCam RMI TBD

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
	<u>Units</u> n/a <u>Location</u> 1) INSTRUMENT_STATE_PARS (Group) <u>X-path</u> TBD <u>Reference</u> TBD	<u>MCZ Cameras</u> TBD <u>SHERLOC</u> TBD <u>PIXL</u> TBD <u>Skycam</u> TBD <u>Type</u> U16
<u>Ops Keyword</u> DETECTOR_FIRST_LINE_SAMPLE <u>Archive Attribute</u> TBD <u>Definition</u> Specifies the starting column from the hardware, such as a charge-coupled device (CCD), that contains data.	<u>Valid Values</u> <u>Eng. Cameras</u> "1" to "960" <u>SuperCam RMI</u> "1" to "2048" <u>MCZ Cameras</u> "TBD" <u>Type</u> integer <u>Units</u> n/a <u>Location</u> INSTRUMENT_STATE_PARS (Group) <u>X-path</u> TBD <u>Reference</u> TBD	<u>Mode</u> DPO in XML format (referenced to APID Name in Appendix A) <u>Field as "<xml name>[:<element>][:<element>]:<field>"</u> <u>Eng. Cameras</u> "<IDPH DPO>:idph:hw_col0" <u>SuperCam RMI</u> TBD <u>MCZ Cameras</u> TBD <u>Type</u> U16
<u>Ops Keyword</u> DETECTOR_LINES <u>Archive Attribute</u> TBD <u>Definition</u> Specifies the number of rows extracted from the hardware, such as a charge-coupled device (CCD), that contain data.	<u>Valid Values</u> <u>Eng. Cameras</u> "0" to "" <u>SuperCam RMI</u> TBD <u>MCZ Cameras</u> TBD <u>Type</u> integer <u>Units</u> n/a <u>Location</u> INSTRUMENT_STATE_PARS (Group) <u>X-path</u> TBD <u>Reference</u> TBD	<u>Mode</u> DPO in XML format (referenced to APID Name in Appendix A) <u>Field as "<xml name>[:<element>][:<element>]:<field>"</u> <u>Eng. Cameras</u> "<IDPH DPO>:idph:hw_rows" <u>SuperCam RMI</u> TBD <u>MCZ Cameras</u> TBD <u>Type</u> U16

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
Ops Keyword DETECTOR_LINE_SAMPLES Archive Attribute TBD Definition Specifies the number of samples extracted from the hardware, such as a charge-coupled device (CCD), that contain data.	Valid Values - Eng. Cameras "0" to "" - MCZ Cameras "UNK" Type integer Units n/a Location INSTRUMENT_STATE_PARAMS (Group) X-path TBD Reference TBD	Mode DPO in XML format (referenced to APID Name in Appendix A) Field as "<xml name>:[<element>]:<field>" - Eng. Cameras "<IDPH DPO>:idph:hw_cols" - SuperCam RMI TBD - MCZ Cameras TBD Type U16
Ops Keyword DETECTOR_TO_IMAGE_ROTATION Archive Attribute TBD Definition Specifies the clockwise rotation, in degrees, that was applied to an image along its optical path through an instrument, from detector to final image orientation.	Valid Values - Eng. Cameras 0 = IMG_ROT_0 = "0.0" 1 = IMG_ROT_90 = "90.0" 2 = IMG_ROT_180 = "180.0" 3 = IMG_ROT_270 = "270.0" - MCZ Cameras "UNK" Type float Units Deg Location INSTRUMENT_STATE_PARAMS (Group) X-path TBD Reference TBD	Mode DPO in XML format (referenced to APID Name in Appendix A), Table Lookup Field as "<xml name>:[<element>]:<field>" Eng. Cameras "<IDPH DPO>:idph:rotation" Type enum
Ops Keyword DOWNLOAD_PRIORITY Archive Attribute TBD Definition Specifies which data to downlink/transmit, based on order of importance. The lower numerical priority (higher-ranked number) data products are transmitted before higher numerical priority (lower-ranked number) data products.	Valid Values "0" to "101" Type integer Units n/a Location - Eng. Cameras - TELEMETRY (Class) - IMAGE_REQUEST_PARAMS (Group) - THUMBNAIL_REQUEST_PARAMS (Group)	Mode DPO in XML format (referenced to APID Name in Appendix A) Field as "<xml name>:[<element>]:<field>" - Eng. Cameras "M20EarthProductMetadata:M20ProductMetadata:Product_Priority" "<IDPH DPO>:idph:params:img_prio" "<IDPH DPO>:idph:params:thumb_prio" "<IDPH DPO>:idph:params:col_prio" "<IDPH DPO>:idph:params:row_prio" "<IDPH DPO>:idph:params:hist_prio"

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
For example, an image with a downlink priority of 25 will be transmitted from the rover before an image with a downlink priority of 50. Value of "0" specifies use of on-board default.	d. COLUMN_SUM_REQUEST_PARMs (Group) e. ROW_SUM_REQUEST_PARMs (Group) f. HISTOGRAM_REQUEST_PARMs (Group) • <u>SuperCam</u> a. TELEMETRY(Group) b. THUMBNAIL_REQUEST_PARMs (Group) c. REFERENCE_PIXEL_REQUEST_PARMs (Group) • <u>MCZ</u> a. TELEMETRY (Class) b. IMAGE_REQUEST_PARMs (Group) <u>X-path</u> TBD <u>Reference</u> TBD	• <u>SuperCam</u> TBD a. "M20EarthProductMetadata:M20ProductMetadata:Product_Priority" b. "<Ancillary DPO>:cmd_parameters:Thumbnail_DP_Priority" c. "<Ancillary DPO>:cmd_parameters:Reference_Pix_DP_Priority" • <u>MCZ</u> a. "M20EarthProductMetadata:M20ProductMetadata:Product_Priority" b. "<Ancillary DPO>:cmd_arguments_image:image_data_priority" <u>Type</u> U8
<u>Ops Keyword</u> DOWNSAMPLE_METHOD <u>Archive Attribute</u> TBD <u>Definition</u> Specifies whether or not hardware downsampling was applied to an image.	<u>Valid Values</u> • <u>Eng. Cameras</u> "HARDWARE", "SOFTWARE", "BOTH", "NONE" • <u>MCZ Cameras</u> "HARDWARE", "NONE" <u>Type</u> string <u>Units</u> n/a <u>Location</u> INSTRUMENT_STATE_PARMs (Group) <u>X-path</u> TBD <u>Reference</u> TBD	<u>Mode</u> - DPO in XML format (referenced to APID Name in Appendix A), Calculation - EMD in XML format <u>Field as "<xml name>:[<element>]:[<element>]:<field>"</u> • <u>Eng. Cameras</u> a. "<IDPH DPO>:idph:hw_res" b. "<IDPH DPO>:idph:res_rows" c. "<IDPH DPO>:idph:res_cols" • <u>MCZ Cameras</u> "M20EarthProductMetadata:M20ProductMetadata:DataFileName" NOTES: • For Eng. Cameras, if res_rows=1 and res_cols=1, then "NONE". • For Eng. Cameras, if hw_res is 2, 4, or 8 and res_rows = res_cols = hw_res, then "HARDWARE" • For Eng. Cameras, if hw_res is 1 and res_rows or res_cols > 1, then "SOFTWARE". • For Eng. Cameras, if hw_res is 2, 4, or 8 and res_rows and res_cols != hw_res, then "BOTH". • For MCZ Cameras, if product filename is Thumbnail, then "HARDWARE". • For MCZ Cameras, if product filename is not Thumbnail, then "NONE". <u>Type</u> a. boolean b. enum c. U16 d. U16 - n/a

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
Ops Keyword DP_CMPRS_MODE Archive Attribute TBD Definition Indicates if the data product is compressed.	Valid Values Surface only "0" = Uncompressed "1" = LZO compressed "2" = GZIP compressed Type integer Units n/a Location TELEMETRY (Class) X-path TBD Reference TBD	Mode EMD in XML format Field as "<xml name>:[<element>]:<field>" "M20EarthProductMetadata:M20ProductMetadata:CompressionIndicator" Type Unsigned integer
Ops Keyword DP_CREATION_TIME Archive Attribute TBD Definition SCLK value corresponding to the time the DP generation command was sent by the RCE. 16 bits for seconds, 8 bits sub- seconds.	Valid Values "n/a" Type string Units n/a Location TELEMETRY (Class) X-path TBD Reference TBD	Mode EMD in XML format Field as "<xml name>:[<element>]:<field>" "M20EarthProductMetadata:M20ProductMetadata:CommandDispatchTime" Type Unsigned integer
Ops Keyword EARLY_IMAGE_RETURN_FLAG Archive Attribute TBD Definition Specifies the deferral of on-board post processing of the image and the returns the image early to an onboard client.	Valid Values Eng. Cameras 0 = "FALSE" 1 = "TRUE" Type string Units n/a Location OBSERVATION_REQUEST_PARMS (Group) X-path TBD Reference TBD	Mode DPO in XML format, Table Lookup Field as "<xml name>:[<element>]:<field>" Eng. Cameras "<IDPH DPO>:idph:params:img_early" Type boolean
Ops Keyword EARLY_PIXEL_SCALE_FLAG	Valid Values Eng. Cameras (B)	Mode DPO in XML format, Table Lookup

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
Archive Attribute TBD Definition Specifies the scaling of pixels. If TRUE, pixels are scaled early (from 12 to 8 bits).	0 = "FALSE" 1 = "TRUE" Type string Units n/a Location OBSERVATION_REQUEST_PARMS (Group) X-path TBD Reference TBD	Field as "<xml name>[:<element>]:<field>" Eng. Cameras "<IDPH DPO>:idph:params:scale_early" Type boolean
Ops Keyword EARTH_RECEIVED_START_TIME Archive Attribute TBD Definition Specifies the beginning time at which telemetry was received during a time period of interest. This should be represented in UTC system format.	Valid Values <YYYY>-<MM>-<DD>T<hh>:<mm>:<ss>[.<fff>] Type time Units n/a Location TELEMETRY (Class) X-path TBD Reference TBD	Mode EMD in XML format, Calculation Field as "<xml name>[:<element>]:<field>" "M20EarthProductMetadata:M20ProductMetadata:PartList:Part:Ert" NOTES: Value from field "Ert" is extracted for all Parts in PartList. The resulting list is sorted. The lowest value is used as Start time. Type n/a
Ops Keyword EARTH_RECEIVED_STOP_TIME Archive Attribute TBD Definition Specifies the ending time for receiving telemetry during a time period of interest. This should be represented in UTC system format.	Valid Values <YYYY>-<MM>-<DD>T<hh>:<mm>:<ss>[.<fff>] Type time Units n/a Location TELEMETRY (Class) X-path TBD Reference TBD	Mode EMD in XML format, Calculation Field as "<xml name>[:<element>]:<field>" "M20EarthProductMetadata:M20ProductMetadata:PartList:Part:Ert" NOTES: Value from field "Ert" is extracted for all Parts in PartList. The resulting list is sorted. The highest value is used as Stop time. Type n/a
Ops Keyword ELEVATION_FOV Archive Attribute TBD Definition	Valid Values n/a Type float Units deg	Mode 1) DPO in XML format (referenced to APID Name in Appendix A) 2) DPO in XML format (referenced to APID Name in Appendix A) or Image DPO mini-header - Calculation:

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
Specifies the angular measure of the vertical field of view of an imaged scene. NOTE: For MER, this was computed as IFOV times the number of vertical pixels. For M2020, it is computed by projecting rays from the top and bottom edges of the image at the center through the camera model, and computing the angle subtended by those rays.	Location INSTRUMENT_STATE_PARMs (Group) X-path TBD Reference TBD	- Line/sample is translated into an origin and "look direction" vectors (left and right). The vectors are then used to calculate Elevation field of view. Field as "<xml name>:[<element>]:[<field>]" Eng. Cameras a. "<IDPH DPO>:idph:cols" b. "<IDPH DPO>:idph:rows" c. "<IDPH DPO>:idph:res_cols" d. "<IDPH DPO>:idph:res_rows" MCZ Cameras a. "<Ancillary DPO>:cmd_arguments_image>window_width" b. "<Ancillary DPO>:cmd_arguments_image>window_height" c. "MCZ_Image_Mini_Header[22]" d. "MCZ_Image_Mini_Header[23]" NOTES: - For MCZ non-recovered data products (cases "a" and "b"), float values come from the Ancillary DPO. - For MCZ recovered data products (cases "c" and "d"), values analogous to cases "a" (window_width) and "b" (window_height) are each comprised of one byte coming from Image DPO mini-header at byte offsets 22 and 23, respectively. - For MCZ, parm "window_width" is number of image samples. - For MCZ, parm "window_height" is number of image lines. - For MCZ non-Thumbnail data products, multiply by factor of 8 to convert to correct line/sample value. Type 1) U16 2) U8
Ops Keyword ERROR_PIXELS Archive Attribute TBD Definition Specifies the number of pixels that are outside a valid DN range, after all decompression and post decompression processing has been completed.	Valid Values n/a Type integer Units n/a Location COMPRESSION_PARMs (Group) X-path TBD Reference TBD	Mode Calculation based on telemetry Field as "<xml name>:[<element>]:[<field>]" n/a Type n/a

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
Ops Keyword EXPECTED_PACKETS Archive Attribute TBD Definition Specifies the total number of telemetry packets which constitute a complete data product, i.e., a data product without missing data. NOTE: For M2020, telemetry data processing does not track "packets", but instead data product "parts".	Valid Values n/a Type integer Units n/a Location TELEMETRY (Class) X-path TBD Reference TBD	Mode EMD in XML format Field as "<xml name>:[<element>]:[<element>]:<field>" "M20EarthProductMetadata:M20ProductMetadata:PartList:TotalExpected" Type n/a
Ops Keyword EXPECTED_TRANSMISSION_PATH Archive Attribute TBD Definition Specifies the planned transmission paths (routes) for the Data Product. Routing control at time of MPDU (Metadata PDU) generation. See also COMMUNICATION_SESSION_ID.	Valid Values n/a Type string Units n/a Location TELEMETRY (Class) X-path TBD Reference TBD	Mode EMD in XML format Field as "<xml name>:[<element>]:[<element>]:<field>" "M20EarthProductMetadata:M20ProductMetadata:TransmissionControlCriterion" Type n/a
Ops Keyword EXPOSURE_COADD IMAGE_COUNT already exists in PDS3 Archive Attribute TBD Definition Specifies the total number of images summed (co-added) together to obtain the final image. Co-adding increases the signal-to-noise ratio.	Valid Values - Eng. Cameras (S) "0", "2", "4", "8", "16" Type integer Units n/a Location - Eng. Cameras - OBSERVATION_REQUEST_PARMS (Group) - INSTRUMENT_STATE_PARMS (Group) X-path TBD Reference TBD	Mode DPO in XML format (referenced to APID Name in Appendix A) Field as "<xml name>:[<element>]:[<element>]:<field>" - Eng. Cameras "<IDPH DPO>:idph:params:exp_coadd" "<IDPH DPO>:idph:exp_coadd" Type U8
Ops Keyword EXPOSURE_COUNT Archive Attribute	Valid Values "0" to "255" Type	Mode DPO in XML format (referenced to APID Name in Appendix A)

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
TBD Definition Specifies the maximum number of exposures taken during a specified interval. The value is dependent on exposure type.	integer <u>Units</u> n/a <u>Location</u> INSTRUMENT_STATE_PARMs (Group) <u>X-path</u> TBD <u>Reference</u> TBD	Field as “<xml name>:[<element>]:[<element>]:<field>” “<IDPH DPO>:idph:exp_count” <u>Type</u> U16
Ops Keyword EXPOSURE_DURATION Archive Attribute TBD Definition Specifies the value of the time between the opening and closing of an instrument aperture (such as a camera shutter). For M2020, there are no mechanical shutters. Instead, an “electronic shutter” concept was adopted whereby the detectors accumulate charge for EXPOSURE_DURATION amount of time and then that charge is flushed to a masked frame transfer area for readout and digitization.	Valid Values 1) <u>Eng. Cameras</u> “0” to “335539.2” 2) <u>MCZ</u> “0” to “6553.5” 3) <u>SuperCam RMI</u> “0” to “65535” 4) <u>PIXL MCC</u> a. 0 = “125” 1 = “250” 2 = “500” 3 = “1000” 4 = “2000” 5 = “4000” 6 = “8000” 7 = “16000” b. “0” to “n” <u>Type</u> float <u>Units</u> ms (<ms> unit tag required) <u>Location</u> 1) <u>Eng. Cameras</u> INSTRUMENT_STATE_PARMs (Group) 2) <u>SuperCam RMI</u> INSTRUMENT_STATE_PARMs (Group) MINI_HEADER (Group) 3) <u>MCZ</u> INSTRUMENT_STATE_PARMs (Group) 4) <u>PIXL</u> INSTRUMENT_STATE_PARMs (Group) 5) <u>Skycam</u> INSTRUMENT_STATE_PARMs (Group) <u>X-path</u> TBD <u>Reference</u> TBD	Mode 1) DPO in XML format (referenced to APID Name in Appendix A) 2) MCZ Image DPO mini-header 3) SCAM Image DPO mini-header 5) Skycam DPO Image Header - Calculation: 1) ECAM = exp_time * 50 us 2) MCZ = exp_time * 0.1 ms 3) SCAM = exp_time 4) PIXL a. table look up (values in ms) b. exp_time / 1000 5) Skycam = exp_time * 5.333333 ms Field as “<xml name>:[<element>]:[<element>]:<field>” 1) <u>Eng. Cameras</u> “<IDPH DPO>:idph:exp_time” 2) <u>SCAM Cameras</u> “SCAM_Image_Mini_Header:exposure_time” 3) <u>MCZ Cameras</u> a. “MMM_Image_Mini_Header:cmd1[exp_time]” b. “MMM_Image_Mini_Header[28]” 4) <u>PIXL</u> a. “MCC_Image_Mini_Header[integration_time]” b. “MCC_Image_Meta_Data:Image_Meta_Data_Parameters[shuttime]” 5) <u>Skycam</u> “MEDA Image Header:exp_time_lowbyte_data” + “MEDA Image Header:exp_time_highbyte_data” NOTES: - ECAM and MCZ values of “exp_time” are in raw counts. - SCAM value of “exp_time” is in milliseconds. - PIXL values of “exp_time”: a. milliseconds b. microseconds.

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
		- For MCZ non-Z-stack data products (case "a"), a value analogous to "exp_time" is comprised of three bytes coming from Image DPO mini-header at offsets 17, 18 and 19. - For MCZ Z-stack data products (case "b"), a value analogous to "stack depth count" is comprised of one byte coming from Image DPO mini-header at offset 28. - For SCAM RMI products, "exp_time" is comprised of two bytes coming from Image DPO mini-header at offset 5. - For Eng. Cameras, each raw count translates to 50 us. So, calculation is: value = exp_time * 50 us - For MCZ Cameras, each raw count translates to 0.1 ms. So, calculation is: value = exp_time * 0.1 - For PIXL, there are two cases (a and b). If the "image meta data available" flag in the MCC image mini-header (byte 9, bit 7) is false (0) then use case a, which is comprised of one byte at offset 5 in the "mini-header". If the flag is true (1) then use case b, which is comprised of four bytes starting at offset X in the "Image Meta Data Parameter Struct". - For Skycam units are in raw tick counts. 1 tick = 5.333333 ms. Type - Eng. Cameras - SCAM - MCZ - PIXL U16 - Skycam U8
Ops Keyword EXPOSURE_DURATION_COUNT Archive Attribute TBD Definition Specifies the value, in raw counts, of the time interval between the opening and closing of an instrument aperture (such as a camera shutter). This is a raw value taken directly from telemetry, as opposed to EXPOSURE_DURATION, which has been converted to engineering units.	Valid Values "0" to "65535" (B) Type integer Units ms (<ms> unit tag required) Location - Eng. Cameras - OBSERVATION_REQUEST_PARMS (Group) - INSTRUMENT_STATE_PARMS (Group)	Mode DPO in XML format (referenced to APID Name in Appendix A) Field as "<xml name>:[<element>]:[<element>]:<field>" - Eng. Cameras "<IDPH DPO>:idph:params:exp_time" "<IDPH DPO>:idph:exp_time" - MCZ - SHERLOC "<Ancillary DPO>:cmd_arguments image:exp_time"

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
	<u>MCZ</u> <u>SHERLOC</u> a. OBSERVATION_REQUEST_PARMS (Group) b. MINI_HEADER (Group) <u>X-path</u> TBD <u>Reference</u> TBD	b. "MMM_Image_Mini_Header:cmd[1][exp_time]" <u>Type</u> U16
<u>Ops Keyword</u> EXPOSURE_READOUT_COUNT <u>Archive Attribute</u> TBD <u>Definition</u> Specifies the number of times an exposure, or part of an exposure (tile), that have occurred.	<u>Valid Values</u> <u>Eng. Cameras</u> (S) "0" to "65535" <u>Type</u> integer <u>Units</u> n/a <u>Location</u> <u>Eng. Cameras</u> INSTRUMENT_STATE_PARMS (Group) <u>X-path</u> TBD <u>Reference</u> TBD	<u>Mode</u> DPO in XML format (referenced to APID Name in Appendix A) <u>Field as "<xml name>:[<element>]:<field>"</u> <u>Eng. Cameras</u> "<IDPH DPO>:idph:itrans" <u>Type</u> U16
<u>Ops Keyword</u> EXPOSURE_SCALE_FACTOR <u>Archive Attribute</u> TBD <u>Definition</u> Specifies a multiplier to the base exposure time. The base exposure time is either user-commanded or is read from the onboard exposure time table. The resulting number is used by the cameras as the actual commanded exposure time. This scale factor is commonly used during multi-spectral imaging, when the base exposure time is known for one filter and EXPOSURE_SCALE_FACTOR is used to scale the exposure time to levels appropriate for the other filters.	<u>Valid Values</u> <u>Eng. Cameras</u> (B) n/a <u>Type</u> float <u>Units</u> n/a <u>Location</u> OBSERVATION_REQUEST_PARMS (Group) <u>X-path</u> TBD <u>Reference</u> TBD	<u>Mode</u> DPO in XML format (referenced to APID Name in Appendix A) <u>Field as "<xml name>:[<element>]:<field>"</u> <u>Eng. Cameras</u> "<IDPH DPO>:idph:params:exp_time_scale" <u>Type</u> F32
<u>Ops Keyword</u> EXPOSURE_TABLE_ID <u>Archive Attribute</u> TBD <u>Definition</u> Specifies a description for the exposure count value.	<u>Valid Values</u> <u>Eng. Cameras</u> (B) 0 = "NONE" 1 = "FHAZCAM_L" 2 = "FHAZCAM_R" 3 = "RHAZCAM_L" 4 = "RHAZCAM_R" 5 = "NAVCAM_L" 6 = "NAVCAM_R"	<u>Mode</u> DPO in XML format (referenced to APID Name in Appendix A), Table Lookup <u>Field as "<xml name>:[<element>]:<field>"</u> <u>Eng. Cameras</u> "<IDPH DPO>:idph:params:exp_table"

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
	7 = "CACHECAM" 8 = "CAMCONFIG_IDS" <u>Type</u> string <u>Units</u> n/a <u>Location</u> OBSERVATION_REQUEST_PARMS (Group) <u>X-path</u> TBD <u>Reference</u> TBD	<u>Type</u> enum
<u>Ops Keyword</u> EXPOSURE_TBL_UPDATE_FLAG <u>Archive Attribute</u> TBD <u>Definition</u> Specifies whether or not an exposure table entry was updated.	<u>Valid Values</u> <u>Eng. Cameras</u> (B) 0 = "FALSE" 1 = "TRUE" <u>Type</u> string <u>Units</u> n/a <u>Location</u> OBSERVATION_REQUEST_PARMS (Group) <u>X-path</u> TBD <u>Reference</u> TBD	<u>Mode</u> DPO in XML format (referenced to APID Name in Appendix A) <u>Field as "<xml name>:[<element>]:<field>"</u> <u>Eng. Cameras</u> "<IDPH DPO>:idph:params:exp_update" <u>Type</u> boolean
<u>Ops Keyword</u> EXPOSURE_TYPE <u>Archive Attribute</u> TBD <u>Definition</u> Specifies the exposure mode used for image acquisition.	<u>Valid Values</u> <u>Eng. Cameras</u> (B) 0 = "NONE" 1 = "MANUAL" 2 = "AUTO" 3 = "TEST" <u>SuperCam</u> (S) 0 = "MANUAL" 1 = "AUTO" <u>MCZ</u> (B) 0 = "MANUAL" non-0 = "AUTO" <u>Type</u> string(15) <u>Units</u> n/a <u>Location</u> <u>Eng. Cameras</u> <u>SuperCam</u> OBSERVATION_REQUEST_PARMS (Group)	<u>Mode</u> 1) DPO in XML format (referenced to APID Name in Appendix A), Table Lookup 2) DPO in XML format (referenced to APID Name in Appendix A), Table Lookup 3) Image DPO mini-header <u>Field as "<xml name>:[<element>]:<field>"</u> <u>Eng. Cameras</u> "<IDPH DPO>:idph:params:exposure" <u>SuperCam</u> "<Ancillary DPO>:cmd_arguments:exposure_type" (TBD) <u>MCZ Cameras</u> "MCZ_Image_Mini_Header[28]", "MCZ_Image_Mini_Header[29]", "MCZ_Image_Mini_Header[30]", "MCZ_Image_Mini_Header[31]" NOTES:

Output Metadata (Product Label)		Input Metadata (Source)																																								
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type																																								
	MCZ OBSERVATION_REQUEST_PARMS (Group) MINI_HEADER (Group) <u>X-path</u> TBD <u>Reference</u> TBD	For MCZ Cameras, a value analogous to “exposure_type” is comprised of four bytes coming from Image DPO mini-header at offsets 28, 29, 30 and 31. <u>Type</u> Enum																																								
<u>Ops Keyword</u> FILE_RECORDS <u>Archive Attribute</u> TBD <u>Definition</u> Specifies the number of physical file records, including both label records and data records. NOTE: In the PDS the use of FILE_RECORDS along with other file-related data elements is fully described in the Standards Reference.	<u>Valid Values</u> “0” to n <u>Type</u> integer <u>Units</u> n/a <u>Location</u> FILE_DATA_ELEMENT (Class) <u>X-path</u> TBD <u>Reference</u> TBD	<u>Mode</u> DPO in XML format (referenced to APID Name in Appendix A), Calculation: - Value = <DPO value> + size of PDS and VICAR labels <u>Field as “<xml name>:[<element>]:[<element>]:<field>”</u> “<IDPH DPO>:idph:rows” <u>Type</u> U16																																								
<u>Ops Keyword</u> FILTER_NAME <u>Archive Attribute</u> TBD <u>Definition</u> Specifies the commonly-used name of the instrument filter through which an image or measurement was acquired or which is associated with a given instrument mode. See also FILTER_NUMBER. NOTE: FILTER_NAME is unique, while the FILTER_NUMBER is not.	<u>Valid Values</u> - MCZ Cameras ... <table><thead><tr><th>Header</th><th>Keyword Value (Left Eye)</th></tr></thead><tbody><tr><td>0</td><td>“MASTCAM-Z_L1_BROADBAND”</td></tr><tr><td>1</td><td>“MASTCAM-Z_L2_525NM”</td></tr><tr><td>2</td><td>“MASTCAM-Z_L3_440NM”</td></tr><tr><td>3</td><td>“MASTCAM-Z_L4_750NM”</td></tr><tr><td>4</td><td>“MASTCAM-Z_L5_675NM”</td></tr><tr><td>5</td><td>“MASTCAM-Z_L6_865NM”</td></tr><tr><td>6</td><td>“MASTCAM-Z_L7_1035NM”</td></tr><tr><td>7</td><td>“MASTCAM-Z_L8_880NM_ND5”</td></tr><tr><td>8</td><td>“NO_POS_CHANGE”</td></tr></tbody></table> <table><thead><tr><th>Header</th><th>Keyword Value (Right Eye)</th></tr></thead><tbody><tr><td>0</td><td>“MASTCAM-Z_R1_BROADBAND”</td></tr><tr><td>1</td><td>“MASTCAM-Z_R2_525NM”</td></tr><tr><td>2</td><td>“MASTCAM-Z_R3_440NM”</td></tr><tr><td>3</td><td>“MASTCAM-Z_R4_750NM”</td></tr><tr><td>4</td><td>“MASTCAM-Z_R5_905NM”</td></tr><tr><td>5</td><td>“MASTCAM-Z_R6_935NM”</td></tr><tr><td>6</td><td>“MASTCAM-Z_R7_1035NM”</td></tr><tr><td>7</td><td>“MASTCAM-Z_R8_440NM_ND5”</td></tr><tr><td>8</td><td>“NO_POS_CHANGE”</td></tr></tbody></table> <u>Type</u> string array <u>Units</u> n/a <u>Location</u> - GEOMETRIC_CAMERA_MODEL (Group) - OBSERVATION_REQUEST_PARMS (Group) - INSTRUMENT_STATE_PARMS (Group)	Header	Keyword Value (Left Eye)	0	“MASTCAM-Z_L1_BROADBAND”	1	“MASTCAM-Z_L2_525NM”	2	“MASTCAM-Z_L3_440NM”	3	“MASTCAM-Z_L4_750NM”	4	“MASTCAM-Z_L5_675NM”	5	“MASTCAM-Z_L6_865NM”	6	“MASTCAM-Z_L7_1035NM”	7	“MASTCAM-Z_L8_880NM_ND5”	8	“NO_POS_CHANGE”	Header	Keyword Value (Right Eye)	0	“MASTCAM-Z_R1_BROADBAND”	1	“MASTCAM-Z_R2_525NM”	2	“MASTCAM-Z_R3_440NM”	3	“MASTCAM-Z_R4_750NM”	4	“MASTCAM-Z_R5_905NM”	5	“MASTCAM-Z_R6_935NM”	6	“MASTCAM-Z_R7_1035NM”	7	“MASTCAM-Z_R8_440NM_ND5”	8	“NO_POS_CHANGE”	<u>Mode</u> DPO in XML format (referenced to APID Name in Appendix A) <u>Field as “<xml name>:[<element>]:[<element>]:<field>”</u> MCZ Cameras “<Supplementary DPO>:left_filter”, “<Supplementary DPO>:right_filter” <u>Type</u> I8
Header	Keyword Value (Left Eye)																																									
0	“MASTCAM-Z_L1_BROADBAND”																																									
1	“MASTCAM-Z_L2_525NM”																																									
2	“MASTCAM-Z_L3_440NM”																																									
3	“MASTCAM-Z_L4_750NM”																																									
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5	“MASTCAM-Z_R6_935NM”																																									
6	“MASTCAM-Z_R7_1035NM”																																									
7	“MASTCAM-Z_R8_440NM_ND5”																																									
8	“NO_POS_CHANGE”																																									

Output Metadata (Product Label)		Input Metadata (Source)																				
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type																				
	<u>X-path</u> TBD <u>Reference</u> TBD																					
<u>Ops Keyword</u> FILTER_NUMBER <u>Archive Attribute</u> TBD <u>Definition</u> Specifies the number of an instrument filter through which an image or measurement was acquired or which is associated with a given instrument mode. See also FILTER_NAME. Note: FILTER_NAME is unique, while the FILTER_NUMBER is not.	<u>Valid Values</u> MCZ Cameras ...<table><tr><th>Header</th><th>Keyword Value</th></tr><tr><td>0</td><td>"0"</td></tr><tr><td>1</td><td>"1"</td></tr><tr><td>2</td><td>"2"</td></tr><tr><td>3</td><td>"3"</td></tr><tr><td>4</td><td>"4"</td></tr><tr><td>5</td><td>"5"</td></tr><tr><td>6</td><td>"6"</td></tr><tr><td>7</td><td>"7"</td></tr><tr><td>8</td><td>"8"</td></tr></table> <u>Type</u> integer <u>Units</u> n/a <u>Location</u> a. GEOMETRIC_CAMERA_MODEL (Group) OBSERVATION_REQUEST_PARMS (Group) INSTRUMENT_STATE_PARMS (Group) b. MINI_HEADER (Group) <u>X-path</u> TBD <u>Reference</u> TBD	Header	Keyword Value	0	"0"	1	"1"	2	"2"	3	"3"	4	"4"	5	"5"	6	"6"	7	"7"	8	"8"	<u>Mode</u> DPO in XML format (referenced to APID Name in Appendix A) <u>Field as "<xml name>:[<element>]:[<element>]:<field>"</u> - MCZ Cameras ... "<Supplementary DPO>:left_filter", "<Supplementary DPO>:right_filter" "MCZ_Image_Mini_Header[TBD]" <u>Type</u> I8
Header	Keyword Value																					
0	"0"																					
1	"1"																					
2	"2"																					
3	"3"																					
4	"4"																					
5	"5"																					
6	"6"																					
7	"7"																					
8	"8"																					
<u>Ops Keyword</u> FIRST_LINE <u>Archive Attribute</u> TBD <u>Definition</u> Specifies the line within a source image that corresponds to the first line in a sub-image.	<u>Valid Values</u> <u>Surface</u> - Eng. Cameras (S) "1" to "3840" - SuperCam - MCZ <u>Cruise</u> - Eng. Cameras (C) "1" to "3840" <u>Type</u> integer <u>Units</u> n/a <u>Location</u> - Eng. Cameras - IMAGE (Object)	<u>Mode</u> 1) DPO in XML format (referenced to APID Name in Appendix A) 2) DPO in XML format (referenced to APID Name in Appendix A) 3) DPO in XML format (referenced to APID Name in Appendix A) or Image DPO mini-header <u>Field as "<xml name>:[<element>]:[<element>]:<field>"</u> <u>Surface</u> - Eng. Cameras (S) "<IDPH DPO>:idph:row0" "<IDPH DPO>:idph:params:sub_row0" - SuperCam (S) "<Ancillary DPO>:cmd_arguments:start_c_pixel" TBD - MCZ (S) "<Ancillary DPO>:cmd_arguments_image>window_x" "MCZ_Image_Mini_Header[20]"																				

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
	b. SUBFRAME_REQUEST_PARMS (Group) • <u>SuperCam</u> SUBFRAME_REQUEST_PARMS (Group) • <u>MCZ</u> a. SUBFRAME_REQUEST_PARMS (Group) b. MINI_HEADER (Group) • <u>Skycam</u> SUBFRAME_REQUEST_PARMS (Group) <u>X-path</u> TBD <u>Reference</u> TBD	• <u>Skycam</u> (B) "MEDA Image Header:skycam_start_row_lowbyte_data" + "<MEDA Image Header>:skycam_start_row_highbyte_data" <u>Cruise</u> • <u>Eng. Cameras</u> (C) a. "<IDPH DPO>:idph:voff" + "<IDPH DPO>:idph:row0" b. "<IDPH DPO>:idph:voff" + "<IDPH DPO>:idph:params:sub_row0" NOTES: • Add "1" to the source value so that image lines begin at Line 1 (see Valid Value range). • For EECAM Cruise, FSW uses a fixed grid (tile) readout defined by a lookup table, commanded from the video offset parameter. See OFFSET_MODE_ID table for predefined values. • For MCZ non-recovered non-Z-stack data products (case "a"), value comes from the Ancillary DPO. • For MCZ recovered data products (case "b"), value analogous to case "a" (window_x) is comprised of one byte coming from Image DPO mini-header at byte offset 20. • For MCZ Z-stack data products, the value is unknown. • For MCZ, parm "window_x" is starting line of the window. <u>Type</u> U16
<u>Ops Keyword</u> FIRST_LINE_SAMPLE <u>Archive Attribute</u> TBD <u>Definition</u> Specifies the sample within a source image that corresponds to the first sample in a sub-image.	<u>Valid Values</u> <u>Surface</u> • <u>Eng. Cameras</u> (S) "1" to "5120" <u>Cruise</u> • <u>Eng. Cameras</u> (C) "1" to "5120" <u>Type</u> integer <u>Units</u> n/a <u>Location</u> a. IMAGE (Object) b. SUBFRAME_REQUEST_PARMS (Group) 2) SUBFRAME_REQUEST_PARMS (Group) 3) a. SUBFRAME_REQUEST_PARMS (Group) b. MINI_HEADER (Group) <u>X-path</u>	<u>Mode</u> 1) DPO in XML format (referenced to APID Name in Appendix A) 2) DPO in XML format (referenced to APID Name in Appendix A) 3) DPO in XML format (referenced to APID Name in Appendix A) or Image DPO mini-header <u>Field as "<xml name>:[<element>]:[<element>]:<field>"</u> <u>Surface</u> • <u>Eng. Cameras</u> (S) a. "<IDPH DPO>:idph:col0" b. "<IDPH DPO>:idph:params:sub_col0" • <u>SuperCa</u> (S) "<Ancillary DPO>:cmd_arguments:start_r_pixel" TBD • <u>MCZ</u> (S) a. "<Ancillary DPO>:cmd_arguments_image>window_y" b. "MCZ_Image_Mini_Header[21]"

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
	TBD <u>Reference</u> TBD	<u>Cruise</u> - Eng. Cameras (C) "<IDPH DPO>:idph:voff" + <IDPH DPO>:idph:col0" "<IDPH DPO>:idph:voff" + <IDPH DPO>:idph:params:sub_col0" NOTES: - Add "1" to the source value so that image lines begin at Line 1 (see Valid Value range). - For EECAM Cruise, FSW uses a fixed grid (tile) readout defined by a lookup table, commanded from the video offset parameter. See OFFSET_MODE_ID table for predefined values. - For MCZ non-recovered non-Z-stack data products (case "a"), value comes from the Ancillary DPO. - For MCZ recovered data products (case "b"), value analogous to case "a" (window_y) is comprised of one byte coming from Image DPO mini-header at byte offset 21. - For MCZ Z-stack data products, the value is unknown. - For MCZ, parm "window_y" is starting sample of the window. <u>Type</u> U16
<u>Ops Keyword</u> FLAT_FIELD_CORRECTION_FLAG <u>Archive Attribute</u> TBD <u>Definition</u> Specifies whether or not a flat field correction was applied to an image.	<u>Valid Values</u> - Eng. Cameras (B) 0 = "FALSE" 1 = "TRUE" 0 = "FALSE" - non-0 = "TRUE" <u>Type</u> string(13) <u>Units</u> n/a <u>Location</u> - Eng. Cameras - OBSERVATION_REQUEST_PARMS (Group) - INSTRUMENT_STATE_PARMS (Group) <u>X-path</u> TBD <u>Reference</u> TBD	<u>Mode</u> DPO in XML format (referenced to APID Name in Appendix A) <u>Field as "<xml name>:[<element>]:<field>"</u> - Eng. Cameras "<IDPH DPO>:idph:params:flat" "<IDPH DPO>:idph:flat_params[5]" <u>Type</u> - Eng. Cameras - boolean - F32[5]
<u>Ops Keyword</u> FLAT_FIELD_CORRECTION_PARM <u>Archive Attribute</u> TBD	<u>Valid Values</u> - Eng. Cameras n/a - MCZ Cameras "UNK"	<u>Mode</u> DPO in XML format (referenced to APID Name in Appendix A)

Output Metadata (Product Label)		Input Metadata (Source)								
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type								
Definition Specifies the onboard flat-field coefficients/parameters used in the algorithm to remove the flat field signature. The FLAT_FIELD_CORRECTION_FLAG will indicate if the signature was removed. NOTE: The algorithm used by M2020 follows: new(x,y) = orig(x,y) * ff(x,y) where, r = (x-a)² + (y-b)² ff(x,y) = 1 + c*r + d*r² + e*r³ and, a = array element 1 b = array element 2 c = array element 3 d = array element 4 e = array element 5	Type float array[5] Units n/a Location INSTRUMENT_STATE_PARMS (Group) X-path TBD Reference TBD	Field as “<xml name>:[<element>]:<field>” Eng. Cameras “<IDPH DPO>:idph:flat_params[5]” Type F32[5]								
Ops Keyword FLAT_FIELD_FILE_NAME Archive Attribute TBD Definition Specifies the name, or array of names, of the flat field image(s) file (an image taken in an optical laboratory of a white background or an image taken in the dawn with the intention to have an equally illuminated background for the whole image) which should be used to perform radiometric calibration of the image. The flat field image provides a reference label of the sensitivity of the used optics across the field of view. The shuttered image needs to be divided by the flat field image during calibration. Selection of the appropriate flat field image may be based on time, camera, temperature, readout conditions, light flood, gain and offset.	Valid Values n/a Type string array Units n/a Location DERIVED_IMAGE_PARMS (Group) X-path TBD Reference TBD	Mode RDR-generating software Field as “<xml name>:[<element>]:<field>” n/a Type n/a								
Ops Keyword FLAT_FIELD_FILE_DESC Archive Attribute TBD Definition Specifies a description of the corresponding flat field files listed in FLAT_FIELD_FILE_NAME.	Valid Values “Flat field image.”, “Flat field standard deviation image.” Type string array Units n/a Location DERIVED_IMAGE_PARMS (Group) X-path TBD Reference TBD	Mode RDR-generating software Field as “<xml name>:[<element>]:<field>” n/a Type n/a								
Ops Keyword FLIGHT_SOFTWARE_MODE Archive Attribute TBD	Valid Values <table><tr><td>Keyword Value</td><td>FSW Field Value</td></tr><tr><td>“0”</td><td>UNKNOWN</td></tr><tr><td>“1”</td><td>TEST</td></tr><tr><td>“2”</td><td>PRELAUNCH</td></tr></table>	Keyword Value	FSW Field Value	“0”	UNKNOWN	“1”	TEST	“2”	PRELAUNCH	Mode EMD in XML format Field as “<xml name>:[<element>]:<field>”
Keyword Value	FSW Field Value									
“0”	UNKNOWN									
“1”	TEST									
“2”	PRELAUNCH									

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
Definition Active Flight Software mode at Data Product creation.	“3” LAUNCH “4” ECLIPSE “5” CRUISE “6” EDL_APPROAH “7” EDL_MAIN “8” SURFACE_NOMINAL “9” SURFACE_STANDBY “10” NONPRIME_TEST “11” NONPRIME_PRELAUNCH “12” NONPRIME_LAUNCH “13” NONPRIME_ECLIPSE “14” NONPRIME_CRUISE “15” NONPRIME_EDL_APPROACH “16” NONPRIME_EDL_MAIN “17” NONPRIME_SURFACE_NOMINAL “18” NONPRIME_SURFACE_STANDBY Type string Units n/a Location TELEMETRY (Class) X-path TBD Reference TBD	"M20EarthProductMetadata:M202ProductMetadata:Fsw Mode" Type unsigned integer
Ops Keyword FLIGHT_SOFTWARE_VERSION_ID Archive Attribute TBD Definition Active Flight Software version at Data Product creation. The Flight Software version is an opaque token – there is no arithmetic value associated with the token. The version is defined as the time of the FSW build, in seconds since 12:00:00, Jan. 1, 2000.	Valid Values n/a Type string Units n/a Location TELEMETRY (Class) X-path TBD Reference TBD	Mode EMD in XML format Field as “<xml name>:[<element>]:[<element>]:<field>” "M20EarthProductMetadata:M20ProductMetadata:Fsw Version" Type unsigned integer
Ops Keyword FRAME_ID Archive Attribute TBD Definition Specifies an identification for a particular instrument measurement frame. A frame consists of a sequence of measurements made over a specified time interval, and may include measurements from different instrument	Valid Values - Eng. Cameras & Mastcam-Z (“LEFT”, “RIGHT”, “MONO”, “N/A”) - SuperCam RMI “MONO” - Mastcam-Z (“LEFT”, “RIGHT”, “MONO”, “N/A”) - SHERLOC “MONO” - MEDA	Mode - Static value: 1) for SuperCam RMI, specify “MONO” 2) for SHERLOC, specify “MONO” 3) for MEDA Skycam, specify “MONO” 4) for PIXL MCC, specify “MONO” - EMD in XML format Field as “<xml name>:[<element>]:[<element>]:<field>” Eng. Cameras & Mastcam-Z

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
modes. These sequences repeat from cycle to cycle and sometimes within a cycle. Note that mosaics may contain more than one value in an array.	"MONO" <u>PIXL</u> "MONO" Type string array Units n/a Location IDENTIFICATION (Class) X-path TBD Reference TBD	"M20EarthProductMetadata:M20ProductMetadata:ProductName" NOTES: If ending character in field "ProductName" is "l", valid value is "LEFT". If ending character is "r", valid value is "RIGHT". Type n/a
Ops Keyword FRAME_TYPE Archive Attribute TBD Definition Specifies whether the image was commanded as part of a stereo pair or as a single left or right monoscopic image. If FRAME_TYPE=STEREO, a left and a right image should be present for the same IMAGE_ID	Valid Values <u>SuperCam RMI</u> 1) "MONO" <u>MARDI & SHERLOC-WATSON</u> 2) "MONO" <u>Mastcam-Z</u> 3) "STEREO" <u>Eng. Cameras</u> 4) 0 = "MONO" 1 = "STEREO" Type string(10) Units n/a Location IDENTIFICATION (Class) X-path TBD Reference TBD	Mode - Static value: 1) for SuperCam RMI, specify "MONO" 2) for MARDI & SHERLOC-WATSON, specify "MONO" 3) for Mastcam-Z, specify "STEREO" - DPO in XML format (referenced to APID Name in Appendix A) Field as "<xml name>:[<element>]:[<element>]:<field>" <u>Eng. Cameras</u> 3) "IDPH DPO>:idph:stereo" Type 4) boolean
Ops Keyword GAIN_NUMBER Archive Attribute TBD Definition Specifies the gain value used in the analog to digital conversion. The gain value is a multiplicative factor used in the analog to digital conversion. For SuperCam on M2020, specifies the commanded gain value	Valid Values TBD Type integer Units n/a Location OBSERVATION_REQUEST_PARMS (Group) X-path TBD	Mode DPO in XML format (referenced to APID Name in Appendix A) Field as "<xml name>:[<element>]:[<element>]:<field>" <u>SuperCam</u> TBD Type U8

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
in the SuperCam Mast Unit (CCMU). See also OFFSET_NUMBER.	<u>Reference</u> TBD	
<u>Ops Keyword</u> GEOMETRY_PROJECTION_TYPE <u>Archive Attribute</u> TBD <u>Definition</u> Specifies the state of the pixels in an image before a re-projection has been applied. Describes if or how the pixels have been reprojected. RAW indicates reprojection has not been done; the pixels are as they came from the camera. For M2020, this means the image uses a CAHVOR or one of the CAHVORE camera models. LINEARIZED means that reprojection has been performed to linearize the camera model (thus removing things like lens distortion). This means the image uses a CAHV camera model.	<u>Valid Values</u> "RAW", "LINEARIZED" <u>Type</u> string <u>Units</u> n/a <u>Location</u> IDENTIFICATION (Class) <u>X-path</u> TBD <u>Reference</u> TBD	<u>Mode</u> Dependent on EDR/RDR state: - For EDRs, static (default = "RAW") - For RDRs, RDR-generating software <u>Field as "<xml name>:[<element>]:<field>"</u> n/a <u>Type</u> n/a
<u>Ops Keyword</u> GROUP_APPLICABILITY_FLAG <u>Archive Attribute</u> TBD <u>Definition</u> Specifies whether a group of keywords are valid values. It is present in a Group only when information is received from telemetry. For M2020, when in a REQUEST_PARMS group, it specifies whether or not the activity represented by the group was commanded. If TRUE, the rest of the contents of the group specify the commanded arguments or parameters for that activity. For example, a value of TRUE in a HISTOGRAM_REQUEST_GROUP means a histogram was requested.	<u>Valid Values</u> - Eng. Cameras (B) - SuperCam (B) 0 = "FALSE" non-0 = "TRUE" <u>Type</u> string <u>Units</u> n/a <u>Location</u> - Eng. Cameras a. IMAGE_REQUEST_PARMS (Group) b. SUBFRAME_REQUEST_PARMS (Group) c. THUMBNAİL_REQUEST_PARMS (Group) d. COLUMN_SUM_REQUEST_PARMS (Group) e. ROW_SUM_REQUEST_PARMS (Group) f. HISTOGRAM_REQUEST_PARMS (Group) - SuperCam IDE (Group) <u>X-path</u> TBD <u>Reference</u> TBD	<u>Mode</u> DPO in XML format (referenced to APID Name in Appendix A) <u>Field as "<xml name>:[<element>]:<field>"</u> - Eng. Cameras a. "<IDPH DPO>:idph:params:image" b. "<IDPH DPO>:idph:params:subframe" c. "<IDPH DPO>:idph:params:thumbnail" d. "<IDPH DPO>:idph:params:colsums" e. "<IDPH DPO>:idph:params:rowsums" f. "<IDPH DPO>:idph:params:histogram" - SuperCam "<Ancillary DPO>:cmd_arguments:hdr" <u>Type</u> - Eng. Cameras a. boolean b. enum c. boolean d. boolean e. boolean f. boolean g. Boolean - SuperCam enum
<u>Ops Keyword</u> HDR_CLIP_THRESHOLD <u>Archive Attribute</u> TBD <u>Definition</u>	<u>Valid Values</u> "0" to "4023" <u>Type</u> integer <u>Units</u>	<u>Mode</u> SCAM Image DPO mini-header <u>Field as "<xml name>:[<element>]:<field>"</u> "SCAM_Image_Mini_Header:hdr_clipping_threshold"

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
TBD	n/a <u>Location</u> HIGH_DYNAMIC_RANGE (Group) MINI_HEADER (Group) <u>X-path</u> TBD <u>Reference</u> TBD	<u>Type</u> U16 Notes: For SCAM RMI products, "hdr clipping threshold" is comprised of two bytes coming from the Image DPO mini-header at offset 27.
<u>Ops Keyword</u> HDR_EXPOSURE_STEP <u>Archive Attribute</u> TBD <u>Definition</u> TBD	<u>Valid Values</u> TBD <u>Type</u> integer array[6] <u>Units</u> ms <u>Location</u> HIGH_DYNAMIC_RANGE (Group) MINI_HEADER (Group) <u>X-path</u> TBD <u>Reference</u> TBD	<u>Mode</u> SCAM Image DPO mini-header <u>Field as "<xml name>:[<element>]:<field>"</u> "SCAM_Image_Mini_Header:hdr_exposureStep0", "SCAM_Image_Mini_Header:hdr_exposureStep1", "SCAM_Image_Mini_Header:hdr_exposureStep2", "SCAM_Image_Mini_Header:hdr_exposureStep3", "SCAM_Image_Mini_Header:hdr_exposureStep4", "SCAM_Image_Mini_Header:hdr_exposureStep5" <u>Type</u> U16 Notes: For SCAM RMI products, "hdr exposure steps" 1 - 6 are comprised of 12 bytes coming from the Image DPO mini-header at offset 30.
<u>Ops Keyword</u> HDR_FRAME_NUMBER <u>Archive Attribute</u> TBD <u>Definition</u> TBD	<u>Valid Values</u> "2" to "7" <u>Type</u> integer <u>Units</u> n/a <u>Location</u> HIGH_DYNAMIC_RANGE (Group) MINI_HEADER (Group) <u>X-path</u> TBD <u>Reference</u> TBD	<u>Mode</u> SCAM Image DPO mini-header <u>Field as "<xml name>:[<element>]:<field>"</u> "SCAM_Image_Mini_Header:hdr_frame_number" <u>Type</u> U8 Notes: For SCAM RMI products, "hdr frame number" is comprised of one byte coming from the Image DPO mini-header at offset 29.
<u>Ops Keyword</u> HDR_MODE <u>Archive Attribute</u> TBD <u>Definition</u> TBD	<u>Valid Values</u> 0 = "OFF" 1 = "ON" <u>Type</u> string <u>Units</u> n/a	<u>Mode</u> SCAM Image DPO mini-header <u>Field as "<xml name>:[<element>]:<field>"</u> "SCAM_Image_Mini_Header:hdr_mode" <u>Type</u> U8

Output Metadata (Product Label)		Input Metadata (Source)
• Ops Keyword • Archive Label Attribute • Definition	• Mission Phase C=Cruise, S=Surface, B=Both • Valid Values (quoted) • Type • Units • Keyword Location in Ops Label • Attribute X-path in Archive label • Attribute Reference in Archive Label (bracketed)	• Mission Phase C=Cruise, S=Surface, B=Both • Mode • Metadata Field • Type
	Location HIGH_DYNAMIC_RANGE (Group) MINI_HEADER (Group) X-path TBD Reference TBD	Notes: For SCAM RMI products, "hdr mode" is comprised of one byte coming from the Image DPO mini-header at offset 7.
Ops Keyword HEADER_TYPE Archive Attribute TBD Definition Specifies a specific type of header data structure. For example: FITS, VICAR. NOTE: In the PDS, HEADER_TYPE is used to indicate non-PDS headers.	Valid Values "VICAR2", "ODL" Type string(12) Units n/a Location IMAGE_HEADER (Object) X-path TBD Reference TBD	Mode Static Value Field as "<xml name>:[<element>]:[<element>]:<field>" n/a Type n/a
Ops Keyword HORIZON_MASK_ELEVATION Archive Attribute TBD Definition Specifies the elevation (in degrees) used as the horizontal cutoff in a mask file that, when applied to another image file, prevents the horizon and sky features in the image from being processed. If this keyword is not present in the product label, no horizon mask was used. For M2020, this is in degrees as measured in the Local Level (or, equivalently, Site) frame. See also MASK_DESC_FILE_NAME.	Valid Values n/a Type float Units n/a Location DERIVED_IMAGE_PARMS (Group) X-path TBD Reference TBD	Mode RDR-generating software Field as "<xml name>:[<element>]:[<element>]:<field>" n/a Type n/a
Ops Keyword ^IMAGE Archive Attribute TBD Definition Specifies a pointer to the IMAGE object. See chapter 14 of the PDS Standards Reference for more information on pointer usage.	Valid Values n/a Type NULL Units n/a Location POINTERS X-path TBD	Mode Calculation Field as "<xml name>:[<element>]:[<element>]:<field>" n/a Type n/a

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
	<u>Reference</u> TBD	
<u>Ops Keyword</u> IMAGE_ACQUIRE_MODE <u>Archive Attribute</u> TBD <u>Definition</u> This keyword describes the mode of image acquisition. Valid values are defined as: a) "NONE" - No image acquired b) "SERNO" - No image acquired, camera serial number returned only c) "IMAGE" - The image was acquired For M2020, there are two acquisition modes pertaining to camera data: "IMAGE" and "SERNO". The normal mode ("IMAGE") acquires and transmits the image according to all the command parameter settings. The "SERNO" case is a mode of acquisition that does just enough to get the camera's hardware serial number. Normally, such a mode should result in a data product with a header only and no image. However, the IMG onboard software supercedes during acquisition and acquires data in hardware windowed mode, resulting in a single row of an image. So, ground processing of the data product will yield a value of 1 for "hw_numrows", but the actual image is still set as 0 rows and 0 columns.	<u>Valid Values</u> - Eng. Cameras (B) 0 = "NONE" 1 = "SERNO" 2 = "IMAGE" - MCZ Cameras "IMAGE" - Skycam 0 = "RESET" 1 = "FULL_FRAME" 2 = "WINDOWED" 3 = "ROW_SUMMED" 4 = "FIXED_PATTERN" 5 = "SERNO" <u>Type</u> string <u>Units</u> n/a <u>Location</u> IDENTIFICATION (Class) <u>X-path</u> TBD <u>Reference</u> TBD	<u>Mode</u> DPO in XML format (referenced to APID Name in Appendix A) <u>Field as "<xml name>:[<element>]:<field>"</u> - Eng. Cameras "<IDPH DPO>:idph:params:acquire" - MCZ TBD - Skycam "<MEDA Image Header>:skycam_acq_mode" <u>Type</u> enum
<u>Ops Keyword</u> IMAGE_DATA_SIZE <u>Archive Attribute</u> TBD <u>Definition</u> Specifies the size of the image data product on the MCZ camera DEA.	<u>Valid Values</u> n/a <u>Type</u> integer <u>Units</u> n/a <u>Location</u> TELEMETRY (Class) <u>X-path</u> TBD <u>Reference</u> TBD	<u>Mode</u> DPO in XML format (referenced to APID Name in Appendix A) <u>Field as "<xml name>:[<element>]:<field>"</u> - MCZ "<Ancillary DPO>:image_id_data:image_size" - Skycam "<MEDA Image Header>:image_size" <u>Type</u> U32
<u>Ops Keyword</u> ^IMAGE_HEADER <u>Archive Attribute</u> TBD	<u>Valid Values</u> n/a <u>Type</u> NULL	<u>Mode</u> Calculation <u>Field as "<xml name>:[<element>]:<field>"</u> n/a

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
Definition Specifies a pointer to the IMAGE_HEADER object. See chapter 14 of the PDS Standards Reference for more information on pointer usage.	Units n/a Location POINTERS X-path TBD Reference TBD	Type n/a
Ops Keyword IMAGE_ID Archive Attribute TBD Definition Specifies an image and typically consists of a sequence of characters representing 1) a routinely occurring measure, such as revolution number, 2) a letter identifying the spacecraft, target, or camera, and 3) a representation of a count within the measure, such as picture number within a given revolution. NOTE: See the Imaging FDD for more detailed definition. Examples: Mariner 9 - LevantalthIdentifier - (orbit, camera, pic #, total # of pics in orbit) Viking Orbiter - (orbit #, sc, pic # (FSC/16)) Viking Lander - (sc, camera, mars doy, diode (filter), pic # for that day) Voyager - (pic # for encounter, FDS for cruise)	Valid Values n/a Type string(30) Units n/a Location - Eng. Cameras IDENTIFICATION (Class) - MCZ - IDENTIFICATION (Class) - IDENTIFICATION (Class) - MINI_HEADER (Group) - Skycam IDENTIFICATION X-path TBD Reference TBD	Mode DPO in XML format (referenced to APID Name in Appendix A) Field as “<xml name>:[<element>]:[<element>]:<field>” Surface - Eng. Cameras “<IDPH DPO>.idph:params:imgid1” - MCZ Cameras “<Ancillary DPO>.cmd_arguments_image:image_ID” “<Ancillary DPO>.cmd_arguments_zstack:startng_ID” “MCZ_Image_Mini_Header[TBD]” - Skycam “<Meda Img Header Segment_Id 0>.image_id” Cruise - Eng. Cameras “<IDPH DPO>.idph:params:imgid” NOTES: - For MCZ non-Z-stack data products, case “a”. - For MCZ Z-stack data products, case “b”. Type U32
Ops Keyword IMAGE_ID_2 Archive Attribute TBD Definition Specifies an image and typically consists of a sequence of characters representing 1) a routinely occurring measure, such as revolution number, 2) a letter identifying the spacecraft, target, or camera, and 3) a representation of a count within the measure, such as picture number within a given revolution. NOTE: See the Imaging FDD for more detailed definition. Examples:	Valid Values n/a Type string(30) Units n/a Location 1) IDENTIFICATION (Class) INPi X-path TBD Reference TBD	Mode DPO in XML format (referenced to APID Name in Appendix A) Field as “<xml name>:[<element>]:[<element>]:<field>” Surface Eng. Cameras “<IDPH DPO>.idph:params:imgid2” Type U32

Output Metadata (Product Label)		Input Metadata (Source)						
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type						
Mariner 9 - LevantHalIdentifier - (orbit, camera, pic #, total # of pics in orbit) Viking Orbiter - (orbit #, sc, pic # (FSC/16)) Viking Lander - (sc, camera, mars doy, diode (filter), pic # for that day) Voyager - (pic # for encounter, FDS for cruise)								
Ops Keyword IMAGE_RADIANCE_FACTOR Archive Attribute TBD Definition For a multi-input image (such as a mosaic), specifies the overall multiplicative factor that was applied to each image as the mosaic was being assembled. Together with IMAGE_RADIANCE_OFFSET, this specifies a simple linear adjustment of the form “dn * factor + offset”, that can be used for (limited) radiometric seam correction. Each pixel within a given input image receives the same correction. The order of images is as defined in the input list file (generally delivered along with the mosaic) or INPUT_PRODUCT_ID. Applies to RDRs only, not EDRs. NOTE: This correction is applied after radiometric correction, if any, performed on the inputs.	Valid Values n/a Type float array Units n/a Location DERIVED_IMAGE_PARMS (Group) X-path TBD Reference TBD	Mode RDR-generating software Field as “<xml name>:[<element>]:[<element>]:<field>” n/a Type n/a						
Ops Keyword IMAGE_RADIANCE_OFFSET Archive Attribute TBD Definition For a multi-input image (such as a mosaic), specifies the overall additive offset that was applied to each image as the mosaic was being assembled. Together with IMAGE_RADIANCE_FACTOR, this specifies a simple linear adjustment of the form “dn * factor + offset”, that can be used for (limited) radiometric seam correction. Each pixel within a given input image receives the same correction. Input values of 0 are not adjusted. The order of images is as defined in the input list file (generally delivered along with the mosaic) or INPUT_PRODUCT_ID. Applies to RDRs only, not EDRs. NOTE: This correction is applied after radiometric correction, if any, performed on the inputs.	Valid Values n/a Type float array Units n/a Location DERIVED_IMAGE_PARMS (Group) X-path TBD Reference TBD	Mode RDR-generating software Field as “<xml name>:[<element>]:[<element>]:<field>” n/a Type n/a						
Ops Keyword IMAGE_TYPE Archive Attribute TBD Definition Specifies the type of image acquired. This may be used to describe characteristics that differentiate one	Valid Values <table><tr><td>Eng. Cameras ...</td><td></td></tr><tr><td>Keyword Value “REGULAR”</td><td>APID Names Any string value starting with “ImgImage”</td></tr><tr><td>“THUMBNAIL”</td><td>Any string value starting with “ImgThumb”</td></tr></table>	Eng. Cameras ...		Keyword Value “REGULAR”	APID Names Any string value starting with “ImgImage”	“THUMBNAIL”	Any string value starting with “ImgThumb”	Mode EMD in XML format Field as “<xml name>:[<element>]:[<element>]:<field>” - Eng. Cameras - SCAM - MCZ
Eng. Cameras ...								
Keyword Value “REGULAR”	APID Names Any string value starting with “ImgImage”							
“THUMBNAIL”	Any string value starting with “ImgThumb”							

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
group of images from another such as the nature of the data in the image file, the purpose for which the image was acquired, or the way in which it was acquired. This element is very similar to the older <code>image_observation_type</code> element, but is designed to resolve ambiguities in cases where missions utilize a naming convention for both specific images and more general observations, which consist of multiple images. In those cases, the latter may be described by the <code>observation_type</code> element.	"HISTOGRAM" Any string value starting with "ImgHistogram" "ROW_SUM" Any string value starting with "ImgRow" "COL_SUM" Any string value starting with "ImgCol" - SuperCam ... <u>Keyword Value</u> <u>APID Names</u> "REGULAR" Any string value starting with "ScamRmiImage" "THUMBNAIL" Any string value starting with "ScamRmiThumb" - MCZ Cameras ... <u>Keyword Value</u> <u>APID Names</u> "REGULAR" Any string value that contains "Image" "THUMBNAIL" Any string value that contains "Thumbnail" - Skycam <u>Keyword Value</u> <u>TBD</u> "REGULAR" image_type = bright or standalone "THUMBNAIL" Any string value starting with "MEDAThumbnailImage" "ROW_SUM" image_acq = row_sum <u>Type</u> string(15) <u>Units</u> n/a <u>Location</u> IDENTIFICATION (Class) <u>X-path</u> TBD <u>Reference</u> TBD	"M20EarthProductMetadata:M20ProductMetadata:ProductName" - Skycam a. "<MEDA Image Header>:image_type" b. "<MEDA Image Header>:skycam_acq_mode" <u>Type</u> n/a
<u>Ops Keyword</u> INPUT_PRODUCT_ID <u>Archive Attribute</u> TBD <u>Definition</u> Specifies the product(s) directly used as input to create this product. It may contain either the PRODUCT_ID or the filename of the input products. For M2020, this RDR-only keyword specifies the PRODUCT_IDs of the inputs most directly used to	<u>Valid Values</u> n/a <u>Type</u> string array <u>Units</u> n/a <u>Location</u> DERIVED_IMAGE_PARMS (Group) <u>X-path</u>	<u>Mode</u> RDR-generating software <u>Field as "<xml name>:[<element>]:[<element>]:<field>"</u> n/a <u>Type</u> n/a

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
create this product, which may themselves be RDRs. For example, a linearized XYZ image would list the left and right linearized input images as well as the disparity file. Contrast this with SOURCE_PRODUCT_ID, which lists only the root EDR source; INPUT_PRODUCT_ID lists the proximate inputs (EDR or RDR) for the process used to create this product.	TBD <u>Reference</u> TBD	
Ops Keyword INSTRUMENT_AZIMUTH Archive Attribute TBD Definition Specifies the value for an instrument's rotation in the horizontal direction. It is usually measured from some kind of low hard stop. Although it may be used for any instrument where it makes sense, it is primarily intended for use in surface-based instruments that measure pointing in terms of azimuth and elevation. When in a DERIVED_GEOMETRY group, defines the azimuth (horizontal rotation) at which the instrument is pointed. This value is expressed using the coordinate system referred to by REFERENCE_COORD_SYSTEM_NAME and REFERENCE_COORD_SYSTEM_INDEX contained within the same group. The interpretation of exactly what part of the instrument is being pointed is mission-specific. It could be the boresight, the camera head direction, the CAHV camera model A vector direction, or any of a number of other things. As such, for multimission use this value should be used mostly as an approximation, e.g. identifying scenes which might contain a given object. For M2020, the interpretation is the boresight of the camera, defined as projecting the center of the nominal image (before downsampling or subframing) through the camera model.	Valid Values n/a Type float Units deg (<deg> unit tag required) Location - ROVER_DERIVED_GEOMETRY_PARMs (Group) - SITE_DERIVED_GEOMETRY_PARMs (Group) X-path TBD Reference TBD	Mode Calculation: - The image center is translated into an origin and "look direction" vectors (left and right). The vectors are then used to calculate instrument azimuth. Field as "<xml name>:[<element>]:[<element>]:<field>" n/a Type n/a
Ops Keyword INSTRUMENT_BAND_ID Archive Attribute TBD Definition Specifies an array of strings identifying the instrument represented by the corresponding band in the image. The first entry in the array identifies the instrument for the first band, the second entry for the second band, etc. For M2020 Preload products, the string also indicates	Valid Values - Reachability RDRs ("DRILL", "BITCAR", "SHERLOC-WATSON", "APXS", "SCOOP_TIP") - Preload RDRs ("DRILL_MINIMUM", "DRILL_MAXIMUM") Type string array[5] Units n/a Location	Mode Static Value Field as "<xml name>:[<element>]:[<element>]:<field>" n/a Type n/a

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
whether the value is a minimum or maximum preload. See also CONFIGURATION_BIT_ID.	DERIVED_IMAGE_PARMS (Group) <u>X-path</u> TBD <u>Reference</u> TBD	
Ops Keyword INSTRUMENT_COORD_FRAME_ID Archive Attribute TBD Definition Along with INSTRUMENT_COORD_FRAME_INDEX, specifies the name of the coordinate frame in which the pointing for this instrument is expressed. The values of INSTRUMENT_COORDINATE should be interpreted using this coordinate frame.	Valid Values - Eng. Cameras (B) 0 = "SITE" 1 = "RNAV_SAVED" 2 = "LL_SAVED" 3 = "SITE_SAVED" 4 = "NAV_GOAL" 5 = "NAV_VTT" 6 = "NAV_MAP" 7 = "AEGIS_1" 8 = "AEGIS_2" 9 = "AEGIS_3" 10 = "AEGIS_4" 11 = "AEGIS_5" 12 = "LL" 13 = "RNAV" 14 = "SUN" 15 = "EARTH" 16 = "PHOBOS" 17 = "DEIMOS" 18 = "RMECH" 19 = "RA_BASE" 20 = "TURRET" 21 = "DRILL" 22 = "DRILL_BIT_TIP" 23 = "FCS" 24 = "GDRT" 25 = "TOOL" 26 = "ARM_TGT" 27 = "ARM_EVAL_TOOL" 28 = "WATSON" 29 = "SHERLOC" 30 = "PIXL_BASE" 31 = "PIXL_TOOL" 32 = "DOCKING_POST" 33 = "CUSTOM_TCP" 34 = "CAL_PIXL" 35 = "ACA_ORIGIN" 36 = "BC_DOCK_FF" 37 = "BC_TX_EF" 38 = "VIS_FOCAL" 39 = "SHA_BASE" 40 = "SHA_EE_TIP" 41 = "SHA_EE_AXIS" 42 = "ROCKER_L" 43 = "ROCKER_R" 44 = "BOGIE_L" 45 = "BOGIE_R" 46 = "WHEEL_LF" 47 = "WHEEL_RF" 48 = "WHEEL_LM" 49 = "WHEEL_RM" 50 = "WHEEL_LR" 51 = "WHEEL_RR"	Mode DPO in XML format (referenced to APID Name in Appendix A) Field as "<xml name>:[<element>]:<field>" Surface - Eng. Cameras "IDPH DPO>:idph:params:rsm_frame" - SuperCam "IDPH DPO>:idph:params:rsm_frame" TBD "Ancillary DPO>:cmd_arguments:frame_id" TBD - MCZ Cameras "Supplementary DPO>:frame_id" TBD Cruise - Eng. Cameras "IDPH DPO>:idph:params:rsm_frame" Type Enum

Output Metadata (Product Label)		Input Metadata (Source)
• Ops Keyword • Archive Label Attribute • Definition	• Mission Phase C=Cruise, S=Surface, B=Both • Valid Values (quoted) • Type • Units • Keyword Location in Ops Label • Attribute X-path in Archive label • Attribute Reference in Archive Label (bracketed)	• Mission Phase C=Cruise, S=Surface, B=Both • Mode • Metadata Field • Type
	52 = "RSM_BASE" 53 = "RSM_GIMBAL" 54 = "RSM_HEAD" 55 = "NCAML_MOUNT" 56 = "NCAML" 57 = "NCAML_SUNID" 58 = "NCAMR_MOUNT" 59 = "NCAMR" 60 = "NCAMR_SUNID" 61 = "SCAM_MOUNT" 62 = "RMI" 63 = "LIBS" 64 = "VISIR" 65 = "MCZL_MOUNT" 66 = "MCZL" 67 = "MCZR_MOUNT" 68 = "MCZR" 69 = "HGA_BASE" 70 = "HGA_HEAD" 71 = "HGA_BORESIGHT" 72 = "FID_MOBILITY_LDIFF" 73 = "FID_MOBILITY_LF" 74 = "FID_MOBILITY_LR" 75 = "FID_MOBILITY_RDIFF" 76 = "FID_MOBILITY_RF" 77 = "FID_MOBILITY_RR" 78 = "FHCAML" 79 = "FHCAMR" 80 = "RHCAML" 81 = "RHCAMR" 82 = "LCAM_CMECH" 83 = "MEDA_ATS1" 84 = "MEDA_ATS2" 85 = "MEDA_ATS3" 86 = "MEDA_ATS4" 87 = "MEDA_ATS5" 88 = "MEDA_HS" 89 = "MEDA_PS" 90 = "MEDA_RDS" 91 = "MEDA_TIRS" 92 = "MEDA_WS1" 93 = "MEDA_WS2" 94 = "RFAX_ANT" 95 = "MOXIE_FILTER" 96 = "SCAM_CAL" 97 = "MCZ_CAL_PRIMARY" 98 = "MCZ_CAL_SECONDARY" 99 = "CAL_SHERLOC" 100 = "FID_AFT1" 101 = "FID_AFT2" 102 = "FID_RPFA1" 103 = "FID_RPFA2" 104 = "FID_TDF1" 105 = "FID_TDF2" 106 = "FID_TDF3" 107 = "FID_TDF4" 108 = "FID_TUR_WS_A" 109 = "FID_TUR_WS_B" 110 = "FID_TUR_WS_C" 111 = "FID_TUR_WS_D" 112 = "FID_TUR_WS_E" 113 = "FID_TUR_WS_F" 114 = "FID_TUR_DOCK_G"	

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
	115 = "FID_TUR_DOCK_H" 116 = "FID_TUR_DOCK_I" 117 = "FID_TUR_DOCK_J" 118 = "FID_TUR_DOCK_K" 119 = "FID_TUR_DOCK_L" 120 = "FID_TUR_DOCK_M" 121 = "FID_TUR_DOCK_N" 122 = "FID_BIT_CAROUSEL_1" 123 = "FID_BIT_CAROUSEL_2" 124 = "FID_BIT_CAROUSEL_3" 125 = "NONE" Type string Units n/a Location OBSERVATION_REQUEST_PARMs (Group) X-path TBD Reference TBD	
Ops Keyword INSTRUMENT_COORD_FRAME_INDEX Archive Attribute TBD Definition Along with INSTRUMENT_COORD_FRAME_ID, specifies the index of the coordinate frame in which the pointing for this instrument is expressed. Only the SITE, LL, RSM_JOINTS and FIDUCIAL frames are indexed. A positive value means to use specifically that index; a negative value means an index relative to the current.	Valid Values "N/A" Type integer Units n/a Location a. OBSERVATION_REQUEST_PARMs (Group) b. INSTRUMENT_STATE_PARMs (Group) X-path TBD Reference TBD	Mode DPO in XML format (referenced to APID Name in Appendix A) Field as "<xml name>:[<element>]:<field>" Surface - Eng. Cameras a. "<IDPH DPO>:idph:params:rsm_findex" b. "<IDPH DPO>:idph:rsm_findex" - SuperCam "<Ancillary DPO>:cmd_arguments:frame_index" TBD - MCZ Cameras "<Ancillary DPO>:frame_index" TBD Cruise - Eng. Cameras "<IDPH DPO>:idph:params:rsm_findex" Type I32
Ops Keyword INSTRUMENT_COORDINATE Archive Attribute TBD Definition Specifies an array of coordinate parameters. These values, typically either XYZ or azimuth/elevation values (see INSTRUMENT_COORDINATE_NAME), specify	Valid Values "N/A" Type double array[3] Units radians for following Frames (<rad> unit tag required): "AZIMUTH", "ELEVATION", "AZIMUTH_RELATIVE", "ELEVATION_RELATIVE", "RSM_AZIMUTH",	Mode DPO in XML format (referenced to APID Name in Appendix A) Field as "<xml name>:[<element>]:<field>" Surface - Eng. Cameras "<IDPH DPO>:idph:params:rsm_coord[3]" - SuperCam

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
where the instrument is actually pointed in the frame defined by INSTRUMENT_COORD_FRAME_ID/INDEX.	"RSM_ELEVATION", "RSM_AZIMUTH_RELATIVE", "RSM_ELEVATION_RELATIVE" meters for following Frames (<m> unit tag required): "X", "Y", "Z" <u>Location</u> OBSERVATION_REQUEST_PARMS (Group) <u>X-path</u> TBD <u>Reference</u> TBD	"<Ancillary DPO>:cmd_arguments:coord1", TBD "<Ancillary DPO>:cmd_arguments:coord2", "<Ancillary DPO>:cmd_arguments:coord3" - MCZ "<Supplementary DPO>:coord1", "<Supplementary DPO>:coord2", "<Supplementary DPO>:coord3" <u>Type</u> - Eng. Cameras F64[3] - SuperCam U32 - MCZ F32
<u>Ops Keyword</u> INSTRUMENT_COORDINATE_NAME <u>Archive Attribute</u> TBD <u>Definition</u> Specifies the name of each element in INSTRUMENT_COORDINATE, indicating how to interpret the value. The "X","Y","Z" values are in meters; all others are in radians. The values should appear in sets as specified in Valid Values.	<u>Valid Values</u> - Eng. Cameras (B) 0 = ("X","Y","Z") 1 = ("AZIMUTH", "ELEVATION") 2 = ("AZIMUTH_RELATIVE", "ELEVATION_RELATIVE") 3 = ("AZIMUTH") 4 = ("ELEVATION") 5 = ("RSM_AZIMUTH", "RSM_ELEVATION") 6 = ("RSM_AZIMUTH_RELATIVE", "RSM_ELEVATION_RELATIVE") 7 = ("RSM_AZIMUTH") 8 = ("RSM_ELEVATION") 9 = ("RSM_NO_MOTION") 10 = ("RSM_NO_MOTION_NO_ARBITRATION") - SuperCam 0 = 1 = 2 = 3 = <u>Cruise</u> <u>Type</u> string <u>Units</u> n/a <u>Location</u> OBSERVATION_REQUEST_PARMS (Group) <u>X-path</u> TBD <u>Reference</u> TBD	<u>Mode</u> DPO in XML format (referenced to APID Name in Appendix A) <u>Field as "<xml name>:[<element>]:[<element>]:<field>"</u> - Eng. Cameras "<IDPH DPO>:idph:params:rsm_ctype" - SuperCam 2) "<Ancillary DPO>:cmd_arguments:coord_type" TBD <u>Type</u> enum
<u>Ops Keyword</u> INSTRUMENT_COORDINATE_TYPE <u>Archive Attribute</u> TBD	<u>Valid Values</u> - Eng. Cameras (B) 0 = ("XYZ") 1 = ("AZEL_ABS") 2 = ("AZEL_REL")	<u>Mode</u> DPO in XML format (referenced to APID Name in Appendix A) <u>Field as "<xml name>:[<element>]:[<element>]:<field>"</u>

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
Definition Specifies what kind of coordinates are in the associated INSTRUMENT_COORDINATE keyword. These could be XYZ coordinates, azimuth/elevation values, etc.	3 = ("AZ_ABS") 4 = ("EL_ABS") 5 = ("JOINTS_AZEL_ABS") 6 = ("JOINTS_AZEL_REL") 7 = ("JOINTS_AZ_ABS") 8 = ("JOINTS_EL_ABS") 9 = ("NO_MOTION") 10 = ("NO_MOTION_NO_ARB") Type string Units n/a Location OBSERVATION_REQUEST_PARMS (Group) X-path TBD Reference TBD	- Eng. Cameras and MCZ Cameras "IDPH DPO>:idph:params:rsm_ctype" - SuperCam "Ancillary DPO>:cmd_arguments:coord_type" TBD Type enum
Ops Keyword INSTRUMENT_CURRENT Archive Attribute TBD Definition Specifies the current, in milliamps, of an instrument or some part of an instrument. Note that this may be an array of multiple values for current on different parts of the instrument. (Example: CCD array and sensor head)	Valid Values - PIXL 0b00 = 0 0b01 = 140 0b10 = 280 0b11 = 500 Type - PIXL MCC float array[16] Units mA (unit tag required) Location INSTRUMENT_STATE_PARMS (Group) X-path TBD Reference TBD	Mode DPO in XML format (referenced to APID Name in Appendix A) Field as "<xml name>:[<element>]:[<element>]:<field>" Surface PIXL MCC "PIXL_Image_Mini_Header:Info" "PIXL_Image_Meta_Data:Parameter_Id:FLICFG_A" "PIXL_Image_Meta_Data:Parameter_Id:FLICFG_B" Cruise Type Notes: For PIXL MCC, the values are only valid if the "Image Meta Data available" flag is True and the parameter values are provided. If the flag is False or the parameters 'FLICFG_A B' are not provided, the values will be "n/a".
Ops Keyword INSTRUMENT_CURRENT_NAME Archive Attribute TBD Definition Specifies the current, in milliamps, of an instrument or some part of an instrument. Note that this may be an array of multiple values for current on	Valid Values PIXL MCC ("FLICFG_A1","FLICFG_A2","FLICFG_A3", "FLICFG_A4","FLICFG_A5","FLICFG_A6", "FLICFG_A7","FLICFG_A8", "FLICFG_B1","FLICFG_B2","FLICFG_B3", "FLICFG_B4","FLICFG_B5","FLICFG_B6", "FLICFG_B7","FLICFG_B8") Type	Mode Static Field as "<xml name>:[<element>]:[<element>]:<field>" Surface Cruise Type

Output Metadata (Product Label)		Input Metadata (Source)
• Ops Keyword • Archive Label Attribute • Definition	• Mission Phase C=Cruise, S=Surface, B=Both • Valid Values (quoted) • Type • Units • Keyword Location in Ops Label • Attribute X-path in Archive label • Attribute Reference in Archive Label (bracketed)	• Mission Phase C=Cruise, S=Surface, B=Both • Mode • Metadata Field • Type
different parts of the instrument. (Example: CCD array and sensor head)	string array[16] Units deg C (<degC> unit tag required) Location INSTRUMENT_STATE_PARMS (Group) X-path TBD Reference TBD	
Ops Keyword INSTRUMENT_ELEVATION Archive Attribute TBD Definition Specifies a value for an instrument's rotation in the vertical direction. It is usually measured from some kind of low hard stop. Although it may be used for any instrument where it makes sense, it is primarily intended for use in surface-based instruments that measure pointing in terms of azimuth and elevation. When in a DERIVED_GEOMETRY group, defines the elevation (vertical rotation) at which the instrument is pointed. This value is expressed using the coordinate system referred to by REFERENCE_COORD_SYSTEM_NAME and REFERENCE_COORD_SYSTEM_INDEX contained within the same group. The interpretation of exactly what part of the instrument is being pointed is mission-specific. It could be the boresight, the camera head direction, the CAHV camera model A vector direction, or any of a number of other things. As such, for multimission use this value should be used mostly as an approximation, e.g. identifying scenes which might contain a given object. For M2020, the interpretation is the boresight of the camera, defined as projecting the center of the nominal image (before downsampling or subframing) through the camera model.	Valid Values "-91.0" to "91.0" Type float Units deg (<deg> unit tag required) Location • ROVER_DERIVED_GEOMETRY_PARMS (Group) • SITE_DERIVED_GEOMETRY_PARMS (Group) X-path TBD Reference TBD	Mode Calculation: - The image center is translated into an origin and "look direction" vectors (left and right). The vectors are then used to calculate instrument elevation. Field as "<xml name>:[<element>]:[<element>]:<field>" n/a Type n/a
Ops Keyword INSTRUMENT_ELEVATION Archive Attribute TBD	Valid Values "-91.0" to "91.0" Type float	Mode Calculation: - The image center is translated into an origin and "look direction" vectors (left and right). The vectors are then used

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
Definition Specifies a value for an instrument's rotation in the vertical direction. It is usually measured from some kind of low hard stop. Although it may be used for any instrument where it makes sense, it is primarily intended for use in surface-based instruments that measure pointing in terms of azimuth and elevation. When in a DERIVED_GEOMETRY group, defines the elevation (vertical rotation) at which the instrument is pointed. This value is expressed using the coordinate system referred to by REFERENCE_COORD_SYSTEM_NAME and REFERENCE_COORD_SYSTEM_INDEX contained within the same group. The interpretation of exactly what part of the instrument is being pointed is mission-specific. It could be the boresight, the camera head direction, the CAHV camera model A vector direction, or any of a number of other things. As such, for multimission use this value should be used mostly as an approximation, e.g. identifying scenes which might contain a given object. For M2020, the interpretation is the boresight of the camera, defined as projecting the center of the nominal image (before downsampling or subframing) through the camera model.	Units deg (<deg> unit tag required) Location - ROVER_DERIVED_GEOMETRY_PARMs (Group) - SITE_DERIVED_GEOMETRY_PARMs (Group) X-path TBD Reference TBD	to calculate instrument elevation. Field as "<xml name>:[<element>]:[<element>]:<field>" n/a Type n/a
Ops Keyword INSTRUMENT_FOCUS_DISTANCE Archive Attribute TBD Definition Specifies the distance to the target to use for focusing the instrument. See also INSTRUMENT_FOCUS_MODE. For M2020 SuperCam, specifies the distance to target for "MANUAL" focus, or the seed for "BASELINE" focus.	Valid Values - SuperCam, Mastcam-Z, SHERLOC-WATSON "0" to "65535" - MARDI "UNK" Type integer Units millimeters (<mm> unit tag required) Location OBSERVATION_REQUEST_PARMs (Group) X-path TBD Reference TBD	Mode DPO in XML format (referenced to APID Name in Appendix A) Field as "<xml name>:[<element>]:[<element>]:<field>" - SuperCam RMI "<Ancillary DPO>:cmd_arguments:range" - SuperCam LIBS "<Ancillary DPO>:cmd_arguments:range" - MCZ Cameras a. "<Supplementary DPO>:left_focus_initial_position" b. "<Supplementary DPO>:right_focus_initial_position" c. "<Supplementary DPO>:focus_initial_position" NOTES: - For MCZ Cameras, Mastcam-Z left is case "a". - For MCZ Cameras, Mastcam-Z right is case "b". - For MCZ Cameras, SHERLOC-WATSON is case "c". - For MCZ Cameras, MARDI is "UNK". Type I16
Ops Keyword INSTRUMENT_FOCUS_INIT_FLAG Archive Attribute	Valid Values 0 = "FALSE" 1 = "TRUE"	Mode DPO in XML format (referenced to APID Name in Appendix A)

Output Metadata (Product Label)		Input Metadata (Source)						
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type						
TBD Definition Specifies whether the instrument focus mechanism should be (or was) initialized before use. For M2020 SuperCam, this pertains to RMI observations only (not LIBS). Value "TRUE" brings the focus stage to the limit switch to initialize its position.	Type string Units n/a Location OBSERVATION_REQUEST_PARMS (Group) X-path TBD Reference TBD	Field as “<xml name>:[<element>]:[<element>]:<field>” "<Ancillary DPO>:cmd_parameters:fromLimitSwitch" Type U8						
Ops Keyword INSTRUMENT_FOCUS_MODE Archive Attribute TBD Definition Specifies how an instrument was focused. The values and their interpretation are mission-specific. For M2020 SuperCam, it is the commanded mode of focus during both RMI and LIBS observations. Valid values for SuperCam are defined as: a) "NO_FOCUS" - Doesn't move focus (others mark SCAM unsafe). b) "BASELINE" - Uses CWL to find optimal focus position. c) "MANUAL" - Positions focus based on keyword INSTRUMENT_FOCUS_PARAM value. d) "AF_OFFSET" - Applies RMI offset from last autofocus solution.	Valid Values SuperCam "NO_FOCUS", "BASELINE", "MANUAL", "AF_OFFSET" Type string Units n/a Location 1) OBSERVATION_REQUEST_PARMS (Group) 2) MINI_HEADER X-path TBD Reference TBD	Mode DPO in XML format (referenced to APID Name in Appendix A) Field as “<xml name>:[<element>]:[<element>]:<field>” 1) SuperCam "<Ancillary DPO>:cmd_arguments:focus" 2) MCZ Cameras "MCZ_Image_Mini_Header[TBD]" Type U8						
Ops Keyword INSTRUMENT_FOCUS_POSITION Archive Attribute TBD Definition Specifies the position in motor counts of the focus motor on a camera. For M2020 SuperCam and MCZ, it specifies the actual focus position that was used by the image.	Valid Values MCZ Cameras ... <table><thead><tr><th>Mode</th><th>Value</th></tr></thead><tbody><tr><td>Z-stack</td><td>n/a</td></tr><tr><td>Non-Z-stack</td><td>Comprised of four bytes coming from Image DPO mini-header at offsets 44, 45, 46 and 47</td></tr></tbody></table> Type integer Units n/a Location INSTRUMENT_STATE_PARMS (Group) MINI_HEADER (Group) X-path	Mode	Value	Z-stack	n/a	Non-Z-stack	Comprised of four bytes coming from Image DPO mini-header at offsets 44, 45, 46 and 47	Mode Image DPO mini-header Field as “<xml name>:[<element>]:[<element>]:<field>” MCZ Cameras "MCZ_Image_Mini_Header[44]", "MCZ_Image_Mini_Header[45]", "MCZ_Image_Mini_Header[46]", "MCZ_Image_Mini_Header[47]" Type U32
Mode	Value							
Z-stack	n/a							
Non-Z-stack	Comprised of four bytes coming from Image DPO mini-header at offsets 44, 45, 46 and 47							

Output Metadata (Product Label)		Input Metadata (Source)																																	
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type																																	
	TBD <u>Reference</u> TBD																																		
Ops Keyword INSTRUMENT_FOCUS_STEP_SIZE Archive Attribute TBD Definition For MCZ cameras on M2020, specifies the size in motor counts of each (or the initial) step taken by the focus adjustment mechanism in an autofocus algorithm.	Valid Values n/a Type integer Units n/a Location a. OBSERVATION_REQUEST_PARMS (Group) b. MINI_HEADER (Group) X-path TBD Reference TBD	Mode DPO in XML format (referenced to APID Name in Appendix A) Field as “<xml name>:[<element>]:<field>” <u>MCZ Cameras</u> a. “<IDPH DPO>:idph:step_size” b. “MCZ_Image_Mini_Header[TBD]” Type U16																																	
Ops Keyword INSTRUMENT_FOCUS_STEPS Archive Attribute TBD Definition For MCZ cameras on M2020, specifies the number of steps (images) to be taken by an autofocus algorithm.	Valid Values n/a Type integer Units n/a Location OBSERVATION_REQUEST_PARMS (Group) X-path TBD Reference TBD	Mode DPO in XML format (referenced to APID Name in Appendix A) Field as “<xml name>:[<element>]:<field>” <u>MCZ Cameras</u> “<IDPH DPO>:idph:n_steps” Type U8																																	
Ops Keyword INSTRUMENT_HOST_ID Archive Attribute TBD Definition Specifies a unique identifier for the host where an instrument is located. This host can be either a spacecraft or an earth base (e.g., and observatory or laboratory on the earth). Thus, INSTRUMENT_HOST_ID can contain values which are either SPACECRAFT_ID values or EARTH_BASE_ID values.	Valid Values • All Cameras <table border="1"> <thead> <tr> <th>VenueType</th><th>Testbed Name</th><th>Value</th></tr> </thead> <tbody> <tr> <td>CRUISE</td><td>n/a</td><td>M20</td></tr> <tr> <td>SURFACE</td><td>n/a</td><td>M20</td></tr> <tr> <td>ATLO</td><td>n/a</td><td>M20</td></tr> <tr> <td>TESTBED</td><td>AVSTB</td><td>M20AVSTB</td></tr> <tr> <td>TESTBED</td><td>FSWTB</td><td>M20FSWTB</td></tr> <tr> <td>TESTBED</td><td>MSTB</td><td>M20MSTB</td></tr> <tr> <td>TESTBED</td><td>VSTB</td><td>M20VSTB</td></tr> <tr> <td>TESTSET</td><td>n/a</td><td>M20WSTS</td></tr> <tr> <td>TBD</td><td>TBD</td><td>M20SCARECROW</td></tr> <tr> <td>TBD</td><td>TBD</td><td>M20ROASTT</td></tr> </tbody> </table> Type string array	VenueType	Testbed Name	Value	CRUISE	n/a	M20	SURFACE	n/a	M20	ATLO	n/a	M20	TESTBED	AVSTB	M20AVSTB	TESTBED	FSWTB	M20FSWTB	TESTBED	MSTB	M20MSTB	TESTBED	VSTB	M20VSTB	TESTSET	n/a	M20WSTS	TBD	TBD	M20SCARECROW	TBD	TBD	M20ROASTT	Mode EMD in XML format Field as “<xml name>:[<element>]:<field>” “M20EarthProductMetadata:SessionInformation:Venue: VenueType”, “M20EarthProductMetadata:SessionInformation:Venue: TestbedName” Type n/a
VenueType	Testbed Name	Value																																	
CRUISE	n/a	M20																																	
SURFACE	n/a	M20																																	
ATLO	n/a	M20																																	
TESTBED	AVSTB	M20AVSTB																																	
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TESTBED	MSTB	M20MSTB																																	
TESTBED	VSTB	M20VSTB																																	
TESTSET	n/a	M20WSTS																																	
TBD	TBD	M20SCARECROW																																	
TBD	TBD	M20ROASTT																																	

Output Metadata (Product Label)		Input Metadata (Source)																																	
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type																																	
	<u>Units</u> n/a <u>Location</u> IDENTIFICATION (Class) <u>X-path</u> TBD <u>Reference</u> TBD																																		
<u>Ops Keyword</u> INSTRUMENT_HOST_NAME <u>Archive Attribute</u> TBD <u>Definition</u> Specifies the full name of the host on which an instrument is based. This host can be either a spacecraft or an earth base. Thus, the INSTRUMENT_HOST_NAME element can contain values which are either SPACECRAFT_NAME values or EARTH_BASE_NAME values. Note that mosaics may contain more than one value in an array.	<u>Valid Values</u> All Cameras <table><tr><th>VenueType</th><th>Testbed Name</th><th>Value</th></tr><tr><td>CRUISE</td><td>n/a</td><td>Mars 2020</td></tr><tr><td>SURFACE</td><td>n/a</td><td>Mars 2020</td></tr><tr><td>ATLO</td><td>n/a</td><td>Mars 2020</td></tr><tr><td>TESTBED</td><td>AVSTB</td><td>M20AVSTB</td></tr><tr><td>TESTBED</td><td>FSWTB</td><td>M20FSWTB</td></tr><tr><td>TESTBED</td><td>MSTB</td><td>M20MSTB</td></tr><tr><td>TESTBED</td><td>VSTB</td><td>M20VSTB</td></tr><tr><td>TESTSET</td><td>n/a</td><td>M20WSTS</td></tr><tr><td>TBD</td><td>TBD</td><td>M20SCARECROW</td></tr><tr><td>TBD</td><td>TBD</td><td>M20ROASTT</td></tr></table> <u>Type</u> string array <u>Units</u> n/a <u>Location</u> IDENTIFICATION (Class) <u>X-path</u> TBD <u>Reference</u> TBD	VenueType	Testbed Name	Value	CRUISE	n/a	Mars 2020	SURFACE	n/a	Mars 2020	ATLO	n/a	Mars 2020	TESTBED	AVSTB	M20AVSTB	TESTBED	FSWTB	M20FSWTB	TESTBED	MSTB	M20MSTB	TESTBED	VSTB	M20VSTB	TESTSET	n/a	M20WSTS	TBD	TBD	M20SCARECROW	TBD	TBD	M20ROASTT	<u>Mode</u> EMD in XML format <u>Field as “<xml name>:[<element>]:<field>”</u> “M20EarthProductMetadata:SessionInfomation:Venue: VenueType”, “M20EarthProductMetadata:SessionInfomation:Venue: TestbedName” <u>Type</u> n/a
VenueType	Testbed Name	Value																																	
CRUISE	n/a	Mars 2020																																	
SURFACE	n/a	Mars 2020																																	
ATLO	n/a	Mars 2020																																	
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TBD	TBD	M20SCARECROW																																	
TBD	TBD	M20ROASTT																																	
<u>Ops Keyword</u> INSTRUMENT_ID <u>Archive Attribute</u> TBD <u>Definition</u> Specifies an abbreviated name or acronym which identifies an instrument. NOTE: INSTRUMENT_ID is not a unique identifier for a given instrument. Note also that the associated INSTRUMENT_NAME element provides the full name of the instrument. Example values: IRTM (for Viking Infrared Thermal Mapper), PWS (for plasma wave spectrometer).	<u>Valid Values</u> <u>Surface</u><table><tr><th>Value</th><th>APID Names</th></tr><tr><td>“FRONT_HAZCAM_LEFT_<A B>”</td><td>Any string value starting with “Img” and ending with “Fhl”</td></tr><tr><td>“FRONT_HAZCAM_RIGHT_<A B>”</td><td>Any string value starting with “Img” and ending with “Fhr”</td></tr><tr><td>“REAR_HAZCAM_LEFT”</td><td>Any string value starting with “Img”</td></tr></table>	Value	APID Names	“FRONT_HAZCAM_LEFT_<A B>”	Any string value starting with “Img” and ending with “Fhl”	“FRONT_HAZCAM_RIGHT_<A B>”	Any string value starting with “Img” and ending with “Fhr”	“REAR_HAZCAM_LEFT”	Any string value starting with “Img”	<u>Mode</u> EMD in XML format <u>Field as “<xml name>:[<element>]:<field>”</u> - Front Hazcam “M20EarthProductMetadata:M20ProductMetadata:ProductName”, “M20EarthProductMetadata:M20ProductMetadata:CreationStringId” - All other Eng. Cameras - Scam “M20EarthProductMetadata:M20ProductMetadata:ProductName”																									
Value	APID Names																																		
“FRONT_HAZCAM_LEFT_<A B>”	Any string value starting with “Img” and ending with “Fhl”																																		
“FRONT_HAZCAM_RIGHT_<A B>”	Any string value starting with “Img” and ending with “Fhr”																																		
“REAR_HAZCAM_LEFT”	Any string value starting with “Img”																																		

Output Metadata (Product Label)		Input Metadata (Source)
• Ops Keyword • Archive Label Attribute • Definition	• Mission Phase C=Cruise, S=Surface, B=Both • Valid Values (quoted) • Type • Units • Keyword Location in Ops Label • Attribute X-path in Archive label • Attribute Reference in Archive Label (bracketed)	• Mission Phase C=Cruise, S=Surface, B=Both • Mode • Metadata Field • Type
	“REAR_HAZCAM_RIGHT” “NAVCAM_LEFT” “NAVCAM_RIGHT” “CACHECAM” “SUPERCAM_RMI” “MCZ_LEFT” “MCZ_RIGHT” “SHERLOC_WATSON” “SHERLOC_ACI” “PIXL_MCC” “SKYCAM” “LVSCAM” “EDL_PUCAM1” “EDL_PUCAM2” “EDL_PUCAM3”	and ending with “Rhl” Any string value starting with “Img” and ending with “Rhr” Any string value starting with “Img” and ending with “NI” Any string value starting with “Img” and ending with “Nr” Any string value starting with “Img” and ending with “Cc” Any string value starting with “ScamRmi” Any string value starting with “MczL”. Any string value starting with “MczR”. Any string value starting with “SrlcImg” and MSB of system serial number = 0 Any string value starting with “SrlcImg” and MSB of system serial number = 1 Any string value starting with “Pixl” and PDP Category NN Any string starting with “Meda” and ending with “Image” Any string starting with “Vce” and TBD <u>SHERLOC</u> “SHERLOC_Image_Mini_Header[50 – 52]” <u>Type</u> n/a

Output Metadata (Product Label)		Input Metadata (Source)																										
• Ops Keyword • Archive Label Attribute • Definition	• Mission Phase C=Cruise, S=Surface, B=Both • Valid Values (quoted) • Type • Units • Keyword Location in Ops Label • Attribute X-path in Archive label • Attribute Reference in Archive Label (bracketed)	• Mission Phase C=Cruise, S=Surface, B=Both • Mode • Metadata Field • Type																										
	“EDL_DDCAM” “EDL_RDCAM” “EDL_RUCAM” Helicopter RTE (TBD) Helicopter NAV (TBD) Cruise <table><thead><tr><th>Value</th><th>APID Names</th></tr></thead><tbody><tr><td>“FRONT_HAZCAM_LEFT_<A B>”</td><td>Any string value starting with “Img” and ending with “Fhl”</td></tr><tr><td>“FRONT_HAZCAM_RIGHT_<A B>”</td><td>Any string value starting with “Img” and ending with “Fhr”</td></tr><tr><td>“REAR_HAZCAM_LEFT”</td><td>Any string value starting with “Img” and ending with “Rhl”</td></tr><tr><td>“REAR_HAZCAM_RIGHT”</td><td>Any string value starting with “Img” and ending with “Rhr”</td></tr><tr><td>“NAVCAM_LEFT”</td><td>Any string value starting with “Img” and ending with “NI”</td></tr><tr><td>“NAVCAM_RIGHT”</td><td>Any string value starting with “Img” and ending with “Nr”</td></tr><tr><td>“CACHECAM”</td><td>Any string value starting with “Img” and ending with “Cc”</td></tr><tr><td>“SUPERCAM_RMI”</td><td>Any string value starting with “ScamRmi”</td></tr><tr><td>“MCZ_LEFT”</td><td>Any string value starting with “MczL”. Any string value starting with “MczR”.</td></tr><tr><td>“MCZ_RIGHT”</td><td>Any string value starting with “MczR”.</td></tr><tr><td>“SHERLOC_WATSON”</td><td>Any string value starting with “SrcImg” and MSB of system serial number = 0</td></tr><tr><td>“SHERLOC_ACI”</td><td>Any string value starting</td></tr></tbody></table>	Value	APID Names	“FRONT_HAZCAM_LEFT_<A B>”	Any string value starting with “Img” and ending with “Fhl”	“FRONT_HAZCAM_RIGHT_<A B>”	Any string value starting with “Img” and ending with “Fhr”	“REAR_HAZCAM_LEFT”	Any string value starting with “Img” and ending with “Rhl”	“REAR_HAZCAM_RIGHT”	Any string value starting with “Img” and ending with “Rhr”	“NAVCAM_LEFT”	Any string value starting with “Img” and ending with “NI”	“NAVCAM_RIGHT”	Any string value starting with “Img” and ending with “Nr”	“CACHECAM”	Any string value starting with “Img” and ending with “Cc”	“SUPERCAM_RMI”	Any string value starting with “ScamRmi”	“MCZ_LEFT”	Any string value starting with “MczL”. Any string value starting with “MczR”.	“MCZ_RIGHT”	Any string value starting with “MczR”.	“SHERLOC_WATSON”	Any string value starting with “SrcImg” and MSB of system serial number = 0	“SHERLOC_ACI”	Any string value starting	
Value	APID Names																											
“FRONT_HAZCAM_LEFT_<A B>”	Any string value starting with “Img” and ending with “Fhl”																											
“FRONT_HAZCAM_RIGHT_<A B>”	Any string value starting with “Img” and ending with “Fhr”																											
“REAR_HAZCAM_LEFT”	Any string value starting with “Img” and ending with “Rhl”																											
“REAR_HAZCAM_RIGHT”	Any string value starting with “Img” and ending with “Rhr”																											
“NAVCAM_LEFT”	Any string value starting with “Img” and ending with “NI”																											
“NAVCAM_RIGHT”	Any string value starting with “Img” and ending with “Nr”																											
“CACHECAM”	Any string value starting with “Img” and ending with “Cc”																											
“SUPERCAM_RMI”	Any string value starting with “ScamRmi”																											
“MCZ_LEFT”	Any string value starting with “MczL”. Any string value starting with “MczR”.																											
“MCZ_RIGHT”	Any string value starting with “MczR”.																											
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“SHERLOC_ACI”	Any string value starting																											

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
	with "SrcImg" and MSB of system serial number = 1 Any string value starting with "Pixl" and PDP Category NN Any string starting with "Meda" and ending with "Image" Any string starting with "Vce" and TBD "PIXL_MCC" "SKYCAM" "LVSCAM" "EDL_PUCAM1" "EDL_PUCAM2" "EDL_PUCAM3" "EDL_DDCAM" "EDL_RDCAM" "EDL_RUCAM" Helicopter RTE (TBD) Helicopter NAV (TBD) <u>Type</u> string <u>Units</u> n/a <u>Location</u> IDENTIFICATION (Class) <u>X-path</u> TBD <u>Reference</u> TBD	
Ops Keyword INSTRUMENT_IDLE_TIMEOUT Archive Attribute TBD Definition Specifies the amount of time in seconds the instrument may be idle before powering off the instrument.	Valid Values Eng. Cameras "0" to "4294967295" Type integer Units seconds (<s> unit tag required) Location OBSERVATION_REQUEST_PARMS (Group) X-path TBD	Mode DPO in XML format (referenced to APID Name in Appendix A) Field as "<xml name>:[<element>]:[<element>]:<field>" "<IDPH DPO>:idph:params:power_timeout" Type U32

Output Metadata (Product Label)		Input Metadata (Source)
• Ops Keyword • Archive Label Attribute • Definition	• Mission Phase C=Cruise, S=Surface, B=Both • Valid Values (quoted) • Type • Units • Keyword Location in Ops Label • Attribute X-path in Archive label • Attribute Reference in Archive Label (bracketed)	• Mission Phase C=Cruise, S=Surface, B=Both • Mode • Metadata Field • Type
	<u>Reference</u> TBD	
Ops Keyword INSTRUMENT_MODE_ID Archive Attribute TBD Definition Specifies an instrument-dependent designation of operating mode. This may be simply a number, letter or code, or a word such as 'normal', 'full resolution', 'near encounter', or 'fixed grating'. For EECAM on M2020 there is difference between cruise and surface readout modes. Due to hardware heritage from MSL, FSW will have to read out ECAM images in parts (tiles) before saving the full image to NVM. Cruise FSW will request EECAM tiles from a fixed grid that covers the entire focal plane as defined from a lookup table. Surface FSW will allow a "floating-tile" concept. Mode, tile resolution, color, and size are listed below. Surface (readout modes 0 - 9): 0 = full resolution, bayer, 960 lines x 1280 samples 1 = 2x2 binned, green pixels, 960 lines x 1280 samples 2 = 2x2 binned, red pixels, 960 lines x 1280 samples 3 = 2x2 binned, blue pixels, 960 lines x 1280 samples 4 = 2x2 binned, all pixels, 960 lines x 1280 samples 5 = 4x4 binned, green pixels, 960 lines x 1280 samples 6 = 4x4 binned, red pixels, 960 lines x 1280 samples 7 = 4x4 binned, blue pixels, 960 lines x 1280 samples 8 = 4x4 binned, all pixels, 960 lines x 1280 samples 9 = 8x8 binned, all pixels, 480 lines x 640 samples Cruise (readout modes 0 - 35) 0 - 15 = full resolution, bayer, 960 lines x 1280 samples 16 - 19 = 2x2 binned, green pixels, 960 lines x 1280 samples 20 - 23 = 2x2 binned, red pixels, 960 lines x 1280 samples 24 - 27 = 2x2 binned, blue pixels, 960 lines x 1280 samples 28 - 31 = 2x2 binned, all pixels, 960 lines x 1280 samples 32 = 4x4 binned, green pixels, 960 lines x 1280 samples 33 = 4x4 binned, red pixels, 960 lines x 1280 samples 34 = 4x4 binned, blue pixels, 960 lines x 1280 samples 35 = 4x4 binned, all pixels, 960 lines x 1280 samples	Valid Values Surface • <u>Eng. Cameras</u> (S) "0" – "9" • <u>SuperCam</u> (S) 0 = "UNK" 1 = "ASYNCR" • <u>MCZ</u> (S) TBD Cruise • <u>Eng. Cameras</u> (C) "0" – "35" • <u>SuperCam</u> (C) 0 = "UNK" 1 = "ASYNCR" • <u>MCZ</u> (C) TBD Type string(20) Units n/a Location • <u>Eng. Cameras</u> INSTRUMENT_STATE_PARMs (Group) • <u>SuperCam</u> OBSERVATION_REQUEST_PARMs (Group) • <u>MCZ</u> OBSERVATION_REQUEST_PARMs (Group) MINI_HEADER (Group) X-path TBD Reference TBD	Mode DPO in XML format (referenced to APID Name in Appendix A) Field as "<xml name>:[<element>]:[<element>]:<field>" Surface • <u>Eng. Cameras</u> (S) "IDPH DPO>:idph:hw_mode" • <u>SuperCam</u> (S) "Ancillary DPO>:cmd_parameters:linkToUse" TBD NOTES: - If linkToUse = 0, then "UNK" - If linkToUse = 1, then "ASYNCR" • <u>MCZ</u> (S) "MCZ_Image_Mini_Header[TBD]" Cruise • <u>Eng. Cameras</u> (C) "IDPH DPO>:idph:params:voff" Type Surface • <u>Eng. Cameras</u> • <u>SuperCam</u> U8 • <u>MCZ</u> I32 Cruise • <u>Eng. Cameras</u> U16 • <u>SuperCam</u> U8 • <u>MCZ</u> I32

Output Metadata (Product Label)		Input Metadata (Source)																						
• Ops Keyword • Archive Label Attribute • Definition	• Mission Phase C=Cruise, S=Surface, B=Both • Valid Values (quoted) • Type • Units • Keyword Location in Ops Label • Attribute X-path in Archive label • Attribute Reference in Archive Label (bracketed)	• Mission Phase C=Cruise, S=Surface, B=Both • Mode • Metadata Field • Type																						
For SuperCam on M2020 ... a) for RMI instrument, specifies the link between the SuperCam Mast Unit (MU) and SuperCam Body Unit (BU) to use during RMI image commanding. Value of "1" denotes async (currently unavailable). For MCZ cameras on M2020, this specifies the image or video readout mode, which determines the rate the image is read out.																								
Ops Keyword INSTRUMENT_NAME Archive Attribute TBD Definition Specifies the full name of an instrument. Note that the associated INSTRUMENT_ID element provides an abbreviated name or acronym for the instrument. Example values: FLUXGATE MAGNETOMETER, NEAR_INFRARED MAPPING SPECTROMETER.	Valid Values <table><thead><tr><th>Value</th><th>APID Names</th></tr></thead><tbody><tr><td>"FRONT HAZARD AVOIDANCE CAMERA LEFT STRING <A B>"</td><td>Any string value starting with "Img" and ending with "Fhl"</td></tr><tr><td>"FRONT HAZARD AVOIDANCE CAMERA RIGHT STRING <A B>"</td><td>Any string value starting with "Img" and ending with "Fhr"</td></tr><tr><td>"REAR HAZARD AVOIDANCE CAMERA LEFT"</td><td>Any string value starting with "Img" and ending with "RhI"</td></tr><tr><td>"REAR HAZARD AVOIDANCE CAMERA RIGHT"</td><td>Any string vaue startiring with "Img" and ending with "Rhr"</td></tr><tr><td>"NAVIGATION CAMERA LEFT"</td><td>Any string value starting with "Img" and ending with "NI"</td></tr><tr><td>"NAVIGATION CAMERA RIGHT"</td><td>Any string value starting with "Img" and ending with "Nr"</td></tr><tr><td>"SUPER CAMERA REMOTE MICRO IMAGER"</td><td>Any string value starting with "ScamRmi"</td></tr><tr><td>"MAST CAMERA ZOOM LEFT"</td><td>Any string value starting with "MczL"</td></tr><tr><td>"MAST CAMERA ZOOM RIGHT"</td><td>Any string value starting with "MczR"</td></tr><tr><td>"SHERLOC-WATSON IMAGER"</td><td>Any string value starting with "TBD"</td></tr></tbody></table>	Value	APID Names	"FRONT HAZARD AVOIDANCE CAMERA LEFT STRING <A B>"	Any string value starting with "Img" and ending with "Fhl"	"FRONT HAZARD AVOIDANCE CAMERA RIGHT STRING <A B>"	Any string value starting with "Img" and ending with "Fhr"	"REAR HAZARD AVOIDANCE CAMERA LEFT"	Any string value starting with "Img" and ending with "RhI"	"REAR HAZARD AVOIDANCE CAMERA RIGHT"	Any string vaue startiring with "Img" and ending with "Rhr"	"NAVIGATION CAMERA LEFT"	Any string value starting with "Img" and ending with "NI"	"NAVIGATION CAMERA RIGHT"	Any string value starting with "Img" and ending with "Nr"	"SUPER CAMERA REMOTE MICRO IMAGER"	Any string value starting with "ScamRmi"	"MAST CAMERA ZOOM LEFT"	Any string value starting with "MczL"	"MAST CAMERA ZOOM RIGHT"	Any string value starting with "MczR"	"SHERLOC-WATSON IMAGER"	Any string value starting with "TBD"	Mode EMD in XML format Field as "<xml name>:[<element>]:[<element>]:<field>" "M20EarthProductMetadata:M20ProductMetadata:ProductName", "M20EarthProductMetadata:M20ProductMetadata:CreationStringId" Type n/a
Value	APID Names																							
"FRONT HAZARD AVOIDANCE CAMERA LEFT STRING <A B>"	Any string value starting with "Img" and ending with "Fhl"																							
"FRONT HAZARD AVOIDANCE CAMERA RIGHT STRING <A B>"	Any string value starting with "Img" and ending with "Fhr"																							
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"NAVIGATION CAMERA LEFT"	Any string value starting with "Img" and ending with "NI"																							
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"SUPER CAMERA REMOTE MICRO IMAGER"	Any string value starting with "ScamRmi"																							
"MAST CAMERA ZOOM LEFT"	Any string value starting with "MczL"																							
"MAST CAMERA ZOOM RIGHT"	Any string value starting with "MczR"																							
"SHERLOC-WATSON IMAGER"	Any string value starting with "TBD"																							

Output Metadata (Product Label)				Input Metadata (Source)	
• Ops Keyword • Archive Label Attribute • Definition	• Mission Phase C=Cruise, S=Surface, B=Both • Valid Values (quoted) • Type • Units • Keyword Location in Ops Label • Attribute X-path in Archive label • Attribute Reference in Archive Label (bracketed)			• Mission Phase C=Cruise, S=Surface, B=Both • Mode • Metadata Field • Type	
TBD Definition Specifies the manufacturer's serial number assigned to an instrument. This number may be used to uniquely identify a particular instrument for tracing its components or determining its calibration history, for example.	Instrument Fhaz Left Fhaz Right Rhaz Left Rhaz Right Navcam Left Navcam Right Cache	A TBD TBD TBD TBD TBD TBD	B TBD TBD TBD TBD TBD TBD	A TBD TBD TBD TBD TBD TBD	2) DPO in XML format (referenced to APID Name in Appendix A) or Image DPO mini-header Field as “<xml name>[:<element>]:<element>[:<field>]” Eng. Cameras 1) “<IDPH DPO>:idph:serial_no” SuperCam RMI 1) “<IDPH DPO>:idph:serial_no” (TBD) MCZ Cameras 2) “MCZ_Image_Mini_Header[41]”, “MCZ_Image_Mini_Header[42]”, “MCZ_Image_Mini_Header[43]” NOTES: For MCZ Cameras, value analogous to “serial_no” is comprised of three bytes coming from Image DPO mini-header at byte offsets 41, 42 and 43. Type U8
	SuperCam Instrument RMI		EM TBD	FM TBD	
	Mastcam-Z Instrument Mastcam-Z Left Mastcam-Z Right		EM TBD TBD	FM TBD TBD	
	SHERLOC Instrument WATSON ACI		EM TBD TBD	FM TBD TBD	
	PIXL Instrument MCC		EM TBD	FM TBD	
	MEDA Instrument SkyCam		EM TBD	FM TBD	
	EDL Instrument LVS PUC A PUC B PUC C DDC RDC RUC		EM TBD TBD TBD TBD TBD TBD TBD	FM TBD TBD TBD TBD TBD TBD	
	Type string				
	Units n/a				
	Location IDENTIFICATION (Class) MINI_HEADER (Group)				
X-path TBD					
Reference TBD					
Ops Keyword INSTRUMENT_STATE	Valid Values • Bit 0: TBD ... 0 = “TBD”			Mode DPO in XML format (referenced to APID Name in Appendix A)	

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
Archive Attribute TBD Definition Specifies an array of identifiers for the state of an instrument at a specified time.	1 = " TBD " - Bit 1: TBD ... 0 = "TBD" 1 = "TBD" - Bit 2: TBD ... 0 = "TBD" 1 = "TBD" - Bit 3: TBD ... 0 = "TBD" 1 = "TBD" - Bit 4: TBD ... 0 = "TBD" 1 = "TBD" - Bit 5: TBD ... 0 = "TBD" 1 = "TBD" - Bit 6: TBD ... 0 = "TBD" 1 = "TBD" - Bit 7: TBD ... 0 = "TBD" 1 = "TBD" Type string array[8] Units n/a Location MINI_HEADER (Group) X-path TBD Reference TBD	Field as "<xml name>:[<element>]:[<element>]:<field>" MCZ_Cameras "MCZ_Image_Mini_Header TBD " Type U8
Ops Keyword INSTRUMENT_STATE_NAME Archive Attribute TBD Definition Specifies the possible value that can be contained in the INSTRUMENT_STATE array.	Valid Values ("TBD", "TBD", "TBD", "TBD", "TBD", "TBD", "TBD", "TBD") Type string array[8] Units n/a Location MINI_HEADER (Group) X-path TBD Reference TBD	Mode Static value: - Single value representing array of Bits 0 thru 7 from INSTRUMENT_STATE. Field as "<xml name>:[<element>]:[<element>]:<field>" n/a Type n/a
Ops Keyword INSTRUMENT_TEMPERATURE	Valid Values n/a	Mode DPO in XML format (referenced to APID Name in Appendix A)

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
Archive Attribute TBD Definition Specifies the temperature, in degrees Celsius, of an instrument or some part of an instrument. Note that this may be an array of multiple values for temperatures on different parts of the instrument. (Example: CCD array and sensor head)	Type float array Units deg C (<degC> unit tag required) Location INSTRUMENT_STATE_PARMS (Group) X-path TBD Reference TBD	Field as “<xml name>:[<element>]:<field>” Surface <u>Eng. Cameras</u> “<IDPH DPO>:idph:temp[0] - [22]” <u>SuperCam</u> “<Ancillary DPO>:instrument_temperatures:mu_obox_temp” “<Ancillary DPO>:instrument_temperatures:mu_laser_if_temp” “<Ancillary DPO>:instrument_temperatures:mu_ebox_temp” “<Ancillary DPO>:instrument_temperatures:mu_sec_mirror_temp” “<Ancillary DPO>:instrument_temperatures:ramp_temp” <u>MCZ Cameras</u> TBD – Ancillary DPO spreadsheet doesn’t match FSW <u>Skycam</u> “<MEDA Image Header>:cam_pcb_prt” “<MEDA Image Header>:cam_ccd_prt” Cruise <u>MCZ Cameras</u> “<Ancillary DPO>:instrument_temperatures:dea_temp”, “<Ancillary DPO>:instrument_temperatures:head_fpa_temp”, “<Ancillary DPO>:instrument_temperatures:head_htr_temp_1”, “<Ancillary DPO>:instrument_temperatures:head_htr_temp_2” Type 1) F32[22] 2) F32[5] 3) F32[4] 4) U16 NOTES: - Skycam value need to be converted?
Ops Keyword INSTRUMENT_TEMPERATURE_NAME Archive Attribute TBD Definition Specifies an array of the formal names identifying each of the values used in INSTRUMENT_TEMPERATURE.	Valid Values <u>Eng. Cameras</u> (“A_FRONT_LEFT_HAZ_1”, “ “A_FRONT_LEFT_HAZ_2”, “A_FRONT_RIGHT_HAZ_1”, “A_FRONT_RIGHT_HAZ_2”, “B_FRONT_LEFT_HAZ_1”, “B_FRONT_LEFT_HAZ_2”, “B_FRONT_RIGHT_HAZ_1”, “B_FRONT_RIGHT_HAZ_2”, “REAR_LEFT_HAZ_1”, “REAR_LEFT_HAZ_2”, “REAR_RIGHT_HAZ_1”, “REAR_RIGHT_HAZ_2”, “NAVCAM_LEFT_1”,	Mode Static values Field as “<xml name>:[<element>]:<field>” n/a Type n/a

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
	“NAVCAM_LEFT_2”, “NAVCAM_RIGHT_1”, “NAVCAM_RIGHT_2”, “CACHE_1”, “CACHE_2”, “FRONT_HAZ_LEFT_CAL”, “FRONT_HAZ_RIGHT_CAL”, “NAVCAM_LEFT_CAL”, “NAVCAM_RIGHT_CAL”) <u>SuperCam</u> (“MU_OBOX_TEMP”, “MU_LASER_IF_TEMP”, “MU_EBOX_TEMP”, “MU_SEC_MIRROR_TEMP”, “RAMP_TEMP”) <u>MCZ Cameras</u> (“DEA”, “HEAD_FPA”, “HEAD_HTR_1”, “HEAD_HTR_2”) Type string array Units n/a Location INSTRUMENT_STATE_PARMs (Group) X-path TBD Reference TBD	
Ops Keyword INSTRUMENT_TEMPERATURE_STATUS Archive Attribute TBD Definition For Mastcam-Z on M2020, specifies the status of temperature read from Thermal module. Note that INSTRUMENT_TEMPERATURE_NAME specifies the name associated with each value for this keyword.	Valid Values n/a Type integer array Units n/a Location INSTRUMENT_STATE_PARMs (Group) X-path TBD Reference TBD	Mode DPO in XML format (referenced to APID Name in Appendix A) Field as “<xml name>:[<element>]:<field>” Surface <u>SuperCam</u> TBD <u>MCZ Cameras</u> TBD “<Ancillary DPO>:instrument_power:left_dea_switch” “<Ancillary DPO>:instrument_power:right_dea_switch” “<Ancillary DPO>:instrument_power_left_htr_status” “<Ancillary DPO>:instrument_power_right_htr_status” Cruise <u>MCZ Cameras</u> TBD “<Ancillary DPO>:instrument_temperatures:dea_status”, “<Ancillary DPO>:instrument_temperatures:fpa_status”, “<Ancillary DPO>:instrument_temperatures:htr_status_1”, “<Ancillary DPO>:instrument_temperatures:htr_status_2” Type

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
		<u>SuperCam</u> TBD <u>MCZ Cameras</u> I32
Ops Keyword INSTRUMENT_TYPE Archive Attribute TBD Definition Specifies the type of an instrument. Example values: POLARIMETER, RADIOMETER, REFLECTANCE SPECTROMETER, VIDICON CAMERA. Note that mosaics may contain more than one value in an array.	Valid Values - Eng. Cameras ... Value "IMAGING CAMERA" string "Img". string value "ScamRmi". value value and PDP Type string(15) Units n/a Location IDENTIFICATION (Class) X-path TBD Reference TBD APID Names - For Eng Cameras, any value starting with - For SuperCam, any starting with - For MCZ, any string starting with "McZ". - For PIXL, any string starting with "PixlSci" category XX.	Mode EMD in XML format Field as "<xml name>:[<element>]:<field>" "M20EarthProductMetadata:M20ProductMetadata:ProductName" Type n/a
Ops Keyword INSTRUMENT_VERSION_ID Archive Attribute TBD Definition Specifies the model of an instrument used to obtain data. For example, this keyword could be used to distinguish between an engineering model of a camera used to acquire test data, and a flight model of a camera used to acquire science data during a mission. Identifiers for use in M2020 are: "BB" - Breadboard "EM" - Engineering Model "FM" - Flight Model	Valid Values - Eng. Cameras & SuperCam ... Serial Number Range Value TBD - MCZ Cameras ... Serial Number Range Value TBD Type string(8) Units n/a Location IDENTIFICATION (Class) X-path TBD	Mode 1) DPO in XML format (referenced to APID Name in Appendix A) 2) DPO in XML format (referenced to APID Name in Appendix A) or Image DPO mini-header Field as "<xml name>:[<element>]:<field>" <u>Eng. Cameras & SuperCam</u> 1) "<IDPH DPO>:idph:serial_no" <u>MCZ Cameras</u> 2) "MCZ_Image_Mini_Header[41]", "MCZ_Image_Mini_Header[42]", "MCZ_Image_Mini_Header[43]" NOTES: - For MCZ Cameras, value analogous to "serial_no" is comprised of three bytes coming from Image DPO mini-header at byte

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
	<u>Reference</u> TBD	offsets 41, 42 and 43. <u>Type</u> U8
<u>Ops Keyword</u> INSTRUMENT_ZOOM_POSITION <u>Archive Attribute</u> TBD <u>Definition</u> Specifies the position in motor counts of the focus motor on a camera. For M2020 MCZ, it specifies the actual zoom position that was used by the image.	<u>Valid Values</u> MCZ Cameras ... TBD <u>Type</u> integer <u>Units</u> n/a <u>Location</u> INSTRUMENT_STATE_PARMS (Group) MINI_HEADER (Group) <u>X-path</u> TBD <u>Reference</u> TBD	<u>Mode</u> Image DPO mini-header <u>Field as “<xml name>:[<element>]:<field>”</u> MCZ Cameras “MCZ_Image_Mini_Header[TBD]”, “MCZ_Image_Mini_Header[TBD]”, “MCZ_Image_Mini_Header[TBD]”, “MCZ_Image_Mini_Header[TBD]” <u>Type</u> U32
<u>Ops Keyword</u> INST_CMPRS_COLOR_MODE <u>Archive Attribute</u> none <u>Definition</u> Specifies the color mode used during compression. For M2020, this is the JPEG color subsampling mode. COLOR_MODE_GRAY indicates a grayscale image, COLOR_MODE_422 indicates 4:2:2 chroma subsampling (the typical case), and COLOR_MODE_444 indicates 4:4:4 chroma sampling (no subsampling). The keyword is not applicable if the INST_CMPRS_NAME is not JPEG and may be omitted.	<u>Valid Values</u> 0 or 1 = “N/A” 2 = “COLOR_MODE_GRAY” 3 = “COLOR_MODE_422” 4 = “COLOR_MODE_444” <u>Type</u> string <u>Units</u> n/a <u>Location</u> COMPRESSION_PARMS (Group) <u>X-path</u> TBD <u>Reference</u> TBD	<u>Mode</u> Image DPO mini-header, Table Lookup <u>Field as “<xml name>:[<element>]:<field>”</u> MCZ Cameras “MCZ_Image_Mini_Header[32]”, “MCZ_Image_Mini_Header[33]”, “MCZ_Image_Mini_Header[34]”, “MCZ_Image_Mini_Header[35]”, “MCZ_Image_Mini_Header[36]”, “MCZ_Image_Mini_Header[37]”, “MCZ_Image_Mini_Header[38]”, “MCZ_Image_Mini_Header[39]” NOTES: For MCZ Cameras, value is comprised of eight bytes coming from Image DPO mini-header at byte offsets 32, 33, 34, 35, 36, 37, 38 and 39. <u>Type</u> enum
<u>Ops Keyword</u> INST_CMPRS_DEFERRED_FLAG <u>Archive Attribute</u> none <u>Definition</u> Specifies a flag to indicate whether compression was done at the time of image acquisition, or was deferred until later (typically at downlink time). For M2020, deferred compression is commonly used with MCZ to downlink differently-compressed versions	<u>Valid Values</u> 0 = “FALSE” non-0 = “TRUE” <u>Type</u> string <u>Units</u> n/a <u>Location</u> 1) COMPRESSION_PARMS (Group) 2) a. IMAGE_REQUEST_PARMS (Group)	<u>Mode</u> 1) EMD in XML format, calculation 2) DPO in XML format (referenced to APID Name in Appendix A), Table Lookup 3) DPO in XML format (referenced to APID Name in Appendix A), Table Lookup 4) DPO in XML format (referenced to APID Name in Appendix A) or Image DPO mini-header, Table Lookup

Output Metadata (Product Label)	Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)
of the same image; the image is stored onboard in raw form. Deferred compression is specified but not currently implemented for the engineering cameras.	b. REFERENCE_PIXEL_REQUEST_PARMS (Group) c. THUMBNAI_REQUEST_PARMS (Group) 3) a. IMAGE_REQUEST_PARMS (Group) b. THUMBNAI_REQUEST_PARMS (Group) 4) COMPRESSION_PARMS (Group) <u>X-path</u> TBD <u>Reference</u> TBD
<u>Ops Keyword</u> INST_CMPRS_FILTER <u>Archive Attribute</u> TBD <u>Definition</u> Specifies the wavelet filter used in the ICER compression and decompression algorithm.	<u>Valid Values</u> <u>Surface</u> <u>Eng. Cameras & SuperCam</u> <u>ICER</u> (S) "A" = 0 = WAVELET_A "B" = 1 = WAVELET_B "C" = 2 = WAVELET_C "D" = 3 = WAVELET_D "E" = 4 = WAVELET_E "F" = 5 = WAVELET_F "Q" = 6 = WAVELET_Q <u>LOCO</u> (S) "N/A" <u>JPEG</u> (S) "N/A" <u>MCZ Cameras</u> lossless = "N/A" not lossless = "UNK" <u>Cruise</u> <u>Type</u>
	<u>Field as "<xml name>:[<element>]:[<element>]:<field>"</u> <u>All Cameras</u> 1) "M20EarthProductMetadata:M20ProductMetadata: ProductName" <u>Eng. Cameras TBD</u> 2) a. "<IDPH DPO>:idph:params:comp:compress" b. "<IDPH DPO>:idph:params:ref_comp:compress" c. "<IDPH DPO>:idph:params:thumb_comp:compress" <u>SuperCam RMI TBD</u> 3) a. "<Ancillary DPO>:cmd_arguments:compression" b. "<Ancillary DPO>:cmd_arguments: THUMBNAI_COMPRESSION" <u>MCZ Cameras</u> 4) "MCZ_Image_Mini_Header[32]", "MCZ_Image_Mini_Header[33]", "MCZ_Image_Mini_Header[34]", "MCZ_Image_Mini_Header[35]", "MCZ_Image_Mini_Header[36]", "MCZ_Image_Mini_Header[37]", "MCZ_Image_Mini_Header[38]", "MCZ_Image_Mini_Header[39]" NOTES: • For MCZ Cameras, value analogous to "compress" is comprised of eight bytes coming from Image DPO mini-header at byte offsets 32, 33, 34, 35, 36, 37, 38 and 39. <u>Type</u> enum <u>Mode</u> 1) Compression algorithm, calculation a. Value returned by compression as variable 2) DPO in XML format (referenced to APID Name in Appendix A), Table Lookup <u>Field as "<xml name>:[<element>]:[<element>]:<field>"</u> <u>Eng. Cameras</u> b. "<IDPH DPO>:idph:params:comp:wfilter" c. "<IDPH DPO>:idph:params:thumb_comp:wfilter" <u>SuperCam</u> b. "<Ancillary DPO>:cmd_parameters:icerFilter" <u>Type</u> a. n/a b. enum c. enum

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
	string <u>Units</u> n/a <u>Location</u> a. COMPRESSION_PARMS (Group) b. IMAGE_REQUEST_PARMS (Group) c. THUMBNAIL_REQUEST_PARMS (Group) <u>X-path</u> TBD <u>Reference</u> TBD	
Ops Keyword INST_CMPRS_MODE Archive Attribute TBD Definition Specifies the method used for on-board compression of data. For the M2020 mission, a value of 0 indicates some form of lossless (or no) compression, while non-0 values indicate modes the lossy compressors may use. See INST_CMPRS_NAME for the actual compression type used. See also INST_CMPRS_COLOR_MODE for JPEG.	Valid Values 1) b. "0" = None or Lossless "1" = Lossy 2) "0" = None "0" = 3 = Lossless (LOCO) "0" = 4 = Lossless (LOCO_DEFERRED) "1" = 1 = Lossy (ICER) "1" = 2 = Lossy (ICER_DEFERRED) "2" = 5 = Lossy (JPEG) "2" = 6 = Loss (JPEG_DEFERRED) 3) "0" = None "0" = 1 = Lossless (LOCO) "1" = 2 = Lossy (ICER_1BPP) "2" = 3 = Lossy (ICER_2BPP) "3" = 4 = Lossy (ICER_3BPP) "4" = 5 = Lossy (ICER_4BPP) "5" = 6 = Lossy (ICER_5BPP) "6" = 7 = Lossy (ICER_6BPP) 4) "0" = None "0" = 1 = MSSS Lossless "1" = 2 = JPEG color mode GRAY "2" = 3 = JPEG color mode 422 "3" = 4 = JPEG color mode 444 <u>Type</u> integer <u>Units</u> n/a <u>Location</u> 1) COMPRESSION_PARMS (Group) 2) a. IMAGE_REQUEST_PARMS (Group) b. REFERENCE_PIXEL_REQUEST_PARMS (Group) c. THUMBNAIL_REQUEST_PARMS (Group) 3) a. IMAGE_REQUEST_PARMS (Group) b. THUMBNAIL_REQUEST_PARMS (Group) 4) - IMAGE_REQUEST_PARMS (Group) - THUMBNAIL_REQUEST_PARMS (Group) - COMPRESSION_PARMS (Group) <u>X-path</u> TBD <u>Reference</u>	Mode 1) a. Compression algorithm (where value returned by compression as variable), calculation b. EMD in XML format, calculation 2) DPO in XML format (referenced to APID Name in Appendix A), Table Lookup 3) DPO in XML format (referenced to APID Name in Appendix A), Table Lookup 4) DPO in XML format (referenced to APID Name in Appendix A) or Image DPO mini-header, Table Lookup Field as "<xml name>:[<element>]:[<element>]:<field>" • <u>All Cameras</u> 1) b. "M20EarthProductMetadata:M20ProductMetadata: ProductName" • <u>Eng. Cameras</u> 2) a. "<IDPH DPO>:idph:params:comp:compress" b. "<IDPH DPO>:idph:params:ref_comp:compress" c. "<IDPH DPO>:idph:params:thumb_comp:compress" • <u>SuperCam RMI</u> 3) a. "<Ancillary DPO>:cmd_arguments:compression" b. "<Ancillary DPO>:cmd_arguments: THUMBNAIL_COMPRESSION" • <u>MCZ Cameras</u> 4) a. "<Ancillary DPO>:cmd_arguments_image:comp_quality" b. "<Ancillary DPO>:cmd_arguments_zstack:comp_quality" c. "MCZ_Image_Mini_Header[32]", "MCZ_Image_Mini_Header[33]", "MCZ_Image_Mini_Header[34]", "MCZ_Image_Mini_Header[35]", "MCZ_Image_Mini_Header[36]", "MCZ_Image_Mini_Header[37]", "MCZ_Image_Mini_Header[38]", "MCZ_Image_Mini_Header[39]"

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
	TBD	NOTES: - For MCZ non-recovered (rangemap or zstack) data products, value comes from Ancillary DPO per case "4a". - For MCZ recovered (rangemap or zstack) data products, value comes from Ancillary DPO per case "4b". - For MCZ Cameras, value analogous to "compress" is comprised of eight bytes coming from Image DPO mini-header at byte offsets 32, 33, 34, 35, 36, 37, 38 and 39 per case "4c". Type 1) a. n/a b. enum 2) enum 3) enum 4) enum
Ops Keyword INST_CMPRS_NAME Archive Attribute TBD Definition Specifies the type of on-board compression used for data storage and transmission. Valid values for compression types supported in M2020 are: a) "NONE" - No compression. b) "ICER" - Adaptive Variable-Length Coding, a lossy or lossless compression algorithm developed at JPL. c) "LOCO" - Low-Complexity Lossless Compression, a lossless compression algorithm developed at JPL. d) "JPEG" - Joint Photographic Experts Group, an industry standard lossy compression algorithm. e) "MSSS_LOSSLESS" - Lossless compression algorithm developed by Malin Space Science Systems. NOTE: For the M2020 mission, LOCO is lossless, JPEG is lossy, and ICER can be either (generally lossy). See INST_CMPRS_MODE for lossless/lossy status.	Valid Values 1) b. 0 = "None or Lossless" 1 = "Lossy" 2) 0 = "None" 1 = "ICER" 2 = "ICER" 3 = "LOCO" 4 = "LOCO" 5 = "JPEG" 6 = "JPEG" 3) 0 = "None" 1 = "LOCO" 2 = "ICER" 3 = "ICER" 4 = "ICER" 5 = "ICER" 6 = "ICER" 7 = "ICER" 4) 0 = "None" 1 = "MSSS_LOSSLESS" 2 = "JPEG" 3 = "JPEG" 4 = "JPEG" Type integer Units n/a Location 1) COMPRESSION_PARMS (Group) 2) a. IMAGE_REQUEST_PARMS (Group) b. REFERENCE_PIXEL_REQUEST_PARMS (Group) c. THUMBNAİL_REQUEST_PARMS (Group) 3) a. IMAGE_REQUEST_PARMS (Group) b. THUMBNAİL_REQUEST_PARMS (Group) 4) - IMAGE_REQUEST_PARMS (Group) - THUMBNAİL_REQUEST_PARMS (Group)	Mode 1) a. Compression algorithm (where value returned by compression as variable), calculation b. EMD in XML format, calculation 2) DPO in XML format (referenced to APID Name in Appendix A), Table Lookup 3) DPO in XML format (referenced to APID Name in Appendix A), Table Lookup 4) Image DPO mini-header, Table Lookup 5) DPO in XML format (referenced to APID Name in Appendix A), Table Lookup Field as "<xml name>:[<element>]:[<element>]:[<field>]" All Cameras 1) b. "M20EarthProductMetadata:M20ProductMetadata:ProductName" Eng. Cameras 2) a. "<IDPH DPO>:idph:params:comp:compress" b. "<IDPH DPO>:idph:params:ref_comp:compress" c. "<IDPH DPO>:idph:params:thumb_comp:compress" SuperCam RMI 3) a. "<Ancillary DPO>:cmd_arguments:compression" b. "<Ancillary DPO>:cmd_arguments:THUMBNAİL_COMPRESSION" MCZ Cameras 4) a. "<Ancillary DPO>:cmd_arguments_image:comp_quality" b. "<Ancillary DPO>:cmd_arguments_zstack:comp_quality" c. "MCZ_Image_Mini_Header[32]", "MCZ_Image_Mini_Header[33]",

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
	- COMPRESSION_PARMS (Group) <u>X-path</u> TBD <u>Reference</u> TBD	"MCZ_Image_Mini_Header[34]", "MCZ_Image_Mini_Header[35]", "MCZ_Image_Mini_Header[36]", "MCZ_Image_Mini_Header[37]", "MCZ_Image_Mini_Header[38]", "MCZ_Image_Mini_Header[39]" NOTES: - For MCZ non-recovered (rangemap or zstack) data products, value comes from Ancillary DPO per case "4a". - For MCZ recovered (rangemap or zstack) data products, value comes from Ancillary DPO per case "4b". - For MCZ Cameras, value analogous to "compress" is comprised of eight bytes coming from Image DPO mini-header at byte offsets 32, 33, 34, 35, 36, 37, 38 and 39 per case "4c". <u>Type</u> 1) a. n/a b. enum 2) enum 3) enum 4) enum
<u>Ops Keyword</u> INST_CMPRS_QUALITY <u>Archive Attribute</u> TBD <u>Definition</u> Specifies a JPEG- or ICER-specific variable which identifies the resultant or targeted image quality index for on-board data compression. NOTE: For MCZ, a value of 0 indicates lossless or no Compression; 1-100 represent JPEG quality levels. Specifies companding, spatial sampling, colorspace processing and compression algorithms (meaning described below and related to compression quality factor and/or output bit rate and algorithms used). If a filter for a given image is not set to clear (filter 0), then compression is performed in "narrowband science" mode; otherwise it is performed in "RGB" mode. If multiple images are acquired by this command, they increment in the order i, i+1, i+2, ... i+(n_images-1).	<u>Valid Values</u> <u>Eng. Cameras</u> ICER (B) "0" to "18" - JPEG (S) "0" to "100" <u>MCZ Cameras</u> (B) "0" to "100" <u>Type</u> integer <u>Units</u> n/a <u>Location</u> 1) a. IMAGE_REQUEST_PARMS (Group) b. THUMBNAI_REQUEST_PARMS (Group) 2) COMPRESSION_PARMS (Group) <u>X-path</u> TBD <u>Reference</u> TBD	<u>Mode</u> 1) DPO in XML format (referenced to APID Name in Appendix A) 2) DPO in XML format (referenced to APID Name in Appendix A) or Image DPO mini-header <u>Field as "<xmlname>:[<element>]:[<element>]:<field>"</u> <u>Eng. Cameras</u> 1) ICER a. "<IDPH DPO>:idph:params:comp:min_loss" b. "<IDPH DPO>:idph:params:thumb_comp:min_loss" JPEG a. "<IDPH DPO>:idph:params:comp:quality" b. "<IDPH DPO>:idph:params:thumb_comp:quality" <u>MCZ Cameras</u> 2) a. "<Ancillary DPO>:image_id_data:comp_quality" b. "MCZ_Image_Mini_Header[32]", "MCZ_Image_Mini_Header[33]", "MCZ_Image_Mini_Header[34]", "MCZ_Image_Mini_Header[35]", "MCZ_Image_Mini_Header[36]", "MCZ_Image_Mini_Header[37]", "MCZ_Image_Mini_Header[38]", "MCZ_Image_Mini_Header[39]" NOTES: - For MCZ Cameras, if user-specified processing is not configured to use Mini-header compression bytes, then

Output Metadata (Product Label)		Input Metadata (Source)
• Ops Keyword • Archive Label Attribute • Definition	• Mission Phase C=Cruise, S=Surface, B=Both • Valid Values (quoted) • Type • Units • Keyword Location in Ops Label • Attribute X-path in Archive label • Attribute Reference in Archive Label (bracketed)	• Mission Phase C=Cruise, S=Surface, B=Both • Mode • Metadata Field • Type
Viewing the compression parameters as 8 bytes ABCDEFGH from MSB to LSB, the bytes mean the following: • A, B, and EFG are unused and should be commanded as zero. • C is the color mode used for compression, where 0 is gray scale, 1 is 422 color subsampling mode, and 2 is 444 color mode (no subsampling). Most broadband color images should use C=1. • For D, if zero and C is 0xff, then lossless compression is used. If D is non-zero, then the image is JPEG-compressed by the quality factor D which ranges from 1 (lowest quality) to 100 (highest quality). Quality 75 is "usually indistinguishable from the source image" in the ITU T.81 JPEG specification. • H specifies the 12-to-8 bit companding table to use. 0 is the default, nominally lossless square-root table. 1-16 encode the pixels linearly by dividing by N with saturation at 255; 17-32 encode the pixels linearly without saturation (the low-order 8 bits are simply transmitted). 0xff selects 16-bit calibration mode, which has restrictions on image dimension, may not be compressed, and is not intended to be used in flight. Other values are not yet defined.		value comes from the Ancillary DPO (case "2a"). • For MCZ Cameras, if user-specified processing is configured to use Mini-header compression bytes, then value is comprised of eight bytes coming from Image DPO mini-header at byte offsets 32, 33, 34, 35, 36, 37, 38 and 39 (case "2b"). Type U8
Ops Keyword INST_CMPRS_RATE Archive Attribute TBD Definition Specifies the average number of bits needed to represent a pixel for an on-board compressed image.	Valid Values • <u>Eng. Cameras</u> 1) n/a 2) a. ICER and LOCO "1.0" to "12.0" b. JPEG "n/a" • <u>SuperCam</u> 3) "1.0" = 2 = SCAM_ICER_1BPP "2.0" = 3 = SCAM_ICER_2BPP "3.0" = 4 = SCAM_ICER_3BPP "4.0" = 5 = SCAM_ICER_4BPP "5.0" = 6 = SCAM_ICER_5BPP "6.0" = 7 = SCAM_ICER_6BPP • <u>MCZ Cameras</u> 4) n/a	Mode 1) Calculation: - Value = bits_in/pixels_out 2) DPO in XML format (referenced to APID Name in Appendix A), Table Lookup 3) DPO in XML format (referenced to APID Name in Appendix A), Table Lookup 4) Calculation: - Value = bits_in * 8/pixels Field as "<xml name>[:<element>[:<element>[:<field>"] • <u>Eng. Cameras</u> 2) a. "<IDPH DPO>:idph:params:comp:bpp" b. "<IDPH DPO>:idph:params:thumb_comp:bpp"

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
	<u>Additional value for Location "1"</u> "N/A" = 0 Type float Units n/a Location 1) COMPRESSION_PARMS (Group) 2) a. IMAGE_REQUEST_PARMS (Group) b. THUMBNAI_REQUEST_PARMS (Group) 3) a. IMAGE_REQUEST_PARMS (Group) b. THUMBNAI_REQUEST_PARMS (Group) 4) - IMAGE_REQUEST_PARMS (Group) - THUMBNAI_REQUEST_PARMS (Group) - COMPRESSION_PARMS (Group) X-path TBD Reference TBD	<u>SuperCam</u> 3) a. "<Ancillary DPO>:cmd_arguments:compression" b. "<Ancillary DPO>:cmd_parameters: THUMBNAI_COMPRESSION" Type 1) n/a 2) F32 3) enum 4) n/a
Ops Keyword INST_CMPRS_RATIO Archive Attribute TBD Definition Specifies the ratio of the size, in bytes, of the original uncompressed data file to its compressed form.	Valid Values Uncompressed = "N/A" Type float Units n/a Location COMPRESSION_PARMS (Group) X-path TBD Reference TBD	Mode 1) For Eng. Cameras, Calculation: - Sum of the size of ICER uncompressed image over the compressed segments area (bits_out/bits_in) 2) For MCZ Cameras, Calculation: - Value = pixels/bits_in Field as "<xml name>:[<element>]:[<element>]:<field>" n/a Type n/a
Ops Keyword INST_CMPRS_SEGMENTS Archive Attribute TBD Definition Specifies the number of segments into which the image was partitioned for the error containment purposes. For ICER compression, the data within each segment is compressed independently, so that data loss across segments is compartmentalized or contained across segments.	Valid Values - Eng. Cameras 1) <u>ICER and LOCO</u> (B) "1" to "32" - JPEG (S) "N/A" Type integer Units n/a Location 1) a. COMPRESSION_PARMS (Group) b. IMAGE_REQUEST_PARMS (Group) c. THUMBNAI_REQUEST_PARMS (Group)	Mode - ICER segment 1) a. Extracted from ICER LOCO segment - DPO in XML format (referenced to APID Name in Appendix A), Table Lookup Field as "<xml name>:[<element>]:[<element>]:<field>" 1) b. "<IDPH DPO>:idph:params:comp:n_segs" c. "<IDPH DPO>:idph:params:thumb_comp:n_segs" Type 1) a. n/a b. U8 c. U8

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
	<u>X-path</u> TBD <u>Reference</u> TBD	
Ops Keyword INST_CMPRS_SEGMENT_QUALITY Archive Attribute TBD Definition Specifies the quality level for each ICER segment. For M2020, upon return by the ICER decompress function “deicer_decompress” [Ref 31] for data segment “seg”, the output quantity “data_seg_status[seg]” contains a number indicating the decode status. The decode status may have one or more of the following flags set: - SHORTDATASEG FLAG (bit 0): If this flag is set, then the segment contained so little data that nothing could be reconstructed in the segment. - INCONSISTENTDATA FLAG (bit 1): If this flag is set, then one or more pieces of information in the segment header (specifically, image width, image height, n segs, wavelet filter, n decomp) are inconsistent with the value(s) in the first (valid) segment. ICER will ignore the data in this segment. - DUPLICATESEG FLAG (bit 2): If this flag is set, then the segment index given in the header equals that given by a previous segment. The decompressor will ignore the data in this segment. - BADBITPLANENUMBER FLAG (bit 3): If this flag is set, then an ICER internal parameter in the header for this segment has probably been corrupted. The decompressor will ignore the data in this segment. - BADBITPLANECOUNT FLAG (bit 4): If this flag is set, then an ICER internal parameter in the header for this segment has probably been corrupted. The decompressor will ignore the data in this segment. - BADDATA FLAG (bit 5): If this flag is set, then either the parameter combination given in the header for this	Valid Values - Eng. Cameras Uncompressed or LOCO = “N/A” - MCZ Cameras “N/A” Type float array Units n/a Location COMPRESSION_PARMS (Group) X-path TBD Reference TBD	Mode Extracted from ICER segment Field as “<xml name>:<element>:<element>:<field>” n/a Type n/a

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
segment are not allowed by ICER, or the segment number is bad. This probably indicates corrupted data. The decompressor will ignore the data in this segment.		
Ops Keyword INST_CMPRS_SEGMENT_STATUS Archive Attribute TBD Definition Specifies a bit mask which provides the status of decoding the nth segment. For M2020, upon return by the ICER decompress function "deicer_decompress" [Ref 31] for data segment "seg", the output quantity "seg_quality[seg]" provides an indication of the quality of the reconstructed segment. Specifically, the value returned is a double for which the integer values correspond to attained min loss values, but in general is an interpolation between these values. Thus lower values of seg_quality[seg] correspond to higher reconstructed qualities, and a value of indicates lossless compression. Note that the compressed stream does not directly contain the value of min loss that was given to the compressor, but the decompressor does know how far along in the decompression process it got before it ran out of bits; this information is used to determine seg quality[seg]. In rare circumstances the decompressor may not be able to determine seg quality[seg] for a segment that it decompresses. In this case it sets seg quality[seg] to -1.0. The reconstructed segment might be either lossy or lossless when this occurs. The technical condition under which a quality value is not determined is that the decompressor runs out of the data for the segment before decoding any bit plane information. (The only way such a reconstructed segment could be lossless is if every bit plane of this segment contains all zeros, so there is no bit plane information to decode. This could only happen if every pixel of the segment and all pixels sufficiently close to the segment have exactly the same value.) Note that if segment seg cannot be reconstructed, then nothing is written to real seg num[seg], int seg bound first line[seg], int seg bound first sample[seg], int seg bound n lines[seg], int seg bound n samples[seg], and seg quality[seg]. The decompressor also returns a status word which may have one of the following flags set: BADNUMDATASEG FLAG (bit 0): If this flag is set, then n data seg, the number of data segments, is outside the	Valid Values - Eng. Cameras Uncompressed or LOCO = "N/A" - MCZ Cameras "N/A" Type integer array Units n/a Location COMPRESSION_PARMS (Group) X-path TBD Reference TBD	Mode Returned from ICER decompression routine Field as "<xml name>:[<element>]:[<element>]:<field>" n/a Type n/a

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
allowed range [1, ICER MAX N SEGS]. Decompression could not be performed. NOGOODSEGMENTS FLAG (bit 1): If this flag is set, then decompression could not be performed because none of the segments contained usable data.		
Ops Keyword INST_CMPRS_SEG_FIRST_LINE Archive Attribute TBD Definition Specifies an array of values in which each nth element identifies the line within a source image that corresponds to the first line where the nth compression segment applies. Value of "-1" denotes the indeterminate case when the decompressor cannot process the segment, or cannot determine seq_quality for a segment that it could decompress.	Valid Values - Eng. Cameras a. "-1" to "1024" b. Uncompressed or LOCO = "N/A" - MCZ Cameras "N/A" Type integer array Units n/a Location COMPRESSION_PARMS (Group) X-path TBD Reference TBD	Mode Extracted from ICER segment Field as "<xml name>:[<element>]:[<element>]:<field>" n/a Type n/a
Ops Keyword INST_CMPRS_SEG_FIRST_LINE_SAMP Archive Attribute TBD Definition Specifies an array of values in which each nth element identifies the line sample within a source image that corresponds to the first line sample where the nth compression segment applies. Value of "-1" denotes the indeterminate case when the decompressor cannot process the segment, or cannot determine seq_quality for a segment that it could decompress.	Valid Values - Eng. Cameras a. "-1" to "1024" b. Uncompressed or LOCO = "N/A" - MCZ Cameras "N/A" Type integer array Units n/a Location COMPRESSION_PARMS (Group) X-path TBD Reference TBD	Mode Extracted from ICER segment Field as "<xml name>:[<element>]:[<element>]:<field>" n/a Type n/a
Ops Keyword INST_CMPRS_SEG_LINES Archive Attribute TBD Definition	Valid Values - Eng. Cameras a. "-1" to "1024" b. Uncompressed or LOCO = "N/A" - MCZ Cameras "N/A" Type	Mode Extracted from ICER segment Field as "<xml name>:[<element>]:[<element>]:<field>" n/a Type n/a

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
Specifies an array of elements in which the nth element identifies the total number of data instances along the vertical axis that the nth compression segment defines. Value of “-1” denotes the indeterminate case when the decompressor cannot process the segment, or cannot determine seq_quality for a segment that it could decompress.	integer array <u>Units</u> n/a <u>Location</u> COMPRESSION_PARMS (Group) <u>X-path</u> TBD <u>Reference</u> TBD	
Ops Keyword INST_CMPRS_SEG_MISSING_PIXELS Archive Attribute TBD Definition Specifies an array of elements in which the nth element identifies the total number of missing pixels that the nth compression segment defines.	Valid Values <u>Eng. Cameras</u> Uncompressed or ICER = “N/A” <u>MCZ Cameras</u> “N/A” Type integer array <u>Units</u> n/a <u>Location</u> COMPRESSION_PARMS (Group) <u>X-path</u> TBD <u>Reference</u> TBD	Mode Extracted from LOCO segment Field as “<xml name>[:<element>[:<element>]:<field>” n/a Type n/a
Ops Keyword INST_CMPRS_SEG_SAMPLES Archive Attribute TBD Definition Specifies an array of elements in which the nth element identifies the total number of data instances along the horizontal axis that the nth compression segment defines. Value of “-1” denotes the indeterminate case when the decompressor cannot process the segment, or cannot determine seq_quality for a segment that it could decompress.	Valid Values <u>Eng. Cameras</u> a. “-1” to “1024” b. Uncompressed or LOCO = “N/A” <u>MCZ Cameras</u> “N/A” Type integer array <u>Units</u> n/a <u>Location</u> COMPRESSION_PARMS (Group) <u>X-path</u> TBD <u>Reference</u> TBD	Mode Extracted from ICER segment Field as “<xml name>[:<element>[:<element>]:<field>” n/a Type n/a
Ops Keyword INST_DECOMP_STAGES Archive Attribute	Valid Values <u>Eng. Cameras</u> ICER (B)	Mode ICER segment 1) Extracted from ICER segment

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
TBD Definition Specifies the number of stages of wavelet decompositions.	"1" to "6" JPEG (S) "N/A" - MCZ Cameras "N/A" - Additional value for Location "1" Uncompressed or LOCO = "N/A" Type integer Units n/a Location 1) COMPRESSION_PARMS (Group) 2) IMAGE_REQUEST_PARMS (Group) 3) THUMBNAIL_REQUEST_PARMS (Group) X-path TBD Reference TBD	DPO in XML format (referenced to APID Name in Appendix A), Table Lookup Field as "<xml name>:[<element>]:[<element>]:<field>" 2) "<IDPH DPO>:idph:params:comp:n_decomps" 3) "<IDPH DPO>:idph:params:thumb_comp:n_decomps" Type 1) n/a 2) U8 3) U8 4) U8
Ops Keyword INTERCHANGE_FORMAT Archive Attribute TBD Definition Specifies the manner in which data items are stored.	Valid Values "ASCII", "BINARY" Type string(6) Units n/a Location - IMAGE_HEADER (Object) - IMAGE (Object) X-path TBD Reference TBD	Mode Static Value Field as "<xml name>:[<element>]:[<element>]:<field>" n/a Type n/a
Ops Keyword INTERPOLATION_METHOD Archive Attribute TBD Definition Some camera models exhibit continuous changes in geometric behavior along a dimension(s). For example, the B-side Navcams vary geometrically based on temperature, and the Mastcams vary based on focus setting. In order to model this continuous variation, camera models may be interpolated, which creates camera models "in between" a set of rigorously calibrated models using piecewise linear interpolation. INTERPOLATION_METHOD specifies the	Valid Values See Definition Type string Units n/a Location GEOMETRIC_CAMERA_MODEL_PARMS (Group)	Mode RDR-generating software Field as "<xml name>:[<element>]:[<element>]:<field>" n/a Type n/a

Output Metadata (Product Label)		Input Metadata (Source)								
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type								
dimension(s) along which interpolation has been performed, and/or where it was performed. For MSL, the valid values for EDR's are "ONBOARD" (temperature interpolation done onboard for Navcam), "NONE" (no interpolation), or "OTHER". For RDR's, the above are still valid, additionally the values may be "TEMPERATURE" or "FOCUS" to indicate the dimension over which interpolation was done. Multiple dimensions of interpolation would be indicated by multiple values in INTERPOLATION_METHOD.										
<u>Ops Keyword</u> INTERPOLATION_VALUE <u>Archive Attribute</u> TBD <u>Definition</u> Specifies the value to which the camera models were interpolated (usually, the specific focus or temperature value used). See INTERPOLATION_METHOD. Multiple dimensions of interpolation would lead to multiple values in this label.	<u>Valid Values</u> n/a <u>Type</u> float <u>Units</u> n/a <u>Location</u> GEOMETRIC_CAMERA_MODEL_PARMS (Group)	<u>Mode</u> RDR-generating software <u>Field as “<xml name>:[<element>]:[<element>]:<field>”</u> n/a <u>Type</u> n/a								
<u>Ops Keyword</u> INVALID_CONSTANT <u>Archive Attribute</u> TBD <u>Definition</u> Specifies the value used when the received data are out of the legitimate range of values. For M2020, the value should be 0.0 for most IDS-generated products, with the exception of Surface Roughness RDRs. For SOAS-generated products, the value may be different.	<u>Valid Values</u> <table><tr><td><u>Product</u></td><td><u>Value</u></td></tr><tr><td>most IDS-gen'd products</td><td>“0.0”</td></tr><tr><td>XYZ & Surface Normal</td><td>“(0.0,0.0,0.0)”</td></tr><tr><td>Surface Roughness</td><td>param value (default is “0.1”)</td></tr></table> <u>Type</u> float or float array <u>Units</u> n/a <u>Location</u> IMAGE (Object) <u>X-path</u> TBD <u>Reference</u> TBD	<u>Product</u>	<u>Value</u>	most IDS-gen'd products	“0.0”	XYZ & Surface Normal	“(0.0,0.0,0.0)”	Surface Roughness	param value (default is “0.1”)	<u>Mode</u> Static Value <u>Field as “<xml name>:[<element>]:[<element>]:<field>”</u> n/a <u>Type</u> n/a
<u>Product</u>	<u>Value</u>									
most IDS-gen'd products	“0.0”									
XYZ & Surface Normal	“(0.0,0.0,0.0)”									
Surface Roughness	param value (default is “0.1”)									
<u>Ops Keyword</u> INVERSE_LUT_FILE_NAME <u>Archive Attribute</u> M2020:INVERSE_LUT_FILE_NAME <u>Definition</u> Specifies the name of the inverse-lookup-table file used in generating the RDR. NOTE: This keyword only applies if SAMPLE_BIT_MODE_ID in Group INSTRUMENT_STATE_PARMS starts with “LUT”, i.e. if an inverse lookup table file was	<u>Valid Values</u> “N/A” <u>Type</u> string <u>Units</u> n/a <u>Location</u> DERIVED_IMAGE_PARMS (Group) <u>X-path</u> TBD	<u>Mode</u> RDR-generating software <u>Field as “<xml name>:[<element>]:[<element>]:<field>”</u> n/a <u>Type</u> n/a								

Output Metadata (Product Label)		Input Metadata (Source)
• Ops Keyword • Archive Label Attribute • Definition	• Mission Phase C=Cruise, S=Surface, B=Both • Valid Values (quoted) • Type • Units • Keyword Location in Ops Label • Attribute X-path in Archive label • Attribute Reference in Archive Label (bracketed)	• Mission Phase C=Cruise, S=Surface, B=Both • Mode • Metadata Field • Type
used.	<u>Reference</u> TBD	
<u>Ops Keyword</u> LABEL_RECORDS <u>Archive Attribute</u> TBD <u>Definition</u> Specifies the number of physical file records that contain only (ODL or PDS) label information. The number of data records in a file is determined by subtracting the value of label_records from the value of file_records. NOTE: In the PDS, the use of label_records along with other file-related data elements is fully described in the Standards Reference.	<u>Valid Values</u> "0" to n <u>Type</u> string array <u>Units</u> n/a <u>Location</u> FILE (Class) <u>X-path</u> TBD <u>Reference</u> TBD	<u>Mode</u> Calculation: - Based on size of PDS label <u>Field as "<xml name>:[<element>]:<field>"</u> n/a <u>Type</u> n/a
<u>Ops Keyword</u> LINEARIZATION_MODE <u>Archive Attribute</u> TBD <u>Definition</u> Specifies the way in which an image has been linearized (see GEOMETRY_PROJECTION_TYPE). It can contain one or two values. The first value specifies what kind of stereo partner was used to linearize the image (the process requires two camera models). A value of "NOMINAL" means that it was linearized using the nominal stereo partner at the same pointing (whether or not the partner image was actually acquired). "ACTUAL" means it was linearized using an actual stereo partner image, which may be at a different pointing or rover location (e.g. long-baseline or re-pointed stereo). The product ID of the actual partner used will be in STEREO_PRODUCT_ID. A value of NONE, means that linearization has not been performed. If the value is not present and linearization is on, "NOMINAL" should be assumed. The second value specifies how the linearized camera model's field of view (FOV) was constructed (corresponding to the "cahv_fov" parameter in MIPL software; see [Ref 27]). "MIN" indicates the FOV is the intersection of the two cameras, which means some data are cut off and typically stretched horizontally, but there are no black areas. "MAX" indicates the FOV is the union of the two cameras, which preserves the edges of the images but typically compresses the image horizontally, and creates black areas. "LINEAR" uses only the CAHV	<u>Valid Values</u> • <u>1st Element in Array</u> "NOMINAL", "ACTUAL", "NONE" • <u>2nd Element in Array</u> "MIN", "MAX", "LINEAR", "NONE" <u>Type</u> string array(2) <u>Units</u> n/a <u>Location</u> DERIVED_IMAGE_PARMS (Group) <u>X-path</u> TBD <u>Reference</u> TBD	<u>Mode</u> RDR-generating software <u>Field as "<xml name>:[<element>]:<field>"</u> n/a <u>Type</u> n/a

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
vectors, which tends to preserve the aspect ratio and scale of the original but both cuts off data and has black areas. "NONE" means no linearization. If the value is not present and linearization is on, "MIN" should be assumed.		
Ops Keyword LINEARIZATION_PRODUCT_ID Archive Attribute TBD Definition Specifies the PRODUCT_ID for the stereo partner used to linearize the image when LINEARIZATION_MODE is "ACTUAL". This keyword may be absent if the mode is not "ACTUAL".	Valid Values n/a Type string Units n/a Location DERIVED_IMAGE_PARMS (Group)	Mode RDR-generating software Field as "<xml name>:[<element>]:[<element>]:<field>" n/a Type n/a
Ops Keyword LINES Archive Attribute TBD Definition Specifies the total number of data instances along the vertical axis of an image. NOTE: In PDS label convention, the number of lines is stored in a 32-bit integer field. The minimum value of 0 indicates no data received.	Valid Values n/a Type integer Units n/a Location <u>Eng. Cameras</u> a. IMAGE (Object) b. SUBFRAME_REQUEST_PARMS (Group) c. THUMBNAIL_REQUEST_PARMS (Group) <u>SuperCam</u> a. MINI_HEADER (Group) b. SUBFRAME_REQUEST_PARMS (Group) <u>MCZ</u> a. SUBFRAME_REQUEST_PARMS (Group) b. MINI-HEADER (Group) <u>Skycam</u> SUBFRAME_REQUEST_PARMS X-path TBD Reference TBD	Mode DPO in XML format (referenced to APID Name in Appendix A) Field as "<xml name>:[<element>]:[<element>]:<field>" Surface <u>Eng. Cameras</u> a. "<IDPH DPO>:idph:rows" b. "<IDPH DPO>:idph:params:sub_rows" c. "<IDPH DPO>:idph:params:thumb_rows" <u>SuperCam</u> a. "SCAM_Image_DPO_Mini_Header[TBD]" b. "<Ancillary DPO>:cmd_arguments:c_height" <u>MCZ</u> a. "<Ancillary DPO>:cmd_arguments_image>window_height" b. "MCZ_Image_Mini_Header[TBD]" <u>Skycam</u> "<MEDA Image Header>:num_rows_lowbyte_data" + "<MEDA Image Header>:num_rows_highbyte_data" Cruise <u>Eng. Cameras (C)</u> a. 960 + ("<IDPH DPO>:idph:rows" - 960) b. 960 + ("<IDPH DPO>:idph:params:sub_rows" - 960) c. 960 + ("<IDPH DPO>:idph:params:thumb_rows" - 960) Type <u>Eng. Cameras</u> <u>SuperCam</u> <u>MCZ</u> U16 <u>Skycam</u> U8

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
Ops Keyword LINE_CAMERA_MODEL_OFFSET Archive Attribute TBD Definition Specifies the location of the image origin with respect to the camera model's origin. For CAHV/CAHVOR models, this origin is not the center of the camera, but is the upper-left corner of the "standard"-size image, which is encoded in the CAHV vectors. (MIPL Projection - Perspective)	Valid Values n/a Type float Units pixel (<pixel> unit tag required) Location SURFACE_PROJECTION_PARMS (Group) X-path TBD Reference TBD	Mode RDR-generating software Field as "<xml name>:[<element>]:[<element>]:<field>" n/a Type n/a
Ops Keyword LINE_PROJECTION_OFFSET Archive Attribute TBD Definition Specifies the line coordinate of the location in the image of the "special" point of the mosaic. For Polar projections, this is the nadir of the polar projection. For Vertical, Orthographic and Orthorectified projections, this is the origin of the projected coordinate system (corresponding to PROJECTION_ORIGIN_VECTOR), and may be off the image.	Valid Values n/a Type float Units pixel (<pixel> unit tag required) Location SURFACE_PROJECTION_PARMS (Group) X-path TBD Reference TBD	Mode RDR-generating software Field as "<xml name>:[<element>]:[<element>]:<field>" n/a Type n/a
Ops Keyword LINE_SAMPLES Archive Attribute TBD Definition Specifies the total number of data instances along the horizontal axis of an image.	Valid Values "1" to "5120" Type integer Units n/a Location - Eng. Cameras - IMAGE (Object) - SUBFRAME_REQUEST_PARMS (Group) - THUMBNAIL_REQUEST_PARMS (Group) - SuperCam - SUBFRAME_REQUEST_PARMS (Group) - THUMBNAIL_REQUEST_PARMS (Group) - MCZ - SUBFRAME_REQUEST_PARMS (Group) - MINI-HEADER (Group) X-path TBD	Mode DPO in XML format (referenced to APID Name in Appendix A) Field as "<xml name>:[<element>]:[<element>]:<field>" Surface - Eng. Cameras "<IDPH DPO>:idph:cols" "<IDPH DPO>:idph:params:sub_cols" "<IDPH DPO>:idph:params:thumb_cols" - SuperCam "<Ancillary DPO>:cmd_arguments:r_height" TBD - MCZ "<Ancillary DPO>:cmd_arguments_image>window_width" "MCZ_Image_Mini_Header[TBD]" - Skycam - Static 1024 Cruise - Eng. Cameras 1280 + ("<IDPH DPO>:idph:cols" - 1280)

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
	<u>Reference</u> TBD	b. 1280 + ("<IDPH DPO>:idph:params:sub_cols" - 1280) c. 1280 + ("<IDPH DPO>:idph:params:thumb_cols" - 1280) <u>Type</u> U16
<u>Ops Keyword</u> LOCAL_MEAN_SOLAR_TIME <u>Archive Attribute</u> TBD <u>Definition</u> Specifies the Local Mean Solar Time, or LMST. It is one of two types of solar time used to express the time of day at a point on the surface of a planetary body. The desire to work with solar days, hours, minutes, and seconds of uniform length led to the concept of the fictitious mean Sun or FMS. The FMS is defined as a point that moves on the celestial equator of a planetary body at a constant rate that represents the average mean motion of the Sun over a planetary year. Local mean solar time is defined, by analogy with Local True Solar Time (LTST), as the difference between the areocentric right ascensions of a point on the surface and of the FMS. The difference between LTST and LMST varies over time. The length of a mean solar day is constant and can be computed from the mean motion of the FMS and the rotation rate of a planet. The mean solar day is also called a 'Sol'. Mean solar hours, minutes, and seconds are defined in the same way as the true solar units. For M2020, the valid value is embedded with a Sol value that can be different than the Sol (see PLANET_DAY_NUMBER) associated with LTST (see LOCAL_TRUE_SOLAR_TIME). The time portion of the valid value is expressed in terms of a 24-hour clock. So, in an example using Sol 27, the valid value range for the 24-hour clock would be represented as "Sol-00027M00:00:00.000" to "Sol-00027M23:59:59.999".	<u>Valid Values</u> Sol-<nnnnn>M<hh>:<mm>:<ss>[.<fff>] NOTE: Value will be uncalibrated if SPICE kernels are unavailable. <u>Type</u> string(30) <u>Units</u> n/a <u>Location</u> IDENTIFICATION (Class) <u>X-path</u> TBD <u>Reference</u> TBD	<u>Mode</u> 1) a. Calculation: - SCLK Kernel - Landing Site Kernel - P Kernel b. DPO in XML format (referenced to APID Name in Appendix A) 2) a. Image DPO mini-header, Calculation: - SCLK Kernel - Landing Site Kernel - P Kernel b. DPO in XML format (referenced to APID Name in Appendix A) <u>Field as "<xml name>[:<element>[:<element>[:<field>"]]</u> 1) <u>Eng. Cameras</u> b. "<IDPH DPO>:idph:scclk_seconds", "<IDPH DPO>:idph:scclk_subseconds" 2) <u>MCZ Cameras</u> a. "MCZ_Image_Mini_Header[8]", "MCZ_Image_Mini_Header[9]", "MCZ_Image_Mini_Header[10]", "MCZ_Image_Mini_Header[11]" b. "<IDPH DPO>:cidph:scclk:seconds" c. "<Ancillary DPO>:scclk:seconds" 3) <u>Skycam</u> "<MEDA Image Header>:time_tag" NOTES: - For MCZ recovered data products, value is comprised of four bytes coming from Image DPO mini-header at byte offsets 8, 9, 10 and 11 per case "2a". - For MCZ non-recovered data products, if IDPH is present, value comes from IDPH DPO per case "2b". - For MCZ non-recovered data products, if IDPH is not present, value comes from Ancillary DPO per case "2c". <u>Type</u> U32
<u>Ops Keyword</u> LOCAL_TRUE_SOLAR_TIME <u>Archive Attribute</u> TBD <u>Definition</u> Specifies the local true solar time, or LTST. It is one of two types of solar time used to express the time of day at a point	<u>Valid Values</u> <hh>:<mm>:<ss>[.<fff>] NOTE: Value will be uncalibrated if SPICE kernels unavailable. <u>Type</u> string(12) <u>Units</u> n/a	<u>Mode</u> 1) a. Calculation: - SCLK Kernel - Landing Site Kernel - P Kernel b. DPO in XML format (referenced to APID Name in Appendix A) 2) a. Image DPO mini-header, Calculation: - SCLK Kernel - Landing Site Kernel - P Kernel

Output Metadata (Product Label)	Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)
on the surface of a planetary body. LTST is measured relative to the true position of the Sun as seen from a point on the planet's surface. The coordinate system used to define LTST has its origin at the center of the planet. Its Z-axis is the north pole vector (or spin axis) of the planet. The X-axis is chosen to point in the direction of the vernal equinox of the planet's orbit. (The vernal or autumnal equinox vectors are found by searching the planetary ephemeris for those times when the vector from the planet's center to the Sun is perpendicular to the planet's north pole vector. The vernal equinox is the time when the Sun appears to rise above the planet's equator.) Positions of points in this frame can be expressed as a radius and areocentric 'right ascension' and 'declination' angles. The areocentric right ascension angle, or ARA, is measured positive eastward in the equatorial plane from the vernal equinox vector to the intersection of the meridian containing the point with the equator. Similarly, the areocentric declination is the angle between the equatorial plane and the vector to the point. LTST is a function of the difference between the ARAs of the vectors to the Sun and to the point on the planet's surface. Specifically, $LTST = (a(P) - a(TS)) * (24 / 360) + 12$ where, - LTST = the local true solar time in true solar hours - a(P) = ARA of the point on the planet's surface in deg - a(TS) = ARA of the true sun in deg The conversion factor of 24/360 is applied to transform the angular measure in decimal degrees into hours-minutes-seconds of arc. This standard representation divides 360 degrees into 24 hours, each hour into 60 minutes, and each minute into 60 seconds of arc. The hours, minutes, and seconds of arc are called 'true solar' hours, minutes, and seconds when used to measure LTST. The constant offset of 12 hours is added to the difference in ARAs to place local noon (12:00:00 in hours, minutes, seconds) at the point where the Sun is directly overhead; at this time, the ARA of the true sun is the same as that of the surface point so that $a(P) - a(TS) = 0.$ The use of 'true solar' time units can be extended to define a true solar day as 24 true solar hours. Due to the eccentricity of planetary orbits and the inclination of orbital planes to equatorial planes (obliquity), the Sun does not move at a uniform rate over the course of a planetary year. Consequently, the number of SI seconds in a true solar day, hour, minute or second is not constant.	b. DPO in XML format (referenced to APID Name in Appendix A) Field as "<xml name>:[<element>]:[<element>]:<field>" <u>Eng. Cameras</u> "<IDPH DPO>:idph:sclk_seconds", "<IDPH DPO>:idph:sclk_subseconds" <u>MCZ Cameras</u> "MCZ_Image_Mini_Header[8]", "MCZ_Image_Mini_Header[9]", "MCZ_Image_Mini_Header[10]", "MCZ_Image_Mini_Header[11]" "<IDPH DPO>:cidph:sclk:seconds" "<Ancillary DPO>:sclk:seconds" <u>Skycam</u> "<MEDA Image Header>:time_tag" NOTES: - For MCZ recovered or Z-stack data products, value is comprised of four bytes coming from Image DPO mini-header at byte offsets 8, 9, 10 and 11 per case "2a". - For MCZ non-recovered and non-Z-stack data products, if IDPH is present, value comes from IDPH DPO per case "2b". - For MCZ non-recovered and non-Z-stack data products, if IDPH is not present, value comes from Ancillary DPO per case "2c". Type U32

Output Metadata (Product Label)		Input Metadata (Source)						
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type						
For M2020, the valid value is expressed in terms of a 24-hour clock, so the acceptable range is “00:00:00.000” to “23:59:59.999”. See also LOCAL_MEAN_SOLAR_TIME and PLANET_DAY_NUMBER.								
Ops Keyword LOCAL_TRUE_SOLAR_TIME_SOL Archive Attribute TBD Definition Specifies the number of solar days elapsed since a reference day (e.g., the day on which a landing vehicle set down) for local true solar time (LTST). Days are measured in rotations of the planet in question from midnight to midnight. For M2020, the reference day is “0”, as Landing day is Sol 0. If before Landing day, then value will be less than or equal to “0” and can be negative. NOTE: Value will be uncalibrated if SPICE kernels are unavailable. See also LOCAL_TRUE_SOLAR_TIME and PLANET_DAY_NUMBER.	Valid Values <table><tr><td><u>Mission Phase</u></td><td><u>Values</u></td></tr><tr><td>Cruise</td><td>less than or equal to 0</td></tr><tr><td>Surface</td><td>“0” to n</td></tr></table> Type integer Units n/a Location IDENTIFICATION (Class) X-path TBD Reference TBD	<u>Mission Phase</u>	<u>Values</u>	Cruise	less than or equal to 0	Surface	“0” to n	Mode - Calculation: - SCLK Kernel - Landing Site Kernel - P Kernel - DPO in XML format (referenced to APID Name in Appendix A) Field as “<xml name>:[<element>]:<field>” 1) <u>Eng. Cameras</u> “<IDPH DPO>:idph:sclk_seconds”, “<IDPH DPO>:idph:sclk_subseconds” 2) <u>MCZ Cameras</u> a. “MCZ_Image_Mini_Header[8]”, “MCZ_Image_Mini_Header[9]”, “MCZ_Image_Mini_Header[10]”, “MCZ_Image_Mini_Header[11]” b. “<IDPH DPO>:cidph:sclk:seconds” c. “<Ancillary DPO>:sclk:seconds” 3) <u>Skycam</u> “<MEDA Image Header>:time_tag” Type U32
<u>Mission Phase</u>	<u>Values</u>							
Cruise	less than or equal to 0							
Surface	“0” to n							
Ops Keyword MAGIC_NUMBERS Archive Attribute none Definition Specifies constant numbers that should be in a data product to confirm the product is being decoded properly. For MCZ, this is the “magic0” and “magic1” fields of the Mini-header, in Hex format.	Valid Values n/a Type string Units n/a Location MINI_HEADER (Group) X-path TBD Reference TBD	Mode Image DPO mini-header Field as “<xml name>:[<element>]:<field>” <u>MCZ Cameras</u> “MCZ_Image_Mini_Header[TBD]”, “MCZ_Image_Mini_Header[TBD]” Type n/a						
Ops Keyword MAP_PROJECTION_TYPE Archive Attribute TBD Definition Specifies the type of projection characteristic of a given map.	Valid Values “CYLINDRICAL”, “VERTICAL”, “PERSPECTIVE”, “POLAR”, “ORTHOGRAPHIC”, “ORTHORECTIFIED”, “CYLINDRICAL-PERSPECTIVE” Type string Units n/a	Mode RDR-generating software Field as “<xml name>:[<element>]:<field>” n/a Type n/a						

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
When in a SURFACE_PROJECTION group, defines the surface-based map projection used in the image.	<u>Location</u> SURFACE_PROJECTION_PARMS (Group) <u>X-path</u> TBD <u>Reference</u> TBD	
<u>Ops Keyword</u> MAP_RESOLUTION <u>Archive Attribute</u> TBD <u>Definition</u> Specifies the scale of a given map. Please refer to the definition for MAP_SCALE for a more complete definition. When in a SURFACE_PROJECTION group, defines the resolution of the map in pixels/degree. For Cylindrical projections, this is constant throughout. For Polar, this is for the Elevation (radial) direction only. For Perspective and Cylindrical-Perspective, this is at the center of the output camera model. Not applicable to Orthographic, Orthorectified and Vertical projections. NOTE: MAP_RESOLUTION and MAP_SCALE both define The scale of a map except that they are expressed in different units: MAP_RESOLUTION is in pixels/deg and MAP_SCALE is in meters/pixel. If two values are present, the first measures in the line direction while the second measures in the sample direction.	<u>Valid Values</u> n/a <u>Type</u> float array[2] <u>Units</u> pixels/deg (<pix/deg> unit tag required) <u>Location</u> SURFACE_PROJECTION_PARMS (Group) <u>X-path</u> TBD <u>Reference</u> TBD	<u>Mode</u> RDR-generating software <u>Field as “<xml name>:[<element>]:[<element>]:<field>”</u> n/a <u>Type</u> n/a
<u>Ops Keyword</u> MAP_SCALE <u>Archive Attribute</u> TBD <u>Definition</u> Specifies the scale of a given map. The scale is defined as the ratio of the actual distance between two points on the surface of the target body to the distance between the corresponding points on the map. MAP_SCALE references the scale of a map at a certain reference point or line. Certain map projections vary in scale throughout the map. When in a SURFACE_PROJECTION group, defines the scale of the map in meters/pixel. Applicable to Vertical,	<u>Valid Values</u> n/a <u>Type</u> float array[2] <u>Units</u> m/pixel (<m/pixel> unit tag required) <u>Location</u> SURFACE_PROJECTION_PARMS (Group) <u>X-path</u> TBD <u>Reference</u> TBD	<u>Mode</u> RDR-generating software <u>Field as “<xml name>:[<element>]:[<element>]:<field>”</u> n/a <u>Type</u> n/a

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
Orthographic and Orthorectified projections only. NOTE: MAP_RESOLUTION and MAP_SCALE both define the scale of a map except that they are expressed in different units: MAP_RESOLUTION is in pixels/deg and MAP_SCALE is in meters/pixel. If two values are present, the first measures in the line direction while the second measures in the sample direction.		
Ops Keyword MARKER_PAD Archive Attribute TBD Definition Static value used by the Body Unit (BU).	Valid Values "0xFDD2" Type string Units n/a Location MINI_HEADER_GROUP X-path TBD Reference TBD	SCAM Image DPO mini-header Field as "<xml name>:[<element>]:[<element>]:<field>" "SCAM_Image_Mini_Header:marker_pad" Type U16 Notes: For SCAM RMI products, "marker pad" is comprised of two bytes coming from the Image DPO mini-header at offset 178.
Ops Keyword MASK_DESC_FILE_NAME Archive Attribute TBD Definition Specifies the name of a file or files containing the parameters used to create a binary mask file that, when applied to another image file, prevents specific areas in the image from being processed. For products generated by MIPL, these files are XML formatted and describe each of the mask components. They are serviceable to program "marsfilter". For products generated by other teams, different formats and methods may apply. See also HORIZON_MASK_ELEVATION.	Valid Values n/a Type string Units n/a Location DERIVED_IMAGE_PARMS (Group) X-path TBD Reference TBD	Mode RDR-generating software Field as "<xml name>:[<element>]:[<element>]:<field>" n/a Type n/a
Ops Keyword MAXIMUM Archive Attribute TBD Definition	Valid Values n/a Type float Units	Mode Calculation Field as "<xml name>:[<element>]:[<element>]:<field>" n/a

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
Specifies the largest value occurring in a given instance of the data object. Note that for PDS applications -- because of the unconventional data type of this data element, the element should appear in labels only within an explicit object, i.e., anywhere between an 'OBJECT =' and an 'END_OBJECT'.	n/a <u>Location</u> IMAGE (Object) <u>X-path</u> TBD <u>Reference</u> TBD	<u>Type</u> n/a
<u>Ops Keyword</u> MAXIMUM_ELEVATION <u>Archive Attribute</u> TBD <u>Definition</u> Specifies the elevation (as defined by the coordinate system) of the first line of the image. For the Polar projection, specifies the highest elevation used, i.e. the elevation of the outermost circle of pixels. Applies to MIPL projections Cylindrical, Polar, Sinusoidal, Perspective and Cylindrical-Perspective.	<u>Valid Values</u> n/a <u>Type</u> float <u>Units</u> deg (<deg> unit tag required) <u>Location</u> SURFACE_PROJECTION_PARAMS (Group) <u>X-path</u> TBD <u>Reference</u> TBD	<u>Mode</u> RDR-generating software <u>Field as “<xml name>:[<element>]:[<element>]:<field>”</u> n/a <u>Type</u> n/a
<u>Ops Keyword</u> MAX_AUTO_EXPOS_ITERATION_COUNT <u>Archive Attribute</u> TBD <u>Definition</u> Specifies the maximum number of exposure iterations the instrument will perform in order to obtain the requested exposure when operating in an autonomous mode.	<u>Valid Values</u> - Eng. Cameras (B) - MCZ (B) “0” to “10” <u>Type</u> unsigned integer <u>Units</u> n/a <u>Location</u> - Eng. Cameras - OBSERVATION_REQUEST_PARAMS (Group) - MCZ a. OBSERVATION_REQUEST_PARAMS (Group) b. MINI-HEADER (Group) <u>X-path</u> TBD <u>Reference</u> TBD	<u>Mode</u> DPO in XML format (referenced to APID Name in Appendix A) <u>Field as “<xml name>:[<element>]:[<element>]:<field>”</u> 1) Eng. Cameras “<IDPH DPO>:idph:params:exp_auto_iter” 2) MCZ Cameras a. “<IDPH DPO>:idph:params:exp_auto_iter” b. “MCZ_Image_Mini_Header[TBD]” <u>Type</u> U8
<u>Ops Keyword</u> MEAN <u>Archive Attribute</u> TBD <u>Definition</u> Specifies the average of the DN values in the image array.	<u>Valid Values</u> n/a <u>Type</u> float <u>Units</u> n/a	<u>Mode</u> Calculation <u>Field as “<xml name>:[<element>]:[<element>]:<field>”</u> n/a <u>Type</u> n/a

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
	<u>Location</u> IMAGE (Object) <u>X-path</u> TBD <u>Reference</u> TBD	
<u>Ops Keyword</u> MEDIAN <u>Archive Attribute</u> TBD <u>Definition</u> Specifies the median value (middle value) occurring in a given instance of the data object. Because of the unconventional data type of this data element, the element should appear in labels only within an explicit object, i.e., anywhere between an 'OBJECT =' and an 'END OBJECT'.	<u>Valid Values</u> n/a <u>Type</u> float <u>Units</u> n/a <u>Location</u> IMAGE (Object) <u>X-path</u> TBD <u>Reference</u> TBD	<u>Mode</u> Calculation <u>Field as “<xml name>:[<element>]:[<element>]:<field>”</u> n/a <u>Type</u> n/a
<u>Ops Keyword</u> MEMORY_BANK <u>Archive Attribute</u> TBD <u>Definition</u> For SuperCam RMI this specifies the Mast Unit (MU) memory bank used to hold image before transfer to RCE. There are 64 memory banks in the MU.	<u>Valid Values</u> “0” to “63” <u>Type</u> integer <u>Units</u> n/a <u>Location</u> MINI_HEADER_GROUP <u>X-path</u> TBD <u>Reference</u> TBD	SCAM Image DPO mini-header <u>Field as “<xml name>:[<element>]:[<element>]:<field>”</u> “SCAM_Image_Mini_Header:memory_bank” <u>Type</u> U8 Notes: For SCAM RMI products, “memory bank” is comprised of one byte coming from the Image DPO mini-header at offset 12.
<u>Ops Keyword</u> MILLISECOND_COUNT <u>Archive Attribute</u> TBD <u>Definition</u> Millisecond count from RCE time sync when image received from mast unit.	<u>Valid Values</u> <u>Type</u> integer <u>Units</u> seconds <u>Location</u> MINI_HEADER_GROUP <u>X-path</u> TBD	SCAM Image DPO mini-header <u>Field as “<xml name>:[<element>]:[<element>]:<field>”</u> “SCAM_Image_Mini_Header[184]” <u>Type</u> U32 Notes: For SCAM RMI products, “millisecond count” is comprised of 4 bytes coming from the Image DPO mini-header at offset 184.

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
	<u>Reference</u> TBD	
<u>Ops Keyword</u> MINIMUM <u>Archive Attribute</u> TBD <u>Definition</u> Specifies the smallest value occurring in a given instance of the data object. Note that for PDS applications -- because of the unconventional data type of this data element, the element should appear in labels only within an explicit object, i.e., anywhere between an 'OBJECT =' and an 'END_OBJECT'.	<u>Valid Values</u> n/a <u>Type</u> float <u>Units</u> n/a <u>Location</u> IMAGE (Object) <u>X-path</u> TBD <u>Reference</u> TBD	<u>Mode</u> Calculation <u>Field as “<xml name>:[<element>]:[<element>]:<field>”</u> n/a <u>Type</u> n/a
<u>Ops Keyword</u> MINIMUM_ELEVATION <u>Archive Attribute</u> TBD <u>Definition</u> Specifies the elevation (as defined by the coordinate system) of the last line of the image for Cylindrical map projections. Applies to Cylindrical, Perspective and Cylindrical-Perspective projections.	<u>Valid Values</u> n/a <u>Type</u> float <u>Units</u> deg (<deg> unit tag required) <u>Location</u> SURFACE_PROJECTION_PARMs (Group) <u>X-path</u> TBD <u>Reference</u> TBD	<u>Mode</u> Calculation <u>Field as “<xml name>:[<element>]:[<element>]:<field>”</u> n/a <u>Type</u> n/a
<u>Ops Keyword</u> MISSING_CONSTANT <u>Archive Attribute</u> TBD <u>Definition</u> Specifies the flag value used in the image to indicate that no science data are available for any given pixel. See the specific product definitions for standard values used for each product.	<u>Valid Values</u> <u>Nominal</u> “0.0” <u>XYZ</u> “(0.0, 0.0, 0.0)” <u>Surface Normal (UVW)</u> “(0.0, 0.0, 0.0)” <u>Type</u> float or float array <u>Units</u> n/a <u>Location</u> IMAGE (Object) <u>X-path</u> TBD	<u>Mode</u> Static Value <u>Field as “<xml name>:[<element>]:[<element>]:<field>”</u> n/a <u>Type</u> n/a

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
	<u>Reference</u> TBD	
<u>Ops Keyword</u> MISSION_NAME <u>Archive Attribute</u> TBD <u>Definition</u> Specifies a major planetary mission or project. A given planetary mission may be associated with one or more spacecraft. Note that mosaics may contain more than one value in an array.	<u>Valid Values</u> "MARS 2020" TBC <u>Type</u> string array <u>Units</u> n/a <u>Location</u> IDENTIFICATION (Class) <u>X-path</u> TBD <u>Reference</u> TBD	<u>Mode</u> Static Value <u>Field as "<xml name>:[<element>]:[<element>]:<field>"</u> n/a <u>Type</u> n/a
<u>Ops Keyword</u> MISSION_PHASE_NAME <u>Archive Attribute</u> TBD <u>Definition</u> Specifies the commonly-used identifier of a mission phase.	<u>Valid Values</u> "DEVELOPMENT", "LAUNCH", "CRUISE AND APPROACH", "ENTRY DESCENT AND LANDING", "PRIMARY SURFACE MISSION", "EXTENDED SURFACE MISSION", "TEST" <u>Type</u> string(30) <u>Units</u> n/a <u>Location</u> IDENTIFICATION (Class) <u>X-path</u> TBD <u>Reference</u> TBD	<u>Mode</u> User specified parameter value <u>Field as "<xml name>:[<element>]:[<element>]:<field>"</u> n/a <u>Type</u> n/a
<u>Ops Keyword</u> MODEL_COMPONENT_1 <u>Archive Attribute</u> TBD <u>Definition</u> Specifies a set of values representing the first component of a model. The significance (or meaning) of this array of values is indicated by the first value of the MODEL_COMPONENT_ID and/or MODEL_COMPONENT_NAME elements. The interpretation of the values themselves depends on the model but they commonly represent a vector, a set of polynomial coefficients, or a simple numeric parameter. For example, for a geometric camera model with a value of "CAHV" for MODEL_TYPE, the first value of the MODEL_COMPONENT_NAME data element is CENTER, meaning that the MODEL_COMPONENT_1	<u>Valid Values</u> n/a <u>Type</u> float array <u>Units</u> n/a <u>Location</u> GEOMETRIC_CAMERA_MODEL (Group) <u>X-path</u> TBD <u>Reference</u> TBD	<u>Mode</u> DPO in XML format (referenced to APID Name in Appendix A) <u>Field as "<xml name>:[<element>]:[<element>]:<field>"</u> <u>Eng. Cameras</u> "<IDPH DPO>:idph:cmod_c[3]" <u>MCZ Cameras</u> "<IDPH DPO>:cidph:camera_model:c[0]", "<IDPH DPO>:cidph:camera_model:c[1]", "<IDPH DPO>:cidph:camera_model:c[2]" <u>Type</u> F64[3]

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
is a focal center vector. The three items in this vector provide X, Y, and Z coordinates of the focal point of the camera. The exact details about each model component vector are provided in MODEL_DESC.		
Ops Keyword MODEL_COMPONENT_2 Archive Attribute TBD Definition Specifies the value of the component of the MODEL_COMPONENT_ID for the second element.	Valid Values n/a Type float array Units n/a Location GEOMETRIC_CAMERA_MODEL (Group) X-path TBD Reference TBD	Mode DPO in XML format (referenced to APID Name in Appendix A) Field as "<xml name>:[<element>]:[<element>]:<field>" Eng. Cameras "<IDPH DPO>:idph:cmmod_a[3]" MCZ Cameras "<IDPH DPO>:cidph:camera_model:a[0]", "<IDPH DPO>:cidph:camera_model:a[1]", "<IDPH DPO>:cidph:camera_model:a[2]" Type F64[3]
Ops Keyword MODEL_COMPONENT_3 Archive Attribute TBD Definition Specifies the value of the component of the MODEL_COMPONENT_ID for the third element.	Valid Values n/a Type float array Units n/a Location GEOMETRIC_CAMERA_MODEL (Group) X-path TBD Reference TBD	Mode DPO in XML format (referenced to APID Name in Appendix A) Field as "<xml name>:[<element>]:[<element>]:<field>" Eng. Cameras "<IDPH DPO>:idph:cmmod_h[3]" MCZ Cameras "<IDPH DPO>:cidph:camera_model:h[0]", "<IDPH DPO>:cidph:camera_model:h[1]", "<IDPH DPO>:cidph:camera_model:h[2]" Type F64[3]
Ops Keyword MODEL_COMPONENT_4 Archive Attribute TBD Definition Specifies the value of the component of the MODEL_COMPONENT_ID for the fourth element.	Valid Values n/a Type float array Units n/a Location GEOMETRIC_CAMERA_MODEL (Group) X-path TBD Reference TBD	Mode DPO in XML format (referenced to APID Name in Appendix A) Field as "<xml name>:[<element>]:[<element>]:<field>" Eng. Cameras "<IDPH DPO>:idph:cmmod_v[3]" MCZ Cameras "<IDPH DPO>:cidph:camera_model:v[0]", "<IDPH DPO>:cidph:camera_model:v[1]", "<IDPH DPO>:cidph:camera_model:v[2]" Type F64[3]
Ops Keyword MODEL_COMPONENT_5	Valid Values n/a	Mode

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
Archive Attribute TBD Definition Specifies the value of the component of the MODEL_COMPONENT_ID for the fifth element.	Type float array Units n/a Location GEOMETRIC_CAMERA_MODEL (Group) X-path TBD Reference TBD	DPO in XML format (referenced to APID Name in Appendix A) Field as “<xml name>:[<element>]:<field>” <u>Eng. Cameras</u> “<IDPH DPO>:idph:cmo_d_o[3]” <u>MCZ Cameras</u> “<IDPH DPO>:cidph:camera_model:o[0]”, “<IDPH DPO>:cidph:camera_model:o[1]”, “<IDPH DPO>:cidph:camera_model:o[2]” Type F64[3]
Ops Keyword MODEL_COMPONENT_6 Archive Attribute TBD Definition Specifies the value of the component of the MODEL_COMPONENT_ID for the sixth element.	Valid Values n/a Type float array Units n/a Location GEOMETRIC_CAMERA_MODEL (Group) X-path TBD Reference TBD	Mode DPO in XML format (referenced to APID Name in Appendix A) Field as “<xml name>:[<element>]:<field>” <u>Eng. Cameras</u> “<IDPH DPO>:idph:cmo_d_r[3]” <u>MCZ Cameras</u> “<IDPH DPO>:cidph:camera_model:r[0]”, “<IDPH DPO>:cidph:camera_model:r[1]”, “<IDPH DPO>:cidph:camera_model:r[2]” Type F64[3]
Ops Keyword MODEL_COMPONENT_7 Archive Attribute TBD Definition Specifies the value of the component of the MODEL_COMPONENT_ID for the seventh element.	Valid Values n/a Type float array Units n/a Location GEOMETRIC_CAMERA_MODEL (Group) X-path TBD Reference TBD	Mode DPO in XML format (referenced to APID Name in Appendix A) Field as “<xml name>:[<element>]:<field>” <u>Eng. Cameras</u> “<IDPH DPO>:idph:cmo_d_e[3]” <u>MCZ Cameras</u> “<IDPH DPO>:cidph:camera_model:e[0]”, “<IDPH DPO>:cidph:camera_model:e[1]”, “<IDPH DPO>:cidph:camera_model:e[2]” Type F64[3]
Ops Keyword MODEL_COMPONENT_8 Archive Attribute TBD Definition Specifies the value of the component of the MODEL_COMPONENT_ID for the eighth element.	Valid Values n/a Type float array Units n/a Location	Mode DPO in XML format (referenced to APID Name in Appendix A) Field as “<xml name>:[<element>]:<field>” <u>Eng. Cameras</u> “<IDPH DPO>:idph:cmo_d_mtype” <u>MCZ Cameras</u>

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
	GEOMETRIC_CAMERA_MODEL (Group) <u>X-path</u> TBD <u>Reference</u> TBD	"<IDPH DPO>:cidph:camera_model:mtype" <u>Type</u> U8
<u>Ops Keyword</u> MODEL_COMPONENT_9 <u>Archive Attribute</u> TBD <u>Definition</u> Specifies the value of the component of the MODEL_COMPONENT_ID for the ninth element.	<u>Valid Values</u> n/a <u>Type</u> float array <u>Units</u> n/a <u>Location</u> GEOMETRIC_CAMERA_MODEL (Group) <u>X-path</u> TBD <u>Reference</u> TBD	<u>Mode</u> DPO in XML format (referenced to APID Name in Appendix A) <u>Field as "<xml name>:[<element>]:<field>"</u> <u>Eng. Cameras</u> "<IDPH DPO>:idph:cmod_mparm" <u>MCZ Cameras</u> "<IDPH DPO>:cidph:camera_model:mparm" <u>Type</u> F64
<u>Ops Keyword</u> MODEL_COMPONENT_ID <u>Archive Attribute</u> TBD <u>Definition</u> Specifies a sequence of identifiers (usually 1 character), where each identifier corresponds to a model component vector. It is used in conjunction with the MODEL_COMPONENT_n elements, where "n" is a number. The first id in the sequence corresponds to MODEL_COMPONENT_1, the second corresponds to MODEL_COMPONENT_2, etc. For example, for a geometric camera model with a value of "CAHV" for MODEL_TYPE, the MODEL_COMPONENT_ID would be (C, A, H, V). Please see the MODEL_COMPONENT_NAME data element for more details.	<u>Valid Values</u> 0 = "NONE" 1 = "(C,A,H,V)" 2 = "(C,A,H,V,O,R)" 3 = "(C,A,H,V,O,R,E,T,P)" <u>Type</u> string array <u>Units</u> n/a <u>Location</u> GEOMETRIC_CAMERA_MODEL (Group) <u>X-path</u> TBD <u>Reference</u> TBD	<u>Mode</u> DPO in XML format (referenced to APID Name in Appendix A) <u>Field as "<xml name>:[<element>]:<field>"</u> <u>Eng. Cameras</u> "<IDPH DPO>:idph:cmod_mclass" <u>MCZ Cameras</u> "<IDPH DPO>:cidph:camera_model:mclass" <u>Type</u> enum
<u>Ops Keyword</u> MODEL_COMPONENT_NAME <u>Archive Attribute</u> TBD <u>Definition</u> Specifies a sequence of names, where each name identifies its corresponding model component vector. It is used in conjunction with the MODEL_COMPONENT_n elements, where "n" is a number. The first name in the sequence identifies	<u>Valid Values</u> 0 = "NONE" 1 = ("CENTER", "AXIS", "HORIZONTAL", "VERTICAL") 2 = ("CENTER", "AXIS", "HORIZONTAL", "VERTICAL", "OPTICAL", "RADIAL") 3 = ("CENTER", "AXIS", "HORIZONTAL", "VERTICAL", "OPTICAL", "RADIAL", "ENTRANCE", "MTYPE", "MPARM") <u>Type</u> string array <u>Units</u> n/a	<u>Mode</u> DPO in XML format (referenced to APID Name in Appendix A) <u>Field as "<xml name>:[<element>]:<field>"</u> <u>Eng. Cameras</u> "<IDPH DPO>:idph:cmod_mclass" <u>MCZ Cameras</u> "<IDPH DPO>:cidph:camera_model:mclass" <u>Type</u> enum

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
MODEL_COMPONENT_1, the second identifies the MODEL_COMPONENT_2, etc. For example, for a geometric camera model with "CAHV" for MODEL_TYPE, the value for MODEL_COMPONENT_NAME would be (CENTER, AXIS, HORIZONTAL, VERTICAL). The three values of MODEL_COMPONENT_1 would describe the focal center vector; the three values of MODEL_COMPONENT_2 would describe the pointing direction (axis) vector; the three values of MODEL_COMPONENT_3 would describe the horizontal image plane vector, and the three values of the MODEL_COMPONENT_4 would describe the vertical image plane vector.	Location GEOMETRIC_CAMERA_MODEL (Group) X-path TBD Reference TBD	
Ops Keyword ^MODEL_DESC Archive Attribute TBD Definition Specifies a textual description of a model (or a pointer to a file containing the description). This is not intended to be a brief summary, but rather a detailed description of the model; at minimum, it should include a reference to a detailed description of the model in published literature. While other data elements such as CALIBRATION_SOURCE_ID, SOLUTION_ID, REFERENCE_COORD_SYSTEM_NAME, and MODEL_COMPONENT_NAME provide quick identifiers that distinguish how this model was generated, the details and data behind each of these identifiers should be explicitly included in the model description.	Valid Values "GEOMETRIC_CM.TXT" Type string Units n/a Location GEOMETRIC_CAMERA_MODEL (Group) X-path TBD Reference TBD	Mode Static Value Field as "<xml name>:[<element>]:[<element>]:<field>" n/a Type n/a
Ops Keyword MODEL_TRANSFORM_QUATERNION Archive Attribute none Definition Specifies, along with MODEL_TRANSFORM_VECTOR, the transform used for the camera model in this image. Camera models created by the calibration process have associated with them a pose, comprised of the position (offset) and orientation (quaternion) of the camera at the time it was calibrated. The model is transformed ("pointed") for a specific image by computing, generally using articulation device kinematics, a final pose for the image. The camera model is then translated and rotated from the calibration to final pose. This keyword specifies the quaternion portion of the final pose.	Valid Values n/a Type float array Units n/a Location GEOMETRIC_CAMERA_MODEL (Group) X-path TBD Reference TBD	Mode DPO in XML format (referenced to APID Name in Appendix A) Field as "<xml name>:[<element>]:[<element>]:<field>" 1) <u>Eng. Cameras & SuperCam RMI</u> "<IDPH DPO>:idph:cmmod_trans:quaternion[3]", "<IDPH DPO>:idph:cmmod_trans:quaternion[0]", "<IDPH DPO>:idph:cmmod_trans:quaternion[1]", "<IDPH DPO>:idph:cmmod_trans:quaternion[2]" 2) <u>Supercam</u> TBD 3) <u>MCZ Cameras</u> "<IDPH DPO>:cidph:transform:quaternion[3]", "<IDPH DPO>:cidph:transform:quaternion[0]", "<IDPH DPO>:cidph:transform:quaternion[1]", "<IDPH DPO>:cidph:transform:quaternion[2]" Type 1) F64[4] 2) TBD

Output Metadata (Product Label)		Input Metadata (Source)
• Ops Keyword • Archive Label Attribute • Definition	• Mission Phase C=Cruise, S=Surface, B=Both • Valid Values (quoted) • Type • Units • Keyword Location in Ops Label • Attribute X-path in Archive label • Attribute Reference in Archive Label (bracketed)	• Mission Phase C=Cruise, S=Surface, B=Both • Mode • Metadata Field • Type
NOTE: Due to a flight software bug, the above description does not apply to early images. Prior to FSW version 10.6, these fields contained not the final pose, but the calibration pose. This discrepancy is reflected in the PDS labels. If keyword FLIGHT_SOFTWARE_VERSION_ID contains a value less than or equal to 141801503 (which corresponds to FSW 10.5.7), this keyword contains the calibration pose. The final pose can be computed for mast-mounted instruments using pointing correction procedures. Reverse-engineering the final value may also be possible for non-mast instruments given the calibration value and the camera model (or vice-versa). See keyword POINTING_MODEL_NAME.		3) F32[4]
Ops Keyword MODEL_TRANSFORM_VECTOR Archive Attribute none Definition Specifies, along with MODEL_TRANSFORM_QUATERNION, the transform used for the camera model in this image. Camera models created by the calibration process have associated with them a pose, comprised of the position (offset) and orientation (quaternion) of the camera at the time it was calibrated. The model is transformed ("pointed") for a specific image by computing, generally using articulation device kinematics, a final pose for the image. The camera model is then translated and rotated from the calibration to final pose. This keyword specifies the offset portion of the final pose.	Valid Values n/a Type float array Units n/a Location GEOMETRIC_CAMERA_MODEL (Group) X-path TBD Reference TBD	Mode DPO in XML format (referenced to APID Name in Appendix A) Field as "<xml name>:[<element>]:<field>" 1) <u>Eng. Cameras</u> "<IDPH DPO>:idph:cmod_trans:position[0]", "<IDPH DPO>:idph:cmod_trans:position[1]", "<IDPH DPO>:idph:cmod_trans:position[2]" 2) <u>Supercam RMI</u> "<IDPH DPO>:idph:cmod_trans:position[0]", "<IDPH DPO>:idph:cmod_trans:position[1]", "<IDPH DPO>:idph:cmod_trans:position[2]" 3) <u>MCZ Cameras</u> "<IDPH DPO>:cidph:transform:position[0]", "<IDPH DPO>:cidph:transform:position[1]", "<IDPH DPO>:cidph:transform:position[2]" Type 1) F64[3] 2) F32[3] 3) F32[3]
Ops Keyword MODEL_TYPE Archive Attribute TBD	Valid Values 0 = "NONE" 1 = "CAHV" 2 = "CAHVOR" 3 = "CAHVORE"	Mode DPO in XML format (referenced to APID Name in Appendix A) Field as "<xml name>:[<element>]:<field>"

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
Definition Specifies an identifier for the type or kind of model. The value should be one of a well defined set, providing an application program with sufficient information to know how to handle the rest of the parameters within the model. (CAHVORE is the only one that uses model component vectors 1-9.) For details on the definitions of the valid camera model types, see [Ref 21] through [Ref 27].	Type string(63) Units n/a Location GEOMETRIC_CAMERA_MODEL (Group) X-path TBD Reference TBD	- Eng. Cameras "<IDPH DPO>:idph:cmmod_mclass" - MCZ Cameras "<IDPH DPO>:cidph:camera_model:mclass" Type enum
Ops Keyword ^MOSAIC_DESC Archive Attribute TBD Definition Specifies a brief textual description of a mosaic.	Valid Values n/a Type string Units n/a Location DERIVED_IMAGE_PARS (Group) X-path TBD Reference TBD	Mode Static Value Field as "<xml name>:[<element>]:[<element>]:<field>" n/a Type n/a
Ops Keyword NUM_SOFTWARE_KEYWORDS Archive Attribute TBD Definition Specifies the number of keywords that were supplied to the primary generating software named in SOFTWARE_MODULE_NAME. The Archive Attributes that contain the software keyword names, values and types (if any) will be provided in SOFTWARE_KEYWORD_NAME, SOFTWARE_KEYWORD_TYPE, and SOFTWARE_KEYWORD_VALUE and are required for each software keyword supplied. For M2020, this keyword may be placed in a xxx_HISTORY_PARS group, where "xxx" is the task name. There may be multiple groups to accommodate multiple software tasks.	Valid Values n/a Type integer Units n/a Location <xxx>_HISTORY_PARS (Group) where, xxx = RDR software name X-path TBD Reference TBD	Mode RDR-generating software Field as "<xml name>:[<element>]:[<element>]:<field>" n/a Type n/a
Ops Keyword NUM_SOFTWARE_PARAMETERS Archive Attribute TBD Definition	Valid Values n/a Type integer Units n/a	Mode RDR-generating software Field as "<xml name>:[<element>]:[<element>]:<field>" n/a Type

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
Specifies the number of parameters that were supplied to the primary generating software named in SOFTWARE_MODULE_NAME. The Archive Attributes that contain the software parameter names, values and types (if any) will be provided in SOFTWARE_PARAMETER_NAME, SOFTWARE_PARAMETER_TYPE, and SOFTWARE_PARAMETER_VALUE and are required for each software keyword supplied. For M2020, this keyword may be placed in a xxx_HISTORY_PARMS group, where "xxx" is the task name. There may be multiple groups to accommodate multiple software tasks.	Location <xxx>_HISTORY_PARMS (Group) where, xxx = RDR software name X-path TBD Reference TBD	n/a
Ops Keyword NUM_IMAGES_TRANSMITTED Archive Attribute TBD Definition Number of images transmitted to the RCE.	Valid Values n/a Type integer Units n/a Location MINI_HEADER_GROUP X-path TBD Reference TBD	SCAM Image DPO mini-header Field as "<xml name>:[<element>]:<field>" "SCAM_Image_Mini_Header[188]" Type U32 Notes: For SCAM RMI products, "num images transmitted" is comprised of 4 bytes coming from the Image DPO mini-header at offset 188.
Ops Keyword OBS_FROM_LIMIT_SWITCH Archive Attribute none Definition For SuperCam on M2020, when set to "1", specifies that the mast unit (MU) focus motor move to the limit switch position before moving to the specified focus position in the MANUAL or BASELINE focus options. For the AF_OFFSET focus option, it is ignored.	Valid Values n/a Type integer Units n/a Location OBSERVATION_REQUEST_PARMS (Group) X-path TBD Reference TBD	Mode DPO in XML format (referenced to APID Name in Appendix A) Field as "<xml name>:[<element>]:<field>" "<Ancillary DPO>:cmd_parameters:obsFromLimitSwitch" Type U8
Ops Keyword OBSERVATION_ID Archive Attribute TBD Definition Specifies a unique identifier for a scientific observation within	Valid Values "UNK" Type string Units n/a Location	Mode DPO in XML format (referenced to APID Name in Appendix A) Field as "<xml name>:[<element>]:<field>" n/a Type n/a

Output Metadata (Product Label)						Input Metadata (Source)					
- Ops Keyword - Archive Label Attribute - Definition						- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)					
a data set. It is set via the data product context ID - which doesn't necessarily map to a specific object - it's just used to group various instrument data sets together via a common keyword.						IDENTIFICATION (Class) <u>X-path</u> TBD <u>Reference</u> TBD					
Ops Keyword OFFSET_MODE_ID Archive Attribute TBD Definition Specifies the analog value that is subtracted from the video signal prior to the analog/digital converters. For M2020, the value is a digital number (DN) which represents the offset used in a camera-specific manner. For M2020 SuperCam RMI, it specifies the commanded analog offset in the SuperCam Mast Unit (CCMU), in the range 0-255. See also GAIN_NUMBER. For M2020 MCZ cameras, it is the DC Offset as reported in the MCZ mini-header. For M2020 engineering cameras, this value corresponds to a lookup table that defines a specific readout mode and portion on the detector. The table has a range of 0-35 as defined below. Each tile is 960 lines by 1280 samples.						Valid Values <u>Eng. Cameras</u> (C) "0" to "35" <u>SuperCam</u> (S) "0" to "4095" <u>MCZ Cameras</u> (B) a. "UNK" b. "0" to "4095" Type string Units n/a Location <u>Eng. Cameras</u> INSTRUMENT_STATE_PARMs (Group) <u>SuperCam</u> OBSERVATION_REQUEST_PARMs (Group) <u>MCZ</u> a. INSTRUMENT_STATE_PARMs (Group) b. MINI_HEADER (Group) <u>Skycam</u> INSTRUMENT_STATE_PARMs (Group) <u>X-path</u> TBD <u>Reference</u> TBD					
Table ID	FPGA Mode	Bayer Channel	Binning	Start Row	Stop Row	Mode 1) DPO in XML format (referenced to APID Name in Appendix A) 2) DPO in XML format (referenced to APID Name in Appendix A) 3) Image DPO mini-header Field as "<xml name>:[<element>]:<field>" <u>Eng. Cameras</u> "<IDPH DPO>:idph:voff" <u>SuperCam</u> "<Ancillary DPO>:cmd_parameters:AD_offset" TBD <u>MCZ Cameras</u> "MCZ_Image_Mini_Header[52]", "MCZ_Image_Mini_Header[53]", "MCZ_Image_Mini_Header[54]", "MCZ_Image_Mini_Header[55]" <u>Skycam</u> "<MEDA Image Header>:skycam_vid_offset_lowbyte_data" + "<MEDA Image Header>:skycam_vid_offset_highbyte_data" NOTES: - For MCZ Cameras, value analogous to "DC_offset" is comprised of four bytes coming from Image DPO mini-header at byte offsets 52, 53, 54 and 55. - For EECAM Cruise, voff values map to a predefined tiling and readout mode. Type <u>Eng. Cameras</u> U16 <u>SuperCam</u> <u>MCZ</u> <u>Skycam</u> U8					
0	0	Bayer	None	0	959						
1	0	Bayer	None	0	959						
2	0	Bayer	None	0	959						
3	0	Bayer	None	0	959						
4	0	Bayer	None	960	1919						
5	0	Bayer	None	960	1919						
6	0	Bayer	None	960	1919						
7	0	Bayer	None	960	1919						
8	0	Bayer	None	1920	2879						
9	0	Bayer	None	1920	2879						
10	0	Bayer	None	1920	2879						
11	0	Bayer	None	1920	2879						
12	0	Bayer	None	2880	3839						
13	0	Bayer	None	2880	3839						
14	0	Bayer	None	2880	3839						
15	0	Bayer	None	2880	3839						
16	2	Red	2x2	0	959						
17	2	Red	2x2	0	959						
18	2	Red	2x2	1920	2879						
19	2	Red	2x2	1920	2879						
20	1	Green	2x2	0	959						
21	1	Green	2x2	0	959						
22	1	Green	2x2	1920	2879						
23	1	Green	2x2	1920	2879						
24	3	Blue	2x2	0	959						
25	3	Blue	2x2	0	959						
26	3	Blue	2x2	1920	2879						
27	3	Blue	2x2	1920	2879						
28	4	Panchro	2x2	0	959						
29	4	Panchro	2x2	0	959						

Output Metadata (Product Label)						Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition						- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
30	4	Panchro	2x2	1920	0	
31	4	Panchro	2x2	1920	2560	
32	5	Red	4x4	0	0	
33	6	Green	4x4	0	0	
34	7	Blue	4x4	0	0	
35	8	Panchro	4x4	0	0	
Ops Keyword ODL_VERSION_ID Archive Attribute TBD Definition Specifies the version number of the PDS standards document used for an ODL format file. ODL3 files conform to the PDS3 standards document structurally and syntactically, the only difference being that keywords do not necessarily need to be defined in a PDS Data Dictionary Examples: ODL3						Valid Values "ODL<version>" Type string[6] Units n/a Location IDENTIFICATION (Class) X-path TBD Reference TBD
Ops Keyword ORIGIN_OFFSET_VECTOR Archive Attribute TBD Definition Specifies the offset from the reference coordinate system's origin to the origin of the coordinate system being defined by the enclosing COORDINATE_SYSTEM group. In other words, it is the location of the current system's origin as measured in the reference system. For M2020, here is an example: In the case of the RSM_COORDINATE_SYSTEM group, ORIGIN_OFFSET_VECTOR describes the rotation of the RSM (camera head) boresight (about the ORIGIN_OFFSET_VECTOR) relative to the Rover frame.						Valid Values n/a Type float array[3] Units meters Location - For EDRs ... - SITE_COORDINATE_SYSTEM (Group) - ROVER_COORDINATE_SYSTEM (Group) - RSM_COORDINATE_SYSTEM (Group) - ARM_COORDINATE_SYSTEM (Group) - TURRET_COORDINATE_SYSTEM (Group) - INITIAL_STATE_PARMS (Group) - TOOL_COORDINATE_SYSTEM (Group) X-path TBD Reference TBD
						Mode - DPO in XML format (referenced to APID Name in Appendix A) Field as "<xml name>:[<element>]:<field>" Surface - Eng. Cameras (S) "<IDPH DPO:idph:site_p[0]" "<IDPH DPO:idph:site_p[1]" "<IDPH DPO:idph:site_p[2]" "<IDPH DPO>:idph:rvr_p[0]" "<IDPH DPO>:idph:rvr_p[1]" "<IDPH DPO>:idph:rvr_p[2]" "<IDPH DPO>:idph:rsm_p[0]" "<IDPH DPO>:idph:rsm_p[1]" "<IDPH DPO>:idph:rsm_p[2]" "<IDPH DPO>:idph:arm:tool_pos[0]" "<IDPH DPO>:idph:arm:tool_pos[1]" "<IDPH DPO>:idph:arm:tool_pos[2]" "<IDPH DPO>:idph:arm:turret_pos[0]" "<IDPH DPO>:idph:arm:turret_pos[1]" "<IDPH DPO>:idph:arm:turret_pos[2]" "<IDPH DPO>:idph:rvr_p0[0]" "<IDPH DPO>:idph:rvr_p0[1]" "<IDPH DPO>:idph:rvr_p0[2]" "<IDPH DPO>:idph:arm:tool_pos[0]" "<IDPH DPO>:idph:arm:tool_pos[1]" "<IDPH DPO>:idph:arm:tool_pos[2]" - SuperCam (S) - MCZ (S) - SHERLOC (S) - MEDA SkyCam (S) - PIXL (S)

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
		2) "<IDPH DPO>:idph:system:sapp_p[0]" "<IDPH DPO>:idph:system:sapp_p[1]" "<IDPH DPO>:idph:system:sapp_p[2]" 3) "<IDPH DPO>:idph:rsm:p[0]" "<IDPH DPO>:idph:rsm:p[1]" "<IDPH DPO>:idph:rsm:p[2]" 4) "<IDPH DPO>:idph:arm:tool_pos[0]" "<IDPH DPO>:idph:arm:tool_pos[1]" "<IDPH DPO>:idph:arm:tool_pos[2]" 5) "<IDPH DPO>:idph:arm:turret_pos[0]" "<IDPH DPO>:idph:arm:turret_pos[1]" "<IDPH DPO>:idph:arm:turret_pos[2]" 7) "<IDPH DPO>:idph:arm:tool_pos[0]" "<IDPH DPO>:idph:arm:tool_pos[1]" "<IDPH DPO>:idph:arm:tool_pos[2]" Cruise MCZ (C) 2) "<IDPH DPO>:cidph:rvr:position[3]", 3) "<IDPH DPO>:cidph:rsm_state:position[3]", Type F32[3]
Ops Keyword ORIGIN_ROTATION_QUATERNION Archive Attribute TBD Definition Specifies an array of four values that specifies the rotation of the coordinate system being defined by the enclosing COORDINATE_SYSTEM group, relative to the reference system. Mathematically this can be expressed as follows: Given a vector expressed in the current frame, multiplication by this quaternion will give the same vector as expressed in the reference frame. Quaternions are expressed as a set of four numbers in the order: (s, v1, v2, v3) where, $s = \cos(\theta/2)$ $v(n) = \sin(\theta/2) \cdot a(n)$ theta = the angle of rotation a = (x,y,z) vector around which rotation occurs Note that quaternions have different component order conventions between flight and ground software. They are received in the order "(v1, v2, v3, s)". However, the ground order convention is "(s, v1, v2, v3)", and all values are converted to the ground order before being stored in the label.	Valid Values n/a Type float array[4] Units meters Location - EDRs a. SITE_COORDINATE_SYSTEM (Group) b. ROVER_COORDINATE_SYSTEM (Group) c. RSM_COORDINATE_SYSTEM (Group) d. ARM_COORDINATE_SYSTEM (Group) e. INITIAL_STATE_PARMS (Group) f. TURRET_COORDINATE_SYSTEM (Group) g. TOOL_COORDINATE_SYSTSEM (Group) X-path TBD Reference TBD	Mode - DPO in XML format (referenced to APID Name in Appendix A) Field as "<xml name>[:<element>]:<field>" Surface Eng. Cameras (S) a. "<IDPH DPO>:idph:site_q[3]" "<IDPH DPO>:idph:site_q[0]" "<IDPH DPO>:idph:site_q[1]" "<IDPH DPO>:idph:site_q[2]" b. "<IDPH DPO>:idph:rvr_q[3]" "<IDPH DPO>:idph:rvr_q[0]" "<IDPH DPO>:idph:rvr_q[1]" "<IDPH DPO>:idph:rvr_q[2]" c. "<IDPH DPO>:idph:rsm_q[3]" "<IDPH DPO>:idph:rsm_q[0]" "<IDPH DPO>:idph:rsm_q[1]" "<IDPH DPO>:idph:rsm_q[2]" d. "<IDPH DPO>:idph:arm:tool_quat[3]" "<IDPH DPO>:idph:arm:tool_quat[0]" "<IDPH DPO>:idph:arm:tool_quat[1]" "<IDPH DPO>:idph:arm:tool_quat[2]" e. "<IDPH DPO>:idph:rvr_q0[3]" "<IDPH DPO>:idph:rvr_q0[0]" "<IDPH DPO>:idph:rvr_q0[1]" "<IDPH DPO>:idph:rvr_q0[2]" f. "<IDPH DPO>:idph:arm:turret_quat[3]" "<IDPH DPO>:idph:arm:turret_quat[0]" "<IDPH DPO>:idph:arm:turret_quat[1]" "<IDPH DPO>:idph:arm:turret_quat[2]" g. "<IDPH DPO>:idph:arm:tool_quat[3]" "<IDPH DPO>:idph:arm:tool_quat[0]" "<IDPH DPO>:idph:arm:tool_quat[1]" "<IDPH DPO>:idph:arm:tool_quat[2]"

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
For M2020, the value for ORIGIN_ROTATION_QUATERNION that defines a coordinate frame like Rover frame is computed with respect to only the orientations of the frame's axes... regardless of whether POSITIVE_ELEVATION_DIRECTION is declared to be "UP" or "DOWN". For M2020, here is an example: In the case of the RSM_COORDINATE_SYSTEM group, ORIGIN_ROTATION_QUATERNION describes the rotation of the RSM (camera head on Mast) boresight (about the ORIGIN_OFFSET_VECTOR) relative to the Rover frame.		<u>Supercam</u> (S) <u>MCZ</u> (S) <u>SHERLOC</u> (S) <u>MEDA_SkyCam</u> (S) <u>PIXL_MCC</u> (S) b. "<IDPH DPO>:idph:sapp_q[3]" "<IDPH DPO>:idph:sapp_q[0]" "<IDPH DPO>:idph:sapp_q[1]" "<IDPH DPO>:idph:sapp_q[2]" c. "<IDPH DPO>:idph:rsm_q[3]" "<IDPH DPO>:idph:rsm_q[0]" "<IDPH DPO>:idph:rsm_q[1]" "<IDPH DPO>:idph:rsm_q[2]" d. "<IDPH DPO>:idph:arm:tool_quat[3]" "<IDPH DPO>:idph:arm:tool_quat[0]" "<IDPH DPO>:idph:arm:tool_quat[1]" "<IDPH DPO>:idph:arm:tool_quat[2]" e. "<IDPH DPO>:idph:rvr_q0[3]" "<IDPH DPO>:idph:rvr_q0[0]" "<IDPH DPO>:idph:rvr_q0[1]" "<IDPH DPO>:idph:rvr_q0[2]" f. "<IDPH DPO>:idph:arm:turret_quat[3]" "<IDPH DPO>:idph:arm:turret_quat[0]" "<IDPH DPO>:idph:arm:turret_quat[1]" "<IDPH DPO>:idph:arm:turret_quat[2]" g. "<IDPH DPO>:idph:arm:tool_quat[3]" "<IDPH DPO>:idph:arm:tool_quat[0]" "<IDPH DPO>:idph:arm:tool_quat[1]" "<IDPH DPO>:idph:arm:tool_quat[2]" Cruise <u>MCZ Cameras</u> b. "<IDPH DPO>:cidph:rvr:quaternion[3]", "<IDPH DPO>:cidph:rvr:quaternion[0]", "<IDPH DPO>:cidph:rvr:quaternion[1]", "<IDPH DPO>:cidph:rvr:quaternion[2]" c. "<IDPH DPO>:cidph:rsm_state:quaternion[3]", "<IDPH DPO>:cidph:rsm_state:quaternion[0]", "<IDPH DPO>:cidph:rsm_state:quaternion[1]", "<IDPH DPO>:cidph:rsm_state:quaternion[2]" Type 1) F32[4] 2) F32[4] 3) F32[4]
Ops Keyword PDS_VERSION_ID Archive Attribute TBD Definition Specifies the version number of the PDS standards document that is valid when a data product label is created. Values for the PDS_version_id are formed by appending the integer for the latest version number to the letters 'PDS'.	Valid Values "PDS<version>" Type string[6] Units n/a Location IDENTIFICATION (Class) X-path TBD	Mode PDS Field as "<xml name>:[<element>]:[<element>]:<field>" n/a Type n/a

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
Examples: PDS3	<u>Reference</u> TBD	
<u>Ops Keyword</u> PIXEL_AVERAGING_HEIGHT <u>Archive Attribute</u> TBD <u>Definition</u> Specifies the vertical dimension, in pixels, of the area over which pixels were averaged prior to image compression. For M2020 EECAM, the value is determined by the hardware mode and the reported downsampling in the idph. See INSTRUMENT_MODE_ID (Surface) and OFFSET_MODE_ID (Cruise) for the full description of hardware readout modes.	<u>Valid Values</u> <u>Eng. Cameras</u> (B) 1) "1" to "64" <u>MCZ Cameras</u> (B) 2) a. "UNK" b. non-Thumbnail = "1" Thumbnail = "8" <u>Type</u> integer <u>Units</u> pixel <u>Location</u> 1) a. IMAGE_REQUEST_PARMs (Group) b. INSTRUMENT_STATE_PARMs (Group) 2) a. IMAGE_REQUEST_PARMs (Group) b. INSTRUMENT_STATE_PARMs (Group) <u>X-path</u> TBD <u>Reference</u> TBD	<u>Mode</u> 1) DPO in XML format (referenced to APID Name in Appendix A) 2) EMD in XML format, calculation <u>Field as "<xml name>[:<element>[:<element>[:<field>"] Surface</u> <u>Eng. Cameras</u> 1) a. "<IDPH DPO>:idph:params:res_rows" b. "<IDPH DPO>:idph:res_rows" <u>MCZ Cameras</u> 2) "M20EarthProductMetadata:M20ProductMetadata:ProductName" <u>Cruise</u> <u>Eng. Cameras</u> 1) a. "<IDPH DPO>:idph:params:res_rows" b. "<IDPH DPO>:idph:res_rows" * "<IDPH DPO>:idph:voff (binning value)" NOTES: - For MCZ Cameras, if product type is Thumbnail, set value to "8". Otherwise, set value to "1". <u>Type</u> U16
<u>Ops Keyword</u> PIXEL_AVERAGING_WIDTH <u>Archive Attribute</u> TBD <u>Definition</u> Specifies the horizontal dimension, in pixels, of the area over which pixels were averaged prior to image compression. For M2020 EECAM, the value is determined by the hardware mode and the reported downsampling in the idph. See INSTRUMENT_MODE_ID (Surface) and OFFSET_MODE_ID (Cruise) for the full description of hardware readout modes.	<u>Valid Values</u> <u>Eng. Cameras</u> 1) "1" to "64" <u>MCZ Cameras</u> 2) a. "UNK" b. non-Thumbnail = "1" Thumbnail = "8" <u>Type</u> integer <u>Units</u> pixel <u>Location</u> 1) a. IMAGE_REQUEST_PARMs (Group) b. INSTRUMENT_STATE_PARMs (Group) 2) a. IMAGE_REQUEST_PARMs (Group) b. INSTRUMENT_STATE_PARMs (Group) <u>X-path</u> TBD <u>Reference</u> TBD	<u>Mode</u> 1) DPO in XML format (referenced to APID Name in Appendix A) 2) EMD in XML format, calculation <u>Field as "<xml name>[:<element>[:<element>[:<field>"] Surface</u> <u>Eng. Cameras</u> 1) a. "<IDPH DPO>:idph:params:res_cols" * b. "<IDPH DPO>:idph:res_cols" * <u>MCZ Cameras</u> 2) "M20EarthProductMetadata:M20ProductMetadata:ProductName" <u>Cruise</u> <u>Eng. Cameras</u> 1) a. "<IDPH DPO>:idph:params:res_cols" b. "<IDPH DPO>:idph:res_cols" * "<IDPH DPO>:idph:voff (binning value)" NOTES: - For MCZ Cameras, if product type is Thumbnail, set value to "8". Otherwise, set value to "1". <u>Type</u>

Output Metadata (Product Label)		Input Metadata (Source)						
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type						
		U16						
Ops Keyword PIXEL_DOWNSAMPLE_OPTION Archive Attribute TBD Definition Specifies whether to downsample the image(s), and if so, which pixel resolution downsample method to use. Note that for M2020, the camera hardware could downsample entire rows 4-to-1, but software had to be used to do additional row-wise downsampling and any column downsampling. For M2020, the meaning of the valid values follows: a) "SW_MEAN" - Downsampling done in <u>software</u> by calculation of the mean. b) "HW_COND" - Use <u>hardware</u> binning if downsampling (by mean calculation) and subframe arguments are consistent. c) "HW_SW" - Use <u>hardware</u> binning by changing the commanded downsampling and subframe arguments to be consistent with hardware binning. Any subsequent downsampling is done in <u>software</u> by calculation of the mean. d) "SW_OUTREJ" -<u>Software</u> pixel averaging with outlier rejection. The pixel whose value lies farthest away from the mean of the sample is rejected. e) "SW_MEDIAN" - <u>Software</u> downsampling done by calculation of the median rather than the mean.	Valid Values <u>Eng. Cameras</u> (B) 0 = "NONE" 1 = "SW_MEAN" 2 = "HW_COND" 3 = "HW_SW" 4 = "SW_OUTREJ" 5 = "SW_MEDIAN" <u>MCZ Cameras</u> "UNK" Type string Units n/a Location IMAGE_REQUEST_PARMS (Group) X-path TBD Reference TBD	Mode DPO in XML format (referenced to APID Name in Appendix A) Field as "<xml name>:[<element>]:<field>" <u>Eng. Cameras</u> "<IDPH DPO>:idph:params:resolution" TBD Type enum						
Ops Keyword PLANET_DAY_NUMBER Archive Attribute TBD Definition Specifies the number of solar days elapsed since a reference day (e.g., the day on which a landing vehicle set down) for local mean solar time (LMST). Days are measured in rotations of the planet in question from midnight to midnight. For M2020, the reference day is "0", as Landing day is Sol 0. If before Landing day, value will be less than or equal to "0". See also LOCAL_MEAN_SOLAR_TIME and LOCAL_TRUE_SOLAR_TIME_SOL.	Valid Values <table><thead><tr><th>Mission Phase</th><th>Values</th></tr></thead><tbody><tr><td>Cruise</td><td>less than or equal to 0</td></tr><tr><td>Surface</td><td>"0" to n</td></tr></tbody></table> NOTE: Value will be uncalibrated if SPICE kernels are unavailable. Type integer Units n/a Location IDENTIFICATION (Class) X-path TBD Reference	Mission Phase	Values	Cruise	less than or equal to 0	Surface	"0" to n	Mode 1) a. Calculation: - SCLK Kernel b. DPO in XML format (referenced to APID Name in Appendix A) 2) a. Image DPO mini-header, Calculation: - SCLK Kernel b. DPO in XML format (referenced to APID Name in Appendix A) Field as "<xml name>:[<element>]:<field>" <u>Eng. Cameras</u> 1) b. "<IDPH DPO>:idph:sclk_seconds", "<IDPH DPO>:idph:sclk_subseconds" <u>MCZ Cameras</u> 2) a. "MCZ_Image_Mini_Header[8]", "MCZ_Image_Mini_Header[9]", "MCZ_Image_Mini_Header[10]", "MCZ_Image_Mini_Header[11]" b. "<IDPH DPO>:cidph:sclk:seconds"
Mission Phase	Values							
Cruise	less than or equal to 0							
Surface	"0" to n							

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
	TBD	c. "<Ancillary DPO>:sclk:seconds" NOTES: - For MCZ recovered or Z-stack data products, value is comprised of four bytes coming from Image DPO mini-header at byte offsets 8, 9, 10 and 11 per case "2a". - For MCZ non-recovered and non-Z-stack data products, if IDPH is present, value comes from IDPH DPO per case "2b". - For MCZ non-recovered and non-Z-stack data products, if IDPH is not present, value comes from Ancillary DPO per case "2c". Type U32
Ops Keyword POINTING_CORRECTION_FILE_NAME Archive Attribute TBD Definition Specifies the name of the file containing pointing correction parameters for each input file of a mosaic. The file is in XML format, and is often called a "nav" file in operations. See also SOLUTION_ID.	Valid Values n/a Type string Units n/a Location DERIVED_IMAGE_PARMS (Group) X-path TBD Reference TBD	Mode RDR-generating software Field as "<xml name>:[<element>]:[<element>]:<field>" n/a Type n/a
Ops Keyword POINTING_MODEL_NAME Archive Attribute TBD Definition Specifies which of several "pointing models" were used to transform the camera model based on updated pointing information. These updates are typically derived from mosaic seam corrections. This keyword and the associated POINTING_MODEL_PARAMS keyword define what the updated pointing information is, providing enough information to re-create the camera model from calibration data. If present, this keyword overrides the default pointing based on telemetry. The special value "NONE" shall be interpreted the same as if the keyword is absent (i.e. the default pointing model should be used).	Valid Values TBD Type string Units n/a Location DERIVED_IMAGE_PARMS (Group) X-path TBD Reference TBD	Mode RDR-generating software Field as "<xml name>:[<element>]:[<element>]:<field>" n/a Type n/a

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
New model names can be created at any time; the models themselves should be described in the camera model ancillary file "M2020_pointing_models.txt" included in the Archive Volume. See also SOLUTION_ID.		
Ops Keyword POINTING_MODEL_PARAMS Archive Attribute TBD Definition Specifies the numeric parameters needed by the pointing model identified in POINTING_MODEL_NAME. The meaning of any given parameter is defined by the pointing model.	Valid Values n/a Type float Units n/a Location DERIVED_IMAGE_PARAMS (Group) X-path TBD Reference TBD	Mode RDR-generating software Field as "<xml name>:[<element>]:<field>" n/a Type n/a
Ops Keyword POINTING_MODEL_PARAMS_NAME Archive Attribute TBD Definition This keyword specifies the name assigned by the pointing model for each parameter specified in POINTING_MODEL_PARAMS.	Valid Values Pointing Model order) TBD Type string array Units n/a Location DERIVED_IMAGE_PARAMS (Group) X-path TBD Reference TBD	Mode RDR-generating software Field as "<xml name>:[<element>]:<field>" n/a Type n/a
Ops Keyword POSITIVE_AZIMUTH_DIRECTION Archive Attribute TBD Definition Specifies the direction in which azimuth is measured in positive degrees for an observer on the surface of a body. The azimuth is measured with respect to the elevational reference plane. A value of CW indicates that azimuth is measured positively Clockwise, and CCW indicates that azimuth increases positively Counter-clockwise. For the M2020 operational coordinate frames, which follow the Mars Pathfinder convention, increasing	Valid Values "CLOCKWISE" Type string Units n/a Location - For EDRs ... - ROVER_COORDINATE_SYSTEM (Group) - RSM_COORDINATE_SYSTEM (Group) - ARM_COORDINATE_SYSTEM (Group) - SITE_DERIVED_GEOMETRY_PARAMS (Group) - For RDRs ...	Mode Static Value: - Determined by Coordinate Frame definitions Field as "<xml name>:[<element>]:<field>" n/a Type n/a

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
azimuth moves in a clockwise ("CLOCKWISE") direction as viewed from above. NOTE: For M2020, early in the mission the value for only the instance of keyword SOLAR_AZIMUTH in label group SITE_DERIVED_GEOMETRY_PARMS was being incorrectly reported as the inverse of what it should have been. For the Ops (ODL) label, the value is flagged "correct" if POSITIVE_AZIMUTH_DIRECTION resides in the SITE_DERIVED_GEOMETRY_PARMS label group. For the PDS label, the value is correct for all files as of archive Volume 2. The value for the instance of SOLAR_AZIMUTH in the label group ROVER_DERIVED_GEOMETRY_PARMS is always correct for the Ops and PDS labels.	- ROVER_COORDINATE_SYSTEM (Group) - RSM_COORDINATE_SYSTEM (Group) - ARM_COORDINATE_SYSTEM (Group) - SITE_COORDINATE_SYSTEM (Group) - LOCAL_LEVEL_COORDINATE_SYSTEM (Group) - SITE_DERIVED_GEOMETRY_PARMS (Group) X-path TBD Reference TBD	
Ops Keyword POSITIVE_ELEVATION_DIRECTION Archive Attribute TBD Definition Specifies the direction in which elevation is measured in positive degrees for an observer on the surface of a body. The elevation is measured with respect to the azimuthal reference plane. A value of "UP" indicates that elevation is measured positively upwards, i.e., the zenith point would be at +90 degrees and the nadir point at -90 degrees. "DOWN" indicates that the elevation is measured positively downwards; the zenith point would be at -90 degrees and the nadir point at +90 degrees. For the M2020 operational coordinate frames, which follow the Mars Pathfinder convention, increasing elevation ("UP") moves towards the negative Z axis.	Valid Values "UP" Type string Units n/a Location - For EDRs ... - ROVER_COORDINATE_SYSTEM (Group) - RSM_COORDINATE_SYSTEM (Group) - ARM_COORDINATE_SYSTEM (Group) - For RDRs ... - ROVER_COORDINATE_SYSTEM (Group) - RSM_COORDINATE_SYSTEM (Group) - ARM_COORDINATE_SYSTEM (Group) - SITE_COORDINATE_SYSTEM (Group) - LOCAL_LEVEL_COORDINATE_SYSTEM (Group) X-path TBD Reference TBD	Mode Static Value: - Determined by Coordinate Frame definitions Field as "<xml name>:[<element>]:[<element>]:<field>" n/a Type n/a
Ops Keyword PROCESSING_HISTORY_TEXT Archive Attribute TBD Definition Specifies an entry for each processing step and program used in generating a particular data file.	Valid Values "CODMAC LEVEL 1 TO LEVEL 2 CONVERSION VIA JPL/MIPL M2020EDRGEN" Type string Units n/a	Mode Dependent on EDR/RDR state: - For EDRs, static - For RDRs, RDR-generating software Field as "<xml name>:[<element>]:[<element>]:<field>" n/a

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
	Location - For EDRs ... - M2020EDRGEN_HISTORY_PARMS (Group) - For RDRs ... <xxx>_HISTORY_PARMS (Group) - where, xxx = RDR software name X-path TBD Reference TBD	Type n/a
Ops Keyword PROCESSING_INFO Archive Attribute TBD Definition Specifies information about the processing used to generate the RDR that is not covered by other PDS label keywords.	Valid Values n/a Type string array Units n/a Location DERIVED_IMAGE_PARMS (Group) X-path TBD Reference TBD	Mode RDR-generating software Field as “<xml name>:[<element>]:[<field>” n/a Type n/a
Ops Keyword PRODUCER_INSTITUTION_NAME Archive Attribute TBD Definition Specifies the identity of a university, research center, NASA center or other institution associated with the production of a data set. This would generally be an institution associated with the element PRODUCER_FULL_NAME.	Valid Values “MULTIMISSION INSTRUMENT PROCESSING LAB, JET PROPULSION LAB” Type string(60) Units n/a Location IDENTIFICATION (Class) X-path TBD Reference TBD	Mode Static Value Field as “<xml name>:[<element>]:[<field>” n/a Type n/a
Ops Keyword PRODUCT_COMPLETION_STATUS Archive Attribute TBD Definition Specifies the completion status of a product, specifying for example, if all portions have been downlinked and received correctly, if it is a partial product, or if it contains transmission errors. The specific valid values	Valid Values “PARTIAL”, “PARTIAL_CHECKSUM_FAIL”, “COMPLETE_CHECKSUM_PASS”, “COMPLETE_NO_CHECKSUM”, “COMPLETE_CHECKSUM_FAIL” Type string Units n/a	Mode EMD in XML format Field as “<xml name>:[<element>]:[<field>” “M20EarthProductMetadata:M20ProductMetadata:GroundStatus” Type n/a

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
may be mission-dependent. For M2020, the valid values indicate whether it was a complete or partial product as it came out of MPCS, and whether the checksum passed, failed, or was missing.	<u>Location</u> TELEMETRY (Class) <u>X-path</u> TBD <u>Reference</u> TBD	
<u>Ops Keyword</u> PRODUCT_CREATION_TIME <u>Archive Attribute</u> TBD <u>Definition</u> Specifies the UTC system format for the time when a product was created.	<u>Valid Values</u> <YYYY>-<MM>-<DD>T<hh>:<mm>:<ss>[.<fff>] <u>Type</u> time <u>Units</u> n/a <u>Location</u> IDENTIFICATION (Class) <u>X-path</u> TBD <u>Reference</u> TBD	<u>Mode</u> Calculation <u>Field as “<xml name>[:<element>[:<element>]:<field>”</u> n/a <u>Type</u> n/a
<u>Ops Keyword</u> PRODUCT_ID <u>Archive Attribute</u> TBD <u>Definition</u> Specifies a permanent, unique identifier assigned to a data product by its producer. For M2020, it is the filename minus the extension. NOTES: In the PDS, the value assigned to product_id must be unique within its data set. The PRODUCT_ID can describe the lowest-level Data object that has a PDS label.	<u>Valid Values</u> n/a <u>Type</u> string(40) <u>Units</u> n/a <u>Location</u> IDENTIFICATION (Class) <u>X-path</u> TBD <u>Reference</u> TBD	<u>Mode</u> - EDR-generating software: - Filename of output EDR product, minus extension - RDR-generating software: - Filename of output RDR product, minus extension <u>Field as “<xml name>[:<element>[:<element>]:<field>”</u> n/a <u>Type</u> n/a
<u>Ops Keyword</u> PRODUCT_TAG <u>Archive Attribute</u> TBD <u>Definition</u> Data Product Tag. NOTE: For M2020, use of this tag is defined separately for Individual product types. It is anticipated that this tag	<u>Valid Values</u> n/a <u>Type</u> string <u>Units</u> n/a <u>Location</u> TELEMETRY (Class) <u>X-path</u> TBD	<u>Mode</u> EMD in XML format <u>Field as “<xml name>[:<element>[:<element>]:<field>”</u> “M20EarthProductMetadata:M20ProductMetadata:ProductTag” <u>Type</u> n/a

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
may be used to associate multiple products for later processing; it may also be used to indicate instrument FSW versions, or other uses.	<u>Reference</u> TBD	
Ops Keyword PRODUCT_VERSION_ID Archive Attribute TBD Definition Specifies the version of an individual product within a data set. PRODUCT_VERSION_ID is intended for use within AMMOS to identify separate iterations of a given product, which will also have a unique FILE_NAME. For M2020, this is a Version field that begins with "V" followed by a decimal number denoting the product's iteration (i.e., version). Example: "V2.0" When appearing in the attached ODL (operations) label, keyword value represents the version incorporated in the Version field of the product's filename. This does <u>not</u> correspond to the keyword value appearing in the detached PDS (archive) label.	Valid Values "V<vernum>" Type string(12) Units n/a Location IDENTIFICATION (Class) X-path TBD Reference TBD	Mode User specified parameter value Field as "<xml name>:[<element>]:[<element>]:<field>" n/a Type n/a
Ops Keyword PROJECTION_AXIS_OFFSET Archive Attribute TBD Definition Specifies an offset from a projection axis in a map projection. For the Cylindrical-Perspective projection, this is the radius of a circle which represents the rotation around the projection origin of the synthetic camera used to calculate each column.	Valid Values n/a Type float Units meters (<m> unit tag required) Location SURFACE_PROJECTION_PARMS (Group) X-path TBD Reference TBD	Mode RDR-generating software Field as "<xml name>:[<element>]:[<element>]:<field>" n/a Type n/a
Ops Keyword PROJECTION_AZIMUTH Archive Attribute TBD Definition	Valid Values n/a Type float Units deg (<deg> unit tag required)	Mode RDR-generating software Field as "<xml name>:[<element>]:[<element>]:<field>" n/a Type n/a

Output Metadata (Product Label)		Input Metadata (Source)
• Ops Keyword • Archive Label Attribute • Definition	• Mission Phase C=Cruise, S=Surface, B=Both • Valid Values (quoted) • Type • Units • Keyword Location in Ops Label • Attribute X-path in Archive label • Attribute Reference in Archive Label (bracketed)	• Mission Phase C=Cruise, S=Surface, B=Both • Mode • Metadata Field • Type
Specifies the azimuth, in degrees, of the horizontal center of projection for the PERSPECTIVE projection (loosely, where the camera model is pointing).	<u>Location</u> SURFACE_PROJECTION_PARMS (Group) <u>X-path</u> TBD <u>Reference</u> TBD	
<u>Ops Keyword</u> PROJECTION_ELEVATION <u>Archive Attribute</u> TBD <u>Definition</u> Specifies the elevation, in degrees, of the vertical center of projection (loosely, where the camera is pointing). For Perspective projections, this applies to the single output camera model; for Cylindrical-Perspective it applies to each column's output camera model, before the rotation specified by PROJECTION_AXIS_Z_VECTOR.	<u>Valid Values</u> n/a <u>Type</u> float <u>Units</u> deg (<deg> unit tag required) <u>Location</u> SURFACE_PROJECTION_PARMS (Group) <u>X-path</u> TBD <u>Reference</u> TBD	<u>Mode</u> RDR-generating software <u>Field as “<xml name>:<element>:<element>:<field>”</u> n/a <u>Type</u> n/a
<u>Ops Keyword</u> PROJECTION_ELEVATION_LINE <u>Archive Attribute</u> TBD <u>Definition</u> Specifies the image line which corresponds to PROJECTION_ELEVATION for each column of the Cylindrical-Perspective projection, before the rotation specified by PROJECTION_AXIS_Z_VECTOR.	<u>Valid Values</u> n/a <u>Type</u> float <u>Units</u> pixel (<pixel> unit tag required) <u>Location</u> SURFACE_PROJECTION_PARMS (Group) <u>X-path</u> TBD <u>Reference</u> TBD	<u>Mode</u> RDR-generating software <u>Field as “<xml name>:<element>:<element>:<field>”</u> n/a <u>Type</u> n/a
<u>Ops Keyword</u> PROJECTION_ORIGIN_VECTOR <u>Archive Attribute</u> TBD <u>Definition</u> Specifies the location of the origin of the projection. For Polar and Cylindrical projections, this is the XYZ point from which all the azimuth/elevation rays emanate. For the Cylindrical-Perspective projection, this defines the center of the circle around which the synthetic camera orbits.	<u>Valid Values</u> n/a <u>Type</u> float array[3] <u>Units</u> meters (<m> unit tag required) <u>Location</u> SURFACE_PROJECTION_PARMS (Group) <u>X-path</u> TBD <u>Reference</u> TBD	<u>Mode</u> RDR-generating software <u>Field as “<xml name>:<element>:<element>:<field>”</u> n/a <u>Type</u> n/a

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
For Orthographic, Orthorectified, and Vertical projections, this optional keyword specifies the point on the projection plane that serves as the origin of the projection (i.e. all points on a line through this point in the direction of PROJECTION_Z_AXIS_VECTOR will be located at X=Y=0 in the projection). If not present, (0,0,0) should be assumed. This translation is generally not necessary and not often used; the (X Y)_AXIS_MINIMUM and (X Y)_AXIS_MAXIMUM fields allow the mosaic to be located arbitrarily in the projection plane.		
Ops Keyword PROJECTION_X_AXIS_VECTOR Archive Attribute TBD Definition Specifies a unit vector defining the X-axis for a given projection. For Orthographic, Orthorectified, and Vertical projections, this vector defines how the * axis in the mosaic is oriented in space. The X and Y axis vectors together define the rotation of the projection plane around the projection axis.	Valid Values n/a Type float array[3] Units n/a Location SURFACE_PROJECTION_PARMS (Group) X-path TBD Reference TBD	Mode RDR-generating software Field as “<xml name>:[<element>]:[<element>]:<field>” n/a Type n/a
Ops Keyword PROJECTION_Y_AXIS_VECTOR Archive Attribute TBD Definition Specifies a unit vector defining the Y-axis for a given projection. For Orthographic, Orthorectified, and Vertical projections, this vector defines how the * axis in the mosaic is oriented in space. The X and Y axis vectors together define the rotation of the projection plane around the projection axis.	Valid Values n/a Type float array[3] Units n/a Location SURFACE_PROJECTION_PARMS (Group) X-path TBD Reference TBD	Mode RDR-generating software Field as “<xml name>:[<element>]:[<element>]:<field>” n/a Type n/a
Ops Keyword PROJECTION_Z_AXIS_VECTOR Archive Attribute TBD Definition Specifies a unit vector defining the Z axis for a given projection. For Orthographic, Orthorectified, and Vertical projections, this vector defines the projection axis for the mosaic. All points along a line parallel to this axis	Valid Values n/a Type float array[3] Units n/a Location SURFACE_PROJECTION_PARMS (Group) X-path	Mode RDR-generating software Field as “<xml name>:[<element>]:[<element>]:<field>” n/a Type n/a

Output Metadata (Product Label)	Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)
are projected to the same spot in the projection plane. For the Cylindrical-Perspective projections, this defines the new axis of the circle around which the synthetic camera orbits (i.e. the normal to the circle), after the cameras have been rotated to correct for rover tilt. CAMERA_ROTATION_AXIS_VECTOR contains the axis before rotation; the difference in these two indicate the rotation amount.	TBD <u>Reference</u> TBD
Ops Keyword QUATERNION_MEASUREMENT_METHOD Archive Attribute TBD Definition Specifies the quality of the rover orientation estimate. Valid values are: a) "UNKNOWN" - The attitude should simply not be trusted. This is the initial grade given on Landing, for example. b) "TILT_ONLY" - The attitude estimate is only good for tilt determination (2-axis knowledge). Activities which require azimuth knowledge should be careful. c) "FINE" - Sun identification or other attitude determination has completed successfully, and the attitude estimate is sufficient for pointing HGA (or "ThreeAxisFine").	Valid Values - ALL Cameras 0 = "UNKNOWN" 1 = "TILT_ONLY" 2 = "FINE" - All Cameras no IDPH = "UNK" Type string Units n/a Location a. ROVER_COORDINATE_SYSTEM (Group) b. INITIAL_STATE_PARMs (Group) X-path TBD Reference TBD
Ops Keyword RADIANCE_OFFSET Archive Attribute TBD Definition Specifies the constant value by which a stored radiance is added. Expressed as an equation as follows: $\text{true_radiance_value} = \text{radiance_offset} + \text{radiance_scaling_factor} * \text{stored_radiance_value}$ There are 3 types of radiometric corrections: <u>Radiance-calibrated RDRs</u> These "RAD" (and "RADL") RDRs have been scaled to absolute radiance units using either pre-flight radiometric calibration coefficients or calibration coefficients derived from in-flight observations of the calibration target. The units on these files are (W/m²/sr/nm).	Valid Values n/a Type float Units - RAD WATT* M**-2* SR**-1* NM**-1 - IOF Unitless I/F - CCD DN Location DERIVED_IMAGE_PARMs (Group) X-path TBD Reference TBD
	Mode DPO in XML format (referenced to APID Name in Appendix A), Table Lookup Field as "<xml name>:[<element>]:[<element>]:[<field>]" - Eng. Cameras (B) <IDPH DPO>:idph:rvr_quality <IDPH DPO>:idph:rvr_quality0 - SuperCam - MCZ - SHERLOC - MEDA SkyCam - PIXL MCC <IDPH DPO>:idph:system:sapp_quality Cruise - MCZ (C) <IDPH DPO>:cidph:rvr:quality Type I32 Mode RDR-generating software Field as "<xml name>:[<element>]:[<element>]:[<field>]" n/a Type n/a

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
<u>Radiance factor-calibrated RDRs</u> These "IOF" (and "IOL") RDRs are unitless but have values in the range of 0.0 to 1.0 (for example, average bright Mars soils exhibit I/F ~ 0.35 at 750 nm and I/F ~ 0.05 at 410 nm). <u>Instrumentally-calibrated RDRs</u> These "CCD" (and "CCL") RDRs have had no radiance scaling applied, so the units on these files are "corrected" DN.		
<u>Ops Keyword</u> RADIANCE_TYPE <u>Archive Attribute</u> TBD <u>Definition</u> Specifies the type of radiance product: <u>RADIANCE</u> Need description <u>SPECTRAL RADIANCE</u> Need description This radiance is normalized for the filter bandwidth used to acquire the image. <u>SCALED SPECTRAL RADIANCE</u> Need description Scaling factor applied to the image that is a combination of solar elevation and tau.	<u>Valid Values</u> "RADIANCE", "SPECTRAL RADIANCE", "SCALED SPECTRAL RADIANCE" <u>Type</u> string <u>Units</u> n/a <u>Location</u> • DERIVED_IMAGE_PARMs (Group) <u>X-path</u> TBD <u>Reference</u> TBD	<u>Mode</u> RDR-generating software <u>Field as "<xml name>:[<element>]:<field>"</u> n/a <u>Type</u> n/a
<u>Ops Keyword</u> RADIANCE_ZENITH_SCALING_FACTOR <u>Archive Attribute</u> TBD <u>Definition</u> Specifies the factor used to convert radiance to scaled radiance.	<u>Valid Values</u> n/a <u>Type</u> float <u>Units</u> n/a <u>Location</u> • DERIVED_IMAGE_PARMs (Group) <u>X-path</u> TBD <u>Reference</u> TBD	<u>Mode</u> RDR-generating software <u>Field as "<xml name>:[<element>]:<field>"</u> n/a <u>Type</u> n/a
<u>Ops Keyword</u> RADIANCE_SCALING_FACTOR <u>Archive Attribute</u> TBD <u>Definition</u> Specifies the constant value by which a stored radiance is multiplied.	<u>Valid Values</u> n/a <u>Type</u> float <u>Units</u> n/a	<u>Mode</u> RDR-generating software <u>Field as "<xml name>:[<element>]:<field>"</u> n/a <u>Type</u> n/a

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
NOTE: Expressed as an equation: $\text{true_radiance_value} = \text{radiance_offset} + \text{radiance_scaling_factor} * \text{stored_radiance value}$ There are 3 types of radiometric corrections: <u>Radiance-calibrated RDRs</u> These "RAD" (and "RAL") RDRs have been scaled to absolute radiance units using either pre-flight radiometric calibration coefficients or calibration coefficients derived from in-flight observations of the calibration target. The units on these files are (W/m²/sr/nm). <u>Radiance factor-calibrated RDRs</u> These "IOF" (and "IOL") RDRs are unitless but have values in the range of 0.0 to 1.0 (for example, average bright Mars soils exhibit I/F ~ 0.35 at 750 nm and I/F ~ 0.05 at 410 nm). <u>Instrumentally-calibrated RDRs</u> These "CCD" (and "CCL") RDRs have had no radiance scaling applied, so the units on these files are "corrected" DN.	<u>Location</u> DERIVED_IMAGE_PARMS (Group) <u>X-path</u> TBD <u>Reference</u> TBD	
<u>Ops Keyword</u> RADIOMETRIC_CORRECTION_TYPE <u>Archive Attribute</u> TBD <u>Definition</u> Identifies the method used for radiometric correction. Values include ""CHEMRAD" for the correction done by the SuperCam team and "MIPLRAD" for the MIPL correction (flat-field, exposure and temperature only), or "NONE" for the case when no radiometric correction has been performed.	<u>Valid Values</u> "CHEMRAD", "MIPLRAD", "NONE" <u>Type</u> string <u>Units</u> n/a <u>Location</u> DERIVED_IMAGE_PARMS (Group) <u>X-path</u> TBD <u>Reference</u> TBD	<u>Mode</u> RDR-generating software <u>Field as "<xml name>:[<element>]:[<element>]:<field>"</u> n/a <u>Type</u> n/a
<u>Ops Keyword</u> RANGE_ORIGIN_VECTOR <u>Archive Attribute</u> TBD <u>Definition</u> Specifies the 3-D space from which the Range values are measured in a Range RDR. This will normally be the same as the C point of the camera. It is expressed in the coordinate system specified by the REFERENCE_COORD_SYSTEM_* keywords in the enclosing DERIVED_IMAGE_PARMS group.	<u>Valid Values</u> n/a <u>Type</u> float array[3] <u>Units</u> meters (<m> unit tag required) <u>Location</u> DERIVED_IMAGE_PARMS (Group) <u>X-path</u> TBD <u>Reference</u>	<u>Mode</u> DPO in XML format (referenced to APID Name in Appendix A) <u>Field as "<xml name>:[<element>]:[<element>]:<field>"</u> "<IDPH DPO>:idph:cmod_c[3]" <u>Type</u> F64[3]

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
	TBD	
Ops Keyword RCE_TIME_SYNC Archive Attribute TBD Definition For SuperCam RMI this specifies the Mast Unit (MU) memory bank used to hold image before transfer to RCE. There are 64 memory banks in the MU.	Valid Values n/a Type integer Units seconds Location MINI_HEADER_GROUP X-path TBD Reference TBD	SCAM Image DPO mini-header Field as “<xml name>:[<element>]:[<element>]:<field>” “SCAM_Image_Mini_Header[180]” Type U32 Notes: For SCAM RMI products, “rce_time_sync” is comprised of 4 bytes coming from the Image DPO mini-header at offset 180.
Ops Keyword RECEIVED_PACKETS Archive Attribute TBD Definition Specifies the total number of telemetry packets which constitute a reconstructed data product. NOTE: For M2020, telemetry data processing does not track “packets”, but instead data product “parts”.	Valid Values n/a Type integer Units n/a Location TELEMETRY (Class) X-path TBD Reference TBD	Mode EMD in XML format Field as “<xml name>:[<element>]:[<element>]:<field>” “M20EarthProductMetadata:M20ProductMetadata:PartL ist: TotalReceived” Type n/a
Ops Keyword RECORD_BYTES Archive Attribute TBD Definition Specifies the number of bytes in a physical file record, including record terminators and separators. NOTE: In the PDS, the use of record_bytes, along with other file-related data elements is fully described in the Standards Reference.	Valid Values “0” to n Type integer Units n/a Location FILE (Class) X-path TBD Reference TBD	Mode Calculation Field as “<xml name>:[<element>]:[<element>]:<field>” n/a Type n/a
Ops Keyword RECORD_TYPE Archive Attribute	Valid Values “FIXED_LENGTH” Type	Mode Calculation

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
TBD Definition Specifies the record format of a file. NOTE: In the PDS, when record_type is used in a detached label file it always describes its corresponding detached data file, not the label file itself. The use of record_type along with other file-related data elements is fully described in the PDS Standards Reference.	string(20) Units n/a Location FILE (Class) X-path TBD Reference TBD	Field as “<xml name>:[<element>]:<field>” n/a Type n/a
Ops Keyword REFERENCE_AZIMUTH Archive Attribute TBD Definition Specifies the azimuth of the line extending from the center of the image to the top center of the image with respect to a polar projection.	Valid Values n/a Type float Units deg (<deg> unit tag required) Location SURFACE_PROJECTION_PARS (Group) X-path TBD Reference TBD	Mode RDR-generating software Field as “<xml name>:[<element>]:<field>” n/a Type n/a
Ops Keyword REFERENCE_COORD_SYSTEM_INDEX Archive Attribute TBD Definition Specifies which instance of the coordinate system named by REFERENCE_COORD_SYSTEM_NAME is the reference coordinate system for the group in which the keyword occurs. This index is a set of integers which serve to identify coordinate system instances in a mission-specific manner. For M2020, these indices are based on the ROVER_MOTION_COUNTER and are in the same order as specified by ROVER_MOTION_COUNTER_NAME. - EDRs will contain 1 or 10, depending on the group in which the keyword occurs. - For RDRs, the number of indices can be anything from 1 (used for SITE_FRAME) up to 10; however only 1, 2, 3, and 10 indices are common in RDRs. See also REFERENCE_COORD_SYSTEM_NAME and	Valid Values n/a Type integer array[10] Units n/a Location - For EDRs ... - SITE_COORDINATE_SYSTEM (Group) - ROVER_COORDINATE_SYSTEM (Group) - RSM_COORDINATE_SYSTEM (Group) - ARM_COORDINATE_SYSTEM (Group) - GEOMETRIC_CAMERA_MODEL (Group) - ROVER_DERIVED_GEOMETRY_PARS (Group) - SITE_DERIVED_GEOMETRY_PARS (Group) - TURRET_COORDINATE_SYSTEM (Group) - TOOL_COORDINATE_SYSTEM (Group) Note: The SITE_COORDINATE_SYSTEM group is only set if the SITE Index is greater than 1 and the Site Quaternion is not 0,0,0,0 (unknown). X-path TBD Reference	Mode DPO in XML format (referenced to APID Name in Appendix A), Calculation 1) If rmc_site > 1 then: reference_coord_system_index = rmc_site - 1 Field as “<xml name>:[<element>]:<field>” Surface - Eng. Cameras (B) “<IDPH DPO>:idph:rmc_site” “<IDPH DPO>:idph:rmc_site” “<IDPH DPO>:idph:rmc_site”, “<IDPH DPO>:idph:rmc_drive”, “<IDPH DPO>:idph:rmc_pose”, “<IDPH DPO>:idph:rmc_arm”, “<IDPH DPO>:idph:rmc_sha”, “<IDPH DPO>:idph:rmc_drill”, “<IDPH DPO>:idph:rmc_rsm”, “<IDPH DPO>:idph:rmc_hga”, “<IDPH DPO>:idph:rmc_bitcar”, “<IDPH DPO>:idph:rmc_seal” - same as #3 above - same as #3 above - same as #3 above - same as #1 above - same as #3 above - same as #3 above

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• Ops Keyword • Archive Label Attribute • Definition	• Mission Phase C=Cruise, S=Surface, B=Both • Valid Values (quoted) • Type • Units • Keyword Location in Ops Label • Attribute X-path in Archive label • Attribute Reference in Archive Label (bracketed)
COORDINATE_SYSTEM_INDEX. The distinction is that COORDINATE_SYSTEM_INDEX is used when defining a coordinate system group to indicate what is being defined, while REFERENCE_COORD_SYSTEM_INDEX is used when referring to one.	TBD • <u>SuperCam</u> (S) • <u>MCZ</u> (S) • <u>SHERLOC</u> (S) • <u>PIXL MCC</u> (S) 2) "<IDPH DPO>:idph:system:rmc:site" 3) "<IDPH DPO>:idph:system:rmc:site", "<IDPH DPO>:idph:system:rmc:drive", "<IDPH DPO>:idph:system:rmc:pose", "<IDPH DPO>:idph:system:rmc:arm", "<IDPH DPO>:idph:system:rmc:sha", "<IDPH DPO>:idph:system:rmc:drill", "<IDPH DPO>:idph:system:rmc:rsm", "<IDPH DPO>:idph:system:rmc:hga", "<IDPH DPO>:idph:system:rmc:bitcar", "<IDPH DPO>:idph:system:rmc:seal" 4) same as #3 above 5) same as #3 above 6) same as #3 above 7) same as #1 above 8) same as #3 above 9) same as #3 above Cruise • <u>MCZ Cameras</u> 1) "<IDPH DPO>:cidph:rmc:site" 2) "<IDPH DPO>:cidph:rmc:site", "<IDPH DPO>:cidph:rmc:drive", "<IDPH DPO>:cidph:rmc:pose", "<IDPH DPO>:cidph:rmc:arm", "<IDPH DPO>:cidph:rmc:sha", "<IDPH DPO>:cidph:rmc:drill", "<IDPH DPO>:cidph:rmc:rsm", "<IDPH DPO>:cidph:rmc:hga", "<IDPH DPO>:cidph:rmc:bitcar", "<IDPH DPO>:cidph:rmc:seal" 3) same as #2 above 4) same as #2 above 5) same as #2 above 6) same as #2 above 7) same as #1 above Type U16
Ops Keyword REFERENCE_COORD_SYSTEM_NAME Archive Attribute TBD Definition Specifies the full name of the reference coordinate system (CS) for the group in which the keyword occurs. All vectors and positions relating to 3-D space within the enclosing group are expressed using this reference coordinate system.	Valid Values • For EDRs ... 1) "SITE_FRAME" 2) "SITE_FRAME" 3) "ROVER_NAV_FRAME" 4) "ROVER_NAV_FRAME" 5) "ROVER_NAV_FRAME" 6) "ROVER_NAV_FRAME" 7) "SITE_FRAME" 8) "ROVER_NAV_FRAME" 9) "ROVER_NAV_FRAME" • For RDRs ... Any value from COORDINATE_SYSTEM_NAME is valid. Mode Static values: 1) for 1 RMC element defining Site Frame 2) for 1 RMC elements defining Rover Navigation Frame 3) for 10 RMC elements defining Rover Navigation Frame 4) for 10 RMC elements defining Rover Navigation Frame 5) for 10 RMC elements defining Rover Navigation Frame 6) for 10 RMC elements defining Rover Navigation Frame 7) for 1 RMC element defining Site Frame 8) for 10 RMC elements defining Rover Navigation Frame

Output Metadata (Product Label)	Input Metadata (Source)
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For rover or lander missions with non-unique coordinate systems (such as "SITE"), the CS name and index (see (REFERENCE_COORD_SYSTEM_INDEX) together, along with the solution id (see REFERENCE_COORD_SYSTEM_SOLN_ID), completely specify the reference CS. The values appearing in a reference index depend on what the reference name is. The valid coordinate frame names vary by project, but are generally the same as for COORDINATE_SYSTEM_FRAME. For M2020, EDRs use a standard, predefined frame name for each occurrence. However, RDRs can use any value available in COORDINATE_SYSTEM_NAME. Despite that, only a few frame names are commonly used. "SITE_FRAME" is used for most SITE, ROVER, and LOCAL_LEVEL CS definitions, as well as for XYZ data and many mosaics. "ROVER_NAV_FRAME" is used for most other CS definitions, surface normals, camera models, and some mosaics. "LOCAL_LEVEL_FRAME" is used for some mosaics. See also COORDINATE_SYSTEM_NAME. The distinction is that COORDINATE_SYSTEM_NAME is used when defining a coordinate system group to indicate what is being defined, while REFERENCE_COORD_SYSTEM_NAME is used when referring to one.	Type string(20) Units n/a Location - For EDRs ... - SITE_COORDINATE_SYSTEM (Group) - ROVER_COORDINATE_SYSTEM (Group) - RSM_COORDINATE_SYSTEM (Group) - ARM_COORDINATE_SYSTEM (Group) - GEOMETRIC_CAMERA_MODEL (Group) - ROVER_DERIVED_GEOMETRY_PARMS (Group) - SITE_DERIVED_GEOMETRY_PARMS (Group) - TURRET_COORDINATE_SYSTEM (Group) - TOOL_COORDINATE_SYSTEM (Group) X-path TBD Reference TBD
Ops Keyword REFERENCE_COORD_SYSTEM_SOLN_ID Archive Attribute TBD Definition See SOLUTION_ID. SOLUTION_ID is used when defining a coordinate system group to indicate what is being defined, while REFERENCE_COORD_SYSTEM_SOLN_ID is used when referring to one. Appears in labels of RDRs only, not EDR products.	Valid Values n/a Type string Units n/a Location For RDRs only ... - ROVER_COORDINATE_SYSTEM (Group) - RSM_COORDINATE_SYSTEM (Group) - ARM_COORDINATE_SYSTEM (Group) - TURRET_COORDINATE_SYSTEM (Group) - TOOL_COORDINATE_SYSTEM (Group) - GEOMETRIC_CAMERA_MODEL (Group) - ROVER_DERIVED_GEOMETRY_PARMS (Group) - SITE_DERIVED_GEOMETRY_PARMS (Group) - SITE_COORDINATE_SYSTEM (Group) - LOCAL_LEVEL_COORDINATE_SYSTEM (Group) - DERIVED_IMAGE_PARMS (Group) - SURFACE_PROJECTION_PARMS (Group) - SURFACE_MODEL_PARMS (Group) X-path TBD Reference TBD
	9) for 10 RMC elements defining Rover Navigation Frame Field as "<xml name>:[<element>]:[<element>]:<field>" n/a Type n/a
	Mode Software dependent Field as "<xml name>:[<element>]:[<element>]:<field>" n/a Type n/a

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
Ops Keyword REFERENCE_PIXEL_IMAGE_ID Archive Attribute M2020:REFERENCE_PIXEL_IMAGE_ID Definition Specifies the value of IMAGE_ID or PRODUCT_ID for the reference pixel EDR that was used to remove bias in generating the RDR. NOTE: If the model rather than a reference pixel EDR was used to remove the bias then this keyword is not included in the PDS label.	Valid Values n/a Type string Units n/a Location DERIVED_IMAGE_PARMS (Group) X-path TBD Reference TBD	Mode RDR-generating software Field as “<xml name>:<element>:<element>:<field>” n/a Type n/a
Ops Keyword RELEASE_ID Archive Attribute TBD Definition Specifies the unique identifier associated with the release to the public of all or part of a data set. The release number is associated with the data set, not the mission. When a data set is released incrementally, such as every three months during a mission, the RELEASE_ID is updated each time part of the data set is released. The first release of a data set in the mission should have a value of “0001”. For example, on M2020 the first release of the EDR data set will have RELEASE_ID = “0001”. The next EDR release will have RELEASE_ID = “0002”.	Valid Values n/a Type string Units n/a Location IDENTIFICATION (Class) X-path TBD Reference TBD	Mode User parameter input Field as “<xml name>:<element>:<element>:<field>” n/a Type n/a
Ops Keyword REQUEST_ID Archive Attribute TBD Definition Specifies the ground-assigned Request ID associated with the data product. The general operational usage of REQUEST_ID is to group related datasets together by science or engineering application or theme, such as frames in a mosaic.	Valid Values n/a Type string Units Location IDENTIFICATION (Class) X-path TBD Reference TBD	Mode EMD in XML format Field as “<xml name>:<element>:<element>:<field>” “M20EarthProductMetadata:M20ProductMetadata:RequestId” Type U16
Ops Keyword RESERVED_1 Archive Attribute	Valid Values n/a (B) Type	Mode DPO in XML format (referenced to APID Name in Appendix A)

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
TBD <u>Definition</u> Reserved for future use.	integer <u>Units</u> n/a <u>Location</u> MINI_HEADER (Group) <u>X-path</u> TBD <u>Reference</u> TBD	Field as “<xml name>:[<element>]:<field>” “SCAM_Image_Mini_Header[189]” <u>Type</u> U8 Notes: For SCAM RMI products, “rmi_header_size” is comprised of one byte coming from the Image DPO mini-header at offset 189.
<u>Ops Keyword</u> RESERVED_2 <u>Archive Attribute</u> TBD <u>Definition</u> Reserved for future use.	<u>Valid Values</u> n/a (B) <u>Type</u> integer <u>Units</u> n/a <u>Location</u> MINI_HEADER (Group) <u>X-path</u> TBD <u>Reference</u> TBD	<u>Mode</u> DPO in XML format (referenced to APID Name in Appendix A) Field as “<xml name>:[<element>]:<field>” “SCAM_Image_Mini_Header[193]” <u>Type</u> U32 Notes: For SCAM RMI products, “rmi_header_size” is comprised of one byte coming from the Image DPO mini-header at offset 193.
<u>Ops Keyword</u> RESPONSIVITY_CONSTANTS <u>Archive Attribute</u> TBD <u>Definition</u> Specifies the array of responsivity constants used in generating the RDR.	<u>Valid Values</u> n/a <u>Type</u> float array <u>Units</u> n/a <u>Location</u> DERIVED_IMAGE_PARDS (Group) <u>X-path</u> TBD <u>Reference</u> TBD	<u>Mode</u> RDR-generating software Field as “<xml name>:[<element>]:<field>” n/a <u>Type</u> n/a
<u>Ops Keyword</u> RESPONSIVITY_CONSTANTS_FILE <u>Archive Attribute</u> TBD <u>Definition</u> Specifies the name of the responsivity constants file used in generating the RDR.	<u>Valid Values</u> n/a <u>Type</u> string <u>Units</u> n/a <u>Location</u> DERIVED_IMAGE_PARDS (Group)	<u>Mode</u> RDR-generating software Field as “<xml name>:[<element>]:<field>” n/a <u>Type</u> n/a

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
	<u>X-path</u> TBD <u>Reference</u> TBD	
Ops Keyword RMI_CMOS_REGISTERS Archive Attribute TBD Definition Dump of all RMI hardware registers.	Valid Values n/a (B) Type Integer Units n/a Location MINI_HEADER (Group) <u>X-path</u> TBD <u>Reference</u> TBD	Mode DPO in XML format (referenced to APID Name in Appendix A) Field as “<xml name>[:<element>]:<field>” “SCAM_Image_Mini_Header[42]” Type U8[] Notes: For SCAM RMI products, “rmi_header_size” is comprised of 128 bytes coming from the Image DPO mini-header at offset 42.
Ops Keyword RMI_FPGA_REGISTERS Archive Attribute TBD Definition Dump of all RMI FPGA registers.	Valid Values n/a (B) Type Integer Units n/a Location MINI_HEADER (Group) <u>X-path</u> TBD <u>Reference</u> TBD	Mode DPO in XML format (referenced to APID Name in Appendix A) Field as “<xml name>[:<element>]:<field>” “SCAM_Image_Mini_Header[170]” Type U8[] Notes: For SCAM RMI products, “rmi_header_size” is comprised of 8 bytes coming from the Image DPO mini-header at offset 170.
Ops Keyword RMI_HEADER_SIZE Archive Attribute TBD Definition Total number of bytes in MU and BU headers	Valid Values “0 to “ 200” (B) Type integer Units bytes Location MINI_HEADER (Group) <u>X-path</u> TBD <u>Reference</u> TBD	Mode DPO in XML format (referenced to APID Name in Appendix A) Field as “<xml name>[:<element>]:<field>” “SCAM_Image_Mini_Header[0]” Type U32 Notes: For SCAM RMI products, “rmi_header_size” is comprised of 4 bytes coming from the Image DPO mini-header at offset 0.

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
Ops Keyword ROI_SIZE Archive Attribute TBD Definition For SuperCam RMI this specifies the number of pixels on an edge of an image.	Valid Values "6" to "2048" Type integer Units n/a Location MINI_HEADER_GROUP X-path TBD Reference TBD	SCAM Image DPO mini-header Field as "<xml name>:[<element>]:[<element>]:<field>" "SCAM_Image_Mini_Header[8]" Type U16 Notes: For SCAM RMI products, "roi_size" is comprised of two bytes coming from the Image DPO mini-header at offset 8.
Ops Keyword ROI_Y_POSITION Archive Attribute TBD Definition For SuperCam RMI this specifies the pixel location for the region of interest upper left corner.	Valid Values "0" to "2042" Type integer Units n/a Location MINI_HEADER_GROUP X-path TBD Reference TBD	SCAM Image DPO mini-header Field as "<xml name>:[<element>]:[<element>]:<field>" "SCAM_Image_Mini_Header:roi_y_position" Type U16 Notes: For SCAM RMI products, "roi_y_position" is comprised of two bytes coming from the Image DPO mini-header at offset 10.
Ops Keyword ROVER_MOTION_COUNTER Archive Attribute TBD Definition Specifies a set of integers which describe a (potentially) unique location (position/orientation) for a rover. Each time something happens that moves, or could potentially move, the rover, a new motion counter value is created. This includes intentional motion due to drive commands, as well as potential motion due to other articulating devices, such as arms or antennae. This motion counter (or part of it) is used as a reference to define instances of coordinate systems which can move such as SITE or ROVER frames. The motion counter is defined in a mission-specific manner. Although the original intent was to have incrementing indices (e.g. MER), the motion counter could also contain any integer values which conform to the above definition, such as time or spacecraft clock	Valid Values n/a Type integer array[10] Units n/a Location 1) IDENTIFICATION (Class) 2) INITIAL_STATE_PARMs (Group) X-path TBD Reference TBD	Mode DPO in XML format (referenced to APID Name in Appendix A) Field as "<xml name>:[<element>]:[<element>]:<field>" Surface - Eng. Cameras (B) "<IDPH DPO>:idph:rmc_site", "<IDPH DPO>:idph:rmc_drive", "<IDPH DPO>:idph:rmc_pose", "<IDPH DPO>:idph:rmc_arm", "<IDPH DPO>:idph:rmc_sha", "<IDPH DPO>:idph:rmc_drill", "<IDPH DPO>:idph:rmc_rsm", "<IDPH DPO>:idph:rmc_hga", "<IDPH DPO>:idph:rmc_bitcar", "<IDPH DPO>:idph:rmc_seal" "<IDPH DPO>:idph:rmc_site0", "<IDPH DPO>:idph:rmc_drive0", "<IDPH DPO>:idph:rmc_pose0",

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
values. For M2020, the motion counter consists of ten values. In order, they are "Site", "Drive", "Pose", "Arm", "Sha", "Drill", "RSM", "HGA", "Bitcar", and "Seal". The Site value increments whenever a new major Site frame is declared. The Drive value increments any time intentional driving is done. Each of those, along with Pose, resets all later indices to 0 when they increment. The Arm, SHA, DRILL, RSM, HGA, BITCAR, and SEAL increment whenever the corresponding articulation device moves. These all increment independently of each other; they are reset to zero only when the SITE or DRIVE or POSE changes.		"<IDPH DPO>:idph:rmc_arm0", "<IDPH DPO>:idph:rmc_sha0", "<IDPH DPO>:idph:rmc_drill0", "<IDPH DPO>:idph:rmc_rsm0", "<IDPH DPO>:idph:rmc_hga0", "<IDPH DPO>:idph:rmc_bitcar0", "<IDPH DPO>:idph:rmc_seal0" - SuperCam (S) - MCZ (S) - SHERLOC (S) "<IDPH DPO>:idph:system:rmc:site", "<IDPH DPO>:idph:system:rmc:drive", "<IDPH DPO>:idph:system:rmc:pose", "<IDPH DPO>:idph:system:rmc:arm", "<IDPH DPO>:idph:system:rmc:sha", "<IDPH DPO>:idph:system:rmc:drill", "<IDPH DPO>:idph:system:rmc:rsm", "<IDPH DPO>:idph:system:rmc:hga", "<IDPH DPO>:idph:system:rmc:bitcar", "<IDPH DPO>:idph:system:rmc:seal" Cruise - MCZ Cameras (C) "<IDPH DPO>:cidph:rmc:site", "<IDPH DPO>:cidph:rmc:drive", "<IDPH DPO>:cidph:rmc:pose", "<IDPH DPO>:cidph:rmc:arm", "<IDPH DPO>:cidph:rmc:sha", "<IDPH DPO>:cidph:rmc:drill", "<IDPH DPO>:cidph:rmc:rsm", "<IDPH DPO>:cidph:rmc:hga", "<IDPH DPO>:cidph:rmc:bitcar", "<IDPH DPO>:cidph:rmc:seal" Type U16
Ops Keyword ROVER_MOTION_COUNTER_NAME Archive Attribute TBD Definition Specifies an array that provides the formal names identifying each integer in ROVER_MOTION_COUNTER.	Valid Values ("SITE", "DRIVE", "POSE", "ARM", "SHA", "DRILL", "RSM", "HGA", "BITCAR", "SEAL") Type string array[10] Units n/a Location - IDENTIFICATION (Class) - INITIAL_STATE_PARMS (Group) X-path TBD Reference TBD	Mode Static value: - Single value representing array of 10 RMC elements from ROVER_MOTION_COUNTER. Field as "<xml name>:[<element>]:<field>" n/a Type n/a
Ops Keyword SAMPLE_BITS	Valid Values "1", "2", "4", "8", "16", "32", "64"	Mode

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
Archive Attribute TBD Definition Specifies the stored total number of bits, or units of binary information, contained in a LINE_SAMPLE value. As an example, a SAMPLE_BIT_MASK value of "2#0000111111111111", expressing 12-bit data in the case of an unscaled EDR or ILUT RDR (see SAMPLE_BIT_METHOD), has a SAMPLE_BITS value of "16".	Type integer Units n/a Location IMAGE (Object) X-path TBD Reference TBD	DPO in XML format (referenced to APID Name in Appendix A), Calculation Field as "<xml name>:[<element>]:[<element>]:<field>" "<IDPH DPO>:idph:scale" Type enum
Ops Keyword SAMPLE_BIT_MASK Archive Attribute TBD Definition Specifies the active bits in a sample. Any bit mask is valid in an RDR. However, only a few are generated for products described in this SIS. Any 8-bit product, whether a scaled EDR or an RDR, will have the value "2#11111111" and be stored in one byte. Any 12-bit product, whether an unscaled EDR, or an ILUT RDR (see SAMPLE_BIT_METHOD), will have the value "2#0000111111111111" and be stored in two bytes. A 15-bit product (e.g. Radiometrically-corrected RDR product type) will have the value "2#0111111111111111" and be stored in two bytes. Any 32-bit integer product (e.g. Histogram EDR) will have the value "2#11111111111111111111111111111111" and be stored in four bytes. For floating-point data, SAMPLE_BIT_MASK is not valid and may be absent. If present, it should be ignored. NOTE: In the PDS, the domain of SAMPLE_BIT_MASK is dependent upon the currently-described value in the SAMPLE_BITS element and only applies to integer values.	Valid Values - For EDRs ... - Eng. Cameras "2#11111111#", "2#0000111111111111#", "2#11111111111111111111111111111111" - MCZ Cameras "2#11111111#" - For RDRs ... - Value Any combination of 0's and 1's is valid Type non-decimal Units n/a Location IMAGE (Object) X-path TBD Reference TBD	Mode DPO in XML format (referenced to APID Name in Appendix A), Calculation Field as "<xml name>:[<element>]:[<element>]:<field>" Eng. Cameras "<IDPH DPO>:idph:scale" NOTES: - For EDRs ... - Eng. Cameras: - If data type is BYTE ... scale = 1 thru 5, 7-11 or 127 and valid value is "2#11111111#" - If data type is HALF ... scale = 0 and valid value is "2#0000111111111111#" - If data type is FULL ... valid value is "2#11111111111111111111111111111111#" Type enum
Ops Keyword SAMPLE_BIT_METHOD Archive Attribute	Valid Values - For EDRs ... - Eng. Cameras & RMI "NONE", "HARDWARE", "SOFTWARE"	Mode DPO in XML format (referenced to APID Name in Appendix A), Calculation

Output Metadata (Product Label)	Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)
TBD Definition Specifies the method in which bit scaling is performed. For M2020, the bit scaling is a 12-bit to 8-bit scaling and can be performed onboard via hardware and/or software , or on the ground in an inverse operation. As valid values, "SOFTWARE_INVERTED" and "HARDWARE_INVERTED" indicate that an Inverse Lookup Table (ILUT) was applied during ground processing to 8-bit data, scaling the 8 bits in the EDR to the lowest 12 bits in the signed 16-bit integer RDR. This characterizes the Inverse LUT RDR (IDS), which is the inverse of the "12 to 8-bit" scaled version of onboard 12-bit data (via LUT).	- <u>MCZ Cameras</u> "NONE", "HARDWARE" - For RDRs ... <u>Eng. Cameras & RMI</u> "NONE", "HARDWARE", "SOFTWARE", "SOFTWARE_INVERTED", "HARDWARE_INVERTED" <u>MCZ Cameras</u> "NONE", "HARDWARE", "HARDWARE_INVERTED" Type string Units n/a Location INSTRUMENT_STATE_PARMs (Group) X-path TBD Reference TBD Field as "<xml name>:[<element>]:<field>" <u>Eng. Cameras & RMI</u> "<IDPH DPO>:idph:scale", "<IDPH DPO>:idph:hw_scale" <u>MCZ Cameras</u> "<Ancillary DPO>:image_id_data:comp_companing_mode" NOTES: - For EDRs ... - Eng. Cameras: a. If scale = 0, value is "NONE" b. If scale > 0 and hw_scale=0 (false), value is "SOFTWARE" c. If scale > 0 and hw_scale=1 (true), value is "HARDWARE" - MCZ Cameras: a. If comp_companing_mode <= 32, value is "HARDWARE" b. If comp_companing_mode = 255, value is "NONE" c. If comp_companing_mode not "a" or "b", value is "UNK" - For RDRs ... - Any of the EDR values are valid. - If ILUT (inverse LUT) operation, then append "_INVERTED" to the EDR value, resulting in "SOFTWARE_INVERTED" or "HARDWARE_INVERTED" as the case may be. Type enum
Ops Keyword SAMPLE_BIT_MODE_ID Archive Attribute TBD Definition Specifies the type of pixel scaling performed. For M2020, pixel scaling is accomplished by using onboard lookup tables or by shifting a specified bit into the most significant bit. Descriptions of the valid values for Eng. Cameras and RMI follow: "NONE" - No scaling; keep as 12-bit (if available) "LUT1" - Use lookup table 1 "LUT2" - Use lookup table 2 "LUT3" - Use lookup table 3 "LUT4" - Use lookup table 4 "LUT5" - Use lookup table 5 "MSB_BIT7" - No scaling; make bit 7 most significant bit "MSB_BIT8" - Shift to make bit 8 most significant bit "MSB_BIT9" - Shift to make bit 9 most significant bit	Valid Values <u>Eng. Cameras (B)</u> 0 = "NONE" 1 = "LUT1" 2 = "LUT2" 3 = "LUT3" 4 = "LUT4" 5 = "LUT5" 7 = "MSB_BIT7" 8 = "MSB_BIT8" 9 = "MSB_BIT9" 10 = "MSB_BIT10" 11 = "MSB_BIT11" 127 = "AUTOSHIFT" <u>MCZ Cameras (B)</u> "MCZ_LUT_<n>", "NONE", "UNK" where, n = 0 to 32 Type string Units n/a Location <u>Eng. Cameras</u> Mode DPO in XML format (referenced to APID Name in Appendix A), Calculation Field as "<xml name>:[<element>]:<field>" <u>Eng. Cameras</u> a. "<IDPH DPO>:idph:scale" b. "<IDPH DPO>:idph:params:scale" c. "<IDPH DPO>:idph:params:thumb_scale" <u>MCZ Cameras</u> a. "<Ancillary DPO>:image_id_data:comp_companing_mode" NOTES: - For MCZ Cameras ... a. If comp_companing_mode <= 32, value is "MCZ_LUT_<n>" b. If comp_companing_mode = 255, value is "NONE" c. If comp_companing_mode not "a" or "b", value is "UNK" Type enum

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
j) "MSB_BIT10" - Shift to make bit 10 most significant bit k) "MSB_BIT11" - Shift to make bit 11 most significant bit l) "AUTOSHIFT" - Auto-shift to keep highest value Descriptions of the valid values for MCZ Cameras follow: a) "NONE" - no scaling b) "MCZ_LUT_<n>" - valid LUT, with n = 0 to 32 NOTE: For M2020, the instance of SAMPLE_BIT_MODE_ID in the PDS label Group INSTRUMENT_STATE_PARMS will never have the value "AUTOSHIFT".	a. INSTRUMENT_STATE_PARMS (Group) b. IMAGE_REQUEST_PARMS (Group) c. THUMBNAIL_REQUEST_PARMS (Group) - MCZ a. INSTRUMENT_STATE_PARMS (Group) <u>X-path</u> TBD <u>Reference</u> TBD	
Ops Keyword SAMPLE_CAMERA_MODEL_OFFSET Archive Attribute TBD Definition Specifies the location of the image origin with respect to the camera model's origin. For CAHV/CAHVOR models, this origin is not the center of the camera, but is the upper-left corner of the "standard"-size image, which is encoded in the CAHV vectors. (MIPL Projections - Perspective)	Valid Values n/a Type float Units pixel (<pixel> unit tag required) Location SURFACE_PROJECTION_PARMS (Group) X-path TBD Reference TBD	Mode RDR-generating software Field as "<xml name>:[<element>]:[<element>]:<field>" n/a Type n/a
Ops Keyword SAMPLE_PROJECTION_OFFSET Archive Attribute TBD Definition Specifies the sample coordinate of the location in the image of the "special" point of the mosaic. For Polar projections, this is the nadir of the polar projection. For Vertical, Orthographic and Orthorectified projections, this is the origin of the projected coordinate system (corresponding to PROJECTION_ORIGIN_VECTOR), and may be off the image. Not applicable to other projections.	Valid Values n/a Type float Units pixel (<pixel> unit tag required) Location SURFACE_PROJECTION_PARMS (Group) X-path TBD Reference TBD	Mode RDR-generating software Field as "<xml name>:[<element>]:[<element>]:<field>" n/a Type n/a
Ops Keyword SAMPLE_TYPE Archive Attribute TBD Definition Specifies the data storage representation of sample value.	Valid Values "IEEE_REAL", "MSB_INTEGER", "MSB_UNSIGNED_INTEGER", "PC_REAL", "LSB_INTEGER", "LSB_UNSIGNED_INTEGER" Type string(30) Units n/a	Mode DPO in XML format (referenced to APID Name in Appendix A), depends on host platform type Field as "<xml name>:[<element>]:[<element>]:<field>" 1) "<IDPH DPO>:idph:scale" 2) "<IDPH DPO>:idph:hw_scale" Type

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
The valid values are platform dependent. JAVA, Suns and PPC-based Macs or Intel-based Macs are MSB and IEEE_REAL. Intel-based machines usually running Windows or Linux are LSB integers and PC_REAL.	<u>Location</u> IMAGE (Object) <u>X-path</u> TBD <u>Reference</u> TBD	1) enum 2) boolean
<u>Ops Keyword</u> SEQUENCE_EXECUTION_COUNT <u>Archive Attribute</u> TBD <u>Definition</u> Specifies how many times this sequence has executed since the last reset of the flight computer. For M2020, this means RCE (Rover Compute Element) start-up.	<u>Valid Values</u> n/a <u>Type</u> integer <u>Units</u> n/a <u>Location</u> TELEMETRY (Class) <u>X-path</u> TBD <u>Reference</u> TBD	<u>Mode</u> EMD in XML format <u>Field as "<xml name>:[<element>]:[<element>]:<field>"</u> "M20EarthProductMetadata:M20ProductMetadata:SequenceExecutionCounter" <u>Type</u> n/a
<u>Ops Keyword</u> SEQUENCE_ID <u>Archive Attribute</u> TBD <u>Definition</u> Specifies the spacecraft sequence associated with the given product.	<u>Valid Values</u> n/a <u>Type</u> string(30) <u>Units</u> n/a <u>Location</u> IDENTIFICATION (Class) <u>X-path</u> TBD <u>Reference</u> TBD	<u>Mode</u> EMD in XML format <u>Field as "<xml name>:[<element>]:[<element>]:<field>"</u> "M20EarthProductMetadata:M20ProductMetadata:SequenceId" <u>Type</u> n/a
<u>Ops Keyword</u> SEQUENCE_VERSION_ID <u>Archive Attribute</u> TBD <u>Definition</u> Specifies the version identifier for a particular observation sequence used during planning or data processing.	<u>Valid Values</u> n/a <u>Type</u> string(30) <u>Units</u> n/a <u>Location</u> IDENTIFICATION (Class) <u>X-path</u> TBD <u>Reference</u>	<u>Mode</u> EMD in XML format <u>Field as "<xml name>:[<element>]:[<element>]:<field>"</u> "M20EarthProductMetadata:M20ProductMetadata:SequenceVersion" <u>Type</u> n/a

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
	TBD	
Ops Keyword SHUTTER_CORRECTION_MODE_ID Archive Attribute TBD Definition Specifies whether shutter subtraction will be performed.	Valid Values Eng. Cameras (B) 0 = "NONE" 1 = "CONDITIONAL" 2 = "ALWAYS" Type string Units n/a Location OBSERVATION_REQUEST_PARMS (Group) X-path TBD Reference TBD	Mode DPO in XML format (referenced to APID Name in Appendix A) Field as "<xml name>:[<element>]:[<element>]:<field>" "<IDPH DPO>:idph:params:shutter" Type enum
Ops Keyword SHUTTER_CORRECT_THRESH_COUNT Archive Attribute TBD Definition Specifies the exposure time threshold for conditional shutter subtraction. For M2020, the count was in increments of 50 μs.	Valid Values Eng. Cameras (B) "0" to "65535" Type unsigned integer Units n/a Location OBSERVATION_REQUEST_PARMS (Group) X-path TBD Reference TBD	Mode DPO in XML format (referenced to APID Name in Appendix A) Field as "<xml name>:[<element>]:[<element>]:<field>" "<IDPH DPO>:idph:params:shutter_thresh" Type U16
Ops Keyword SHUTTER_EFFECT_CORRECTION_FLAG Archive Attribute TBD Definition Specifies whether or not a shutter effect correction was applied to the image. The shutter effect correction involves the removal from the image of the shutter, or fixed-pattern.	Valid Values Eng. Cameras (B) 0 = "FALSE" 1 = "TRUE" Type string(5) Units n/a Location INSTRUMENT_STATE_PARMS (Group) X-path TBD Reference TBD	Mode DPO in XML format (referenced to APID Name in Appendix A) Field as "<xml name>:[<element>]:[<element>]:<field>" "<IDPH DPO>:idph:shutter" Type boolean

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
Ops Keyword SOFTWARE_KEYWORD_NAME Archive Attribute M2020:SOFTWARE_KEYWORD_NAME Definition Specifies the value of all keyword names used as input to the primary generating software named in SOFTWARE_MODULE_NAME. For M2020, this keyword may be placed in a xxx_HISTORY_PARMS group, where “xxx” is the task name. There may be multiple groups to accommodate multiple software tasks.	Valid Values n/a Type string array Units n/a Location <xxx>_HISTORY_PARMS (Group) where, xxx = RDR software name X-path TBD Reference TBD	Mode RDR-generating software Field as “<xml name>:[<element>]:[<element>]:<field>” n/a Type n/a
Ops Keyword SOFTWARE_KEYWORD_TYPE Archive Attribute TBD Definition Specifies the value of a keyword type used as input to the primary generating software named in SOFTWARE_MODULE_NAME. For M2020, this keyword may be placed in a xxx_HISTORY_PARMS group, where “xxx” is the task name. There may be multiple groups to accommodate multiple software tasks. Use of this keyword is optional. NOTE: The value of this keyword will be relevant to SOFTWARE_LANGUAGE.	Valid Values n/a Type string array Units n/a Location <xxx>_HISTORY_PARMS (Group) where, xxx = RDR software name X-path TBD Reference TBD	Mode RDR-generating software Field as “<xml name>:[<element>]:[<element>]:<field>” n/a Type n/a
Ops Keyword SOFTWARE_KEYWORD_VALUE Archive Attribute TBD Definition Specifies the value of a keyword value used as input to the primary generating software named in SOFTWARE_MODULE_NAME. Array values can be nested. Example: “SOFTWARE_KEYWORD_VALUE = (value 1, (value2A, value2b), value3).” For M2020, this keyword may be placed in a xxx_HISTORY_PARMS group, where “xxx” is the task name. There may be multiple groups to accommodate multiple software tasks.	Valid Values n/a Type string array Units n/a Location <xxx>_HISTORY_PARMS (Group) where, xxx = RDR software name X-path TBD Reference TBD	Mode RDR-generating software Field as “<xml name>:[<element>]:[<element>]:<field>” n/a Type n/a
Ops Keyword SOFTWARE_LANGUAGE Archive Attribute	Valid Values n/a Type	Mode RDR-generating software

Output Metadata (Product Label)		Input Metadata (Source)
• Ops Keyword • Archive Label Attribute • Definition	• Mission Phase C=Cruise, S=Surface, B=Both • Valid Values (quoted) • Type • Units • Keyword Location in Ops Label • Attribute X-path in Archive label • Attribute Reference in Archive Label (bracketed)	• Mission Phase C=Cruise, S=Surface, B=Both • Mode • Metadata Field • Type
TBD Definition Specifies the programming language that the primary RDR generating software is written in (eg: IDL)	string <u>Units</u> n/a <u>Location</u> <xxx>_HISTORY_PARDS (Group) where, xxx = RDR software name <u>X-path</u> TBD <u>Reference</u> TBD	Field as “<xml name>:[<element>]:<field>” n/a <u>Type</u> n/a
<u>Ops Keyword</u> SOFTWARE_MODULE_NAME <u>Archive Attribute</u> TBD Definition Specifies the name of the primary software module used to generate this product. This is the module to which the PDS label entries SOFTWARE_PARAMETER_* and SOFTWARE_KEYWORD_* apply. For M2020, this keyword may be placed in a xxx_HISTORY_PARDS group, where “xxx” is the task name. There may be multiple groups to accommodate multiple software tasks.	<u>Valid Values</u> n/a <u>Type</u> string <u>Units</u> n/a <u>Location</u> <xxx>_HISTORY_PARDS (Group) where, xxx = RDR software name <u>X-path</u> TBD <u>Reference</u> TBD	<u>Mode</u> RDR-generating software Field as “<xml name>:[<element>]:<field>” n/a <u>Type</u> n/a
<u>Ops Keyword</u> SOFTWARE_MODULE_TYPE <u>Archive Attribute</u> TBD Definition Specifies the type of the primary software module named in SOFTWARE_MODULE_NAME.	<u>Valid Values</u> “PROCEDURE”, “FUNCTION”, “SCRIPT” <u>Type</u> string <u>Units</u> n/a <u>Location</u> <xxx>_HISTORY_PARDS (Group) where, xxx = RDR software name <u>X-path</u> TBD <u>Reference</u> TBD	<u>Mode</u> RDR-generating software Field as “<xml name>:[<element>]:<field>” n/a <u>Type</u> n/a
<u>Ops Keyword</u> SOFTWARE_NAME <u>Archive Attribute</u> TBD Definition Specifies the name of data processing software such as a program or a program library.	<u>Valid Values</u> “M2020EDRGEN”, other <u>Type</u> string(60) <u>Units</u> n/a	<u>Mode</u> Dependent on EDR/RDR state: - For EDRs, static - For RDRs, RDR-generating software Field as “<xml name>:[<element>]:<field>” n/a

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
	Location - For EDRs ... - M2020EDRGEN_HISTORY_PARMS (Group) - For RDRs ... <xxx>_HISTORY_PARMS (Group) where, xxx = RDR software name X-path TBD Reference TBD	Type n/a
Ops Keyword SOFTWARE_PARAMETER_NAME Archive Attribute TBD Definition Specifies the value of a parameter name used as input to the primary generating software named in SOFTWARE_MODULE_NAME. For M2020, this keyword may be placed in a xxx_HISTORY_PARMS group, where "xxx" is the task name. There may be multiple groups to accommodate multiple software tasks.	Valid Values n/a Type string array Units n/a Location <xxx>_HISTORY_PARMS (Group) where, xxx = RDR software name X-path TBD Reference TBD	Mode RDR-generating software Field as "<xml name>:[<element>]:<field>" n/a Type n/a
Ops Keyword SOFTWARE_PARAMETER_TYPE Archive Attribute M2020:SOFTWARE_PARAMETER_TYPE Definition Specifies the value of a parameter type used as input to the primary generating software named in SOFTWARE_MODULE_NAME. For M2020, this keyword may be placed in a xxx_HISTORY_PARMS group, where "xxx" is the task name. There may be multiple groups to accommodate multiple software tasks. Use of this keyword is optional. NOTE: The value of this keyword will be relevant to SOFTWARE_LANGUAGE.	Valid Values n/a Type string array Units n/a Location <xxx>_HISTORY_PARMS (Group) where, xxx = RDR software name X-path TBD Reference TBD	Mode RDR-generating software Field as "<xml name>:[<element>]:<field>" n/a Type n/a
Ops Keyword SOFTWARE_PARAMETER_VALUE Archive Attribute M2020:SOFTWARE_PARAMETER_VALUE Definition Specifies the value of a parameter value used as input to the primary generating software named in	Valid Values n/a Type array Units n/a Location <xxx>_HISTORY_PARMS (Group)	Mode RDR-generating software Field as "<xml name>:[<element>]:<field>" n/a Type n/a

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
SOFTWARE_MODULE_NAME. Array values can be nested. Example: "SOFTWARE_PARAMETER_VALUE = (value 1, (value2A, value2b),value3)." For M2020, this keyword may be placed in a xxx_HISTORY_PARMS group, where "xxx" is the task name. There may be multiple groups to accommodate multiple software tasks.	where, xxx = RDR software name <u>X-path</u> TBD <u>Reference</u> TBD	
<u>Ops Keyword</u> SOFTWARE_VERSION_ID <u>Archive Attribute</u> TBD <u>Definition</u> Specifies the version (development level) of a program or a program library.	<u>Valid Values</u> n/a <u>Type</u> string(20) <u>Units</u> n/a <u>Location</u> - For EDRs ... - M2020EDRGEN_HISTORY_PARMS (Group) - For RDRs ... <xxx>_HISTORY_PARMS (Group) where, xxx = RDR software name <u>X-path</u> TBD <u>Reference</u> TBD	<u>Mode</u> Dependent on EDR/RDR state: - For EDRs, static - For RDRs, RDR-generating software <u>Field as "<xml name>:[<element>]:[<field>]"</u> n/a <u>Type</u> n/a
<u>Ops Keyword</u> SOLAR_AZIMUTH <u>Archive Attribute</u> TBD <u>Definition</u> Specifies one of two angular measurements indicating the direction to the Sun as measured from a specific point on the surface of a planet (ex., from a lander or rover). The positive direction of the elevation is set by the POSITIVE_AZIMUTH_DIRECTION data element. The azimuth is measured positively in the clockwise direction (as viewed from above) with the meridian passing through the positive spin axis of the planet (i.e., the north pole) defining the zero reference. 0 <= SOLAR_AZIMUTH <= 360. Units are degrees. For M2020, the value in the SITE_DERIVED_GEOMETRY group is calculated using SPICE based on the time of the observation. The value in the ROVER_DERIVED_GEOMETRY group reflects what was sent in telemetry (as az/el, converted from the telemetered unit vector). Thus, even after they are converted to a common frame, the values will likely differ by a small	<u>Valid Values</u> "0.0" to "359.99" "N/A" if any SPICE kernel is unavailable. <u>Type</u> float <u>Units</u> deg (<deg> unit tag required) <u>Location</u> - SITE_DERIVED_GEOMETRY_PARMS (Group) - ROVER_DERIVED_GEOMETRY_PARMS (Group) <u>X-path</u> TBD <u>Reference</u> TBD	<u>Mode</u> a. Calculation using SPICE: - EK Kernel - SCLK Kernel - Leapsecond Kernel - SPK Kernel - PCK Kernel - Surface Kernel b. DPO in XML format (referenced to APID Name in Appendix D), Calculation 2) DPO in XML format (referenced to APID Name in Appendix A) <u>Field as "<xml name>:[<element>]:[<field>]"</u> b. "<IDPH DPO>:idph:sclk_seconds", "<IDPH DPO>:idph:sclk_subseconds" "<IDPH DPO>:idph:sun_dir" <u>Type</u> U32

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
amount, representing the difference between the rover's knowledge and the (more accurate) SPICE computation. NOTE: For M2020, early in the mission the value for only the instance of this keyword in the label group SITE_DERIVED_GEOMETRY_PARMS was being incorrectly reported as the inverse of what it should have been. For the Ops (ODL) label, the value is flagged as correct by the presence of another keyword (see POSITIVE_AZIMUTH_DIRECTION) in group SITE_DERIVED_GEOMETRY_PARMS. For the PDS label, the value is correct for all files as of archive Volume 2. The instance of this keyword in group ROVER_DERIVED_GEOMETRY_PARMS is always correct for the Ops and PDS labels.		
Ops Keyword SOLAR_ELEVATION Archive Attribute TBD Definition Specifies one of two angular measurements indicating the direction to the Sun as measured from a specific point on the surface of a planet (ex., from a lander or rover). The positive direction of the elevation is set by the POSITIVE_ELEVATION_DIRECTION data element. The elevation is measured from the plane which is normal to the line passing between the surface point and the planet's center of mass, and which intersects the surface point. -90 <= SOLAR_ELEVATION <= 90. Units are degrees. For M2020, the value in the SITE_DERIVED_GEOMETRY group is calculated using SPICE based on the time of the observation. The value in the ROVER_DERIVED_GEOMETRY group reflects what was sent in telemetry (as az/el, converted from the telemetered unit vector). Thus, even after they are converted to a common frame, the values will likely differ by a small amount, representing the difference between the rover's knowledge and the (more accurate) SPICE computation.	Valid Values "-90.0" to "90.0" "N/A" if any SPICE kernel is unavailable. Type float Units deg (<deg> unit tag required) Location 1) SITE_DERIVED_GEOMETRY_PARMS (Group) 2) ROVER_DERIVED_GEOMETRY_PARMS (Group) X-path TBD Reference TBD	Mode 1) a. Calculation using SPICE: - EK Kernel - SCLK Kernel - Leapsecond Kernel - SPK Kernel - PCK Kernel - Surface Kernel b. DPO in XML format (referenced to APID Name in Appendix D), Calculation 2) DPO in XML format (referenced to APID Name in Appendix A) Field as "<xml name>:[<element>]:[<element>]:[<field>]" 1) b. "<IDPH DPO>:idph:sclk_seconds", "<IDPH DPO>:idph:sclk_subseconds" 2) "<IDPH DPO>:idph:sun_dir" Type U32
Ops Keyword SOLAR_LONGITUDE Archive Attribute TBD	Valid Values "0.0" to "359.99" "N/A" if any SPICE kernel is unavailable. Type	Mode • Calculation: - SCLK Kernel - Landing Site Kernel - P Kernel

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
Definition Specifies the value of the angle between the body_Sun line at the time of interest and the body_Sun line at the vernal equinox. This provides a measure of season on a target body, with values of 0 to 90 degrees representing northern spring, 90 to 180 degrees representing northern summer, 180 to 270 degrees representing northern autumn and 270 to 360 degrees representing northern winter. Example: For IRAS, the geocentric ecliptic longitude (B1950) of the Sun at the start of a scan.	float Units deg (<deg> unit tag required) Location IDENTIFICATION (Class) X-path TBD Reference TBD	DPO in XML format (referenced to APID Name in Appendix A), Calculation Field as "<xml name>:[<element>]:[<element>]:<field>" "<IDPH DPO>:idph:scik_seconds", "<IDPH DPO>:idph:scik_subseconds" Type U32
Ops Keyword SOLUTION_ID Archive Attribute TBD Definition Specifies the unique identifier for the solution set to which the values in the group belong. For certain kinds of information, such as pointing correction (pointing models) and rover localization (coordinate system definitions), the "true" value is unknown and only estimates of the true value exist. Thus, more than one set of estimates may exist simultaneously, each valid for its intended purpose. Each of these sets is called a "solution" to the unknown true value. SOLUTION_ID is used to identify which solution is being expressed by the containing group. No specific naming convention is defined here, however it is recommended that projects adopt one. The intent is to be able to identify who created the solution, and why. Possible components of the naming convention include user, institution, purpose, request ID, version, program, date/time. The value is not case-sensitive. For M2020, when in a COORDINATE_SYSTEM group, the SOLUTION_ID specifies the ID of the coordinate system being defined in that group. It must be globally unique across all coordinate system instances, i.e. it cannot be reused to define the same coordinate system instance differently. Different coordinate system instances (for example, different values of the RMC index) may share the same SOLUTION_ID. See also REFERENCE_COORD_SYSTEM_NAME and REFERENCE_COORD_SYSTEM_SOLN_ID. The SOLUTION_ID should be the same identifier used in the PLACES rover localization database. The special name "telemetry" is used for values telemetered from the rover. If SOLUTION_ID is absent, "telemetry" should be assumed.	Valid Values - EDR-generating software "TELEMETRY" - RDR-generating software any string value Type string Units n/a Location - For EDRs ... - ROVER_COORDINATE_SYSTEM (Group) - RSM_COORDINATE_SYSTEM (Group) - ARM_COORDINATE_SYSTEM (Group) - For RDRs ... - ROVER_COORDINATE_SYSTEM (Group) - RSM_COORDINATE_SYSTEM (Group) - ARM_COORDINATE_SYSTEM (Group) - SITE_COORDINATE_SYSTEM (Group) - LOCAL_LEVEL_COORDINATE_SYSTEM (Group) - DERIVED_IMAGE_PARMS (Group) X-path TBD Reference TBD	Mode - EDR-generating software: - Static Value - RDR-generating software: - Determined Field as "<xml name>:[<element>]:[<element>]:<field>" n/a Type n/a

Output Metadata (Product Label)		Input Metadata (Source)																																																			
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type																																																			
For M2020, when in the DERIVED_GEOMETRY_PARMS group, SOLUTION_ID specifies the identifier of the pointing correction solution used to derive the model specified via POINTING_MODEL_PARAMS et al. This identifier should also appear in the pointing correction file referenced by POINTING_CORRECTION_FILE_NAME. If there is only one identifier in the correction file, then SOLUTION_ID may be omitted. SOLUTION_ID may be reused in the context of pointing corrections, although uniqueness is recommended. The pointing correction solution ID namespace is separate from the coordinate system namespace.																																																					
Ops Keyword SOURCE_ID Archive Attribute TBD Definition Specifies an identifier for the source. For M2020, it identifies the FSW element that requested the image, i.e. what was the source of the command. The field is based on the value for IMAGE_ID and the mappings are per convention; there is no guarantee that the mappings are used in this way during operations.	Valid Values <table> <tr> <th>Value</th><th>Range</th><th>Source</th></tr> <tr> <td>"GND"</td><td>0 - 99999999</td><td>Ground</td></tr> <tr> <td>"NAVF"</td><td>100000000 - 199999999</td><td>Navigation, Front (left/right)</td></tr> <tr> <td>"NAVR"</td><td>200000000 - 299999999</td><td>Navigation, Rear (left/right)</td></tr> <tr> <td>"NAVS"</td><td>300000000 - 399999999</td><td>Navigation, Steerable (left/right)</td></tr> <tr> <td>"HAFIQ"</td><td>400000000 - 499999999</td><td>Hazard Avoidance, Fault Image Queue</td></tr> <tr> <td>"SUN"</td><td>500000000 - 599999999</td><td>Sun-Finding</td></tr> <tr> <td>"FAULT"</td><td>600000000 - 699999999</td><td>Fault Protection/Response</td></tr> <tr> <td>"VTT"</td><td>700000000 - 799999999</td><td>Target Tracking (left/right)</td></tr> <tr> <td>"ARMC"</td><td>800000000 - 899999999</td><td>Touch and Go, coarse res (Penultimate left/right)</td></tr> <tr> <td>"ARMF"</td><td>900000000 - 999999999</td><td>Touch and Go, fine res (Final left/right)</td></tr> <tr> <td>"WATCH"</td><td>1000000000 - 1099999999</td><td>Dust-Devil detection</td></tr> <tr> <td>"VISODOM"</td><td>1100000000 - 1199999999</td><td>Visual Odometry (left/right)</td></tr> <tr> <td>"SPARE1"</td><td>1200000000 - 1299999999</td><td>Spare group 1</td></tr> <tr> <td>"SPARE2"</td><td>1300000000 - 1399999999</td><td>Spare group 2</td></tr> <tr> <td>"SPARE3"</td><td>1400000000 - 1499999999</td><td>Spare group 3</td></tr> <tr> <td>"SPARE4"</td><td>1500000000 - 1599999999</td><td>Spare group 4</td></tr> </table>	Value	Range	Source	"GND"	0 - 99999999	Ground	"NAVF"	100000000 - 199999999	Navigation, Front (left/right)	"NAVR"	200000000 - 299999999	Navigation, Rear (left/right)	"NAVS"	300000000 - 399999999	Navigation, Steerable (left/right)	"HAFIQ"	400000000 - 499999999	Hazard Avoidance, Fault Image Queue	"SUN"	500000000 - 599999999	Sun-Finding	"FAULT"	600000000 - 699999999	Fault Protection/Response	"VTT"	700000000 - 799999999	Target Tracking (left/right)	"ARMC"	800000000 - 899999999	Touch and Go, coarse res (Penultimate left/right)	"ARMF"	900000000 - 999999999	Touch and Go, fine res (Final left/right)	"WATCH"	1000000000 - 1099999999	Dust-Devil detection	"VISODOM"	1100000000 - 1199999999	Visual Odometry (left/right)	"SPARE1"	1200000000 - 1299999999	Spare group 1	"SPARE2"	1300000000 - 1399999999	Spare group 2	"SPARE3"	1400000000 - 1499999999	Spare group 3	"SPARE4"	1500000000 - 1599999999	Spare group 4	Mode DPO in XML format (referenced to APID Name in Appendix A) Field as "<xml name>:[<element>]:[<element>]:<field>" "<IDPH DPO>:idph.params:imgid1" TBD Type U32
Value	Range	Source																																																			
"GND"	0 - 99999999	Ground																																																			
"NAVF"	100000000 - 199999999	Navigation, Front (left/right)																																																			
"NAVR"	200000000 - 299999999	Navigation, Rear (left/right)																																																			
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"SUN"	500000000 - 599999999	Sun-Finding																																																			
"FAULT"	600000000 - 699999999	Fault Protection/Response																																																			
"VTT"	700000000 - 799999999	Target Tracking (left/right)																																																			
"ARMC"	800000000 - 899999999	Touch and Go, coarse res (Penultimate left/right)																																																			
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"WATCH"	1000000000 - 1099999999	Dust-Devil detection																																																			
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"SPARE1"	1200000000 - 1299999999	Spare group 1																																																			
"SPARE2"	1300000000 - 1399999999	Spare group 2																																																			
"SPARE3"	1400000000 - 1499999999	Spare group 3																																																			
"SPARE4"	1500000000 - 1599999999	Spare group 4																																																			

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
	Type string Units n/a Location - OBSERVATION_REQUEST_PARMs (Group) - IMAGE_REQUEST_PARMs (Group) - SUBFRAME_REQUEST_PARMs (Group) - THUMBNAI_REQUEST_PARMs (Group) - REFERENCE_PIXEL_REQUEST_PARMs (Group) - ROW_SUM_REQUEST_PARMs (Group) - COLUMN_SUM_REQUEST_PARMs (Group) - HISTOGRAM_REQUEST_PARMs (Group) X-path TBD Reference TBD	
Ops Keyword SOURCE_PRODUCT_ID Archive Attribute TBD Definition Identifies a product used as input to create a new product. The source_product_id may be based on a file name. See also: PRODUCT_ID. For M2020, this keyword indicates the PRODUCT_ID (filename minus extension) of the EDRs (not RDRs) that were used to create this product. In an EDR, this keyword exists and refers to itself; i.e. it is equivalent to PRODUCT_ID. See also INPUT_PRODUCT_ID.	Valid Values n/a Type string array Units n/a Location IDENTIFICATION (Class) X-path TBD Reference TBD	Mode - EDR-generating software: - Filename of output EDR product, minus extension - RDR-generating software: - Filename of root input EDR product, minus extension Field as “<xml name>:[<element>]:[<element>]:<field>” n/a Type n/a
Ops Keyword SOURCE_PRODUCT_WAVELENGTH Archive Attribute M2020:SOURCE_PRODUCT_WAVELENGTH Definition Specifies the effective wavelength of the corresponding images listed in SOURCE_PRODUCT_ID.	Valid Values n/a Type integer array Units n/a Location DERIVED_IMAGE_PARMs (Group) X-path TBD Reference TBD	Mode RDR-generating software Field as “<xml name>:[<element>]:[<element>]:<field>” n/a Type n/a
Ops Keyword SPACECRAFT_CLOCK_CNT_PARTITION	Valid Values “1”	Mode Static Value

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
Archive Attribute TBD Definition Specifies the clock partition active for the SPACECRAFT_CLOCK_START_COUNT and SPACECRAFT_CLOCK_STOP_COUNT elements.	Type integer Units n/a Location IDENTIFICATION (Class) X-path TBD Reference TBD	Field as “<xml name>:[<element>]:[<element>]:<field>” n/a Type n/a
Ops Keyword SPACECRAFT_CLOCK_START_COUNT Archive Attribute TBD Definition Specifies the value of the spacecraft clock at the beginning of a time period of interest. Format is “sssssssss.MCZ”, stored as a floating point number where, “sssssssss” = seconds converted from clock’s coarse counter “MCZ” = milliseconds converted from clock’s fine counter. For M2020, the time period of interest is the beginning of data acquisition. The fractional component “MCZ” is computed as follows: $[(\text{shift right 12 bits}) / 2^{**20}] * 1000$	Valid Values ssssssssss.MCZ Type string(30) Units n/a Location 1) IDENTIFICATION (Class) 2) IDENTIFICATION (Class) 3) a. IDENTIFICATION (Class) b. IDENTIFICATION (Class) c. MINI_HEADER (Group) 4) IDENTIFICATION X-path TBD Reference TBD	Mode 1) DPO in XML format (referenced to APID Name in Appendix A) 2) DPO in XML format (referenced to APID Name in Appendix A) 3) DPO in XML format (referenced to APID Name in Appendix A) or Image DPO mini-header Field as “<xml name>:[<element>]:[<element>]:<field>” Surface 1) <u>Eng. Cameras</u> “<IDPH DPO>:idph:sclk_seconds”, “<IDPH DPO>:idph:sclk_subseconds” 2) <u>SuperCam</u> “<IDPH DPO>:system:sclk_sec”, “<IDPH DPO>:system:sclk_subsec” 3) <u>MCZ Cameras</u> a. “<IDPH DPO>:system:sclk_sec”, “<IDPH DPO>:system:sclk_subsec” b. “<Ancillary DPO>:sclk:seconds” c. “MCZ_Image_Mini_Header:sclk” 4) <u>Skycam</u> “<MEDA Image Header>:time_tag” Cruise 1) <u>Eng. Cameras</u> “<IDPH DPO>:idph:sclk_seconds”, “<IDPH DPO>:idph:sclk_subseconds” 2) <u>SuperCam</u> TBD 3) <u>MCZ Cameras</u> a. “<IDPH DPO>:cidph:sclk:seconds” b. “<Ancillary DPO>:sclk:seconds” c. “MCZ_Image_Mini_Header:sclk” NOTES: For MCZ recovered or Z-stack data products, value is comprised of four bytes coming from Image DPO mini-header at byte offsets 8, 9, 10 and 11 per case “3a”.

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
		- For MCZ non-recovered and non-Z-stack data products, if IDPH is present, value comes from IDPH DPO per case "3a". - For MCZ non-recovered and non-Z-stack data products, if IDPH is not present, value comes from Ancillary DPO per case "3b". Type U32
Ops Keyword SPACECRAFT_CLOCK_STOP_COUNT Archive Attribute TBD Definition Specifies the value of the spacecraft clock at the end of a time period of interest. Format is "ssssssss.MCZ", stored as a floating point number where, "ssssssss" = seconds converted from clock's coarse counter "MCZ" = milliseconds converted from clock's fine counter. For M2020, the time period of interest is the end of data acquisition. The fractional component "MCZ" is computed as follows: $[(\text{shift right 12 bits}) / 2^{**20}] * 1000$	Valid Values ssssssss.MCZ Type string(30) Units n/a Location IDENTIFICATION (Class) X-path TBD Reference TBD	Mode - DPO in XML format (referenced to APID Name in Appendix A), Calculation: - Stop SCLK Count = Start SCLK Count + (exp_time * 5.12) - DPO in XML format (referenced to APID Name in Appendix A) or Image DPO mini-header, Calculation: - Stop SCLK Count = Start SCLK Count + (exp_time / 1000) - DPO in XML format (referenced to APID Name in Appendix A), Calculation: - Stop SCLK Count = Start SCLK Count + (exp_time * 5.333333) Field as "<xml name>:[<element>]:[<element>]:<field>" Eng. Cameras & SuperCam "<IDPH DPO>:idph:sclk_seconds", "<IDPH DPO>:idph:sclk_subseconds", "<IDPH DPO>:idph:exp_time" MCZ Cameras "<IDPH DPO>:cidph:sclk:seconds" "<Ancillary DPO>:sclk:seconds" Skycam "<MEDA Image Header>:time_tag" "<MEDA Image Header>:skycam_exp_time_lowbyte_data" "<MEDA Image Header>:skycam_exp_time_highbyte_data" NOTES: - For Eng. Cameras and SuperCam, "exp_time" is in raw counts, with each count translating to 5.12 ms. - For MCZ non-recovered and non-Z-stack data products, if IDPH is present, value comes from IDPH DPO per case "2b". - For MCZ non-recovered and non-Z-stack data products, if IDPH is not present, value comes from Ancillary DPO per case "2c". - For Skycam, "exp_time" is in raw counts, with each count translating to 5.333333 ms. Type U32
Ops Keyword	Valid Values	Mode

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
SPECIAL_LINE <u>Archive Attribute</u> none <u>Definition</u> For SuperCam on M2020, specifies the line number in a particular RMI image that has a special meaning. The context of the meaning is defined by SPECIAL_NAME.	"497" <u>Type</u> integer <u>Units</u> n/a <u>Location</u> INSTRUMENT_STATE_PARMS (Group) <u>X-path</u> TBD <u>Reference</u> TBD	Static value <u>Field as "<xml name>:[<element>]:[<element>]:<field>"</u> n/a <u>Type</u> U8
<u>Ops Keyword</u> SPECIAL_NAME <u>Archive Attribute</u> none <u>Definition</u> Specifies how the coordinate defined as "(SPECIAL_LINE, SPECIAL_SAMPLE)" is to be interpreted; i.e. why that coordinate is special. For SuperCam on M2020, identifies the LIBS laser "hot spot" in a particular RMI image. The text string used is "Location of the LIBS laser spot in the RMI image described by the camera model".	<u>Valid Values</u> "Location of the LIBS laser spot in the RMI image described by the camera model" <u>Type</u> string <u>Units</u> n/a <u>Location</u> INSTRUMENT_STATE_PARMS (Group) <u>X-path</u> TBD <u>Reference</u> TBD	<u>Mode</u> Static value <u>Field as "<xml name>:[<element>]:[<element>]:<field>"</u> n/a <u>Type</u> U8
<u>Ops Keyword</u> SPECIAL_SAMPLE <u>Archive Attribute</u> none <u>Definition</u> For SuperCam on M2020, specifies the sample number in a particular RMI image that has a special meaning. The context of the meaning is defined by SPECIAL_NAME.	<u>Valid Values</u> "532" <u>Type</u> integer <u>Units</u> n/a <u>Location</u> INSTRUMENT_STATE_PARMS (Group) <u>X-path</u> TBD <u>Reference</u> TBD	<u>Mode</u> Static value <u>Field as "<xml name>:[<element>]:[<element>]:<field>"</u> n/a <u>Type</u> U8
<u>Ops Keyword</u> SPICE_FILE_ID <u>Archive Attribute</u> TBD <u>Definition</u>	<u>Valid Values</u> n/a <u>Type</u> string <u>Units</u> n/a	<u>Mode</u> User parameter input <u>Field as "<xml name>:[<element>]:[<element>]:<field>"</u> n/a <u>Type</u>

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
Specifies an abbreviated name or acronym which identifies a particular SPICE file.	<u>Location</u> TELEMETRY (Class) <u>X-path</u> TBD <u>Reference</u> TBD	n/a
<u>Ops Keyword</u> SPICE_FILE_NAME <u>Archive Attribute</u> TBD <u>Definition</u> Specifies the names of the SPICE files used in processing the data.	<u>Valid Values</u> n/a <u>Type</u> string(180) <u>Units</u> n/a <u>Location</u> TELEMETRY (Class) <u>X-path</u> TBD <u>Reference</u> TBD	<u>Mode</u> User parameter input <u>Field as "<xml name>:[<element>]:[<element>]:<field>"</u> n/a <u>Type</u> n/a
<u>Ops Keyword</u> STACK_1_LEVEL <u>Archive Attribute</u> none <u>Definition</u> For SuperCam on M2020, specifies the maximum current stack level for Stack 1 for the LIBS instrument.	<u>Valid Values</u> n/a <u>Type</u> integer <u>Units</u> n/a <u>Location</u> OBSERVATION_REQUEST_PARMS (Group) <u>X-path</u> TBD <u>Reference</u> TBD	<u>Mode</u> DPO in XML format (referenced to APID Name in Appendix A) <u>Field as "<xml name>:[<element>]:[<element>]:<field>"</u> "<Ancillary DPO>:cmd_parameters:stack1Level" <u>Type</u> F32
<u>Ops Keyword</u> STACK_2_LEVEL <u>Archive Attribute</u> none <u>Definition</u> For SuperCam on M2020, specifies the maximum current stack level for Stack 2 for the LIBS instrument.	<u>Valid Values</u> n/a <u>Type</u> integer <u>Units</u> n/a <u>Location</u> OBSERVATION_REQUEST_PARMS (Group) <u>X-path</u> TBD <u>Reference</u>	<u>Mode</u> DPO in XML format (referenced to APID Name in Appendix A) <u>Field as "<xml name>:[<element>]:[<element>]:<field>"</u> "<Ancillary DPO>:cmd_parameters:stack2Level" <u>Type</u> F32

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
	TBD	
Ops Keyword STACK_3_LEVEL Archive Attribute none Definition For SuperCam on M2020, specifies the maximum current stack level for Stack 3 for the LIBS instrument.	Valid Values n/a Type integer Units n/a Location OBSERVATION_REQUEST_PARMS (Group) X-path TBD Reference TBD	Mode DPO in XML format (referenced to APID Name in Appendix A) Field as “<xml name>:[<element>]:[<element>]:<field>” “<Ancillary DPO>:cmd_parameters:stack3Level” Type F32
Ops Keyword STACK_DURATION Archive Attribute none Definition For SuperCam on M2020, specifies the stack duration (in µsec) for the LIBS instrument.	Valid Values n/a Type integer Units n/a Location OBSERVATION_REQUEST_PARMS (Group) X-path TBD Reference TBD	Mode DPO in XML format (referenced to APID Name in Appendix A) Field as “<xml name>:[<element>]:[<element>]:<field>” “<Ancillary DPO>:cmd_parameters:stackDuration” Type U8
Ops Keyword STANDARD_DEVIATION Archive Attribute TBD Definition Specifies the standard deviation of the DN values in the image array.	Valid Values n/a Type float Units n/a Location IMAGE (Object) X-path TBD Reference TBD	Mode Calculation Field as “<xml name>:[<element>]:[<element>]:<field>” n/a Type n/a
Ops Keyword START_AZIMUTH Archive Attribute TBD	Valid Values “0” to “360” Type float	Mode RDR-generating software Field as “<xml name>:[<element>]:[<element>]:<field>” n/a

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
Definition Specifies the angular distance from a fixed reference position at which an image or observation starts. Azimuth is measured in a spherical coordinate system, in a plane normal to the principal axis. Azimuth values increase according to the right hand rule relative to the positive direction of the principal axis of the spherical coordinate system. When in a SURFACE_PROJECTION or SITE_DERIVED_GEOMETRY group, specifies the azimuth of the left edge of the output map. Applies to Cylindrical and Cylindrical-Perspective projections only.	Units deg (<deg> unit tag required) Location - SURFACE_PROJECTION_PARMs (Group) - SITE_DERIVED_GEOMETRY_PARMs (Group) X-path TBD Reference TBD	Type n/a
Ops Keyword START_IMAGE_ID Archive Attribute TBD Definition For SuperCam on M2020, specifies the commanded image start in case the SRAM on the SuperCam Mast Unit (SCMU) goes bad.	Valid Values n/a Type integer Units n/a Location OBSERVATION_REQUEST_PARMs (Group) X-path TBD Reference TBD	Mode DPO in XML format (referenced to APID Name in Appendix A) Field as “<xml name>:[<element>]:[<field>” “<IDPH DPO>:cmd_parameters:startImageID” TBD Type U8
Ops Keyword START_ROW_UV Archive Attribute none Definition For SuperCam on M2020, specifies the beginning row of the SuperCam CCD over which UV integration occurs.	Valid Values n/a Type integer Units n/a Location OBSERVATION_REQUEST_PARMs (Group) X-path TBD Reference TBD	Mode DPO in XML format (referenced to APID Name in Appendix A) Field as “<xml name>:[<element>]:[<field>” “<Ancillary DPO>:cmd_parameters:startRowUV” Type U16
Ops Keyword START_ROW_VIS Archive Attribute none Definition For SuperCam on M2020, specifies the beginning row of the SuperCam CCD over which VIS integration occurs.	Valid Values n/a Type integer Units n/a Location OBSERVATION_REQUEST_PARMs (Group)	Mode DPO in XML format (referenced to APID Name in Appendix A) Field as “<xml name>:[<element>]:[<field>” “<Ancillary DPO>:cmd_parameters:startRowVIS” Type U16

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
	<u>X-path</u> TBD <u>Reference</u> TBD	
<u>Ops Keyword</u> START_ROW_VNIR <u>Archive Attribute</u> TBD <u>Definition</u> For SuperCam on M2020, specifies the beginning row of the SuperCam CCD over which VNIR integration occurs.	<u>Valid Values</u> n/a <u>Type</u> integer <u>Units</u> n/a <u>Location</u> OBSERVATION_REQUEST_PARMS (Group) <u>X-path</u> TBD <u>Reference</u> TBD	<u>Mode</u> DPO in XML format (referenced to APID Name in Appendix A) <u>Field as “<xml name>:[<element>]:[<element>]:<field>”</u> “<Ancillary DPO>:cmd_parameters:startRowVNIR” <u>Type</u> U16
<u>Ops Keyword</u> START_TIME <u>Archive Attribute</u> TBD <u>Definition</u> Specifies the date and time of the beginning of an event or observation (whether it be a spacecraft, ground-based, or system event) in UTC system format. For M2020, the time period of interest is returned from SPICE subroutines and based on the beginning of data acquisition.	<u>Valid Values</u> <YYYY>-<MM>-<DD>T<hh>:<mm>:<ss>[.<fff>] NOTE: Value will be uncalibrated if SPICE kernels unavailable. <u>Type</u> time <u>Units</u> n/a <u>Location</u> IDENTIFICATION (Class) <u>X-path</u> TBD <u>Reference</u> TBD	<u>Mode</u> 1) a. DPO in XML format (referenced to APID Name in Appendix E), Calculation using SPICE Kernels: - SCLK b. EMD in XML format, Calculation using SPICE Kernels: - SCLK 2) DPO in XML format (referenced to APID Name in Appendix A), Calculation using SPICE Kernels: - SCLK 3) DPO in XML format (referenced to APID Name in Appendix A) or Image DPO mini-header, Calculation using SPICE Kernels: - SCLK 4) DPO in XML format (referenced to APID Name in Appendix A), Calculation using SPICE Kernels: - SCLK <u>Field as “<xml name>:[<element>]:[<element>]:<field>”</u> <u>Eng. Cameras</u> a. “<IDPH DPO>:idph:sclk_seconds”, “<IDPH DPO>:idph:sclk_subseconds” b. “M20EarthProductMetadata:M20ProductMetadata:OnboardCreationTime” <u>SuperCam</u> 2) “<Ancillary DPO>:image_sclk”, “<Ancillary DPO>:soh_sclk” <u>MCZ Cameras</u> 3) a. “MCZ_Image_Mini_Header[8]”, “MCZ_Image_Mini_Header[9]”, “MCZ_Image_Mini_Header[10]”,

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
		"MCZ_Image_Mini_Header[11]" b. "<IDPH DPO>:cidph:scik:seconds" c. "<Ancillary DPO>:scik:seconds" - Skycam 4) "<MEDA Image Header>:time_tag" NOTES: - For SuperCam, if APID Name contains string "Soh" in it, use field "soh_scik". Otherwise, use field "image_scik". - For MCZ recovered or Z-stack data products, value is comprised of four bytes coming from Image DPO mini-header at byte offsets 8, 9, 10 and 11 per case "3a". - For MCZ non-recovered and non-Z-stack data products, if IDPH is present, value comes from IDPH DPO per case "3b". - For MCZ non-recovered and non-Z-stack data products, if IDPH is not present, value comes from Ancillary DPO per case "3c". Type 1) a. U32 b. n/a 2) U32 3) U32
Ops Keyword STEREO_BASELINE Archive Attribute TBD Definition Specifies the separation, in meters, between the two cameras used for processing of the stereo image.	Valid Values n/a Type float Units n/a Location DERIVED_IMAGE_PARMS (Group) X-path TBD Reference TBD	Mode RDR-generating software Field as "<xml name>:[<element>]:[<element>]:<field>" n/a Type n/a
Ops Keyword STEREO_PRODUCT_ID Archive Attribute TBD Definition Specifies the PRODUCT_ID for the stereo partner used to linearize the image when LINEARIZATION_MODE is "ACTUAL". This keyword may be absent if the mode is not "ACTUAL".	Valid Values n/a Type string Units n/a Location DERIVED_IMAGE_PARMS (Group) X-path TBD	Mode RDR-generating software Field as "<xml name>:[<element>]:[<element>]:<field>" n/a Type n/a

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
	<u>Reference</u> TBD	
<u>Ops Keyword</u> STOP_AZIMUTH <u>Archive Attribute</u> TBD <u>Definition</u> Specifies the angular distance from a fixed reference position at which an image or observation stops. Azimuth is measured in a spherical coordinate system, in a plane normal to the principal axis. Azimuth values increase according to the right hand rule relative to the positive direction of the principal axis of the spherical coordinate system. When in a SURFACE_PROJECTION or SITE_DERIVED_GEOMETRY group, specifies the azimuth of the right edge of the output map. Applies to Cylindrical and Cylindrical-Perspective projections only.	<u>Valid Values</u> "0" to "360" <u>Type</u> float <u>Units</u> deg (<deg> unit tag required) <u>Location</u> - SURFACE_PROJECTION_PARMs (Group) - SITE_DERIVED_GEOMETRY_PARMs (Group) <u>X-path</u> TBD <u>Reference</u> TBD	<u>Mode</u> RDR-generating software <u>Field as "<xml name>:[<element>]:[<element>]:<field>"</u> n/a <u>Type</u> n/a
<u>Ops Keyword</u> STOP_ROW_UV <u>Archive Attribute</u> none <u>Definition</u> For SuperCam on M2020, specifies the end row of the SuperCam CCD over which UV integration occurs.	<u>Valid Values</u> n/a <u>Type</u> integer <u>Units</u> n/a <u>Location</u> OBSERVATION_REQUEST_PARMs (Group) <u>X-path</u> TBD <u>Reference</u> TBD	<u>Mode</u> DPO in XML format (referenced to APID Name in Appendix A) <u>Field as "<xml name>:[<element>]:[<element>]:<field>"</u> "<Ancillary DPO>:cmd_parameters:stopRowUV" <u>Type</u> U16
<u>Ops Keyword</u> STOP_ROW_VIS <u>Archive Attribute</u> none <u>Definition</u> For SuperCam on M2020, specifies the end row of the SuperCam CCD over which VIS integration occurs.	<u>Valid Values</u> n/a <u>Type</u> integer <u>Units</u> n/a <u>Location</u> OBSERVATION_REQUEST_PARMs (Group) <u>X-path</u> TBD <u>Reference</u> TBD	<u>Mode</u> DPO in XML format (referenced to APID Name in Appendix A) <u>Field as "<xml name>:[<element>]:[<element>]:<field>"</u> "<Ancillary DPO>:cmd_parameters:stopRowVIS" <u>Type</u> U16
<u>Ops Keyword</u> STOP_ROW_VNIR	<u>Valid Values</u> n/a	<u>Mode</u>

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
Archive Attribute none Definition For SuperCam on M2020, specifies the end row of the SuperCam CCD over which VNIR integration occurs.	Type integer Units n/a Location OBSERVATION_REQUEST_PARMS (Group) X-path TBD Reference TBD	DPO in XML format (referenced to APID Name in Appendix A) Field as “<xml name>:[<element>]:<field>” “<Ancillary DPO>:cmd_parameters:stopRowVNIR” Type U16
Ops Keyword STOP_TIME Archive Attribute TBD Definition Specifies the date and time of the end of an event or observation (whether it be a spacecraft, ground-based, or system event) in UTC system format.	Valid Values <YYYY>-<MM>-<DD>T<hh>:<mm>:<ss>[.<fff>] NOTE: Value will be uncalibrated if SPICE kernels unavailable. Type time Units n/a Location IDENTIFICATION (Class) X-path TBD Reference TBD	Mode 1) DPO in XML format (referenced to APID Name in Appendix A), Calculation using SPICE Kernels: - SCLK 2) DPO in XML format (referenced to APID Name in Appendix A) or Image DPO mini-header, Calculation using SPICE Kernels: - SCLK Field as “<xml name>:[<element>]:<field>” Eng. Cameras & SuperCam “<IDPH DPO>:idph:sclk_seconds” “<IDPH DPO>:idph:sclk_subseconds” “<IDPH DPO>:idph:exp_time” MCZ Cameras “MCZ_Image_Mini_Header[8]”, “MCZ_Image_Mini_Header[9]”, “MCZ_Image_Mini_Header[10]”, “MCZ_Image_Mini_Header[11]” “<IDPH DPO>:cidph:sclk:seconds” “<Ancillary DPO>:sclk:seconds” NOTES: - For Eng. Cameras and SuperCam, “exp_time” is in raw counts, with each count translating to 5.12 ms. - For MCZ recovered or Z-stack data products, value is comprised of four bytes coming from Image DPO mini-header at byte offsets 8, 9, 10 and 11 per case “2a”. - For MCZ non-recovered and non-Z-stack data products, if IDPH is present, value comes from IDPH DPO per case “2b”. - For MCZ non-recovered and non-Z-stack data products, if IDPH is not present, value comes from Ancillary DPO per case “2c”. Type 1) a. U32

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
		b. U32 c. U16 2) U32
Ops Keyword STRIPING_COUNT Archive Attribute TBD Definition Specifies the number of stripes (N) used during dark current mitigation within image acquisition. Image "striping" is comprised of reading out the image in N different parts ("stripes") using hardware windowing mode, (where N is a commanded number from 0 to 1024), using N separate exposures (with identical exposure times). These successive stripes correspond to physically different locations on the detector. As part of the striping exposure technique, also include a commandable overlap (M rows) to allow each successive row to "cover" the image pixels towards the readout region.	Valid Values Eng. Cameras (C) "0" Type integer Units n/a Location TELEMETRY (Class) X-path TBD Reference TBD	Mode DPO in XML format (referenced to APID Name in Appendix A) Field as "<xml name>:[<element>]:<field>" Eng. Cameras "<IDPH DPO>:idph:stripes" Type U16
Ops Keyword STRIPING_OVERLAP_ROWS Archive Attribute TBD Definition Specifies the number of rows (M) of striping overlap used during dark current mitigation within image acquisition. Image "striping" is comprised of reading out the image in N different parts ("stripes") using hardware windowing mode, (where N is a commanded number from 0 to 1024), using N separate exposures (with identical exposure times). These successive stripes correspond to physically different locations on the detector. As part of the striping exposure technique, also include a commandable overlap (M rows) to allow each successive row to "cover" the image pixels towards the readout region.	Valid Values Eng. Cameras (C) "0" Type integer Units n/a Location TELEMETRY (Class) X-path TBD Reference TBD	Mode DPO in XML format (referenced to APID Name in Appendix A) Field as "<xml name>:[<element>]:<field>" Eng. Cameras "<IDPH DPO>:idph:overlap" Type U16
Ops Keyword SUBFRAME_TYPE Archive Attribute TBD Definition Specifies the method of subframing performed on the image. Descriptions of the valid values follow: "NONE" - No subframing requested. "SW_ONLY" - Software processing only. "HW_COND" - Use hardware only if compatible. "HW_SW" - Use hardware then software. "SUN_NO_IMG" - If the sun is found, send a subframed	Valid Values - Eng. Cameras (B) 0 = "NONE" 1 = "SW_ONLY" 2 = "HW_COND" 3 = "HW_SW" 4 = "SUN_NO_IMG" 5 = "SUN_FULL" - MCZ Cameras (B) - If width is 1648 and height is 1200, then "NONE" - Otherwise, "HW_ONLY" Type string Units	Mode - DPO in XML format (referenced to APID Name in Appendix A) - DPO in XML format (referenced to APID Name in Appendix A) or Image DPO mini-header Field as "<xml name>:[<element>]:<field>" - Eng. Cameras "<IDPH DPO>:idph:params:subframe" - MCZ Cameras a. "<Ancillary DPO>:cmd_arguments_image>window_width" b. "<Ancillary DPO>:cmd_arguments_image>window_height"

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
image of the sun. If sun is not found, send back no image. f) "SUN_FULL" - If the sun is found, send a subframed image of the sun. If the sun is not found, send back the entire image.	n/a Location SUBFRAME_REQUEST_PARMS (Group) X-path TBD Reference TBD	c. "MCZ_Image_Mini_Header[22]" d. "MCZ_Image_Mini_Header[23]" NOTES: - For MCZ non-recovered data products (cases "a" and "b"), float values come from the Ancillary DPO. - For MCZ recovered data products (cases "c" and "d"), values analogous to cases "a" (window_width) and "b" (window_height) are each comprised of one byte coming from Image DPO mini-header at byte offsets 22 and 23, respectively. - For MCZ, parm "window_width" is number of image samples. - For MCZ, parm "window_height" is number of image lines. - For MCZ non-Thumbnail data products, multiply by factor of 8 to convert to correct line/sample value. Type 1) enum 2) U8
Ops Keyword SUN_VIEW_DIRECTION Archive Attribute TBD Definition Specifies a unit vector identifying the sun viewing direction.	Valid Values n/a Type float array[3] Units n/a Location ROVER_DERIVED_GEOMETRY_PARMS (Group) X-path TBD Reference TBD	Mode DPO in XML format (referenced to APID Name in Appendix A) Field as "<xml name>:[<element>]:[<element>]:<field>" Eng. Cameras & SuperCam "<IDPH DPO>:idph:sun_dir[3]" Type F32[3]
Ops Keyword SURFACE_GROUND_LOCATION Archive Attribute TBD Definition Specifies any point on the surface (for SURFACE_MODEL_TYPE of "PLANE"), or the center of the sphere (for the three "SPHERE" types). This point is measured in the coordinates specified by the REFERENCE_COORD_SYSTEM_* keywords in the same group.	Valid Values n/a Type float array[3] Units meters Location SURFACE_MODEL_PARMS (Group) X-path TBD Reference TBD	Mode RDR-generating software Field as "<xml name>:[<element>]:[<element>]:<field>" n/a Type n/a
Ops Keyword	Valid Values	Mode

Output Metadata (Product Label)		Input Metadata (Source)
• Ops Keyword • Archive Label Attribute • Definition	• Mission Phase C=Cruise, S=Surface, B=Both • Valid Values (quoted) • Type • Units • Keyword Location in Ops Label • Attribute X-path in Archive label • Attribute Reference in Archive Label (bracketed)	• Mission Phase C=Cruise, S=Surface, B=Both • Mode • Metadata Field • Type
SURFACE_MODEL_FILE_NAME Archive Attribute TBD Definition Specifies the name of an XYZ, Z-component or other RDR used as a digital elevation model onto which the data were projected.	n/a Type string Units n/a Location SURFACE_MODEL_PARMS (Group) X-path TBD Reference TBD	RDR-generating software Field as “<xml name>:[<element>]:[<element>]:<field>” n/a Type n/a
Ops Keyword SURFACE_MODEL_TYPE Archive Attribute TBD Definition Specifies the type of surface used for the reprojection performed during the mosaicking process. The valid values are defined as: a) "INFINITY" - refers to an infinitely distant "surface" in all directions and has no parameters. b) "PLANE" - refers to a flat plane and require the SURFACE_NORMAL_VECTOR and SURFACE_GROUND_LOCATION keywords as parameters. c) "SPHERE" – refers to a spherical model where the camera is at the center of the sphere. The origin is specified by SURFACE_GROUND_LOCATION, and the radius by the first element of SURFACE_NORMAL_VECTOR. d) "SPHERE1" – refers to a general sphere model, whose center is defined by SURFACE_GROUND_LOCATION, and radius by the first element of SURFACE_NORMAL_VECTOR. If the camera is outside the sphere, the first intersection with the sphere is used; this makes it useful for modeling hills or rocks. e) "SPHERE2" – just like SPHERE1, except the second intersection with the sphere is used; this makes it useful for modeling craters.	Valid Values "INFINITY", "PLANE", "SPHERE", "SPHERE1", "SPHERE2" Type string Units n/a Location SURFACE_MODEL_PARMS (Group) X-path TBD Reference TBD	Mode RDR-generating software Field as “<xml name>:[<element>]:[<element>]:<field>” n/a Type n/a
Ops Keyword SURFACE_NORMAL_VECTOR Archive Attribute	Valid Values n/a Type	Mode RDR-generating software

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
TBD Definition Specifies a vector normal to the surface (for SURFACE_MODEL_TYPE of "PLANE"). This vector is measured in the coordinates specified by the REFERENCE_COORD_SYSTEM_* keywords in the same group. For the "SPHERE" surface model types, the first element is used to specify the radius; the other two elements are unused. This is a misuse of this keyword's definition, which is retained for historical reasons.	float array[3] <u>Units</u> n/a <u>Location</u> SURFACE_MODEL_PARMS (Group) <u>X-path</u> TBD <u>Reference</u> TBD	Field as "<xml name>:[<element>]:<field>" n/a <u>Type</u> n/a
Ops Keyword TARGET_NAME Archive Attribute TBD Definition Specifies a target. The target may be a planet, satellite, ring, region, feature, asteroid or comet. See TARGET_TYPE.	Valid Values For EDRs ... <u>Value</u> "MARS", "CALIBRATION" For RDRs ... <u>Value</u> any <u>Type</u> string(30) <u>Units</u> n/a <u>Location</u> IDENTIFICATION (Class) <u>X-path</u> TBD <u>Reference</u> TBD	Mode Calculation: - By algorithm to determine if looking at calibration target ... if not, then value is "MARS" Field as "<xml name>:[<element>]:<field>" n/a <u>Type</u> n/a
Ops Keyword TARGET_TYPE Archive Attribute TBD Definition Specifies the type of a named target.	Valid Values "CALIBRATION", "DUST", "SUN", "PLANET", "SATELLITE", "N/A" <u>Type</u> string <u>Units</u> n/a <u>Location</u> IDENTIFICATION (Class) <u>X-path</u> TBD <u>Reference</u> TBD	Mode Static Value Field as "<xml name>:[<element>]:<field>" n/a <u>Type</u> n/a
Ops Keyword TELEMETRY_PROVIDER_ID Archive Attribute TBD	Valid Values "MPCS_M2020_DP" <u>Type</u> string	Mode User parameter input Field as "<xml name>:[<element>]:<field>"

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
Definition Specifies the provider and version of the telemetry data used in the generation of this data.	Units n/a Location TELEMETRY (Class) X-path TBD Reference TBD	n/a Type n/a
Ops Keyword TELEMETRY_SOURCE_CHECKSUM Archive Attribute TBD Definition Checksum for the source product from which this product was derived. For M2020, it is the sum of each (unsigned) byte in the data areas of all DPOs. It does not include the DPO headers.	Valid Values n/a Type integer Units n/a Location TELEMETRY (Class) X-path TBD Reference TBD	Mode EMD in XML format Field as “<xml name>:[<element>]:<field>” “M20EarthProductMetadata:M20ProductMetadata:ProductChecksum” Type n/a
Ops Keyword TELEMETRY_SOURCE_HOST_NAME Archive Attribute TBD Definition Specifies the name of the host venue that provides the telemetry source used in creation of this data set. For M2020, example is “M2020mstbgds1”. See also TELEMETRY_SOURCE_NAME.	Valid Values n/a Type string Units n/a Location TELEMETRY (Class) X-path TBD Reference TBD	Mode EMD in XML format Field as “<xml name>:[<element>]:<field>” “M20EarthProductMetadata:SessionInformation:Venue:Host” Type n/a
Ops Keyword TELEMETRY_SOURCE_NAME Archive Attribute TBD Definition Specifies the name of the telemetry source used in creation of this data set. For M2020, example is “MczRScidata_0123456789-00000-1.dat”.	Valid Values n/a Type string Units n/a Location TELEMETRY (Class) X-path	Mode EMD in XML format Field as “<xml name>:[<element>]:<field>” “M20EarthProductMetadata:M20ProductMetadata:DataFileName” Type n/a

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
	TBD <u>Reference</u> TBD	
Ops Keyword TELEMETRY_SOURCE_SCLK_START Archive Attribute TBD Definition Specifies the value of the spacecraft clock (in seconds) at the creation time of the source product from which this product was derived. This differs from SPACECRAFT_CLOCK_START_COUNT, which is the time the instrument acquired the data. For M2020, it refers to the creation time (DVT) of the onboard DPO and comes from the secondary packet header. Note that this is the SCLK used by Data Management operationally to identify data products.	Valid Values ssssssss.MCZ Type string(30) Units n/a Location TELEMETRY (Class) X-path TBD Reference TBD	Mode EMD in XML format Field as “<xml name>:[<element>]:[<element>]:<field>” “M20EarthProductMetadata:M20ProductMetadata:DvtCoarse”, “M20EarthProductMetadata:M20ProductMetadata:DvtFine” Type n/a
Ops Keyword TELEMETRY_SOURCE_SIZE Archive Attribute TBD Definition Specifies the length in bytes of the source product from which this product was derived. For M2020, it is the length of the user portion of the Data Product Object (DPO).	Valid Values n/a Type integer Units n/a Location TELEMETRY (Class) X-path TBD Reference TBD	Mode EMD in XML format Field as “<xml name>:[<element>]:[<element>]:<field>” “M20EarthProductMetadata:M20ProductMetadata:ProductFileSize” Type n/a
Ops Keyword TELEMETRY_SOURCE_START_TIME Archive Attribute TBD Definition Specifies the creation time of the source product from which this product was derived. It is the same as TELEMETRY_SOURCE_SCLK_START converted to Spacecraft Event Time (SCET).	Valid Values <YYYY>-<DDD>T<hh>:<mm>:<ss>[.<fff>] Type time Units n/a Location IDENTIFICATION (Class) X-path TBD Reference TBD	Mode EMD in XML format Field as “<xml name>:[<element>]:[<element>]:<field>” “M20EarthProductMetadata:M20ProductMetadata:PartIdentifier:Part:Scet” Type n/a
Ops Keyword TELEMETRY_SOURCE_TYPE	Valid Values “DATA PRODUCT”	Mode User parameter input

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
Archive Attribute TBD Definition Specifies the classification of the source of the telemetry used in creating this data set.	Type string(12) Units n/a Location TELEMETRY (Class) X-path TBD Reference TBD	Field as “<xml name>:[<element>]:[<element>]:<field>” n/a Type n/a
Ops Keyword TRANSMISSION_PATH Archive Attribute TBD Definition Routing status at time of MPDU (Metadata PDU) generation. Indicates the actual transmission paths (routes) of the Data Product.	Valid Values n/a Type string Units n/a Location TELEMETRY (Class) X-path TBD Reference TBD	Mode EMD in XML format Field as “<xml name>:[<element>]:[<element>]:<field>” “M20EarthProductMetadata:M20ProductMetadata:TransmissionStatus” Type n/a
Ops Keyword VALID_MAXIMUM_PIXEL Archive Attribute TBD Definition Specifies the maximum pixel DN value that is considered valid for some use. For M2020 SuperCam, specifies the threshold for pixels used by the autoexposure algorithm.	Valid Values “0” to “1023” Type integer Units n/a Location OBSERVATION_REQUEST_PARMS (Group) X-path TBD Reference TBD	Mode DPO in XML format (referenced to APID Name in Appendix A) Field as “<xml name>:[<element>]:[<element>]:<field>” SuperCam “<Ancillary DPO>:cmd_parameters:upperThreshold” Type U16
Ops Keyword VALID_MINIMUM_PIXEL Archive Attribute TBD Definition Specifies the minimum pixel DN value that is considered valid for some use.	Valid Values “0” to “1023” Type integer Units n/a Location	Mode DPO in XML format (referenced to APID Name in Appendix A) Field as “<xml name>:[<element>]:[<element>]:<field>” SuperCam “<Ancillary DPO>:cmd_parameters:lowerThreshold” Type

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
For M2020 SuperCam, specifies the threshold for pixels used by the autoexposure algorithm.	OBSERVATION_REQUEST_PARMS (Group) <u>X-path</u> TBD <u>Reference</u> TBD	U16
<u>Ops Keyword</u> VIRTUAL_CHANNEL_ID <u>Archive Attribute</u> TBD <u>Definition</u> The Virtual Channel Identifier is used by M2020 to identify the RCE string generating the Transfer Frame, and to indicate the type of data flowing in the telemetry virtual channel. RCE String A is indicated by all Virtual Channel Identifier values having a '0' as the high bit (e.g., virtual channels 0 to 31); RCE String B is indicated by all Virtual Channel Identifier values having a '1' for the high bit (e.g., virtual channels 32 to 63).	<u>Valid Values</u> "0" to "63" <u>Type</u> string <u>Units</u> n/a <u>Location</u> TELEMETRY (Class) <u>X-path</u> TBD <u>Reference</u> TBD	<u>Mode</u> EMD in XML format <u>Field as "<xml name>:[<element>]:<field>"</u> "M20EarthProductMetadata:M20ProductMetadata:Vcid" <u>Type</u> n/a
<u>Ops Keyword</u> X_AXIS_MAXIMUM <u>Archive Attribute</u> TBD <u>Definition</u> Specifies the value of the X coordinate (measured in the projection frame) of a Vertical, Orthographic or Orthorectified projection at the top of the image. Note that +X is at the top of the image and +Y is at the right, so +X corresponds to North in the Vertical projection.	<u>Valid Values</u> n/a <u>Type</u> float <u>Units</u> meters (<m> unit tag required) <u>Location</u> SURFACE_PROJECTION_PARMS (Group) <u>X-path</u> TBD <u>Reference</u> TBD	<u>Mode</u> RDR-generating software <u>Field as "<xml name>:[<element>]:<field>"</u> n/a <u>Type</u> n/a
<u>Ops Keyword</u> X_AXIS_MINIMUM <u>Archive Attribute</u> TBD <u>Definition</u> Specifies the value of the X coordinate (measured in the projection frame) of a Vertical, Orthographic or Orthorectified projection at the bottom of the image.	<u>Valid Values</u> n/a <u>Type</u> float <u>Units</u> meters (<m> unit tag required) <u>Location</u> SURFACE_PROJECTION_PARMS (Group) <u>X-path</u> TBD <u>Reference</u> TBD	<u>Mode</u> RDR-generating software <u>Field as "<xml name>:[<element>]:<field>"</u> n/a <u>Type</u> n/a

Output Metadata (Product Label)		Input Metadata (Source)
- Ops Keyword - Archive Label Attribute - Definition	- Mission Phase C=Cruise, S=Surface, B=Both - Valid Values (quoted) - Type - Units - Keyword Location in Ops Label - Attribute X-path in Archive label - Attribute Reference in Archive Label (bracketed)	- Mission Phase C=Cruise, S=Surface, B=Both - Mode - Metadata Field - Type
Ops Keyword Y_AXIS_MAXIMUM Archive Attribute TBD Definition Specifies the value of the Y coordinate (measured in the projection frame) of a Vertical, Orthographic or Orthorectified projection at the right edge of the image.	Valid Values n/a Type float Units meters (<m> unit tag required) Location SURFACE_PROJECTION_PARMS (Group) X-path TBD Reference TBD	Mode RDR-generating software Field as “<xml!name>:[<element>]:[<element>]:<field>” n/a Type n/a
Ops Keyword Y_AXIS_MINIMUM Archive Attribute TBD Definition Specifies the value of the Y coordinate (measured in the projection frame) of a Vertical, Orthographic or Orthorectified projection at the left edge of the image.	Valid Values n/a Type float Units meters (<m> unit tag required) Location SURFACE_PROJECTION_PARMS (Group) X-path TBD Reference TBD	Mode RDR-generating software Field as “<xml!name>:[<element>]:[<element>]:<field>” n/a Type n/a
Ops Keyword ZERO_ELEVATION_LINE Archive Attribute TBD Definition Specifies the image line representing 0.0 degree elevation. Applies to Cylindrical projections.	Valid Values n/a Type float Units pixel (<pixel> unit tag required) Location SURFACE_PROJECTION_PARMS (Group) X-path TBD Reference TBD	Mode RDR-generating software Field as “<xml!name>:[<element>]:[<element>]:<field>” n/a Type n/a
Ops Keyword ZERO_EXPOSURE_IMAGE_ID Archive Attribute TBD Definition	Valid Values n/a Type string Units n/a	Mode RDR-generating software Field as “<xml!name>:[<element>]:[<element>]:<field>” n/a Type

Output Metadata (Product Label)		Input Metadata (Source)
• Ops Keyword • Archive Label Attribute • Definition	• Mission Phase C=Cruise, S=Surface, B=Both • Valid Values (quoted) • Type • Units • Keyword Location in Ops Label • Attribute X-path in Archive label • Attribute Reference in Archive Label (bracketed)	• Mission Phase C=Cruise, S=Surface, B=Both • Mode • Metadata Field • Type
Specifies the value of PRODUCT_ID for the zero-exposure EDR that is subtracted during RDR generation to account for shutter smear and masked-region dark current. NOTE: This keyword is only placed in the PDS label if a zero-exposure image EDR product was used during on-ground calibration. It is not set when on-board shutter-subtraction is done.	<u>Location</u> DERIVED_IMAGE_PARMs (Group) <u>X-path</u> TBD <u>Reference</u> TBD	n/a

